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Good morning, World Tours.

My name is Jamie.

How can I help you?

Good morning.

I want some information on self-drive tours in the USA.

Could you send me a brochure?

Of course.

Could I have your name, please?

Andrea Brown.

Thank you.

And your address?

24 Arley Road.

Can you spell that?

A R D L E I G H Road.

Post code BH520 P.

Thanks.

And can I have your phone number?

Is a mobile.

All right.

Fine.

It's 078643 091.

Thank you.

And can I ask you where you heard about World Tours?

From a friend or did you see an advert somewhere?

No, I read about you in the newspaper.

Okay, I'll get the brochures and the post to you, but can I give you some information over the phone?

Uh, what kinds of things do you want to do on your holiday?

I'm interested in going to California with my family.

I've got two children, and we want to hire a car.

Okay.

We have a couple of self-drive tours there, visiting different places of interest in California.

The first one begins in Los Angeles, and there's plenty of time to visit some of the theme parks there.

Ah, that's something on my children's list.

So, I'd want to include that.

Good.

Uh, then you drive to San Francisco.

From San Francisco, you can drive to Yusede Park where you spend a couple of nights.

You can choose to stay in a lodge or on the campsite.

I don't like the idea of staying in a tent.

It would be too hot.

Right.

And the tour ends in Las Vegas.

Okay.

The other trip we can arrange is slightly different.

It starts in San Francisco.

Then you drive south to Cambria.

Someone told me there's a really nice castle near Cambria.

Will we go near that?

Hurst Castle is on that road, so you could stop there.

Good.

I'd like to do that.

Does this trip also go into the desert?

No, it continues to Santa Monica, where most people like to stop and do some shopping.

We have enough of that at home, so that doesn't interest us.

Okay.

Well, you could go straight onto San Diego.

That's good for beaches, isn't it?

That's right.

That's a good place to relax.

And your children might like to visit the zoo before flying home.

I don't think so.

We want some time for sunbathing and swimming.

So, how many days are the trips?

And how much do they cost?

The first one I told you about is a self-drive tour through California, which lasts 12 days and covers 2,20 km.

The shortest journey is 206 km and the longest is 632 km.

The cost is £525 per person.

That includes accommodation, car rental, and a flight, but no meals.

Okay.

And the other trip that lasts 9 days, but you spend only 3 days on the road.

You cover about 980 km altogether.

So, is that cheaper then?

Yes, it's almost £100 cheaper.

It's £429 per person, which is a good deal.

So, that covers accommodation and car hire.

What about flights?

They aren't included, but these hotels offer dinner in the price.

Okay.

Well, thank you very much.

I'll be in touch when I've had a chance to look at the brochure.

I'm pleased to help.

Goodbye.

Goodbye.

On behalf of LP Clubs, I'd like to welcome you all here today.

My name's Sandy Fischer and I'm one of the fitness managers here.

Before we start our tour of the club, I'll just run through some basic information about the facilities we have here, including recent improvements, and explain the types of membership available.

Our greatest asset is probably our swimming pool, which at 25 m isn't Olympic sized, but now we've expanded it to eight lanes.

It's much wider.

This means there are rarely more than a couple of people at a time in each lane.

Unfortunately, there isn't space for an outdoor pool here, but the glass roof on the swimming pool is partly retractable, which means you can enjoy something of the open air experience on warmer days.

Our recently refurbished fitness suite has all the latest exercise equipment, including 10 new running machines and a wide range of weight training machines.

Each member is given full training in how to operate the equipment, and there is always a trainer on duty to offer help and advice.

Although we do have adult only times after 6 and at certain times at weekends, children are well catered for.

Older children continue to benefit from a wide range of tuition.

Anything from trampolining to yoga.

One thing all our members appreciate about us is that we take very good care of them.

This starts on day one with your personal assessment.

You [snorts] are asked to fill in a questionnaire giving details of any health problems.

One of our personal trainers will then go through this with you.

The trainer will then take you through the safety rules for using the equipment in the fitness suite.

[snorts] During your next exercise session, a personal trainer will work with you to make sure you understand these.

It's very important to do this because we really do want to avoid having any sports injuries.

There's a lot more to looking after yourself than simply lifting weights.

At the end of the personal assessment, the trainer will draw up a plan outlining what you should try to achieve within a six-week period.

This will then be reviewed at the end of the 6 weeks.

Now, I'll just quickly run through the types of membership we have available.

All members must pay a joining fee of £90 in addition to the rates for the monthly membership fees.

Gold membership entitles you to free entry at all LP clubs.

There are now LP clubs in all major cities and towns.

So, if you travel a lot, this will be a great advantage.

Individual gold membership costs £50 a month and joint membership for you and your partner will cost £75.

Premium membership is for professional people whose work commitments make it difficult for them to use the club during the day and so LP gives booking preferences to premier members at peak times.

[snorts] This means you'll find it easier to book the sessions at times that suit you.

Reciprocal arrangements with other LP clubs are available to premier members.

Premier membership is for individuals only, but you'll be sent passes for guests every month.

The monthly fee is £65.

You don't have to have any special clothes or equipment when you visit the club.

We provide robes and hair dryers in the changing rooms, but it's very important to remember your photo card because you won't be able to get in without it.

For people who aren't working during the day, um, hello, professor.

I'm John Wishart.

I'm working on my entry for the global design competition.

My tutor said you might be able to help me with it.

Ah, yes.

I got a copy of your drawings.

come in and tell me about it.

What sort of competition is it?

Well, it's an international design competition and we have to come up with a new design for a typical domestic kitchen appliance.

I see.

And are there any special conditions?

Does it have to save energy, for example?

Actually, that was the focus in last year's competition.

This year's different.

We have to adopt an innovative approach to existing technology, using it in a way that hasn't been thought of before.

I see.
That sounds tricky.
And what kitchen appliance have you chosen?
Well, I decided to choose the dishwasher.
Interesting.
What made you choose that?
Well, they're an everyday kitchen appliance in most Australian houses, but they're all pretty boring and almost identical to each other.
I think some people will be prepared to pay a little extra for something that looks different.
It's a nice idea.
I see you've called your design the rockpool.
Why is that?
basically because it looks like the rock pools you find on a beach.
The top is made of glass so that you can look down into it.
And there's a stone at the bottom.
Is that just for decoration?
Actually, it does have a function.
Instead of [clears throat] pushing a button, you turn the stone.
So, it's really just a novel way of starting the dishwasher.
That's right.
It's a really nice design, but what makes it innovative?
Well, I decided to make a dishwasher that uses carbon dioxide in place of water and detergent.
How will you manage that?
The idea is to pressurize the carbon dioxide so that it becomes a liquid.
The fluid is then released into the dishwasher where it cleans the dishes all by itself.
Sounds like a brilliant idea.
Your system will totally do away with the need for strong detergents.
So, what happens once the dishes are clean?
Well, to allow them to dry, the liquid carbon dioxide and the waste materials all go to an area called the holding chamber.
That's where the liquid is depressurized, and so it reverts to a gas.
Then the oil and grease are separated out and sent to the waste system.
It sounds like you've thought it all out very thoroughly.
So, what happens to the carbon dioxide once the process is complete?
Not wasted, I hope.
Actually, that's where the real savings are made.
The carbon dioxide is sent back to the cylinder and can be used again and again.
What a terrific idea.
Do you think it will ever be built?
Probably not, but that's okay.
Well, I'm sure a lot of positive things will come out of your design.
Now, you seem to have thought about everything.
So, what exactly do you need me to help you with?
Well, my design has made it to the final stage of the competition, and in a few months time, I have to give a presentation, and that's the part I was hoping you could help me with.
Right.
Well, that should be easy enough.
What have you managed to do so far?
Well, I've got detailed drawings to show how it will work.
And I've also written a 500word paper on it.
I see.
Well, if you want to stand a good chance of winning, you really need a model of the machine.
Yes, I thought I might, but I'm having a few problems.
What's the main difficulty so far?
Uh, let me guess.
Is it the materials?
Yes.
I want it to look professional, but everything that's top quality is also very expensive.
Look, projects like this are very important to us.
They really help lift our profile.
So, why don't you talk to the university about a grant?
Oh, I can help you fill out the application forms if you like.
That would be great.
You'd better show me this paper you've written as well.
For a global competition such as this, you need to make sure the technical details you've given are accurate and thorough.
Oh, that would be a great help.
Is there anything else I can do?
Well, I'm really nervous.
Today, we continue our series on ecology and conservation with a look at a particularly endangered member of the black bear family.
One in 10 black bears is actually born with a white coat, which is the result of a special gene that surfaces in a few.
Local people have named it the spirit bear.
And according to the legends of these communities, its snowy fur brings with it a special power.

Because of this, it has always been highly regarded by them.
so much that they do not speak of seeing it to anyone else.
It is their way of protecting it when strangers visit the area.
The white bear's habitat is quite interesting.
The bear's strong relationship with the old growth rainforest is a complex one.
The white bear relies on the huge centuries old trees in the forest in many ways.
For example, the old growth trees have extremely long roots that help prevent erosion of the soil along the banks of the many fish streams.
Keeping these banks intact is important because these streams are home to salmon, which are the bear's main food source.
In return, the bear's feeding habits nurture the forest.
As the bears eat the salmon, they discard the skin and bones in great amounts on the forest floor, which provide vital nutrients.
These produce lush vegetation that sustains thousands of other types of life forms, from birds to insects and more.
Today, the spirit bear lives off the coast of the province of British Columbia on a few islands.
There is great concern for their survival since it is estimated that less than 200 of these white bears remain.
The best way to protect them is to make every effort to preserve the delicate balance of their forest environment.
In other words, their ecosystem.
The greatest threat to the bear's existence is the loss of its habitat.
Over many years, logging companies have stripped the land by cutting down a large number of trees.
In addition, they have built roads which have fractured the areas where the bear usually feeds, and many hibernation sites have also been lost.
The logging of the trees along the streams has damaged the places where the bears fish.
To make matters worse, the number of salmon in those streams is declining because there is no legal limit on fishing at the moment.
All these influences have a negative impact on the spirit bear's very existence, which is made all the more fragile by the fact that reproduction among these bears has always been disappointingly low.
And so what's the situation going forward?
Community organizations, environmental groups, and the British Columbia government are now working together on the problem.
The government is now requiring logging companies to adopt a better logging method, which is a positive step.
However, these measures alone may not be sufficient to ensure a healthy population of the spirit bear in the future.
Other steps also need to be taken.
While it is important to maintain the spirit bear's habitat, there also needs to be more emphasis on its expansion.
The move is justified as it will also create space for other bears that are losing their homes to dam
Excuse me.
Yes.
I wonder if you could spare a few minutes to do a survey on transport.
It won't take long.
Uh, no, that's fine.
Oh, lovely.
The survey is on behalf of the local council.
They'd like to know about what transport you use and any suggestions for improvement.
Uh, can I start by asking you how you traveled to town today?
Sure.
I came on the bus.
Great.
Now, can I get a few details about yourself?
What's your name?
It's Louisa.
Hardy.
Can you spell that, please?
Yes, it's H A R D I E.
Great.
Thanks.
And can I have your address?
[clears throat] It's 19 Whit Stone Road.
Oh, right.
I know that area.
It's Bradfield, isn't it?
Is the post code GT7?
It's actually GT82 LC.
Great.
And could I ask what your job is?
Are you a student?
I've actually just finished my training.
I'm a hairdresser.
Oh, right.
And one more question in this section.
What is the reason for you coming into town today?
Actually, it's not for shopping today, which would be my normal reason, uh, but to see the dentist.
Right.
Thanks.
Now, in this last section, I'd like you to give us some ideas about the facilities and arrangements in the city for getting to and from work.
Um, any suggestions you have for improvements?

Well, something I've thought about for some time is that when I do walk and I'm doing a later shift, I think the lighting should be better. Yes, good point.

And of course, I think it's a real shame they've been cutting down on the number of footpaths.

They should have more of those.

Then people would walk more.

Yes.

Right.

And I don't think there are enough trains.

That's why I don't use them.

You have to wait so long.

Thanks.

And finally, I'd like to ask your opinion on cycling.

As you may know, there's a drive in the city to get more people to cycle to work.

Right.

But we realize that there are things which the council, but also employers might do to help encourage workers to cycle to work.

Yeah.

Well, I have thought about it.

And where I work, there are no safe places to leave your bikes.

And also, I'd have to cycle uphill.

And on a hot day, I'd arrive at work pretty sweaty.

So, I think I'd need a shower somewhere at work, right?

And I suppose the last thing is that I wouldn't be all that confident about cycling on such busy roads.

I think I'd like to see you offering training for that.

You know, I'd feel a lot better about starting if that was the case.

Well, that's very helpful.

Uh, thank you very much for your time.

No problem.

Bye.

Good morning.

I'm very pleased to have this opportunity to say a little about two exciting new developments in the city.

The Brackenside open air swimming pool and the children's adventure playground in Central Park.

As many of you may know, the idea for these initiatives came from you, the public, in the extensive consultation exercise which the city council conducted last year.

and they have been realized using money from the SWRDC, the Southwest Regional Development Commission.

First of all, Brackenside Pool.

As many of the older members of the audience will remember, there used to be a wonderful open air pool on the seafront 30 years ago, but it had to close when it was judged to be unsafe.

For the design of this new heated pool, we were very happy to secure the talents of internationally renowned architect Ellen Wendon, who has managed to combine a charming 1930s design, which fits in so well with many of the other buildings in the area with up to the minute features such as a recycling system, the only one of its kind in the world, which enables seawater to be used in the pool.

Now, there's been quite a bit of discussion in the local press about whether there would be enough room for the number of visitors we're hoping to attract, but the design is deceptive and there have been rigorous checks about capacity.

Also, just in case you were wondering, we're on schedule for a June 15th opening date and well within budget, a testimony to the excellent work of local contractors Hickman's.

We hope that as many people as possible will be there on June the 15th.

We [snorts] have engaged award-winning actress Coral White to declare the pool open, and there'll be drinks and snacks available at the poolside.

There'll also be a competition for the public to decide on the sculpture we plan to have at the entrance.

You will decide which famous historical figure from the city we should have.

And now, moving on to the Central Park playground, which we're pleased to announce has just won the Douglas Award for safety.

The news came through only last week.

The unique design is based on the concept of the global village with a playground being divided into six areas showing different parts of the world each with a representative feature.

For example, there is a section on Asia and this is represented by rides and equipment in the shape of snakes, orangutans, tigers and so on.

Fauna native to the forests of the region.

Moving south to the Antarctic, we couldn't run to an ice rink, I'm afraid, but opted instead for climbing blocks in the shape of mountains.

I thought they could have had slides for the glaciers, but the designers did want to avoid being too literal.

Then onto South America, and here the theme is El Dorado games replicating the search for mines full of precious stones.

And then moving up to North America, here there was considerable debate.

I know the contribution of cinema and jazz was considered, but the designers finally opted for rockets and the international space station.

Eastwards to Europe then and perhaps the most traditional choice of all the areas, medieval castles and other fortifications.

Then last but not least, moving south to Africa and a whole set of wonderful mosaics and trails to represent the great rivers of this fascinating and varied continent.

Now, the opening date for our global playground is the 10th of July.

And again, we'd love to see you there.

So, make a date and come and see this magnificent original new amenity right in the heart of the city.

Right.

Well, for our presentation, shall I start with the early life of Toriadal?

Sure.

Why don't you begin with describing the type of boy he was, especially his passion for collecting things?

That's right.

He had his own little museum.

And I think it's unusual for children to develop their own values and not join in their parents' hobbies.

I'm thinking of how Hayadal wouldn't go hunting with his dad, for example.

Yeah.

He preferred to learn about nature by listening to his mother read to him.

And quite early on, he knew he wanted to become an explorer when he grew up.

That came from his camping trips he went on in Norway, I think.

No, it was climbing that he spent his time on as a young man.

Oh, right.

After university, he married a classmate and together they decided to experience living on a small island to find out how harsh weather conditions shaped people's lifestyles.

As part of their preparation, before they left home, they learned basic survival skills like building a shelter.

I guess they needed that knowledge in order to live wild in a remote location with few inhabitants cut off by the sea, which is what they were aiming to do.

An important part of your talk should be the radical theory Hyodal formed from examining mysterious ancient carvings that he happened to find on the island.

I think you should finish with that.

All right, Victor.

So, after your part, I'll talk about to Hayodal's adult life.

Continuing from the theory he had about Polynesian migration.

Up until that time of course academics had believed that humans first migrated to the islands in Polynia from Asia in the west.

Yes.

They thought that travel from the east was impossible because of the huge empty stretch of ocean that lies between the islands and the nearest inhabited land.

Yes.

But Hyadal spent ages studying the cloud movements, ocean currents, and wind patterns to find if it was actually possible.

And another argument was that there was no tradition of large ship building in the communities lying to the east of Polynesia.

But Hyodal knew they made lots of coastal voyages in locally built canoes.

Yes.

Or sailing on rafts, as was shown by the long voyage that Hyodal did next.

It was an incredibly risky journey to undertake.

Sometimes I wonder if he did that trip for private reasons, you know, to show others that he could have spectacular adventures.

What do you think, Olivia?

Well, I think it was more a matter of simply trying out his idea to see if migration from the east was possible.

Yes, that's probably it.

And the poor guy suffered a bit at that time because the war forced him to stop his work for some years.

Yes.

when he got started again and planned his epic voyage.

Do you think it was important to him that he achieve it before anyone else did?

Um, I haven't read anywhere that that was his motivation.

The most important factor seems to have been that he used only ancient techniques and local materials to build his raft.

Yes.

I wonder how fast it went.

Well, it took them 97 days from South America to the Pacific Islands.

M.

And after that, Hyal went to Easter Island, didn't he?

We should mention the purpose of that trip.

I think he sailed there in a boat made out of reeds.

No, that was later on in Egypt, Olivia.

Oh, yes, that's right.

But what he wanted to do was talk to the local people about their old stone carvings and then make one himself to learn more about the process.

I see.

Well, what a great life.

Even though many of his theories have been disproven, he certainly left a lasting impression on many disciplines, didn't he?

To my mind, he was the first person to establish what modern academics call practical archaeology.

I mean that they try to recreate something from the past today, like he did with his raft trip.

It's unfortunate that his ideas about where Polynesians originated from have been completely discredited.

Yes.

Right.

Well, I'll prepare a PowerPoint slide at the end that acknowledges our sources.

I mainly used the life and work of Tor Hayodal by William Oliver.

I thought the research methods he used were very sound, although I must say I found the overall tone somewhat old-fashioned.

I think they need to do a new revised edition.

Yeah, I agree.

What about the subject matter?

I found it really challenging.

Well, it's a complex issue.

I thought the book had lots of good points.

What did you think of the illustration?

Well, I've been talking to managers in a number of businesses and reading surveys about the future of management.

And what I'm going to present in this seminar is a few ideas about how the activity is likely to change in the next 10 years.

It isn't a scientific statistical analysis, just some ideas for us to discuss.

One area I want to mention is business markets.

And I'm sure a really significant development will be a major increase in competition with companies from all around the world trying to sell similar products.

Consumers will have much more choice.

For instance, food products sold in Australia might be manufactured in the USA, China, Finland, and dozens of other countries.

At the same time, mergers and takeovers mean that governments are actually losing power to major global corporations.

We can probably all think of companies that exert a great deal of influence which may be good for consumers.

A third point I want to make about markets is that in the rapidly expanding economies such as India, China, Brazil and Russia, demand is growing very fast.

This is putting pressure on resources all over the world.

I think businesses are becoming more open to external influences.

In particular, companies are consulting customers more and more before making their business decisions.

Companies are finding out what they want and providing it instead of making products and then trying to sell them, which is the model of years ago.

Another influence is that concerns about the environment will force manufacturers to extend product life cycles to reduce the amount of pollution and waste.

And in some cases, regulation will need to be strengthened.

Many societies are much more fluid and democratic, and the structure of companies is changing to reflect that.

I think we're going to see a greater emphasis within companies on teams created with a specific project in mind.

And when they're completed, the teams will be disbanded and new ones formed.

More and more people see work as simply one part of their lifestyle and not the most important one.

And as the workforce is shrinking in some countries, businesses are having to compete for staff instead of being able to choose among a lot of applicants.

Typical examples that will attract and retain staff are traditional ones like flexible hours and something that has been made possible by advances in technology, remote working with people based at their home, abroad, or almost anywhere they choose.

Management styles will almost certainly continue to change.

Senior managers will require a lot more than the efficiency that they've always needed.

Above all, they'll need great skills and leadership so that their organization can initiate and respond to change in a fastmoving world where they face lots of competing requirements and potential conflicts.

In most of the world, the senior managers of large businesses are mainly men in their 50s and 60s.

The predominant style of management will almost certainly become more consultative and collaborative caused above all by more women moving into senior management positions.

Many of the changes are influenced by developments in the wider economy.

The traditional emphasis of business was manufacturing and of course the service sector is very important but we shouldn't overlook the growing financial contribution of IP that is intellectual property.

Some books and films generate enormous sums from the sale of related DVDs, music, games, clothes and so on.

Another point I'd like to make is that although I've been talking about companies, one trend that they have to face is the move away from people working for the same employer for years.

Instead, more and more people are becoming self-employed to gain the freedom and control over their lives that they're unlikely to get from being employed.

Okay.

Well, that's all I want to say.

So, let's open it up for discussion.

Good morning.

Welcome to the early learning child care center.

How may I help you?

Hi, I spoke to you last week about enrolling my daughter for next year.

Oh, yes.

I'll just get some details from you.

So, you're her mother.

And can I have your name?

It's Carol.

Carol Smith.

And your daughter's name?

It's Kate.

Now, we have several groups at the center and we cater for children from 3 to 5 years old.

How old is your daughter?

She's three now, but she turns four next month.

I'll put four down because that's how old she'll be when she starts.

Fine.

She's so excited about her birthday and coming to the center.

Oh, that's good to hear.

And what's your address?

It's 46 Wombat Road.

That's W B A T Woodside 4032.

And what's the phone number?
Oh, it's 9865.
So, have you decided on the days you would like to bring your daughter here?
I'd prefer Monday and Wednesday if possible.
M I'll check.
Monday is fine, but I think the center is already full for Wednesday.
Uh, yes, sorry.
It seems to be a very popular day.
We can offer you a Thursday or a Friday as well.
Oh dear.
I suppose Thursday would be all right because she has swimming on Friday.
Okay, got that.
Because a lot of parents work, we do offer flexible start and finish times.
We're open from 7:30 in the morning until 6:00 at night.
What time would you like your daughter to start?
I need to get to work in the city by 9:00, so I'll drop her off at 8:30.
Mhm.
You're pretty close to the city here, so that should give me plenty of time to get there.
That's fine.
Now, we also need to decide which group she'll be in.
We have two different groups, and they're divided up according to age.
There's the green group, which is for 3 to four year olds, and then there's the red group, which is for four to 5 year olds.
She's quite mature for her age, and she can already write her name and read a little.
Well, I'll put her in the red group and we can always change her to the green one if there are any problems.
That sounds fine.
Okay, let's move on to meals.
We can provide breakfast, lunch, and dinner.
As she's finishing pretty early, she won't need dinner.
Will you give her breakfast before she comes?
Yes, she'll only need lunch.
Now, does she have any medical conditions we need to know about?
Does she have asthma or any hearing problems, for example?
No, but she does need to wear glasses.
Oh, I'll make a note of that.
Yeah, she's pretty good about wearing them.
She can't see much without them.
Right.
Okay.
Now, I also need emergency contact details.
So, what sort of information do you need?
Just the name and number of a friend or family member we can contact in case we can't get hold of you at any time.
Okay, that better be my sister, Jenny Ball.
Uhhuh.
That's B A L.
Her phone number is 33467523.
Great.
So, she is the child's aunt.
Yes, that's right.
I'll make a note of that as well.
Now, is there anything you'd like to ask?
What about payment?
How much are the fees each term?
Well, for 2 days in the hours you've chosen, that will be \$450 altogether.
Okay.
And do I have to pay that now?
No, we send out invoices once the children start at the center.
You can choose to pay at the end of each term or we do offer a slightly discounted rate if you pay every month.
Oh, I'll do that then.
I find it easier to budget that way and I'm not used to the term dates just yet.
Good.
It makes it a lot simpler for us as well.
Well, that's everything.
Would you like me to show you around the center?
We can go today.
We're pleased to have on the show Alice Busel from the Dolphin Conservation Trust.
Tell us about the trust, Alice.
Well, obviously its purpose is to protect dolphins in seas all around the world.
It tries to raise people's awareness of the problems these marine creatures are suffering because of pollution and other threats.
It started 10 years ago and it's one of the fastest growing animal charities in the country.
Although it's still fairly small compared with the big players in animal protection, we're particularly proud of the work we do in education.

Last year, we visited a huge number of schools in different parts of the country, going around to talk to children and young people aged from 5 to 18.

In fact, about 35% of our members are children.

The charity uses its money to support campaigns, for example, for changes in fishing policy and so forth.

It hopes soon to be able to employ its first full-time biologist with dolphin expertise to monitor populations.

Of course, many people give their services on a voluntary basis, and we now have volunteers working in observation, office work, and other things.

I should also tell you about the award we won from the Charity Commission last year for our work in education.

Although it's not meant an enormous amount of money for us, it has made our activities even more widely publicized and understood.

In the long term, it may not bring in extra members, but we're hoping it'll have this effect.

Is it possible to see dolphins in UK waters?

Yes, in several locations and we have a big project in the east part of Scotland.

This has long been a haven for dolphins because it has very little shipping.

However, that may be about to change soon because oil companies want to increase exploration there.

We're campaigning against this because although there'll be little pollution from oil, exploration creates a lot of underwater noise.

It means the dolphins can't rest and socialize.

This is how I became interested in dolphin conservation in the first place.

I had never seen one and I hadn't been particularly interested in them at school.

Then I came across this story about a family of dolphins who had to leave their home in the Mor because of the oil companies and about a child who campaigned to save them.

I couldn't put the book down.

I was hooked.

I'm sure our listeners will want to find out what they can do to help.

You mentioned the adopt a dolphin scheme.

Can you tell us about that?

Of course.

People can choose one of our dolphins to sponsor.

They receive a picture of it and news updates.

I'd like to tell you about four which are currently being adopted by our members, Moon Dancer, Ekko, Kiwi, and Samson.

Unfortunately, Ekko is being rather elusive this year and hasn't yet been cited by our observers, but we remain optimistic that he'll be out there soon.

All the others have been out in force.

Samson and Moon Dancer are often photographed together, but it's Kiwi who's our real character, as she seems to love coming up close for the cameras, and we've captured her on film hundreds of times.

They all have their own personalities.

Moon Dancer is very elegant and curves out and into the water very smoothly.

Whereas Samson has a lot of energy.

He's always leaping out of the water with great vigor.

You'd probably expect him to be the youngest.

He's not quite.

That's Kiwi.

But Samson's the latest of our dolphins to be chosen for the scheme.

Kiwi makes a lot of noise, so we can often pick her out straight away.

Ekko and Moon Dancer are noisy, too, but Moon Dance is easy to find because she has a particularly large fin on her back, which makes her easy to identify.

So, yes, they're all very different.

Well, they sound a fascinating group.

And how would you go back?

Hi, Rob.

How's the course going?

Oh, hi, Mia.

Yeah, great.

I can't believe the first term's nearly over.

H I saw your group's performance last night at the student theater.

It was good.

Really?

Yeah.

But now we have to write a report on the whole thing.

An in-depth analysis.

I don't know where to start.

Like I have to write about the role I played, the doctor.

How I developed the character.

Well, what was your starting point?

Uh my grandfather was a doctor before he retired and I just based it on him.

Okay.

But how uh did you talk to him about it?

He must have all sorts of stories.

But he never says much about his work even now.

He has a sort of authority though.

So how did you manage to capture that?

I'd visualize what he must have been like in the past when he was sitting in his consulting room listening to his patients.

Okay.

So that's what you explain in your report, right?

Then there's the issue of atmosphere.

So in the first scene, we needed to know how boring life was in the doctor's village in the 1950s.

So when the curtain went up on the first scene in the waiting room, there was that long silence before anyone spoke.

And then people kept saying the same thing over and over, like cold, isn't it?

Yes.

And everyone wore gray and brown and just sat in a row.

Yes.

all those details of the production and I have to analyze how I functioned in the group, what I found out about myself.

I know I was so frustrated at times when we couldn't agree.

Yes.

So, did one person emerge as the leader?

Sophia did.

That was okay.

She helped us work out exactly what to do for the production and that made me feel better, I suppose.

When you understood what needed doing?

Yes.

And Sophia did some research too that was useful in developing our approach.

Like what?

Well, she found these articles from the 1950s about how relationships between children and their parents or between the public and people like bank managers or the police were shifting.

Interesting.

And did you have any practical problems to overcome?

Well, in the final rehearsal, everything was going fine until the last scene.

That's where the doctor's first patient appears on stage on his own.

The one in the wheelchair.

Yes.

And he had this really long speech with the stage all dark except for one spotlight and then that stuck somehow.

So it was shining on the wrong side of the stage.

But anyway, we got that fixed.

Thank goodness.

Yes, it was fine on the night.

But while you're here, Mia, I wanted to ask you about the year abroad option.

Would you recommend doing that?

Yes, definitely.

It's a fantastic chance to study in another country for a year.

I think I'd like to do it, but it looks very competitive.

There's only a limited number of places.

Yes.

So, next year when you're in the second year of the course, you need to work really hard in all your theater studies modules.

Only students with good marks get places.

You have to prove that you know your subject really well.

Right.

So, how did you choose where to go?

Well, I decided I wanted a program that would fit in with what I wanted to do after I graduate.

So, I looked for a university with emphasis on acting rather than directing, for example.

It depends on you.

Then, about 6 months before you go, you have to email the scheme coordinator with your top three choices.

I had a friend who missed the deadline and didn't get her first choice.

So, you do need to get a move on at that stage.

You'll find that certain places are very popular with everyone.

And don't you have to write a personal statement at that stage?

Right.

I'll get some of the final year students to give me some tips.

Maybe see if I can read what they wrote.

I think that's a very good idea.

I don't mind showing you what I did.

And while you're abroad, don't make the mistake I made.

I got so involved I forgot all about making arrangements for when I came back here for the final year.

Make sure you stay in touch so they know your choices for the optional modules.

You don't want to miss out doing your preferred specialisms.

Right.

Today I want to talk about self-regulatory focus theory and how the actions of leaders can affect the way followers approach different situations.

Self-regulatory focus theory is a theory developed by Tory Higgins.

He says that a person's focus at any given time is to either approach pleasure or avoid pain.

These are two basic motivations that each and every one of us has and they cause us to have different kinds of goals.

Promotion goals in different life situations emphasize achievement.

Prevention goals are oriented towards the avoidance of punishment.

In a specific situation, our thoughts might focus more on promotion goals or more on prevention. The theory suggests that two factors affect which goals we are focusing on.

First, there is a chronic factor.

This factor is connected to a person's personality and says that each person has a basic tendency to either focus more on promotion goals or focus more on prevention goals as part of his or her personality.

Second, there is a situational factor which means that the context we are in can make us more likely to focus on one set of goals or the other.

For example, we are more likely to be thinking about pleasure and to have promotion goals when we are spending time with a friend.

In contrast, if we are working on an important project for our boss, we are more likely to try to avoid making mistakes and therefore have more prevention goals in our mind.

Research has shown that the goals we are focusing on at a given time affect the way we think.

For example, when focusing on promotion goals, people consider their ideal self, their aspirations and gains.

They don't think about what they can lose.

So, they think in a happier mode.

They feel more inspired to change.

When people are focusing on prevention goals, they think about their ought self.

What are they supposed to be?

What are people expecting from them?

They consider their obligations to others.

As a result, they experience more anxiety and try to avoid situations where they could lose.

Now that I have talked about the two focuses and how they affect people, I want to look at the idea that the way leaders behave or their style of leading can affect the focus that followers adopt in a specific situation.

In talking about leadership, we often mention transformational leaders and transactional leaders.

Transformational leaders when interacting with their followers focus on their development in their words and actions.

Transformational leaders highlight change.

Their speech is passionate and conveys a definitive vision.

All of these things can encourage followers to think about what could be.

In other words, they inspire a promotion focus in their followers.

In contrast, transactional leaders focus on developing clear structures that tell their followers exactly what is expected of them.

While they do explain the rewards people will get for following orders, they emphasize more how a follower will be punished or that a follower won't get rewarded if his or her behavior doesn't change.

In short, they emphasize the consequences of making a mistake.

This emphasis will clearly lead followers to focus on avoiding punishment and problems.

This is clearly a prevention focus.

In conclusion, it is important to understand that one focus is not necessarily better than the other one.

For a designer who works in a field where a lot of innovation is needed, a promotion focus is probably better.

In contrast, a prevention focus, which causes people to work more cautiously and produce higher quality work, might be very appropriate for a job like a surgeon, for example.

The main point of the research though is that the actions of leaders can greatly influence whether people approach a situation with more of a promotion focus or more of a prevention focus.

Thorndike.

Good morning.

Is that Mr.

Thorndike speaking?

How can I help?

I've got quite a few things which need painting and fixing in the flat and I wonder whether you'd be able to do the work.

I'm sure I'd be able to help, but let me take down a few details.

Yes, of course.

Well, uh, firstly, how did you hear about us?

It was my friend May Hampton.

You did some excellent work for her a couple of years ago.

Do you remember?

Oh, yes.

That was in West Park Flats.

Lovely lady.

Yes, she is.

And what's your name, It's Edith Petta.

Edith, can you spell your surname, It's P A R G E T E R.

Double T.

Right.

And do you live in West Park Flats as well?

No, actually it's East Park, Flat 4.

Oh, right.

That's over the road, I seem to remember.

Quite difficult to get to.

Yes, it's at the back of the library.

Right.

I know.

And uh what's your phone 875934.

But I'm out a great deal in the afternoons and evenings.

So, would the best time to ring you be in the morning?

Fine.

I've made a note of that.

Can I just ask?

I'll be in a van and I know parking is rather difficult around your flat.

Where would you recommend?

Well, I always tell people in larger vehicles to park by the post box on the other side of the road from the entrance.

Good.

Thanks.

And will you be able to give me a full itemized quote?

Oh, yes.

I'll list all the jobs separately with individual prices.

That would be a great help.

No problem.

Now, what would you like me to do?

Firstly, and most urgently is in the kitchen.

With all the weather damage, the glass in the door has cracked, and I'd need that fixing.

I presume you mean replacing?

Oh, yes, and as soon as possible.

What I'll do is come around tomorrow morning and do that immediately.

Oh, thank you so much.

The other things aren't so urgent, but now I'll make a note of everything you want doing.

Well, in the kitchen, I'd like some painting doing all the kitchen walls.

Just the area over the cooker.

It's very greasy.

Right.

It does tend to get that way.

Well, if you want a proper job done, what I'd need to do is strip the old paint and plaster it about a week before I paint it.

Of course.

Now, May tells me you also do work in the garden.

That's right.

Well, I'd like you to replace a fence.

Just one?

Yes, at the far end.

Fine.

Shouldn't be a problem.

And that's the lot.

Fine.

Yeah, as I say, I can come around tomorrow morning to look over things with you.

Well, that's great.

Thank you.

So, I look forward to seeing you tomorrow at Welcome to Man Port, where a thousand years of history are brought to life.

All the family can enjoy a day out at Manom.

Visit our copper mine.

See models of the machinery it used.

Have your photo taken in 19th century costume.

Experience at firsthand how people lived at different stages throughout history and especially how children studied, worked, and played.

The port of Manom is located in beautiful and peaceful countryside on a bend in the great river Aaven and developed here because it's the highest navigable point of the A.

Boats can go no higher up this river and proved a handy place to load and unload cargo to and from the sea which is over 23 mi away.

A small port was already established here when about 900 years ago tin was discovered nearby.

Though it wasn't until the industrial revolution when a tremendous need for metals of all kinds developed that Manom expanded to become one of the busiest ports in the country.

And because it was already so busy, prospectors began to look for other minerals.

And by the end of the 19th century, lead, copper, manganese, and arsenic were added to the cargos leaving Manom.

In the early days, the ores had been smelted or processed in the same area they were mined.

But as demand grew, the smelting process required huge factory furnaces or fires to melt the metal from the rock.

And there was not enough coal in the local area.

So the rocks containing minerals had to be shipped long distances.

Sadly in the 20th century the great port of Manom declined and thousands of workers were forced to immigrate out of the area.

The building at the port fell into disrepair and the place became almost forgotten.

But then the Manam Trust was formed to conserve the historical resources of the area.

It organized scores of local volunteers to remove undergrowth to find the original outlines of the installations.

It then brought in paid professionals to match installations with maps of the original port complex and to set about reconstructing it.

Today you can see the results of this ambitious program of restoration.

The intention and we believe this will be realized before the end of the year is to return Manom Port to the condition it reached at its peak as the greatest copper port in the country.

But what can you do and see on your visit today?

Here are just a few highlights.

We suggest you start with the visit to the copper mine.

Travel on converted mining trains and journey into the depths of the mountain along seams once worked by hundreds of miners.

Watch out especially for the great pumping machines which rid the mine of water.

But please be warned that like all mines, ours is very dark and closed in, and we do say that children under five and also dogs should not be taken into the mine.

The next recommended visit is to the village school.

While looking around the classrooms, take a special look at our display of games, which is one of the largest in the world.

And it's recommended that you time your visit to coincide with a guided tour.

This will give you the opportunity to ask lots of questions.

Near the school is the beautiful old sailing catch called the George.

You're welcome to board the boat and look around the cabins.

Look out for the ship's wheel, which was missing until only 5 years ago when it was dredged out of the silt by a local fisherman.

We have no idea how it got there, but it's been polished and proudly restored to its original place on the boat.

Please take care going down the ladders if you wish to visit the lower deck.

We don't recommend you allow young children to use them.

So, we hope you have a memorable visit to Manport and we'll tell your friends all about us.

Hi, Laura.

Could you spare a few minutes to talk about the work placement you did last summer?

I'm thinking of doing one myself.

Hi, Tim.

Sure.

Didn't you do yours at an environmental services company?

That's right.

It's only a very small company and they needed someone to produce a company brochure and I wanted to get some business experience because I'm interested in a career in occupational psychology in a business environment.

It was good because I had overall responsibility for the project.

What kind of skills do you think you developed on the placement?

I mean, apart from the ones you already had, did you have to do all the artwork for the brochure, the layout, and everything?

We hired the services of a professional photographer for that.

I did have to use my IT skills to a certain extent because I cut and pasted text from marketing leaflets, but that didn't involve anything I hadn't done before.

Do you think you got any better at managing your time and prioritizing things?

You always used to say you had trouble with that.

Oh, definitely.

There was so much pressure to meet the project deadline.

And I also got better at explaining things and asserting my opinions because I had to have weekly consultations with the marketing manager and give him a progress report.

It sounds as if you got a lot out of it then.

Absolutely.

It was really worthwhile.

But, you know, the company benefited too.

Yes, they must have done.

After all, if they'd used a professional advertising agency to produce their brochure instead of doing it inhouse, presumably they'd have paid a lot more.

Oh, yes, I worked it out.

It would have been 250% more, and I thought the end result was good, even though we did everything on site.

The company has quite a powerful computer and I managed to borrow some scanning software from the university.

The new brochure looks really professional.

It enhances the image of the company straight away.

So, in the long run, it should help them to attract clients and improve their sales figures.

That's the idea.

Yeah.

Well, all in all, it sounds very positive.

I think I'll go ahead and apply for a placement myself.

How do I go about it?

It's easy enough to do because there's a government agency called STEPP ST that organizes placements for students.

You should start by getting their booklet with all the details.

I expect you can download one from their website.

Actually, they've got copies in the psychology department.

I've seen them there.

I'll just go to the office and pick one up.

Right.

And then if I were you, after I'd looked at it, I'd go over all the options with someone.

I suppose I should ask my tutor's advice.

He knows more about me than anyone.

One of the career officers would be better.

They've got more knowledge about the jobs market than your personal tutor would have.

And then when you know what you want, you can register with STEP.

You'll find their address in the booklet.

And once you've registered, they assign you to a mentor who looks after your application.

And then I suppose you just sit back and wait till you hear something.

They told me at the careers office that it's best to be proactive and get updates yourself by checking the website for new placement alerts.

Your mentor is supposed to keep you informed, but you can't rely on that.

I don't suppose it's a good idea to get in touch with companies directly.

Is it?

Not really.

But it is the company who notifies you if they want you to go for an interview.

You get a letter of invitation or an email from personnel department.

And do I reply directly to them?

Yes, you do.

STEPP only gets involved again once you've been made a job offer.

Right.

So once you've had an interview, you should let your mentor know what the outcome is.

I mean whether you're offered a job and whether you've decided to accept it.

That's right.

They'll inform the careers office once a placement has been agreed.

So you don't have to do that.

Is that all then?

More or less.

Only once you've accepted an offer, you'll probably have to supply a reference because the placement will be conditional on that.

And that's something you should ask your own tutor to provide.

He knows about your academic ability and also about your qualities like reliability.

Well, thanks very much for the information.

I'm starting to look forward to Today we're going to look at an important area of science, namely nanotechnology.

So what is it?

Nano means tiny.

So, it's science and engineering on the scale of atoms and molecules.

The idea is that by controlling and rearranging atoms, you can literally create anything.

However, as we'll see, the science of the small has some big implications affecting us in many ways.

There's no doubt that nanotechnology promises so much for civilization.

However, all new technologies have their teething problems, and with nanotechnology, society often gets the wrong idea about its capabilities.

[snorts] Numerous science fiction books and movies have raised people's fears about nanotechnology with scenarios such as inserting little nano robots into your body that monitor everything you do without you realizing it or self-replicating nano robots that eventually take over the world.

So how do we safeguard such a potentially powerful technology?

Some scientists recommend that nano particles be treated as new chemicals with separate safety tests and clear labeling.

[snorts] They believe that greater care should also be taken with nano particles in laboratories and factories.

Others have called for a withdrawal of new nano products such as cosmetics and a temporary halt to many kinds of nanotech research.

But as far as I'm concerned, there's a need to plow ahead with the discoveries and applications of nanotechnology.

I really believe that most scientists would welcome a way to guard against unethical uses of such technology.

We can't go around thinking that all innovation is bad.

All advancement is bad.

As with the debate about any new technology, it is how you use it that's important.

So let's look at some of its possible uses.

Thanks to nanotechnology, there could be a major breakthrough in the field of transportation with the production of more durable metals.

These could be virtually unbreakable, lighter, and much more pliable, leading to planes that are 50 times lighter than at present.

Those same improved capabilities will dramatically reduce the cost of traveling into space, making it more accessible to ordinary people and opening up a totally new holiday destination.

In terms of technology, the computer industry will be able to shrink computer parts down to minute sizes.

We need nanotechnology in order to create a new generation of computers that will work even faster and will have a million times more memory, but will be about the size of a sugar cube.

Nanotechnology could also revolutionize the way that we generate power.

The cost of solar cells will be drastically reduced, so harnessing this energy will be far more economical than at present.

But nanotechnology has much wider applications than this and could have an enormous impact on our environment.

For instance, tiny airborne nano robots could be programmed to actually rebuild the ozone layer, which could lessen the impact of global warming on our planet.

That's a pretty amazing thought, isn't it?

On a more local scale, this new technology could help with the cleanup of environmental disasters as nanotechnology will allow us to remove oil and other contaminants from the water far more effectively.

And if nanotechnology progresses as expected, as a sort of building block set of about 90 atoms, then you could build anything you wanted from the bottom up.

[snorts] In terms of production, this means that you only use what you need, and so there wouldn't be any waste.

The notion that you could create anything at all has major implications for our health.

It [snorts] means that we'll eventually be able to replicate anything.

This would have a phenomenal effect on our society.

In time, it could even lead to the eradication of famine through the introduction of machines that produce food to feed the hungry.

But it's in the area of medicine that nanotechnology may have its biggest impact.

How we detect disease will change as tiny bio sensors are developed to analyze tests in minutes rather than days.

There's even speculation nano robots could be used to slow the aging process, lengthening life expectancy. As you can see, I'm very excited by the implications that could be available to us in the next few decades. Just how long it'll take, I honestly don't know.

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Hello.

Oh, hello.

I wanted to inquire about hiring a room in the village hall for the evening of September the 1st.

Let me just see.

Uh, yes, we have both rooms available that evening.

There's our main hall that's got seating for 200 people.

Or there's the Charlton room.

Sorry, the Charlton room.

C H A R L T O N.

That's got seating for up to 100.

Well, we're organizing a dinner to raise money for a charity, and we're hoping for at least 150 people.

So, I think we'll go for the main hall.

How much would that cost?

Let's see.

Um, you wanted it for the evening of September the 1st.

Yes, that's a Saturday.

So, from 6:00 p.m.

to midnight, that would be £115.

That's the weekend price.

It's £75 on weekdays.

That's all right.

And I have to tell you, there's also a deposit of £250, which is returnable, of course, as long as there's no damage.

But we do insist that this is paid in cash.

We don't take cards for that.

You can pay the actual rent of the room however you like, though.

Cash, credit card, check.

Oh, well, I suppose that's okay.

So, does the charge include use of tables and chairs and so on?

Oh, yes.

And what about parking?

Yeah, that's all included.

The only thing that isn't included is uh you said you were organizing a dinner.

Yeah.

Well, you'll have to pay extra for the kitchen if you want to use that.

It's £25.

It's got very good facilities, good quality cookers and fridges and so on.

Okay.

Well, I suppose that's all right.

We can cover the cost in our entry charges, right?

So, I'll make a note of that.

Mhm.

Now, there are just one or two things you need to think about before the event.

For example, you'll have to see about getting a license if you're planning to have any music during the meal.

Oh, really?

It's quite straightforward.

I'll give you the details later on.

And about a week or 10 days before your event, you'll need to contact the caretaker, that's Mr.

Evans to make the arrangements for entry.

He'll sort that out with you.

And do I give him the payment as well?

No, you do that directly with me.

Right.

Now, is there anything I need to know about what happens during the event?

Well, as you'll be aware, of course, the building is no smoking throughout.

Of course.

Now, are you having a band?

Yes.

Well, they'll have a lot of equipment, so rather than using the front door, they should park their van around the back and use the stage door there.

You can open that from inside, but don't forget to lock it at the end.

Okay.

And talking of bands, I'm sure I don't need to tell you this, but you must make sure that no one fiddles about with the black box by the fire door.

That's a system that cuts in when the volume reaches a certain level.

It's a legal requirement.

Sure.

Anyway, we want people to be able to talk to one another, so we don't want anything too loud.

Oh, that reminds me.

We'll be having speeches.
Are there any microphones available?
Yep.
Just let the caretaker know.
He'll get those for you.
Right now, when the event is over, we do ask that the premises are left in good condition.
So, there's a locked cupboard, and you'll be informed of the code you need to open that.
It's got all the cleaning equipment, brushes, and detergent, and so on.
Right.
So, what do we need to do after everyone's gone?
Sweep the floors, I suppose.
Well, actually, they have to be washed, not just swept.
Then you'll be provided with black plastic bags, so all the rubbish must be collected up and left outside the door.
Of course, we'll make sure everything's left tidy.
Oh, and I forgot to ask.
I presume we can have decorations in the room.
Yes, but you must take them down afterwards.
Sure.
And the chairs and tables should be stacked up neatly at the back of the hall. I'll make sure I've got a few people to help me.
Welcome to the Fiddy Working Heritage Farm.
This open air museum gives you the experience of agriculture and rural life in the English countryside at the end of the 19th century.
So, you'll see a typical farm of that period.
And like me, all the staff are dressed in clothes of that time.
I must give you some advice and safety tips.
Before we go any further, as it's a working farm, please don't frighten or injure the animals.
We have a lot here, and many of them are breeds that are now quite rare.
And do stay at a safe distance from the tools.
Some of them have sharp points which can be pretty dangerous, so please don't touch them.
We don't want any accidents, do we?
The ground is very uneven, and you might slip if you're wearing sandals, so I'm glad to see you're all wearing shoes.
We always advise people to do that.
Now, children of all ages are very welcome here, and usually even very young children love the ducks and lambs, so do bring them along next time you come.
I don't think any of you have brought dogs with you, but in case you have, I'm afraid they'll have to stay in the car park unless they're guide dogs.
I'm sure you'll understand that they could cause a lot of problems on a farm.
Now, let me give you some idea of the layout of the farm.
The building where you bought your tickets is the new barn immediately to your right.
And we're now at the beginning of the main path to the farmland.
And of course, the car park is on your left.
The scarecrow you can see in the car park in the corner beside the main path is a traditional figure for keeping the birds away from crops.
But our scarecrow is a permanent sculpture.
It's taller than a human being, so you can see it from quite a distance.
If you look ahead of you, you'll see a maze.
It's opposite the new barn beside the side path that branches off to the right just over there.
The maze is made out of hedges which are too tall for young children to see over them, but it's quite small, so you can't get lost in it.
Now, can you see the bridge crossing the fish pool further up the main path?
If you want to go to the cafe, go towards the bridge and turn right just before it.
Walk along the side path and the cafes on the first bend you come to.
The building was originally the schoolhouse and it's well over a 100 years old.
As you may know, we run skills workshops here where you can learn traditional crafts like woodwork and basket making.
You can see examples of the work and talk to someone about the courses in the black barn.
If you take the side path to the right here, just by the new barn, you'll come to the black barn just where the path first bends.
Now, I mustn't forget to tell you about picnicking, as I can see, some of you have brought your lunch with you.
You can picnic in the field, though, do clear up behind you, of course.
Or if you'd prefer a covered picnic area, there's one near the farmyard just after you cross the bridge.
There's a covered picnic spot on the right.
And the last thing to mention is Fiddy House itself.
From here, you can cross the bridge, then walk along the footpath through the field to the left of the farmyard that goes to the house, and it'll give you a lovely view of it.
It's certainly worth a few photographs, but as it's a private home, I'm afraid you can't go inside.
Right.
Well, if you're all ready, we'll set off on our tour of the farm.
Okay, Greg.
So, I finally managed to read the article you mentioned, the one about the study on gender and physics.
About the study of college students done by Akira Miyaki and his team.
Yeah, I was interested that the researchers were actually a mix of psychologists and physicists.
That's an unusual combination.
Yeah, I got a little confused at first about which students the study was based on.
They weren't actually majoring in physics.

They were majoring in what's known as the STEM disciplines.

That's science, technology, engineering, and and math.

Yes.

But they were all doing physics courses as part of their studies.

That's correct.

So as I understood it, Miyaki and Co started from the fact that women are under represented in introductory physics courses at college and also that on average the women who do enroll on these courses perform more poorly than the men.

No one really knows why this is the case.

Yeah.

But what the researchers wanted to find out was basically what they could do about the relatively low level of the women's results.

But in order to find a solution, they needed to find out more about the nature of the problem.

Right now, let's see if I can remember.

It was that in the physics class, the female students thought the male students all assumed that women weren't any good at physics.

Was that it?

And they thought that the men expected them to get poor results in their tests.

That's what the women thought and that made them nervous.

So they did get poor results.

But actually they were wrong.

No one was making any assumptions about the female students at all.

Anyway, what Miyaki's team did was quite simple.

Getting the students to do some writing before they went into the physics class.

What did they call it?

values affirmation.

They had to write an essay focusing on things that were significant to them, not particularly to do with the subject they were studying, but more general things like music or people who mattered to them.

Right?

So, the idea of doing the writing is that this gets the students thinking in a positive way.

And putting these thoughts into words can relax them and help them overcome the psychological factors that lead to poor performance.

Yeah.

But what the researchers in the study hadn't expected was that this one activity raised the women's physics grades from the C to the B range.

A huge change.

Pity it wasn't to an A, but still.

No, but it does suggest that the women were seriously underperforming beforehand in comparison with the men.

Yes, mind you, Miyaki's article left out a lot of details, like did the students do the writing just once or several times?

And had they been told why they were doing the writing, that might have affected the results.

You mean if they know the researchers thought it might help them to improve then they just try to fulfill that expectation.

Exactly.

So anyway, I thought for our project we could do a similar study but investigate whether it really was the writing activity that had that result.

Okay.

So, we could ask them to do a writing task about something completely different, something more factual, like a general knowledge topic, maybe.

Or we could have half the students doing a writing task and half doing something else, like an oral task.

Or even half do the same writing task as in the original research and half do a factual writing task.

Then we'd see if it really is the topic that made the difference or something else.

That's it.

Good.

So, at our meeting with the supervisor on Monday, we can tell him we've decided on our project.

We should have our aims ready by then.

I suppose we need to read the original study.

The article's just a summary.

And there was another article I read by Smolinsky.

It was about her research on how women and men perform in mixed teams in class compared with single sex teams and on their own.

Let me guess, the women were better at teamwork.

That's what I expected.

But actually, the men and the women got the same results whether they were working in teams or on their own.

But I guess it's not that relevant to us.

What worries me anyway is how we're going to get everything done in the time.

We'll be okay now we know what we're doing.

Though I'm not clear how we assess whether the students in our experiment actually make any progress or not.

No, we may need some advice on that.

The main thing is to make sure we have the right size sample, not too big or too small.

That shouldn't be difficult, right?

What do we need to do next?

We could have a look at the timetable for the science classes.

Or perhaps we should just make an appointment to see one of the science professors.

That'd be better.

Great.

And we could even get to observe one of the classes.

What for?

Well, okay.

Maybe let's just go with your idea, right?

Well, I think that's everything for now.

I've been looking at ocean biodiversity.

That's the diversity of species that live in the world's oceans.

About 20 years ago, biologists developed the idea of what they called biodiversity hotspots.

These are the areas which have the greatest mixture of species.

So, one example is Madagascar.

These hotspots are significant because they allow us to locate key areas for focusing efforts at conservation.

Biologists can identify hotspots on land fairly easily, but until recently, very little was known about species distribution and diversity in the oceans, and no one even knew if hotspots existed there.

Then a Canadian biologist called Boris Worm did some research in 2005 on data on ocean species that he got from the fishing industry.

Worm located five hotspots for large ocean predators like sharks and looked at what they had in common.

The main thing he'd expected to find was that they had very high concentrations of food, but to his surprise, that was only true for four of the hot spots.

The remaining hot spot was quite badly off in that regard.

But what he did find was that in all cases the water at the surface of the ocean had relatively high temperatures even when it was cool at greater depths.

So this seemed to be a factor in supporting a diverse range of these large predators.

However, this wasn't enough on its own because he also found that the water needed to have enough oxygen in it.

So these two factors seemed necessary to support the high metabolic rate of these large fish.

A couple of years later in 2007, a researcher called Lisa Balance who was working in California also started looking for ocean hotspots but not for fish.

What she was interested in was marine mammals, things like seals.

And she found three places in the oceans which were hotspots.

And what these had in common was that these hotspots were all located at boundaries between ocean currents.

And this seems to be the sort of place that has lots of the plankton that some of these species feed on.

So now people who want to protect the species that are endangered need to get as much information as possible.

For example, there's an international project called the Census of Marine Life.

They've been surveying oceans all over the world, including the Arctic.

One thing they found there, which stunned other researchers, was that there were large numbers of species which live below the ice, sometimes under a layer up to 20 m thick.

Some of these species had never been seen before.

They've even found species of octopus living in these conditions.

and other scientists working on the same project, but researching very different habitats on the ocean floor have found large numbers of species congregating around volcanoes attracted to them by the warmth and nutrients there.

However, biologists still don't know how serious the threat to their survival is for each individual species.

So, a body called the Global Marine Species Assessment is now creating a list of endangered species on land.

So, they consider things like the size of the population, how many members of one species there are in a particular place and then they look at their distribution in geographical terms, although this is quite difficult when you're looking at fish because they're so mobile.

And then thirdly, they calculate the rate at which the decline of the species is happening.

So far, only 1,500 species have been assessed, but they want to increase this figure to 20,000.

For each one they assess, they use the data they collect on that species to produce a map showing its distribution.

Ultimately, they will be able to use these to figure out not only where most species are located, but also where they are most threatened.

So, finally, what can be done to retain the diversity of species in the world's oceans?

Firstly, we need to set up more reserves in our oceans, places where marine species are protected.

We have some, but not enough.

In addition to preserve species such as leatherback turtles which live out in the high seas but have their nesting sites on the American coast, we need to create corridors for migration so they can get from one area to another safely.

As well as this action needs to be taken to lower the levels of fishing quotas to prevent over fishing of endangered species.

And finally, there's the problem of by catch.

This refers to the catching of unwanted fish by fishing boats.

They're returned to the sea, but they're often dead or dying.

If these commercial fishing boats used equipment which was more selective so that only the fish wanted for consumption were caught, this problem could be overcome.

Okay.

So, does anyone have any questions about Good morning, Youth Council.

Caroline speaking.

Oh, hello.

I'm interested in standing for election to the Youth Council, and I was told to give you a call.

That's good.

Could I have your name, please?

Yes, it's Roger Brown.

Thank you.

I'm Caroline, the youth council administrator.

So, do you know much about what the council does, Roger?

I've talked to Stephanie.

I think she's the chair of the council.

That's right.

And she told me a lot about it.
How it's a way for young people to discuss local issues, for example, and make suggestions to the town council.
That's what made me interested.
Fine.
Well, let me take down some of your details.
First of all, how old are you?
You know, the council is for young people aged from 13 to 18.
I've just turned 18.
And where do you live, Roger?
Uh, well, that's a bit complicated.
At the moment, I'm looking for a flat to rent here.
So, I'm in a hostel from Monday to Friday.
I go back to my parents' place at the weekend.
Okay.
So, where's the best place to send you some information about the council?
Oh, to my parents address, please.
That's 17 Bley Street.
B U C K L E I G H Street Stamford Lincolnshire.
Though you don't really need the county.
Oh, I know Stamford.
It's a lovely town.
And what's the postcode?
P E 9 7 Q T.
Right.
Thank you.
So, are you working here or are you a student?
I started studying at the university a couple of weeks ago and I've got a part-time job for a few hours a week.
What do you do?
Well, I've done several different things.
I've just finished a short-term contract as a courier and now I'm working as a waiter in one of the big hotels.
Uh-huh.
That can't leave you much time for studying.
Oh, it's not too bad.
I managed to fit it all in.
What are you studying?
My ambition is to go into parliament eventually.
So my major subject is politics.
That's partly why I think the youth council is important and want to be a part of it.
And I suppose you're also taking a minor subject, aren't you?
I know a lot of people study economics, too.
I chose history.
To be honest, I'm not finding it as interesting as I expected.
Okay.
So, with your studying and your part-time job, do you have time for any other interests or hobbies?
Well, I spend quite a lot of time cycling, both around town to get to university and to work and also long distance from here to London, for instance.
That's pretty impressive.
Anything else?
For relaxation, I'm also keen on the cinema.
I used to go at least once a week, but I can't manage to go so often now.
Right.
Are you sure you'll have enough time for the youth council?
Yes, I've worked out that I can afford to reduce my hours at work and that will make the So, is there any particular aspect of the youth council's work that appeals to you, Roger?
Well, my sister is blind, so I'm particularly interested in working with disabled young people to try and improve the quality of their lives.
That's great.
Well, the best way to get involved is to be nominated by some people who you know, right?
Can you tell me how to set about organizing that?
You should talk to Jeffrey, our elections officer.
I can arrange a meeting in the council office with him if you like.
Yes, please.
He'll be here next Monday if that suits you.
That's the 14th, isn't it?
Yes.
I can manage late afternoon.
Would you like to suggest a time?
He generally leaves around 5:30.
Well, would 4:30 be okay?
My last class finishes at 4, so I'd have plenty of time to get to your office.
Right, that's fine.

Oh, and could I have a phone number we can contact you on?

Yes, my mobile number's 07813671.

Thank you.

Well, we'll look forward to seeing you next week.

Thanks very much.

Goodbye.

Bye.

Hi.

Great to see you.

I'm Jodie and I'll be looking after both of you for the first month you're working here at the Amisham Theater.

I'll tell you something about the theater now, then take you to meet two of the other staff.

It's an old building and it's been modernized several times.

In fact, as you can see, we're carrying out a major refurbishment at the moment.

The interior has just been repainted and we're about to start on the exterior of the building.

That'll be a big job.

The work's running over budget, so we've had to postpone installing an elevator.

I hope you're happy running up and down stairs.

When the theater was built, people were generally slimmer and shorter than now, and the seats were very close together.

We've replaced them with larger seats with more leg room.

This means fewer seats in total, but we've taken the opportunity to install seats that can easily be moved to create different acting spaces.

We've also turned a few store rooms over to other purposes like using them for meetings.

We try hard to involve members of the public in the theater.

One way is by organizing backstage tours so people can be shown around the building and learn how a theater operates.

These are proving very popular.

What we're finding is that people want to have lunch or a cup of coffee while they're here.

So, we're looking into the possibility of opening a cafe in due course.

We have a bookshop which specializes in books about drama and that attracts plenty of customers.

Then there are two large rooms that will be decorated next month and they'll be available for hire for conferences and private functions such as parties.

We're also considering hiring out costumes to amateur drama clubs.

Now I want to tell you about our workshops.

We recently started a program of workshops that anyone can join.

Eventually, we intend to run courses in acting, but we're waiting until we've got the right people in place as trainers.

That's proving more difficult than we'd expected.

There's a big demand to learn about the technical side of putting on a production, and our lighting workshop has already started with great success.

We're going to start one on sound next month.

A number of people have inquired about workshops on makeup and that's something we're considering for the future.

A surprise success is the workshop on making puppets.

We happen to have someone working here who does it as a hobby and she offered to run a workshop.

It was so popular we're now running them every month.

Now, a word about the layout of the building.

The auditorium, stage, and dressing rooms for the actors are all below ground level.

Here on the ground floor, we have most of the rooms that the public doesn't see.

The majority are internal, so they have windows in the roof to light them.

Standing here in the foyer, you're probably wondering why the box office isn't here where the public would expect to find it.

Well, you might have noticed it on your way in.

Although it's part of this building, it's next door with a separate entrance from the road.

For the theater manager's office, you go across the foyer and through the double doors.

Turn right and it's the room at the end of the corridor with the door on the left.

The lighting box is where the computerized stage lighting is operated and it's at the back of the building.

When you're through the double doors, turn left, turn right at the water cooler and right again at the end.

It's the second room along that corridor.

The lighting box has a window into the auditorium, which of course is below us.

The artistic director's office is through the double doors.

Turn right and it's the first room you come to on the right hand side.

And finally, for the moment, the room where I'll take you next, the relaxation room.

So, if you'd like to come with me, I've brought my notes on our biology field trip to Rocky Bay, Colin, so we can work on our report on the research we did together.

Okay, I've got mine, too.

Let's look at the aims of the trip first.

Right.

What did you have?

I just put something about getting experience of the different sorts of procedures used on a field trip.

But we need something about what causes different organisms to choose particular habitats.

I agree.

And something about finding out how to protect organisms in danger of dying out in our aims, but we weren't really looking at that.

I suppose not.

Okay.

Now, there's the list of equipment we all had to bring on the field trip.
What did they tell us to bring a ruler for?
It was something about measuring the slope of the shore.
But of course, we didn't need it because we were measuring wind direction and we'd brought the compass for that.
But not the piece of string to hold up in the air.
Didn't Mr.
Blake make a fuss about us leaving that behind?
Yeah, he does go on.
Anyway, it was easy to get one from another of the students.
Now, the next section's the procedure.
I sent you the draft of that.
Yeah.
Um, it was clear, but I don't think we need all these details of what time we left and what time we got back and how we divided up the different research tasks.
Okay, I'll look at that again.
Then we have to describe our method of investigation in detail.
So, let's begin with how we measured wave speed.
I was surprised how straightforward that was.
I'd expected us to have some sort of high-tech device, not just stand there and count the number of waves per minute.
Not very precise, but I suppose it was good enough.
But the way we measured the amount of salt was interesting in the water from the rock pools.
Yeah.
Oh, I wanted to check the chemicals we used in the lab when we analyzed those samples.
Was it potassium chromate and silver nitrate?
That's right.
Okay.
And we need the map of the seashore.
You just left that to me.
And I had to do it while the tide was low.
Well, that was okay.
But the place I started it from was down on the beach.
Then I realized I should have gone up higher to get better visibility, so I had to start all over again.
But at least I'd got the squared paper or I'd have had problems drawing it all to scale.
Yeah, it looks good.
We could get a map of the region off the internet and see if we need to make any changes.
I had a look, but I couldn't find anything.
But you took some pictures, didn't you?
Yeah.
I'll email you them if you want.
Okay.
I'll make my amendments using those.
Then I can scan it into our report.
Great.
Now, when we get to our findings, I thought we could divide them up into the different zones we identified on the shore and the problems organisms face in each zone.
So, for the highest area, the splash zone, yeah, we found mostly those tiny shellfish that have strong hard shells that act as protection, but not from other organisms that might eat them, predators.
No, that's not the main danger for them.
But the shells prevent them from drying out because they're in the open air most of the time, right?
And since they're exposed, they need to be able to find some sort of shelter or cover themselves up so they don't get too hot.
Then in the middle and lower zones near the sea, we need to discuss the effects of wave action.
Yes.
And how organisms develop structures to prevent themselves from being swept away or even destroyed by being smashed against the rocks.
H I haven't done anything on the geological changes.
I don't know what to put for that.
M no, we weren't concentrating on that.
Maybe we need to find some websites.
Good idea.
I've got the lecture notes from Mr.
Blake's geology course, but they're too general.
But we could ask him which books on our reading list might be most helpful.
Right.
Okay.
Now I did a draft of the section of sources of possible error in our research but I don't know if you agree for example the size of the sample and whether it's big enough to make any general conclusions from but I thought actually we did have quite a big sample.
We did and our general method of observation seemed quite reliable but we might not be all that accurate as far as the actual numbers go.
Yeah, we might have missed some organisms if they were hiding under a rock, for example.
I wasn't sure about the way we described their habitats.

I decided it was probably okay.

Yeah, and the descriptions we gave of the smaller organisms.

They weren't very detailed, but they were adequate in this context.

I'm not sure we identified all the species correctly, though.

Okay, we better mention that.

Now, how are we going We've been discussing the factors the architect has to consider when designing domestic buildings.

I'm going to move on now to consider the design of public buildings.

And I'll illustrate this by referring to the new Taylor Concert Hall that's recently been completed here in the city.

So, as with a domestic building, when designing a public building, an architect needs to consider the function of the building.

For example, is it to be used primarily for entertainment or for education or for administration.

The second thing the architect needs to think about is the context of the building.

This includes its physical location obviously, but it also includes the social meaning of the building, how it relates to the people it's built for.

And finally, for important public buildings, the architect may also be looking for a central symbolic idea on which to base the design, a sort of metaphor for the building and the way in which it is used.

Let's look at the new Taylor Concert Hall in relation to these ideas.

The location chosen was a site in a run-down district that has been ignored in previous redevelopment plans.

It was occupied by a factory that has been empty for some years.

The whole area was some distance from the high-rise office blocks of the central business district and shopping center, but it was only 1 kilometer from the ring road.

The site itself was bordered to the north by a canal which had once been used by boats bringing in raw materials when the area was used for manufacturing.

The architect chosen for the project was Tom Harrison.

He found the main design challenge was the location of the site in an area that had no neighboring buildings of any importance.

To reflect the fact that the significance of the building in this quite run-down location was as yet unknown, he decided to create a building centered around the idea of a mystery, something whose meaning still has to be discovered.

So, how was this reflected in the design of the building?

Well, Harrison decided to create pedestrian access to the building and to make use of the presence of water on the site.

As people approach the entrance, they therefore have to cross over a bridge.

He wanted to give people a feeling of suspense as they see the building first from a distance and then close up.

And the initial impression he wanted to create from the shape of the building as a whole was that of a box.

The first side that people see, the southern wall, is just a high flat wall uninterrupted by any windows.

This might sound off-putting, but it supports Harrison's concept of the building that the person approaching is intrigued and wonders what will be inside.

And this flat wall also has another purpose.

At nighttime, projectors are switched on and it functions as a huge screen onto which images are projected.

The auditorium itself seats 1,500 people.

The floors supported by 10 massive pads.

These are constructed from rubber and so are able to absorb any vibrations from outside and prevent them from affecting the auditorium.

The walls are made of several layers of honeycombed wood.

All sourced from local beach trees.

In order to improve the acoustic properties of the auditorium and to amplify the sound, they are not straight, they are curved.

The acoustics are also adjustable according to the size of orchestra and the type of music being played.

In order to achieve this, there are nine movable panels in the ceiling above the orchestra, which are all individually motorized.

and the walls also have curtains which can be opened or closed to change the acoustics.

The reaction of the public to the new building has generally been positive.

However, the evaluation of some critics has been less enthusiastic.

In spite of Harrison's efforts to use local materials, they criticize the style of the design as being international rather than local and say it doesn't reflect features of the landscape or society for which it is built.

Good morning.

This is Burnham tourist office.

Martin speaking.

Oh, hello.

I saw a poster about free things to do in the area and it said people should phone you for information.

I'm coming to Burnham with my husband and two children for a few days on June the 27th or possibly the 28th and I'd like some ideas for things to do on the 29th.

Yes, of course.

Okay, then let's start with a couple of events especially for children.

The art gallery is holding an event called family welcome that day when there are activities and trails to use throughout the gallery.

That sounds interesting.

What time does it start?

The gallery opens at 10:00 and the family welcome event runs from 10:30 until 2:00.

The gallery stays open until 5 and several times during the day they're going to show a short film that the gallery has produced.

It demonstrates how ceramics are made and there'll be equipment and materials for children to have a go themselves.

Last time they ran the event, there was a film about painting which went down very well with the children and they're now working on one about sculpture.

I like the sound of that.

And what other events happen in Burnham?

Well, do you all enjoy listening to music?

Well, there are several free concerts taking place at different times.

One or two in the morning, the majority at lunchtime and a couple in the evening.
And they range from pop music to Latin American.
The Latin American could be fun.
What time is that?
It's being repeated several times in different places.
They're performing in the central library at 1:00.
Then at 4 it's in the city museum and in the evening at 7:30 there's a longer concert in the theater.
Right.
I'll suggest that to the rest of the family.
Something else you might be interested in is the boat race along the river.
Oh yes.
Do tell me about that.
The race starts at Offord Marina to the north of Burnham and goes as far as Summer Pool.
The best place to watch it from is Charlesworth Bridge, though that does get rather crowded.
And who's taking part?
Well, local boat clubs, but the standard is very high.
One of them came first in the West of England regional championship in May this year.
It was the first time a team from Bernamas won.
It means that next year they'll be representing the region in the national championship.
Now, I've heard something about Paxton Nature Reserve.
It's a good place for spotting unusual birds, isn't it?
That's right.
throughout the year.
There is a lake there as well as a river and they provide a very attractive habitat.
So, it's a good idea to bring binoculars if you have them.
And just at the moment, you can see various flowers that are pretty unusual.
The soil at Paxton isn't very common.
They're looking good right now, right?
My husband will be particularly interested in that.
And there's going to be a talk and slideshow about mushrooms, and you'll be able to go out and pick some afterwards and study the different varieties.
Uh-huh.
And is it possible for children to swim in the river?
Yes.
Part of it has been fenced off to make it safe for children to swim in.
It's very shallow and there's a lifeguard on duty whenever it's open.
The lake is too deep, so swimming isn't allowed there.
Okay.
We must remember to bring their swimming things in case we go to Paxton.
How long does it take to get there by car from Burnham?
m about 20 minutes, but parking is very limited, so it's usually much easier to go by bus and it takes about the same time.
Right.
Well, I'll discuss the options with the rest of the family.
Thanks very much for all your help.
You're welcome.
Goodbye.
First of all, let me thank you all for coming to this public meeting to discuss the future of our town.
Our first speaker is Shaa Ferguson from Barford Town Council.
Shona, thank you.
First, I'll briefly give you some background information.
Then I'll be asking you for your comments on developments in the town.
Well, as you don't need me to tell you, Barford has changed a great deal in the last 50 years.
These are some of the main changes.
50 years ago, buses linked virtually every part of the town and the neighboring towns and villages.
Most people use them frequently, but not now because the bus companies concentrate on just the routes that attract most passengers.
So parts of the town are no longer served by buses.
[snorts] Even replacing old uncomfortable buses with smart new ones has had little impact on passenger numbers.
It's sometimes said that bus fares are too high, but in relation to average incomes, fares are not much higher than they were 50 years ago.
Changes in the road network are affecting the town.
The center was recently closed to traffic on a trial basis, making it much safer for pedestrians.
The impact of this is being measured.
The new cycle paths separating bikes from cars in most main roads are being used far more than was expected, reducing traffic and improving air quality.
And although the council's attempts to have a bypass constructed have failed, we haven't given up hope of persuading the government to change its mind.
Shopping in the town center has changed over the years.
Many of us can remember when the town was crowded with people going shopping.
Numbers have been falling for several years despite efforts to attract shoppers, for instance, by opening new car parks.
Some people combine shopping with visits to the town's restaurants and cafes.
Most shops are small independent stores, which is good, but many people prefer to use supermarkets and department stores in nearby

large towns as there are so few well-known chain stores here.

Turning now to medical facilities.

The town is served by family doctors in several medical practices, fewer than 50 years ago, but each catering for far more patients.

Our hospital closed 15 years ago, which means journeys to other towns are unavoidable.

On the other hand, there are more dentists than there used to be.

Employment patterns have changed along with almost everything else.

The number of schools and colleges has increased making that the main employment sector.

Services such as website design and accountancy have grown in importance and surprisingly perhaps manufacturing hasn't seen the decline that has affected it in other parts of the country.

Now I'll very quickly outline current plans for some of the town's facilities before asking for your comments.

As you'll know, if you regularly use the car park at the railway station, it's usually full.

The railway company applied for permission to replace it with a multi-story car park, but that was refused.

Instead, the company has bought some adjoining land, and this will be used to increase the number of parking spaces.

The Grand, the old cinema in the high street, will close at the end of the year and reopen on a different site.

You've probably seen the building under construction.

The plan is to have three screens with fewer seats rather than just the one large auditorium in the old cinema.

I expect many of you shop in the indoor market.

It's become more and more shabby looking and because of fears about safety, it was threatened with demolition.

The good news is that it will close for 6 weeks to be made safe and redecorated and the improved building will open in July.

Lots of people use the library, including school and college students who go there to study.

The council has managed to secure funding to keep the library open later into the evening, twice a week.

We would like to enlarge the building in the not too distant future, but this is by no means definite.

There's no limit on access to the nature reserve on the edge of town, and this will continue to be the case.

What will change though is that the council will no longer be in charge of the area.

Instead, it will become the responsibility of a national body that administers most nature reserves in the country.

Okay.

Now, let me ask you if you have Hello, Helen.

Sorry I'm late.

Hi, Jeremy.

No problem.

Well, we'd better work out where we are on our project, I suppose.

Yeah.

I've looked at the drawings you've done for my story, The Forest, and I think they're brilliant.

They really create the atmosphere I had in mind when I was writing it.

I'm glad you like them.

There are just a few suggestions I'd like to make.

Go ahead.

Now, I'm not sure about the drawing of the cave.

It's got trees all around it, which is great, but the drawing's a bit too static, isn't it?

I think it needs some action.

Yes.

There's nothing happening.

Perhaps I should add the boy, Malcolm, isn't it?

He would be walking up to it.

Yes, let's have Malcolm in the drawing.

And what about putting in a tiger?

The one that he makes friends with a bit later.

Maybe it could be sitting under a tree washing itself.

And the tiger stops in the middle of what it's doing when it sees Malcolm walking past.

That's a good idea.

Okay, I'll have a go at that.

Then there's the drawing of the crowd of men and women dancing.

They're just outside the forest and there's a lot going on.

That's right.

You wanted them to be watching a carnival procession, but I thought it would be too crowded.

Do you think it works like this?

Yes, I like what you've done.

The only thing is, could you add Malcolm to it without changing what's already there?

H.

What about having him sitting on the tree trunk on the right of the picture?

Yes, that would be fine.

And do you want him watching the other people?

No, he's been left out of all the fun, so I'd like him to be crying.

That'll contrast nicely with the next picture where he's laughing at the clowns in the carnival.

Right, I'll do that.

And then the drawing of the people ice skating in the forest.

H I wasn't too happy with that one because they're supposed to be skating on grass, aren't they?

That's right.

And it's frozen over at the moment.

It doesn't look quite right.

I see what you mean.

I'll have another go at that.
And I like the wool hats they're wearing.
Maybe you could give each of them a scarf as well.
Yeah, that's easy enough.
They can be streaming out behind the people to suggest they're skating really fast.
Great.
Well, that's all on the drawings, right?
So, you finished writing your story and I just need to finish illustrating it.
And my story and your drawings are done.
So, the next thing is to decide what exactly we need to write about in the report that goes with the stories and how we're going to divide the work.
Right.
Helen, what do you think about including a section on how we planned the project as a whole?
Jeremy, that's probably quite important.
Yeah.
Well, you've had most of the good ideas so far.
How do you feel about drafting something?
Then we can go through it together and discuss it.
Okay, that seems reasonable.
And I could include something on how we came up with the ideas for our two stories, couldn't I?
Well, I've started writing something about that.
So, why don't you do the same and we can include the two things.
Right.
So, what about our interpretation of the stories?
Do we need to write about what we think they show?
Like the value of helping other people or that sort of thing?
That's going to come up later, isn't it?
I think everyone in the class is going to read each other's stories and come up with their own interpretations, which we're going to discuss.
Oh, I missed that.
So, it isn't going to be part of the report at all.
No, but we need to write about the illustrations because they're an essential element of children's experience of reading the stories.
It's probably easiest for you to write that section as you know more about drawing than I do.
Maybe, but I find it quite hard to write about.
I'd be happier if you did it.
Okay.
So, when do you think we can get this?
So what I'm going to talk about to you today is something called ethnography.
This is a type of research aimed at exploring the way human cultures work.
It was first developed for use in anthropology and it's also been used in sociology and communication studies.
So what's it got to do with business you may ask?
Well, businesses are finding that ethnography can offer them deeper insight into the possible needs of customers, either present or future, as well as providing valuable information about their attitudes towards existing products.
And ethnography can also help companies to design new products or services that customers really want.
Let's look at some examples of how ethnographic research works in business.
One team of researchers did a project for a company manufacturing kitchen equipment.
They watched how cooks used measuring cups to measure out things like sugar and flour.
They saw that the cooks had to check and recheck the contents because although the measuring cups had numbers inside them, the cooks couldn't see these easily.
So a new design of cup was developed to overcome this problem and it was a top seller.
Another team of ethnographic researchers looked at how cell phones were used in Uganda in Africa.
They found that people who didn't have their own phones could pay to use the phones of local entrepreneurs.
Because these customers paid in advance for their calls, they were eager to know how much time they'd spent on the call so far.
So the phone company designed phones for use globally with this added feature.
Ethnographic research has also been carried out in computer companies.
In one company, IT systems administrators were observed for several weeks.
It was found that a large amount of their work involved communicating with colleagues in order to solve problems, but that they didn't have a standard way of exchanging information from spreadsheets and so on.
So, the team came up with an idea for software that would help them to do this.
In another piece of research, a team observed and talked to nurses working in hospitals.
This led to the recognition that the nurses needed to access the computer records of their patients no matter where they were.
This led to the development of a portable computer tablet that allowed the nurses to check records in locations throughout the hospital.
Occasionally research can be done even in environments where the researchers can't be present.
For example, in one project done for an airline, respondents used their smartphones to record information during airline trips, in a study aiming at tracking the emotions of passengers during a flight.
So, what makes studies like these different from ordinary research?
Let's look at some of the general principles behind ethnographic research in business.
First of all, the researcher has to be completely open-minded.
He or she hasn't thought up a hypothesis to be tested as is the case in other types of research.
Instead, they wait for the participants in the research to inform them.

As far as choosing the participants themselves is concerned, that's not really all that different from ordinary research. The criteria according to which the participants are chosen may be something as simple as the age bracket they fall into or the researchers may select them according to their income or they might try to find a set of people who all use a particular product for example but it's absolutely crucial to recruit the right people as participants as well as the criteria I've mentioned they have to be comfortable talking about themselves and being watched as they go about their activities. Actually, most researchers say that people open up pretty easily, maybe because they're often in their own home or workplace. So, what makes this type of research special is that it's not just a matter of sending a questionnaire to the participants. Instead, the research is usually based on first-hand observation of what they are doing at the time. But that doesn't mean that the researcher never talks to the participants. However, unlike in traditional research, in this case, it's the participant rather than the researchers who decides what direction the interview will follow. This means that there's less likelihood of the researcher imposing his or her own ideas on the participant. But after they've said goodbye to their participants and got back to their office, the researcher's work isn't finished. Most researchers estimate that 70 to 80% of their time is spent not on the collecting of data, but on its analysis, looking at photos, listening to recordings, and transcribing them, and so on. The researchers may end up with hundreds of pages of notes. And to determine what's significant, they don't focus on the sensational things or the unusual things. Instead, they try to identify a pattern of some sort in all this data and to discern the meaning behind it. This can result in some compelling insights that can in turn feed back to the whole design process. Good morning, Streutten Festival box office. How can I help you? Oh, hello. My family and I are on holiday in the area, and we've seen some posters about the festival this week. Could you tell me about some of the events, please? First of all, are there still tickets available for the jazz band on Saturday? There are, but only £15. The £12 seats have all been sold. Okay. And the venue is the school, isn't it? Yes, that's right. The secondary school. Make sure you don't go to the primary school by mistake. And there's an additional performer who isn't mentioned on the posters. Carolyn Hart is going to play with the band. Oh, I think I've heard her on the radio. Doesn't she play the oboe or flutes or something? Yes, the flute. She usually plays with symphony orchestras. And apparently this is her first time with a jazz band. Well, I'd certainly like to hear her. Then the next thing I want to ask about is the duck races. I saw a poster beside a river. What are they exactly? Well, you buy a yellow plastic duck or as many as you like. They're a pound each and you write your name on each one. There'll be several races depending on the number of ducks taking part. and John Stevens, a champion swimmer who lives locally, is going to start the races. All the ducks will be launched into the river at the back of the cinema. Then they'll float along the river for 500 m as far as the railway bridge. And are there any prizes? Yes. The first duck in each race to arrive at the finishing line wins its owner free tickets for the concert on the last night of the festival. You said you can buy a duck. I'm sure my children will both want one. They're on sale at a stall in the market. You can't miss it. It's got an enormous sign showing a couple of ducks. Okay, I'll go there this afternoon. I remember walking past there yesterday. Now, could you tell me something about the flower show, please? Well, admission is free and the show is being held in Bithweight Hall. Sorry, how do you spell that? B Y T H W A I T E Bait. Is it easy to find? I'm not very familiar with the town yet. Oh, you won't have any problem. It's right in the center of Streathton. It's the only old building in the town, so it's easy to recognize. I know it. I presume it's open all day.

Yes, but if you'd like to see the prizes being awarded for the best flowers, you'll need to be there at 5:00.

The prizes are being given by a famous actor, Kevin Chapplas.

He lives nearby and gets involved in a lot of community events.

Gosh, I've seen him on TV.

I'll definitely go to the prize giving.

Right.

I've seen a list of plays that are being performed this week, and I'd like to know which are suitable for my children and which ones my husband and I might go to.

How old are your children?

Five and seven.

What about the mystery of Moldun?

That's aimed at 5 to 10 year olds.

So, if I take my children, I can expect them to enjoy it more than I do, I think.

So, if you'd like something for yourself and your husband and leave your children with a babysitter, you might like to see Fire and Flood.

It's about events that really happened in Streton 200 years ago, and children might find it rather frightening.

Oh, thanks for the warning.

And finally, what about Silly Sailor?

That's a comedy, and it's for young and old.

In fact, it won an award in the Streton Drama Festival a couple of months ago.

Okay.

Well, goodbye and thanks for all the information.

I'm looking forward to the festival.

Good morning and welcome to the museum.

One with a remarkable range of exhibits which I'm sure you'll enjoy.

My name is Greg and I'll tell you about the various collections as we go around.

But before we go, let me just give you a taste of what we have here.

Well, for one thing, we have a fine collection of 20th and 21st century paintings, many by very well-known artists.

I'm sure you'll recognize several of the paintings.

This is the gallery that attracts the largest number of visitors, so it's best to go in early in the day before the crowds arrive.

Then there are the 19th century paintings.

The museum was opened in the middle of that century, and several of the artists each donated one work to get the museum started, as it were.

So, they're of special interest to us.

We feel closer to them than to other works.

The sculpture gallery has a number of fine exhibits, but I'm afraid it's currently closed for refurbishment.

You'll need to come back next year to see it properly, but a number of the sculptures have been moved to other parts of the museum.

Around the world is a temporary exhibition.

You've probably seen something about it on TV or in the newspapers.

It's created a great deal of interest because it presents objects from every continent and many countries and provides information about their social context, why they were made, who for, and so on.

Then there's the collection of coins.

This is what you might call a focused specialist collection because all the coins come from this country and were produced between 2,000 and a thousand years ago.

and many of them were discovered by ordinary people digging their gardens and donated to the museum.

All our porcelain and glass was left to the museum by its founder when he died in 1878.

And in the terms of his will, we're not allowed to add anything to that collection.

He believed it was perfect in itself and we don't see any reason to disagree.

Okay, that was something about the collections.

And now here's some more practical information in case you need it.

Most of the museum facilities are downstairs in the basement.

So you go down the stairs here.

When you reach the bottom of the stairs, you'll find yourself in a sitting area with comfortable chairs and sofas where you can have a rest before continuing your exploration of the museum.

We have a very good restaurant which serves excellent food all day in a relaxing atmosphere.

To reach it, when you get to the bottom of the stairs, go straight ahead to the far side of the sitting area.

Then turn right into the corridor.

You'll see the door of the restaurant facing you.

If you just want a snack or if you'd like to eat somewhere with facilities for children, we also have a cafe.

When you reach the bottom of the stairs, you'll need to go straight ahead, turn right into the corridor, and the cafe is immediately on the right.

And talking about children, there are baby changing facilities downstairs.

Cross the sitting area.

Continue straight ahead along the corridor on the left and you and your baby will find the facilities on the left hand side.

The cloak room where you should leave coats, umbrellas, and any large bags is on the left hand side of the sitting area.

It's through the last door before you come to the corridor.

There are toilets on every floor, but in the basement, they're the first rooms on the left when you get down there.

Okay.

Now, if you've got anything to leave in the cloak room, please do that now and then we'll start our tour.

Hi, Joanna.

Good to meet you.

Now, before we discuss your new research project, I'd like to hear something about the psychology study you did last year for your master's degree.

So, how did you choose your subject for that?

Well, I had six subjects, all professional musicians and all female.

Three were violinists and there was also a cello player and a pianist and a flute player.

They were all very highly regarded in the music world and they'd done quite extensive tours in different continents and quite a few had won prizes and competitions as well.

And they were quite young, weren't they?

Yes.

between 25 and 29.

Um, the mean was 27.8.

I wasn't specifically looking for artists who'd produced recordings, but this is something that's just taken for granted these days, and they all had.

Right.

Now, you collected your data through telephone interviews, didn't you?

Yes.

I realized if I was going to interview leading musicians, it would only be possible over the phone because they're so busy.

I recorded them using a telephone recording adapter.

I'd been worried about the quality, but it worked out all right.

I managed at least a 30inut interview with each subject, sometimes longer.

Did doing it on the phone make it more stressful?

I thought it might.

Um, it was all quite informal though, and in fact, they seemed very keen to talk.

And I don't think using the phone meant I got less rich data.

Rather the opposite in fact.

Interesting.

And you were looking at how performers dress for concert performances.

That's right.

Uh my research investigated the way players see their role as a musician and how this is linked to the type of clothing they decide to wear.

But that focus didn't emerge immediately.

When I started, I was more interested in trying to investigate the impact of what was worn on those listening and also whether someone like a violinist might adopt a different style of clothing from say someone playing the flute or the trumpet.

H it's interesting that the choice of dress is up to the individual, isn't it?

Yes.

You'd expect there to be rules about it in orchestras, but that's quite rare.

You only had women performers in your study.

Was that because male musicians are less worried about fashion?

I think a lot of the men are very much influenced by fashion, but in social terms, the choices they have are more limited.

They'd really upset audiences if they strayed away from quite narrow boundaries.

Now, popular music has quite different expectations.

Uh, did you read Mike Frost's article about the dress of women performers in popular music?

No.

Well, he points out that a lot of female singers and musicians in popular music tend to dress down in performances and wear less feminine clothes, um, like jeans instead of skirts.

Uh, and he suggests this is because otherwise they'd just be discounted as trivial.

But you could argue they're just wearing what's practical.

I mean, a pop music concert is usually a pretty energetic affair.

Yes, he doesn't make that point, but I think you're probably right.

I was interested by the effect of the audience at a musical performance when it came to the choice of dress.

The subjects I interviewed felt this was really important.

It's all to do with what we understand by performance as a public event.

They believed the audience had certain expectations and it was up to them as performers to fulfill these expectations to show a kind of esteem.

They weren't afraid of looking as if they'd made an effort to look good.

I think in the past the audience would have had those expectations of one another too, but that's not really the case now.

Not in the UK anyway.

No.

And I also got interested in what sports scientists are doing too with regard to clothing.

Musicians are quite vulnerable physically, aren't they?

Because the movements they carry out are very intensive and repetitive.

So, I'd imagine some features of sports clothing could safeguard the players from the potentially dangerous effects of this sort of thing.

Yes, but musicians don't really consider it.

They avoid clothing that obviously restricts their movements, but that's as far as they go.

Anyway, coming back to your own research, do you have any idea where you're going from here?

I was thinking of doing a study using an audience including as we saw in the last lecture, a major cause of climate change is the rapid rise in the level of carbon dioxide in the atmosphere over the last century.

If we could reduce the amount of CO2, perhaps the rate of climate change could also be slowed down.

One potential method involves enhancing the role of the soil that plants grow in with regard to absorbing CO2.

Ratan Laauo, a soil scientist from Ohio State University in the USA, claims that the world's agricultural soils could potentially absorb 13%

of the carbon dioxide in the atmosphere, the equivalent of the amount released in the last 30 years.

And research is going on into how this might be achieved.

La first came to the idea that soil might be valuable in this way not through an interest in climate change but rather out of concern for the land itself and the people dependent on it.

Carbonrich soil is dark, crumbly and fertile and retains some water.

But erosion can occur if soil is dry which is a likely effect if it contains inadequate amounts of carbon.

Erosion is of course bad for people trying to grow crops or breed animals on that terrain.

In the 1970s and 80s, La was studying soils in Africa so devoid of organic matter that the ground had become extremely hard like cement.

There he met a pioneer in the study of global warming who suggested that carbon from the soil had moved into the atmosphere.

This is now looking increasingly likely.

Let me explain.

For millions of years, carbon dioxide levels in the atmosphere have been regulated in part by a natural partnership between plants and microbes, tiny organisms in the soil.

Plants absorb CO₂ from the air and transform it into sugars and other carbon based substances.

While a proportion of these carbon products remain in the plant, some transfer from the roots to fungi and soil microbes which store the carbon in the soil.

The invention of agriculture some 10,000 years ago disrupted these ancient soil building processes and led to the loss of carbon from the soil.

When humans started draining the natural top soil and plowing it up for planting, they exposed the buried carbon to oxygen.

This created carbon dioxide and released it into the air.

And in some places, grazing by domesticated animals has removed all vegetation, releasing carbon into the air.

Tons of carbon have been stripped from the world's soils where it's needed and pumped into the So, what can be done?

Researchers are now coming up with evidence that even modest changes to farming can significantly help to reduce the amount of carbon in the atmosphere.

Some growers have already started using an approach known as regenerative agriculture.

This aims to boost the fertility of soil and keep it moist through established practices.

These include keeping fields planted all year round and increasing the variety of plants being grown.

Strategies like these can significantly increase the amount of carbon stored in the soil.

So, agricultural researchers are now building a case for their use in combating climate change.

One American investigation into the potential for storing CO₂ on agricultural lands is taking place in California.

Soil scientist Wendy Silva of the University of California, Berkeley, is conducting a firstofits-kind study on a large cattle farm in the state.

She and her students are testing the effects on carbon storage of the compost that is created from waste, both agricultural, including manure and corn stalks, and waste produced in gardens such as leaves, branches, and lawn trimmings.

In Australia, soil ecologist Christine Jones is testing another promising soil enrichment strategy.

Jones and 12 farmers are working to build up soil carbon by cultivating grasses that stay green all year round.

Like composting, the approach has already been proved experimentally.

Jones now hopes to show that it can be applied on working farms and that the resulting carbon capture can be accurately measured.

It's hoped in the future that projects such as these will demonstrate the role that farmers and other land managers can play in reducing the harmful effects of greenhouse gases.

For example, in countries like the United States, where most farming operations use large applications of fertilizer, changing such long-standing habits will require a change of system.

Ratan Laauro argues that farmers should receive payment not just for the corn or beef they produce but also for the carbon they can store in their soil.

Another study being carried out

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Hi, can I help you?

I'd like to find out if you have any excursions suitable for families.

Sure.

How about taking your family for a cruise?

We have a steam ship that takes passengers out several times a day.

It's over a 100 years old.

That sounds interesting.

How long is the trip?

About an hour and a half.

And don't forget to take pictures of the mountains.

They're all around you when you're on the boat and they look fantastic.

Okay.

And I assume there's a cafe or something on board.

Sure.

How old are your children?

Uh my daughter's 15 and my son's seven.

Right.

Well, there are various things you can do once you've crossed the lake to make a day of it.

One thing that's very popular is a visit to the country farm.

You're met off the boat by the farmer and he'll take you to the holding pens where the sheep are kept.

Children love feeding them.

My son would love that.

He really likes animals.

Well, there's also a 40minute trek around the farm on a horse if he wants.

Do you think he'd manage it?

He hasn't done that before.

Sure, it's suitable for complete beginners.

Good.

And again, visitors are welcome to explore the farm on their own as long as they take care to close gates and so on.

There are some very beautiful gardens along the side of the lake which also belong to the farm.

They'll be just at their best now.

You could easily spend an hour or two there.

Okay.

Well, that all sounds good.

And can we get lunch there?

You can, and it's very good, though it's not included in the basic cost.

You pay when you get there.

Right.

So, is there anything else to do over on that side of the lake?

Well, what you can do is take a bike over on the ship and then go on a cycling trip.

There's a trail there called the Back Road.

You could easily spend 3 or 4 hours exploring it, and the scenery is wonderful.

They'll give you a map when you get your ticket for the cruise.

There's no extra charge.

What's the trail like in terms of difficulty?

Quite challenging in places.

It wouldn't be suitable for your 7-year-old.

It needs someone who's got a bit more experience.

H Well, my daughter loves cycling and so do I.

So maybe the two of us could go and my wife and son could stay on the farm.

That might work out quite well.

But we don't have bikes here.

Is there somewhere we could rent them?

Yes, there's a place here in the city.

It's called Ratcherson's.

I'll just make a note of that.

Uh, how do you spell it?

R A T C H E S O N S.

It's just by the cruise ship terminal.

Okay.

You'd also need to pick up a repair kit for the bike from there to take along with you.

And you'd need to take along a snack and some water.

It would be best to get those in the city.

Fine, that shouldn't be a problem.

And I assume I can rent a helmet from the bike place.

Sure, you should definitely get that.

It's a great ride, but you want to be well prepared because it's very remote.

You won't see any shops around there or anywhere to stay.

So, you need to get back in time for the last boat.

Yeah.

So, what sort of prices are we looking at here?

Let's see.

That would be one adult and one child for the cruise with farm tour.

That's \$117.

And an adult and a child for the cruise only.

So that's \$214 altogether.

Oh, wait a minute.

How old did you say your daughter was?

15.

Then I'm afraid it's \$267 because she has to pay the adult fair which is \$75 instead of the child fair which is \$22.

Sorry about that.

That's okay.

Uh so how do we find out?

Good morning everyone.

My name's Joy Parkkins and I'm the restaurant manager.

And I understand that none of you have had any previous experience as kitchen assistants.

Well, you might be feeling a bit nervous now, but most of our kitchen assistants say they enjoy the work.

Okay, they might get shouted at sometimes, but it's nothing personal.

And they're pleased that they have so many different things to do, which means they never get bored.

and I'll tell you straight away that if you do well, we might think about moving you up and giving you some more responsibility.

Right.

Well, you've all shown up on time, which is an excellent start.

Now, I'm glad to see none of you have unsuitable footwear, so that's good.

You need to be careful as the floors can get very wet and slippery.

Those of you with long hair, I've got it well out of the way.

But some of you will need to remove your rings and bracelets, just put them somewhere safe for today.

And remember to leave them at home tomorrow as they can be a safety hazard.

Now, it's going to be a busy day for you all today.

We don't have any tables free for this evening and only a few for lunch.

Fortunately, we've got our head chef back.

He was away on holiday all last week, which meant the other chefs had extra work.

Now, I'll tell you a bit more about the job in a minute.

But first, some general regulations.

For all of you, whatever your age, there's some equipment you mustn't use until you've been properly trained, like the waste disposal system, for example, for health and safety reasons.

Then I think there are two of you here who are under 18.

That's Emma and Jake, isn't it?

Right.

So, for you two, the meat slicer is out of bounds.

And of course, none of you are allowed to use the electric mixer until you've been shown how it works.

Now, you may have heard that this can be a stressful job, and I have to say that can be true.

You'll be working an 8-hour day for the first week, though you'll have the chance to do overtime after that as well if you want to.

But however long the hours are, you'll get a break in the middle.

What you will find is that you're on your feet all day long lifting and carrying.

So, if you're not fit now, you soon will be.

You'll find you don't have much chance to take it easy.

When someone tells you to do something, you need to do it straight away.

But at least we do have a very efficient air conditioning system compared with some kitchens.

Now, let me tell you about some of the people you need to know.

So, as I said, I'm Joy Parkkins, and I decide who does what during the day and how long they work for.

I'll be trying to get you to work with as many different people in the kitchen as possible so that you learn while you're on the job.

One person whose name you must remember is David Field.

If you injure yourself at all, even if it's really minor, you must report to him and he'll make sure the incident is recorded and you get the appropriate treatment.

He's trained to give basic treatment to staff himself or he'll send you off somewhere else if necessary.

Then there's Dexter Wills.

He's the person you need to see if you smash a plate or something like that.

Don't just leave it and hope no one will notice.

It's really important to get things noted and replaced or there could be problems later.

And finally, there's Mike Smith.

He's the member of staff who takes care of all the stores of perishables.

So, if you notice we're getting low in flour or sugar or something, make sure you let him know so he can put in an order.

Okay.

Now, the next thing we need to call Okay, Stuart, we need to start planning our paper on public libraries.

Have you thought of an angle yet?

Well, there's so much we could look into.

How libraries have changed over the centuries, for instance, or how different countries organize them.

What do you think, Trudy?

Maybe we should concentrate on this country and try and relate the changes in libraries to external developments, like the fact that far more people can read than a century ago, and that the local population may speak lots of different languages.

We could include something about changes in the source of funding, too.

Yes, but remember we're only supposed to write a short paper, so it's probably best if we don't go into funding in any detail.

Right.

Well, should we just brainstorm a few ideas to get started?

Okay.

We obviously need to look at the impact of new technology, particularly the internet.

Now that lots of books have been digitalized, people can access them from their own computers at home.

And if everyone did that, libraries would be obsolete.

Yes.

But the digitalized books that are available online for free are mostly out of copyright, aren't they?

And copyright in this country lasts for 70 years after the author dies.

So you won't find the latest bestseller or upto-date information.

That's an important point.

Anyway, I find it hard to concentrate when I'm reading a long text on a screen.

I'd much rather read a physical book and it takes longer to read on a screen.

Oh, I prefer it.

I suppose it's just a personal preference.

I expect the libraries will go on evolving in the next few years.

Some have already become centers where community activities take place, like local clubs meeting there.

I think that'll become even more common.

I'd like to think so, and that they'll still be serving their traditional function, but I'm not so sure.

There are financial implications after all.

What I'm afraid will happen is that books and magazines will all disappear and they'll just be rows and rows of computers.

They won't look anything like the libraries we're used to.

Well, we'll see.

I've just had an idea.

Why don't we make an in-depth study of our local public library as background to our paper?

Yes, that'd be interesting and raise all sorts of issues.

Let's make a list of possible things we could ask about, then work out some sort of structure.

For instance, um we could interview some of the staff and find out whether the library has its own budget or if that's controlled by the local council and what their policies are.

I know they don't allow food, but I'd love to find out what types of noise they ban.

There always seems to be a lot of talking, but never music.

I don't know if that's a policy or it just happens.

I've often wondered.

Then there are things like how the library is affected by employment laws.

I suppose there are rules about working hours, facilities for staff and so on, right?

Then there are other issues relating to the design of the building and how customers use it, like what measures does the library take to ensure their safety.

They'd need floor coverings that aren't slippery and emergency exits for instance.

Oh, and another thing, there's the question of the kind of insurance the library needs to have in case anyone gets injured.

Yes, that's something else to find out.

You know, something I've often wondered.

What's that?

Well, you know, they've got an archive of local newspapers going back years.

Well, next to it, they've got the diary of a well-known politician from the late 19th century.

I wonder why it's there.

Do you know what his connection was with this area?

No idea.

Let's add it to our list of things to find out.

Oh, I've just thought, you know, people might ask in the library about local organizations like sports clubs.

Well, I wonder if they keep a database or whether they just look online.

Right.

I quite fancy finding out what the differences are between a library that's open to the public and one that's part of a museum, for example. They must be very different.

Then something else I'd like to know is in public discussion of business, we take certain values for granted.

Today I'm going to talk about four of them.

collaboration, hard work, creativity, and excellence.

Most people would say they're all good things.

I'm going to suggest that's an oversimple view.

The trouble with these values is that they're theoretical concepts removed from the reality of day-to-day business.

Pursue values, by all means, but be prepared for what may happen as a result.

They can actually cause damage which is not at all the intention.

Business leaders generally try to do the right thing but all too often the right thing backfires if those leaders adopt values without understanding and managing the side effects that arise.

The values can easily get in the way of what's actually intended.

Okay.

So the first value I'm going to discuss is collaboration.

Uh let me give you an example.

On a management training course I once attended, we were put into groups and had to construct a bridge across a stream using building blocks that we were given.

The rule was that everyone in the team had to move at least one building block during the construction.

This was intended to encourage teamwork, but it was really a job best done by one person.

The other teams tried to collaborate on building the structure and descended into confusion with everyone getting in each other's way.

Our team leader solved the challenge brilliantly.

She simply asked everyone in the team to move a piece a few cm to comply with the rule and then let the person in the team with an aptitude for puzzles like this build it alone.

We finished before any other team.

My point is that the task wasn't really suited to teamwork.

So why make it one?

Teamwork can also lead to inconsistency, a common cause of poor sales.

In the case of a smartphone that a certain company launched, one director wanted to target the business market and another demanded it was aimed at consumers.

The company wanted both directors to be involved, so gave the product a consumer-friendly name, but marketed it to companies.

The result was that it met the needs of neither group.

It would have been better to let one director or the other have his way, not both.

Now, industriousness or hard work.

It's easy to mock people who say they work hard.

Uh after all, a hamster running around in a wheel is working hard and getting nowhere.

Of course, hard work is valuable, but only when properly targeted.

Otherwise, it wastes the resources that companies value most, time and energy, and that's bad for the organization.

There's a management model that groups people according to four criteria.

clever, hardworking, stupid, and lazy.

Here, lazy means having a rational determination not to carry out unnecessary tasks.

It doesn't mean trying to avoid work altogether.

Most people display two of these characteristics, and the most valuable people are those who are both clever and lazy.

They possess intellectual clarity and they don't rush into making decisions.

They come up with solutions to save the time and energy spent by the stupid and hardworking group.

Instead of throwing more man-hour at a problem, the clever and lazy group looks for a more effective solution.

Next, we come to creativity.

This often works well.

Creating an attentiongrabbing TV commercial, for example, might lead to increased sales, but it isn't always a good thing.

Some advertising campaigns are remembered for their creativity without having any effect on sales.

This happened a few years ago with the launch of a chocolate bar.

Subsequent research showed that plenty of consumers remembered the adverts but had no idea what was being advertised.

The trouble is that the creator derives pleasure from coming up with the idea and wrongly assumes the audience for the campaign will share that feeling.

A company that brings out thousands of new products may seem more creative than a company that only has a few, but it may be too creative and make smaller profits.

Creativity needs to be targeted to solve a problem that the company has identified.

Just coming up with more and more novel products isn't necessarily a good thing.

And finally, excellence.

We all know companies that claim they strive for excellence, but it takes a long time to achieve excellence.

In business, being first with a product is more profitable than having the best product.

A major study of company performance compared pioneers, that is companies bringing out the first version of a particular product with followers, the companies that copied and improved on that product.

The study found that the pioneers commanded an average market share of 29%.

while the followers achieved less than half that only 13% even though their product might have been better.

Insisting on excellence in everything we do is time consuming, wastess energy and leads to losing out on opportunities.

Sometimes second rate work is more worthwhile than excellence.

Make sure it's excellent.

Sounds like a good approach to business.

But the just get started approach is likely to be more successful.

Good morning, Kenton Festival box office.

How can I help you?

Oh, good morning.

I'm coming to Kenton for a few days holiday next month and a friend told me there's a festival.

She gave me this number to find out about it.

That's right.

The festival begins on the 16th of May and goes on till the 19th.

Oh, that's great.

I'll be there from the 15th till the 19th.

So, could you tell me the program, please?

Well, on the first day, there's the opening ceremony in the town center.

People start gathering around 2:00 to get a good place to see from, and the events will start at 2:45 and finish about 5:30.

Okay.

Thanks.

I'll make sure I get there early to get a good spot.

The festival will be officially opened by the mayor.
He'll just speak for a few minutes welcoming everyone to the festival.
All the town councillors will be there and of course lots of other people.
Then there'll be a performance by a band.
Most years we have a children's choir, but this year the local army cadetses offered to perform and they're very good.
Uhhuh.
After that, a community group from the town will perform a play they've written themselves.
Just a short one.
It's about Helen Tugate.
I don't know if you've heard of her.
I certainly have.
She was a scientist years ago.
That's right.
She was born in Kenton exactly 100 years ago.
So, we're celebrating her centenary.
I'm a biologist, so I've always been interested in her.
I didn't realize she came from Kenton.
Yes.
Well, all that will take place in the afternoon.
And later, as the sun sets, there'll be a firework display.
You should go to the park to watch as you'll get the best view from there.
And the display takes place on the opposite side of the river.
It's always one of the most popular events in the festival.
Sounds great.
And what's happening on the other days?
There are several events that go on the whole time.
For example, the students of the art college have produced a number of videos all connected with relationships between children and their grandparents.
That sounds interesting.
It makes a change from children and parents, doesn't it?
Exactly.
Because the art college is in use for classes throughout the festival, the videos are being shown in Hansworth House.
How do you spell the name?
H A N D S W O R T H.
Hansworth House.
It's close to the town hall.
Right now, let me see.
What else can I tell you about?
Are there any displays of ballet dancing?
I'm particularly interested in that as I do it as a hobby.
There isn't any ballet, I'm afraid, but there'll be a demonstration of traditional dances from all around the country.
Well, that would be nice.
Where is that being held?
It's in the market in the town center.
the outdoor one, not the covered market.
And it's on at 2 and 5 every afternoon of the festival, apart from the first day.
Lovely.
I'm interested in all kinds of dancing, so I'm sure I'll enjoy that.
M I'm sure you will.
And I'd really like to go to some concerts if there are any.
Yes, there are several.
Three performed by professionals and one by local children.
And where is it being held?
It's in the library which is in Park Street on the 18th at 6:30 in the evening.
I presume I'll need tickets for that.
Yes, you can book online or you can buy them when you arrive in Kenton, either at the festival box office or from any shops displaying our logo in the windows.
Well, I think that'll keep me busy for the whole of my stay in Kenton.
Thank you so much for all your help.
You're welcome.
I hope you enjoy your stay.
Thank you.
Goodbye.
Right, I've now almost succeeded in finalizing plans for our tour, so I'll bring you up to date with what I know.
As you know, we're flying first to Munich on Monday the 4th.
The flight is at 11:30, so it's too early to have lunch at the airport.
I suggest we meet there for coffee at 10:00, which should give us plenty of time for breakfast before we leave home.
When we arrive in Munich, we'll be met at the airport by Klaus Bower.
Klouse works for a tour operator, and he'll look after us for the time we'll be in Germany.
He's already liaised with the managers of the theaters we're going to visit and he's also arranged for an officer of the National Theater in

Munich to show us around the theater one afternoon during our stay.

Now, last time we discussed this trip, I didn't have the precise cost for hotel rooms, but now I have.

The normal rate at the hotel where we're staying is €150 a night for a double room.

I'd hope to get that down to €120, but in fact, I've been able to negotiate a rate of 110.

That'll be reflected in the final payment which you'll need to make by the end of this week.

On Tuesday, the day after our arrival, I had hoped we could sit in on a rehearsal at one of the theaters, but unfortunately, that's proved very difficult to arrange.

So, instead, we'll have a coach trip to one of the amazing castles in the mountains south of Munich.

On Tuesday evening, we'll all have dinner together in a restaurant near our hotel.

From talking to you all about your preferences, it was clear that a typical local restaurant would be too meat oriented for some of you.

Some of you suggested an Italian restaurant, but I must confess that I decided to book a Lebanese one as we have plenty of opportunities to go to an Italian restaurant at home.

On Wednesday afternoon, the director of the play we're going to see that evening will talk to us at the theater.

She'll describe the whole process of producing a play, including how she chose the actors and as the play we're going to see is a modern one, how she worked with the playwright.

Right now, I'd just like to make a few points about the plays we're going to see.

Partly because it might influence your choice of clothes to take with you.

The play we're seeing on Wednesday evening is a modern one and we're going to the premiere, so it'll be quite a dressy occasion.

Though, of course, you don't have to dress formally.

I gather it's rather a multimedia production with amazing lighting effects and a soundtrack of electronic music, though unfortunately the playwright is ill and is unlikely to be able to attend.

On Thursday, we're seeing a play that was first performed last year when it was commissioned to mark a hundred years since the birth in the town of a well-known scientist.

We're going to see a revival of that production, which aroused a lot of interest.

Friday's play will really make you think hard about what clothes to pack, as it'll be in the garden of a palace.

It's a beautiful setting, but I'd better warn you, there won't be much protection from the wind.

On Saturday, we're going by coach to a theater in another town not far from Munich.

This will be the opening of a drama festival, and the mayor and all the other dignitaries of the town will be attending.

After the performance, the mayor is hosting a reception for all the audience and there'll be a band playing traditional music of the region.

And after having a day off on Sunday, our final play is on Monday, and it's in the stunning setting of the old town hall, which dates back to the 14th century.

The performance marks the 50 years that the lead actor has been on stage and the play is the one where he made his first professional appearance all those years ago.

And the day after that we'll be flying back home.

Now, have you got any questions before I Oh, good morning.

Uh, you must be James.

I'm Beth Cartwright.

Please call me Beth.

Thank you.

Now, as this is your first tutorial since you started on the Scandinavian studies course, I'd like to find out something about you.

Why did you decide to take this course?

Well, my mother is Danish, and although we always lived in England, she used to talk about her home a lot, and that made me want to visit Denmark.

We hardly ever did, though.

My mother usually went on her own.

But whenever her relations or friends were in England, they always came to see us.

I see.

So I assume you already speak Danish, one of the languages you'll be studying.

I can get by when I talk to people, though I'm not terribly accurate.

Now, you probably know that you'll spend the third year of the course abroad.

Have you had any thoughts about that?

I'm really looking forward to it.

And although Denmark seems the obvious place to go because of my family connections, I'd love to spend the time in Iceland.

Oh, I'm sure it can be arranged.

Do you have any plans for when you graduate?

A lot of students go on to take a master's degree.

I think the four years of the undergraduate course will be enough for me.

I'm interested in journalism and I quite like the idea of moving to Scandinavia and writing for magazines.

I'd find that more creative than translating, which I suppose most graduates do.

Okay.

Now, how are you finding the courses you're taking this term, James?

Well, I'm really enjoying the one on Swedish cinema.

That'll continue next term, but the one on Scandinavian literature that's running at the moment will be replaced by more specialized courses.

Oh, and by the way, if you're interested in watching Danish television programs, there's going to be a course on that the term after next.

That sounds good.

Have you started thinking about the literature paper that you have to write in the next few weeks?

Yes.

My first choice would be to do something on the Icelandic sagas.

It's the trouble with that is that a lot of people choose that topic and it can be difficult to get hold of the books you'll need.

Why not leave that for another time?

You might find modern novels or 19th century playwrights interesting.
I've read or seen several plays in translation, so that would be a good idea.
Fine, I'll put you down for that topic.

Right.

So, what would you advise me to aim at in the paper?

First, I suggest you avoid taking one writer and going into a great deal of detail.

That approach certainly has its place, but I think you first need to get an understanding of the literature in the context of the society in which it was produced, who it was written for, how it was published, and so on.

I also think that's more fruitful than placing it within the history of the genre.

Okay, that sounds reasonable.

Could I ask for some advice about writing the paper I'm working on about the Vikings?

I have to do that this week, and I'm a bit stuck.

Of course.

Have you decided yet what to write about?

No, I haven't.

There's so much that seems interesting.

Viking settlement in other countries, trade, mythology.

Well, what I suggest is that you read an assignment a student wrote last year, which is kept in the library.

It's short and well focused, and I'm sure you'll find it helpful.

I'll give you the details in a moment.

Textbooks usually cover so many topics, it can be very difficult to choose just one.

Okay.

I've got a DVD of the film about the Vikings that came out earlier this year.

Should I watch that again?

If it's the one I am thinking of, I'd ignore it.

It's more fantasy than reality.

But I've got a recording of a documentary that you should watch.

It makes some interesting and provocative points which I think will help you to focus your topic.

Right.

So then should I work out an outline?

Yes, just headings for different sections at this stage.

And then you should start looking for suitable articles and books to draw on and take notes which you organize according to those headings.

I see.

Then put short phrases and sentences as bullet points under each heading.

Make sure that this skeleton makes sense and flows properly before writing up the paper in full.

Okay.

Thanks.

That's very helpful.

Over the years, attitudes towards workers have changed considerably.

After all, there was a time when workers had no rights at all and labored in appalling conditions.

Conditions have improved a lot, but conflict in the workplace is still common and human resources managers nowadays need to be able to deal with it when necessary.

What is conflict in the workplace?

Definitions vary, but I'm taking it to refer to a whole range of behaviors that the victim finds unacceptable.

From minor, harmless arguments to, at the opposite, extreme physical violence.

Much of this is covered by the term bullying.

By which I mean one or more people behaving abusively or aggressively against another who is in a weaker position.

Although all behavior like this is a form of conflict, not all conflict can be described in these terms.

As with all human behavior, there are numerous reasons for it, but often it's caused by someone who feels the need to show their superiority over someone else in order to feel that they aren't at the lowest level in a hierarchy or a group of people.

In some cases, one person simply dislikes the other on the basis that the personality of one is in some way incompatible with that of the other person.

A general habit of optimism in one person could make them intolerant of a colleague who's constantly pessimistic.

Not that that justifies treating them badly.

Of course, some conflicts arise when people are more interested in promoting themselves and their team than in the company as a whole.

These conflicts are called structural and could come about for example when a sales team believe they are the only people in the business who do any useful work and look down on behind the scenes administrators.

Conflict obviously affects the individuals concerned.

The situation is likely to be very stressful for victims, resulting in their absence from work, possibly for months.

For the company, if no effort is made to deal with conflict, it can spiral out of control and even lead to the breakdown of the business.

Some interesting work with chief executives, CEOs, has uncovered some of the reasons why they may treat colleagues badly.

Many CEOs combine two opposing characteristics.

confidence.

That is the belief that they're capable of great achievements with a high level of anxiety, a fear of missing targets, whether set by themselves or by the directors of the company.

This combination can make them respond badly to anyone who questions their decisions in a high pressure work environment.

Such characteristics become problematic and it's particularly difficult to tackle the situation where colleagues, managers, and board members are all trying to achieve their own visions.

When they can't agree on strategic issues and on where they see the business going, there are real problems. For managers at lower levels within the organization, it might seem that an autocratic form of management where the chief executive gives orders and everyone else has to obey would see more conflict than others. Interestingly though, a company with a more democratic business model can suffer more when uncertainty about who to report to leads to conflicting demands.

Now, I'll say a little about dealing with the type of conflict that has harmful effects. Of course, the ideal is to prevent it arising in the first place. A good manager at any level will make efforts to earn the respect of the people they work with, particularly those who report to them. That will involve politeness in all communications and treating them as equals who happen to have a different role within the organization. Sometimes of course conflict does occur and can get out of hand. In such cases the human resources department often gets involved. However, if one of the parties in a conflict sees human resources as simply a mouthpiece for the chief executive, then an external mediator might be able to help. By talking to both sides and trying to find the truth of what's been happening, they can build a clear picture of the situation and give feedback that both sides will accept precisely because they're independent.

Hello.

Hi, Susy.

It's Paul here.

How are you enjoying your new job?

You're working at the library, aren't you?

Yes.

I started when the library reopened a month ago.

It's great.

Actually, Carol and I have been meaning to join for a while.

Oh, you should.

It doesn't cost anything.

And the new library has all sorts of facilities.

It's not just a place where you borrow books.

For instance, there's an area with comfortable seats where you can sit and read the magazines they have there.

Some people spend their whole morning there.

Wish I had that amount of time to spend.

Yes.

You must be pretty busy at present with the children and everything.

We are.

Yes.

But we're hoping to get away this summer.

We're thinking of going to Greece.

Well, we've got a much larger section of the library devoted to travel books now.

So, you should come and have a look.

I can't remember if there's anything specifically on Greece, but I should think so.

Okay.

Now, Carol's organizing a project for the history class she teaches at school.

It's about life in the town 100 years ago.

Do you have anything that might be useful?

Yes, actually.

We've now got a new section with materials on the history of the town and surrounding region.

Right.

I'll tell her.

You can't always find that sort of thing on the internet.

Now, in the old library, there used to be a separate room with reference books.

It was a really nice, quiet room.

Yes, we've put those books in the main part of the library now, but we do have a room called the community room.

It can be hired out for meetings, but at other times, people can use it to study.

I might use that.

It's hard to find anywhere quiet at home sometimes.

I can't remember how old your son and daughter are.

We've introduced a special section of fiction written specially for teenagers, but they might be a bit young for that.

Yes, they would be.

Well, we do have lots of activities for younger children.

Yes.

For example, we have a science club.

At the next meeting, they're going to be doing experiments with stuff that everyone has in the kitchen, sugar and flour and so on.

They might be interested.

Yes.

And we have a competition for children called reading challenge that doesn't begin until after the end of term.

They have to read six books and they get a certificate if they manage it.

So that gives them something to do while they're on holiday instead of getting bored.

That's the idea.

And there's special activities for adults, too.

On Friday, we have a local author called Tanya Stre who's going to be talking about her new novel.

It's called Catch the Mouse, and she based the story on a crime that actually took place here years ago.

Right.

We're not free on Friday, but I'll look out for the book.

Now, this probably isn't for you, but we do have IT support available for members.

We get quite a few older people coming along who are wanting to get up to speed with computer technology.

It's on Tuesday mornings.

They don't need to make an appointment or anything.

They just turn up.

Well, my mother might be interested.

I'll let her know.

And there's another service which you wouldn't expect from a library which is a free medical checkup.

The hospital arranges for someone to come along and measure the level of sugar in your blood and they check cholesterol levels at the same time.

Really?

Yes.

But that's only for the over 60s.

So you wouldn't qualify.

Okay.

Well, I'll tell my mother.

She might be interested.

What other information?

Well, we do have a little shop with things like wall charts and greetings cards and also stamps, so you can post the cards straight away, which is really useful.

Yeah.

Well, I'll bring the children round at the weekend and we'll join.

Oh, one more thing.

I'll be bringing the car.

Is there parking available?

Yes, and it's free in the evening and at weekends.

Perfect.

Well, thanks Susie.

See you soon.

In this session in your training day, we're going to look at some of the more specialized holidays we offer at BC Travel.

Now, the travel business is very competitive and it's important to be aware of how the markets changing and developing.

In terms of age groups, the over 65s are an important market and one that's increasing steadily year on year.

The fewest holidays are taken by the 31 to 42 year olds and that figure shows no sign of rising.

The biggest market at present is still the youngest group, the 16 to 30s, but this group's also seen the biggest drop over the last few years.

Whereas there's a noticeable growth in the number of holidays taken by the 55 to 64 year olds.

As far as the 43 to 54 year olds are concerned, bookings there are steady, but I have to say we haven't seen the increase we expected.

One trend we're noticing with nearly all age groups is the growing popularity of holidays in which clients do some kind of specialized activity.

I'm not talking here about adventure holidays where clients take part in high-risk activities like white water rafting just for the thrill of it.

Activity holidays usually involve rather less high-risk sports or things like art and music.

They're not necessarily cheaper than ordinary holidays, often the opposite, in fact, but they do often take place outside the main tourist centers, which gives an opportunity for clients to find out more about the local people and customs, and many say this is one of the most positive features of these holidays.

Of course, they offer the chance to develop a new skill or talent, but clients often say that more than this, it's the chance to create lasting relationships with other like-minded people.

That's the main draw.

Let me give you some examples of BC travel activity holidays.

Our painting holidays take place in four different centers in France and Italy and they're very popular with clients of all abilities from beginners onwards.

We've got an excellent team of artists to lead the classes.

Some of them have been with us from the start and five additional ones will be joining us this year so that we can offer a greater number of classes in each center.

As far as cooking holidays are concerned, I know a lot of agents offer holidays where clients cook recipes related to one particular country, usually the one they're staying in.

But we focus on dishes from a great many different ones.

Apart from that, you'll find the usual emphasis on good quality organic ingredients.

That's more or less a given nowadays.

And there are generally some meat-free recipes included.

Our photography holidays take place in a wide range of countries from Iceland to Vietnam, and clients have the opportunity to see some stunning scenery.

Groups are small, no more than eight, so clients can have one-on-one tuition during the holiday, and excursions are arranged with fully trained guides.

At the end of each holiday, an exhibition is held of the photographs taken so that clients can see one another's work and receive valuable feedback from the tutor.

Finally, let me tell you about our fitness holidays.

In Ireland and Italy, we run one general fitness classes for all ages and levels of fitness.

Clients start the course with a consultation with a trainer and together they draw up an individual program. As well as improving general fitness, clients find that they end up losing much of the stress they've built up in their daily lives. In Greece, we have a two-week holiday for clients who want to do something about their weight. This has all the features you'd expect, like a personalized diet program. But one of its most popular features is that the exercise classes are all held on the beach. People say it's far preferable to being in a gym. Finally, we offer several holidays in Morocco. One very popular one is the mountain biking holiday. Bikes are provided and there are different routes according to people's ability. We offer one which is tailored to the needs of families which is particularly popular.

Okay.

So that's about all the time I have today.

So thank you very much.

Dave, I'm worried about our case study.

I've done a bit of reading, but I'm not sure what's involved in actually writing a case study.

I missed the lecture where Dr. Baker talked us through it.

Okay.

Well, it's quite straightforward.

We've got our focus.

That's tourism at the Horton Castle site.

And you said you'd done some reading about it.

Yes.

I found some articles and made notes of the main points.

Did you remember to keep a record of where you got the information from?

Sure.

I know what a pain it is when you forget that.

Okay.

So, we can compare what we've read.

Then we have to decide on a particular problem or need at our site and then think about who we're going to interview to get more information.

Okay.

So, who's that be?

The people who work there and presumably some of the tourists, too.

Yes, both those groups.

So, we'll have to go to the sites to do that, I suppose.

But we might also do some of our interviewing away from the site.

We could even contact some people here in the city like administrators involved in overseeing tourism.

Okay.

So, we'll need to think about our interview questions and fix times and places for the meetings.

It's all going to take a lot of time.

And if we can, we should ask our interviewees if they can bring along some numerical data that we can add to support our findings and photographs.

I think we have plenty of those already, but Dr. Baker also said we have to establish with our interviewees whether we can identify them in our case study or whether they want to be anonymous.

Oh, I wouldn't have thought of that.

Okay, once we've got all this information, I suppose we have to analyze it.

Yes.

Put it all together and choose what's relevant to the problem we're focusing on and analyze that carefully to find out if we can identify any trends or regularities there.

That's the main thing at this stage rather than concentrating on details or lots of facts.

Okay.

And then once we've analyzed that, what next?

Well, then we need to think about what we do with the data we've selected to make it as clear as possible to our readers. Things like graphs or tables or charts, right?

Then the case study itself is mostly quite standard.

We begin by presenting the problem and giving some background, then go through the main sections.

But the thing that surprised me is that in a normal report, we'd end with some suggestions to deal with the problem or need we identified. But in a case study, we end up with a question or a series of questions to our readers and they decide what ought to be done.

Oh, I hadn't realized that.

So basically the problem we're addressing in our case study of the Horton Castle site is why so few tourists are visiting it. And we'll find out more from our interviews.

But I did find one report on the internet that suggested that one reason might be because as far as transport goes, access is difficult. I read that too.

But that report was actually written 10 years ago when the road there was really bad, but that's been improved now.

And I think there's plenty of fascinating stuff there for a really good day out, but you'd never realize it from the castle website. Maybe that's the problem.

Yes, it's really dry and boring.

I read somewhere a suggestion that what the castle needs is a visitor center.

So, we could have a look for some information about that on the internet.

What would we need to know?

Well, who'd use it for a start?

It'd be good to know what categories the visitors fell into, too, like school parties or retired people, but I think we'd have to talk to staff to get that Okay.

And as we're thinking of suggesting a visitor center, we'd also have to look at potential problems.

I mean, obviously it wouldn't be cheap to set up.

No, but it could be a really good investment.

And as it's on a historical site, it would need to get special planning permission, I expect.

That might be hard, right?

Especially as the only possible place for it would be at the entrance, and that's right in front of the castle.

But it could be a good thing for the town of Horton.

At present, it's a bit of a ghost town.

Once they've left school and got any skills or qualifications, the young people all get out as fast as they can to get jobs in the city.

And the only people left are children and those who've retired, Something else we could investigate would be the potential damage that tourists might cause to the castle site.

I mean, their environmental impact.

At present, the tourists can just wander around wherever they want.

But if numbers increase, there might have to be some restrictions like sticking to marked ways and there'd need to be guides and wardens around to make sure these are enforced.

Yes, we could look at that, too.

Okay.

Well, we certainly Okay, so we've been looking at how man-made changes in our environment can affect wildlife.

Now I'll discuss a particular example.

Let's take a look at Mercury.

Mercury is one of the 120 or so elements that make up all matter and it has the symbol HG.

It's a shiny silvery substance.

You may have seen it in old-fashioned thermometers, but it's not used much for domestic purposes now because it's highly toxic.

But the problem is that the amount of mercury in the environments increasing.

The main reason for this is the power plants used to produce electricity.

The main source of energy that most of them use is still coal.

And when it's burned, it releases mercury into the atmosphere.

Some of this gets deposited into lakes and rivers.

And if it's ingested by a fish, it's not excreted.

It stays in the fish's body and it enters the food chain.

So, it's been known for some time that birds which eat fish may be affected.

But what wasn't known until quite recently is that those that eat insects can also be affected.

So a woman called Claire Varian Ramos is doing some research on how this is affecting birds.

And rather than looking at how many birds are actually killed by mercury poisoning, she's looking for more subtle sub effects.

And these may be to do with the behavior of the birds or with the effect of mercury on the way their brain works.

So whether it leads to problems with memory, for example.

and she's particularly focusing on the effects of mercury on bird song.

Now, the process of song learning happens at a particular stage in the bird's development.

And what you may not know is that a young bird seems to acquire this skill by listening to the songs produced by its father rather than by any other bird.

And Varian Ramos has already found in her research that if young male birds are exposed to mercury, if they eat food contaminated with mercury, then the songs they produce aren't as complex as those produced by other birds.

So quite low-level exposure to mercury is likely to have an impact on male birds in a natural situation because it can mean that they're less attractive to female birds and so it can affect their chances of reproduction.

Now the way she's carrying out this research is worth thinking about.

She's using a mixture of studies using birds kept in laboratories and studies carried out outdoors in the wild.

The lab-based studies have the advantage that you don't get all the variables you would in a natural setting.

So, the experimentter has a much higher level of control and that means they can be more confident about their results in some ways.

And of course, they don't have to worry about going out and finding the birds in order to observe them.

So, what are the implications here for humans?

Well, because many birds are migratory, they may be transporting mercury far from contaminated sites.

For example, it's been found that ducks who'd been feeding at a contaminated site were later shot by hunters over a thousand kilometers away and presumably eaten.

But these birds likely had mercury levels high enough to warrant concern for human consumption.

In addition, going back to song learning by birds, we saw that this may be affected by mercury contamination.

Well, we also know that in humans, mercury causes developmental delays in the acquisition of language.

And in fact, this process is very similar in the brain regions it involves and even the genes that are involved.

But mercury contamination has other important implications for humans as well.

It's now known that an unborn child can be affected if the food eaten by its mother contains high levels of mercury.

And these effects can be quite substantial.

In the end, it comes down to whether more value is placed on human economic well-being or environmental well-being.

It's true there are new regulations for mercury emissions from power plants, but these will need billions of dollars to implement and increase costs for everyone.

Some argue that's too much to pay to protect wildlife, but as we've seen, the issues go beyond that, and I think it's an issue we need to consider very carefully.

Hello, Pemrook Cycling Holidays.

Bob speaking.

Oh, hello.

I've seen your advert for people to lead cycle trips.

Are you the right person to speak to?

Yes, I am.

Could I have your name, It's Margaret Smith.

Are you looking for a permanent job, Margaret?

No, temporary.

I've got a permanent job starting in a few months time, and I want to do something else until then.

What work do you do?

This will probably sound crazy.

I used to be a lawyer and then I made a complete career change and I'm going to be a doctor.

I've just finished my training.

Right.

And have you had any experience of leading cycle trips?

Yes.

I've led several bike tours in Africa.

The trip to India that I had arranged to lead next month has now been cancelled.

So when I saw you were advertising for tour leaders, I decided to apply.

Okay.

Now, we normally have two or three leaders on a trip, depending on the size of the group.

Some tours are for very experienced cyclists, but we've got a tour coming up soon in Spain, which is proving so popular, we need an additional leader.

It's a cycling holiday for families.

Would that suit you?

It certainly would.

I enjoy working with children, and I probably need some more experience before I go on a really challenging trip.

That tour includes several teenagers.

Have you worked with that age group before?

Yes, I'm a volunteer worker in a youth club where I help people to improve their cycling skills.

Before that, I helped out in a cycling club where I taught beginners.

Well, that's great.

Now, the trip I mentioned is just for a fortnight, but there might be the possibility of leading other tours after that.

Would that fit in with your plans?

That would be fine.

I'll be free for 5 months.

My job is due to start on October the 2nd and I'm available from May the 1st until late September.

Good.

Now, is there anything I need to know about the food you eat?

We usually have one or two people in the group who don't eat meat or have some sort of food allergy, so we're always very careful about that.

Yes, I'm allergic to cheese.

Would that be a problem?

No.

As long as we have enough notice, we can deal with that.

That's great.

It sounds really interesting.

Would you like me to fill in an application form?

Yes, please.

Where should I post it to?

Could you send it to 27 RBN place?

A R B U T H N O T Place Dumpries.

And what's the post code, please?

DG74PH.

Was that Pa or B Bravo?

Pa.

Got that?

If you could return the application form by Friday this week, we can interview you on Tuesday next week, say 2, would that be possible for you?

Yes, it's fine.

You're quite a long way from where I live, so I'll drive over on Monday.

Should I bring anything to the interview?

We'll have your application form, of course, but we'll need to see any certificates you've got that are relevant in cycling, first aid, or whatever.

Okay.

And at the interview, we'd like to find out about your experience of being a tour guide.

So, could you prepare a 10-minute talk about that, please?

You don't need slides or any complicated equipment, just some notes.

Right.

I'll start thinking about that straight away.

Good.

Well, we'll look forward to receiving your application form and we'll contact you to confirm the interview.

Thanks very much.

Thank you, Margaret.

Goodbye.

Bye.

Welcome to this podcast about the sheep market, which is one of the oldest parts of the city.

As its name suggests, there was originally a market here where farmers brought their sheep, but now it's been redeveloped into a buzzing, vibrant area of the city, which is also home to one of the city's fastest growing communities.

The nearby university has always meant the areas popular with students who come in to enjoy the lively night life.

But now graduates embarking on careers in the worlds of fashion and design are buying up the new apartments recently built here to replace the small houses where the market workers used to live.

The narrow old side streets are great places for finding original pictures, jewelry, and ceramics which won't break the bank, as well as local produce like fruit and vegetables.

There's also lots of pavement cafes where you can have a coffee and watch tourists from all over the world go by.

The oldest buildings in the area are on the main streets, including the city's first department store built in the 1880s, which is still open today.

The sheet market is a center for fashion, and there's a policy of encouraging new young designers.

The young fashion competition is open to local young people who are passionate about fashion.

This year, they've been asked to design an outfit based on ideas from the music and technology that's part of their everyday life using both natural and man-made fibers.

The garments will be judged by a panel of experts and fashion designers, and the winning entries will be modeled at a special gala evening.

Parking at the sheet market is easy.

There are plenty of pay and display car parking spaces on the roadsides, which are fine if you just want to stay for an hour or two.

But if you want to spend the day there, it's better to park in one of the four underground car parks.

It's not expensive, and if you can present a receipt from one of the local stores, you'll not be charged at all.

After 6:00 p.m., many of the car parks have a flat rate, which varies, but it is usually very reasonable.

The Sheep Market is one of the main centers for art and history in the whole of the country.

If you look at our map, you'll see some of the main attractions there.

Most visitors start from Crawley Road at the bottom of the map.

The Reynolds House is one of the oldest houses in the city and is open to the public.

It's on the north side of Crawley Road next to the footpath that leads to the public gardens.

The area is particularly interesting for its unusual sculptures.

The Thumb is just what its name suggests, but it's about 10 m high.

You'll see it on Hill Road across the road from the bank.

The museum's got a particularly fine collection of New Zealand landscapes.

It's on the east side of the sheep market on City Road.

It's on the other side of the road from the public gardens, immediately facing the junction with Hill Road.

The Contemporary Art Gallery is on a little road that leads off Station Square, not far from the public gardens.

The road ends at the gallery.

It doesn't go anywhere else.

That's open every day except Mondays.

The Warner Gallery specializes in 19th century art.

It's on City Road near the junction with Crawley Road on the same side of the road as the public gardens.

It's open on weekdays from 9:00 to 5:00 and entry is free.

Finally, if you're interested in purchasing highquality artwork, the place to go is Nucleus.

You need to go from Crawley Road up through Station Square and east along Hill Road until you get to a small winding road turning off.

Go up there and it's on your right.

If you get to City Road, you've gone too far.

Joe, you know I'm giving a presentation in our film studies class next week.

Well, could we discuss it?

I could do with getting someone else's opinion.

Of course, Katie.

What are you going to talk about?

It's about film adaptations of Shakespeare's plays.

I've got very interested in all the different approaches that film directors take.

Uh-huh.

So, I thought I'd start with Janetti, who's a professor of film and literature.

And in one of his books, he came up with a straightforward classification of film adaptations based on how faithful they are to the original plays and novels.

I've already made some notes on that, so I just need to sort those out before the presentation.

I thought that next I'd ask the class to come up with the worst examples of Shakespeare adaptations that they've seen and to say why.

That should be more fun than having their favorite versions.

Yes, I can certainly think of a couple.

Right.

Next, I want to talk about Rachel Malco.

I came across something on the internet about her work on film adaptations and I was thinking of showing some film clips to illustrate her ideas.

Will you have enough time though, both to prepare and during the presentation?

After all, I doubt if you'll be able to find all the clips you want.

Perhaps you're right.

Okay.

Well, I'd better do some slides instead.

Saying how various films relate to what she says.

That should encourage discussion.

Next, I want to say something about how plays may be chosen for adaptation because they're concerned with issues of the time when the film is made.

You mean things like patriotism or the role of government?

It's quite tricky, but I've got a few ideas I'd like to discuss.

And finally, I want to talk about a few adaptations that I think illustrate a range of approaches and make some comments on them.

Do you know the Japanese film Ran?

I haven't seen it.

It was based on Shakespeare's King Lear, wasn't it?

That's right.

It was a very loose adaptation using the same situation and story but moving it to 16th century Japan instead of 16th century Britain.

So for example, the king's daughters become sons because in Japanese culture at that time women couldn't succeed to the throne.

Okay.

I hope you're going to talk about the 1993 film of Much Ado About Nothing.

I think that's one of the best Shakespeare films.

It really brings the play to life, doesn't it?

Yes, I agree.

And I think filming it in Italy, where the play is set, makes you see what life was like at the time of the play.

Absolutely.

Right.

What's next?

Uh, next I thought Romeo and Juliet, the 1996 film, which moves the action into the present day.

Yes, it worked really well.

I thought, changing the two feuding families in the original to two competing business empires, even though they're speaking in the English of the original play.

You'd expect it would sound really bizarre, but I found I soon got used to it.

Me, too.

Then I thought I'd include a real Hollywood film, one that's intended to appeal to a mass commercial audience.

There must be quite a number of those.

Yes, but I've picked the 1996 film of Hamlet.

It included every line of the text, but it's more like a typical action hero movie.

There are loads of special effects, but no unifying interpretation of the play.

All show and no substance.

Exactly.

Then there's Prospero's books based on The Tempest.

That was really innovative from a stylistic point of view.

Didn't it include dance and singing and animation as well as live actors?

Yes, it did.

I also want to mention Looking for Richard.

Did you ever see it?

No, but I've read about it.

It was a blend of a documentary with a few scenes from Richard III, wasn't it?

That's right.

It's more a way of looking into how people nowadays connect with the playright.

The play is really just the starting point and that'll be where I finish.

Well, it sounds as though it'll be very interesting.

This lecture will be about the science of acoustics, the study of sound in relation to urban environments such as cities.

As an acoustic engineer myself, I think this is an area where we're likely to see great changes.

In the past, researching urban soundscapes was simple.

We measured levels of sound in decibels.

So I used to take my sound meter and I measured the noise somewhere and then I might ask a sample of people to say at what level the sound became annoying.

With data like this, acoustic engineers have been able to build up what we call noise maps, maps of the sound environment.

But actually these aren't a lot of use.

What they do show is that the highest noise levels are generally on roads.

Well, that's not really very surprising, but there's quite a lot going on that these maps don't show because they can't capture the complex way that sound varies over time.

So, they ignore important issues such as the noise someone might hear from the open windows or gardens of their neighbors.

And this sort of noise can be quite significant in summer.

We don't have any databases on this sort of information.

As well as that, these records of sound levels take no account of the fact that people vary in their perceptions of noise.

So someone like me with years of working in acoustics might be very different from you in that regard.

But anyway, even though these noise maps are fairly crude, they've been useful in providing information and raising awareness that noise matters.

We need to deal with it.
And so, it's a political matter and that's important.
We need rules and regulations because noise can cause all sorts of problems.
Those of you who are city dwellers know that things go on 24 hours a day.
So, city dwellers often suffer from interrupted sleep.
It's also known that noise can lead to a rise in levels of stress due to physical changes in the body affecting the composition of the blood.
And there are other problems as well.
For instance, if school children don't have a quiet place to study, their work will suffer.
Now, one problem with decibel measurement is that it doesn't differentiate between different types of noise.
Some types of sounds that most people would probably think of as nice and relaxing, might well score quite highly in decibel levels.
Think of the sound made by a fountain in a town square, for example.
That's not necessarily something that we'd want to control or reduce.
So maybe researchers should consider these sorts of sounds in urban design.
This is going to be tricky because just measuring decibel levels isn't going to help us here.
Instead, many researchers are using social science techniques, studying people's emotional response to sound by using questionnaires and so on.
So what exactly do people want to hear in an urban environment?
Some recent interdisciplinary research has come out with results that at first sight seem contradictory.
A city needs to have a sense of activity, so it needs to be lively with sounds like the clack of high heels on a pavement or the hiss of a coffee machine.
But these mustn't be too intrusive because at the same time, we need to be able to relax.
One of the major problems in achieving this will be getting architects and town planners to use the research.
Apart from studying the basics of acoustics, these people receive very little training in this area.
But in fact, they should be regarding sound as an opportunity to add to the experience of urban living.
Whereas at present they tend to see it as something to be avoided or reduced as far as possible or something that's just a job for engineers like the street drainage system.
What's needed is for noise in cities to be regarded as an aesthetic quality as something that has the qualities of an art form.
If we acknowledge this then we urgently need to know what governs it and how designers can work with it.
We need to develop a complex understanding of many factors.
What is the relationship between sound and culture?
What can we learn from disciplines such as psychology about the way that sound interacts with human development and social relationships and the way that sound affects our thoughts and feelings?
Can we learn anything from physics about the nature of sound itself?
Today's powerful technologies can also help us to show us their ideas and help us to imagine the effect their buildings will have.
Architects and town planners already use virtual reality.
But these programs are silent.
In the future, such programs could use realistic sounds, meaning that soundscapes could be explored before being built.
So hopefully using the best technology we can lay our hands on, the city of the future will be a pleasure to the ears as well as the eyes.

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Hello, tourist information center.

Mike speaking.

How can I help you?

Oh, hi.

I wanted to find out about cookery classes.

I believe there are some one-day classes for tourists.

Well, they're open to everyone, but tourists are always welcome.

Okay, let me give you some details of what's available.

There are several classes.

One very popular one is at the food studio.

Okay.

They focus on seasonal products and as well as teaching you how to cook them, they also show you how to choose them.

Right, that sounds good.

How big are the classes?

I'm not sure exactly, but they'll be quite small.

And could I get a private lesson there?

I think so.

Let me check.

Yes, they do offer those, though.

In fact, most of the people who attend the classes find it's a nice way of getting to know one another.

I suppose it must be.

Yes.

And this company has a special deal for clients where they offer a discount of 20% if you return for a further class.

Okay.

But you said there were several classes.

That's right.

Another one you might be interested in is Bond's cookery school.

They're quite new.

They just opened 6 months ago, but I've heard good things about them.

They concentrate on teaching you to prepare healthy food, and they have quite a lot of specialist staff.

So, is that food for people on a diet?

and things like that.

I don't know if I'd be interested in that.

Well, I don't think they particularly focus on low calorie diets or weight loss.

It's more to do with recipes that look at specific needs, like including ingredients that will help build up your bones and make them stronger, that sort of thing.

I see.

Well, I might be interested.

I'm not sure.

Do they have a website I could check?

Yes, just key in the name of the school.

it'll come up.

And if you want to know more about them, every Thursday evening they have a lecture at the school.

It's free and you don't need to book or anything.

Just turn up at 7:30.

And that might give you an idea of whether you want to go to an actual class.

Okay.

There's one more place you might be interested in.

That's got a rather strange name.

It's called the Arettza Center.

That's spelled A R E T S A.

Okay.

They've got a very good reputation.

They do a bit of meat and fish cookery, but they mostly specialize in vegetarian dishes.

Right.

That's certainly an area I'd like to learn more about.

I've got lots of friends who don't eat meat.

In fact, I think I might have seen that school today.

Is it just by the market?

That's right.

So, they don't have any problem getting their ingredients.

They're right next door.

And they also offer a special 2-hour course in how to use a knife.

They cover all the different skills, buying them, sharpening, chopping techniques.

It gets booked up quickly, though, so you'd need to check it was available.

Right.

Well, thank you very much.

I'll go and check that out.

Good evening everyone.

My name's Phil Sutton and I'm chairman of the Highways Committee.

We've called this meeting to inform members of the public about the new regulations for traffic and parking we're proposing for Granford.

I'll start by summarizing these changes before we open the meeting to questions.

So, why do we need to make these changes to traffic systems in Granford?

Well, we're very aware that traffic is becoming an increasing problem.

It's been especially noticeable with the increase in heavy traffic while they've been building the new hospital.

But it's the overall rise in the volume of traffic of all kinds that's concerning us.

To date, there's not been any increase in traffic accidents, but that's not something we want to see happen.

Obviously, we recently carried out a survey of local residents, and their responses were interesting.

People were very concerned about the lack of visibility on some roads due to cars parked along the sides of the roads.

We'd expected complaints about the congestion near the school when parents are dropping off their children or picking them up, but this wasn't top of the list and nor were noise and fumes from trucks and lorries, though they were mentioned by some people.

We think these new traffic regulations would make a lot of difference, but we still have a long way to go.

We've managed to keep our proposals within budget just so they can be covered by the council.

But of course, it's no good introducing new regulations if we don't have a way of making sure that everyone obeys them.

And that's an area we're still working on with the help of representatives from the police force.

Okay.

So this slide shows a map of the central area of Granford with the high street in the middle and school road on the right.

Now we already have a set of traffic lights in the high street at the junction with station road, but we're planning to have another set at the other end at the school road junction to regulate the flow of traffic along the high street.

We've decided we definitely need a pedestrian crossing.

We considered putting this on school road just outside the school, but in the end we decided that could lead to a lot of traffic congestion.

So, we decided to locate it on the high street crossing the road in front of the supermarket.

That's a very busy area, so it should help things there.

We are proposing some changes to parking.

At present, parking isn't allowed on the high street outside the library, but we are going to change that and allow parking there, but not at the other end of the high street near school road.

There'll be a new no parking sign on school road just by the entrance to the school forbidding parking for 25 m.

This should improve visibility for drivers and pedestrians, especially on the bend just to the north of the school.

As far as disabled drivers are concerned, at present they have parking outside the supermarket, but lorries also use those spaces.

So, we've got two new disabled parking spaces on the side road up towards the bank.

It's not ideal, but probably better than the present arrangement.

We also plan to widen the pavement on school road.

We think we can manage to get an extra half meter on the bend just before you get to the school on the same side of the road.

Finally, we've introduced new restrictions on loading and unloading for the supermarket.

So, lorries will only be allowed to stop there before 8:00 a.m.

That's the supermarket on school road.

We kept to the existing arrangements with the High Street Supermarket.

Okay, so that's about it.

Now, we've got to choose a topic for our experiment, haven't we, Jack?

Were you thinking of something to do with seeds?

H, that's right.

I thought we could look at seed germination, how a seed begins to grow.

Okay.

Any particular reason?

I know you're hoping to work in plant science eventually.

Yeah, but practically everything we do is going to feed into that.

No, there's an optional module on seed structure and function in the third year that I might do.

So, I thought it might be useful for that.

If I choose that option, I don't have to do a dissertation module.

Good idea.

Hm.

Well, I thought for this experiment we could look at the relationship between seed size and the way the seeds are planted.

So, we could plant different sized seeds in different ways and see which grow best.

Okay, we'd need to allow time for the seeds to come up.

That should be fine if we start now.

A lot of the other possible experiments need quite a bit longer.

So, that would make it a good one to choose.

And I don't suppose it would need much equipment.

We're not doing chemical analysis or anything.

Though, that's not really an issue.

We've got plenty of equipment in the laboratory.

Yeah, we need to have a word with the tutor if we're going to go ahead with it, though.

I'm sure our aim's okay.

It's not very ambitious, but the assignment's only 10% of our final mark, isn't it?

But we need to be sure we're the only ones doing it.

Yeah, it's only 5% actually.

But it would be a bit boring if everyone was doing it.

Did you read that book on seed germination on our reading list?

The one by Graves?

I looked through it for my last experiment, though it wasn't all that relevant there.

It would be for this experiment, though.

I found it quite hard to follow.

Lots about the theory, which I hadn't expected.

Yes, I'd been hoping for something more practical.

It does include references to the recent findings on genetically modified seeds, though.

Yes, that was interesting.

I read an article about seed germination by Lee Hall.

About seeds that lie in the ground for ages and only germinate after a fire.

H, that's the one.

I knew a bit about it already, but not about this research.

His analysis of figures comparing the times of the fires and the proportion of seeds that germinated was done in a lot of detail.

Very impressive.

Was that the article with the illustrations of early stages of plant development?

They were very clear.

I think those diagrams were in another article.

Anyway, shall we have a look at the procedure for our experiment?

We'll need to get going with it quite soon.

Right.

So, the first thing we have to do is find our seeds.

I think vegetable seeds would be best.

And obviously they mustn't all be the same size.

So how many sorts do we need?

About four different ones.

I think that would be enough.

There'll be quite a large number of seeds for each one.

Then for each seed, we need to find out how much it weighs and also measure its dimensions.

And we need to keep a careful record of all that.

That'll be quite timeconsuming.

And we also need to decide how deep we're going to plant the seeds.

Right on the surface, a few millime down or several cm.

Okay.

So, then we get planting.

Do you think we can plant several seeds together in the same plant pot?

No, I think we need a different one for each seed.

Right.

And we'll need to label them.

We can use different colored labels.

Then we wait for the seeds to germinate.

I reckon that'll be about 3 weeks depending on what the weather's like.

Then we see if our plants have come up and write down how tall they've grown.

Then all we have to do is look at our numbers and see if there's any relation between them.

That's right.

So then we get Hi.

Today we're going to be looking at animals in urban environments and I'm going to be telling you about some research on how they're affected by these environments.

Now, in evolutionary terms, urban environments represent huge upheavalss, the sorts of massive changes that usually happen over millions of years.

And we used to think that only a few species could adapt to this new environment.

One species which is well known as being highly adaptable is the crow and there have been various studies about how they manage to learn new skills.

Another successful species is the pigeon because they're able to perch on ledges on the walls of city buildings just like they once perched on cliffs by the sea.

But in fact, we're now finding that these early immigrants were just the start of a more general movement of animals into cities and of adaptation by these animals to city life.

And one thing that researchers are finding especially interesting is the speed with which they're doing this.

We're not talking about gradual evolution here.

These animals are changing fast.

Let me tell you about some of the studies that have been carried out in this area.

So in the University of Minnesota, a biologist called Emily Snelroode and her colleagues looked at specimens of urbanized small mammals such as mice and gophers that had been collected in Minnesota and that are now kept in museums there.

And she looked at specimens that had been collected over the last h 100red years, which is a very short time in evolutionary terms.

And she found that during that time, these small mammals had experienced a jump in brain size when compared to rural mammals.

Now, we can't be sure this means they're more intelligent, but since the sizes of other parts of the body didn't change, it does suggest that something cognitive was going on.

And Snellroode thinks that this change might reflect the cognitive demands of adjusting to city life, having to look in different places to find food, for example, and coping with a whole new set of dangers.

Then over in Germany at the Max Plank Institute, there's another biologist called Katerina Miranda who's done some experiments with blackbirds living in urban and rural areas.

And she's been looking not at their anatomy, but at their behavior.
So, as you might expect, she's found that the urban blackbirds tend to be quite bold.
They're prepared to face up to a lot of threats that would frighten away their country counterparts.
But there's one type of situation that does seem to frighten the urban blackbirds, and that's anything new, anything they haven't experienced before.
And if you think about it, that's quite sensible for a bird living in the city.
Jonathan Atwell in Indiana University is looking at how a range of animals respond to urban environments.
He's found that when they're under stress, their endocrine systems react by reducing the amount of hormones such as corticosterone into their blood.
It's a sensible seeming adaptation.
A rat that gets scared every time a subway train rolls past won't be very successful.
There's just one more study I'd like to mention which is by Sarah Parton and her team and they've been looking at how squirrels communicate in an urban environment and they found that a routine part of their communication is carried out by waving their tails.
You do also see this in the country, but it's much more prevalent in cities, possibly because it's effective in a noisy environment.
So what are the long-term implications of this?
One possibility is that we may see completely new species developing in cities.
But on the other hand, it's possible that not all of these adaptations will be permanent.
Once the animals got accustomed to its new environment, it may no longer need the features it's developed.
So now we've had a look at adaptation.
Hello, South City Cycling Club.
Oh, hi.
Um, I want to find out about joining the club.
Right.
I can help you there.
I'm the club secretary and my name's Jim Hunter.
Oh, hi Jim.
So, are you interested in membership for yourself?
That's right.
Okay.
Well, there are basically two types of adult membership.
If you're pretty serious about cycling, there's the full membership that costs \$260 and that covers you not just for ordinary cycling, but also for races both here in the city and also in other parts of Australia.
Right.
Well, I'm not really up to that standard.
I was more interested in just joining a group to do some cycling in my free time.
Sure.
That's why most people join.
So, in that case, you'd be better with the recreational membership.
That's \$18 if you're over 19 and \$95 if you're under.
I'm 25.
Okay.
It's paid quarterly and you can upgrade it later to the full membership if you want to, of course.
Now, both types of membership include the club fee of \$20.
They also provide insurance in case you have an accident, though we hope you won't need that, of course.
No.
Okay.
Well, I'll go with the recreational membership, I think.
And that allows me to join in the club activities and so on.
That's right.
And once you're a member of the club, you're also permitted to wear our kit when you're out cycling.
It's green and white.
Yes, I've seen cyclists wearing it.
So, can I buy that at the club?
Uh, no.
It's made to order by a company in Brisbane.
You can find them online.
They're called Jerry's.
That's J E R R Y S.
You can use your membership number to put in an order on their website.
Okay.
Now, can you tell me a bit about the rides I can do?
Sure.
So, we have training rides pretty well every morning, and they're a really good way of improving your cycling skills as well as your general level of fitness, but they're different levels.
Level A is pretty fast.
You're looking at about 30 or 35 km an hour.
If you can do about 25 kilometers an hour, you'd probably be level B.
And then level C are the novices who stay at about 15 km per hour, right?
Well, I reckon I'd be level B.
So, when are the sessions for that level?

Uh, there are a couple each week.
They're both early morning sessions.
There's one on Tuesdays and for that one you meet at 5:30 a.m.
and the meeting points the stadium.
Do you know where that is?
Yes, it's quite near my home in fact.
Okay.
And how about the other one?
That's on Thursdays.
It starts at the same time, but they meet at the main gate to the park.
Is that the one just past the shopping mall?
That's it.
So, how long are the rides?
Uh, they're about an hour and a half, so if you have a job, it's easy to fit in before you go to work.
And the members often go somewhere for coffee afterwards, so it's quite a social event.
Okay, that sounds good.
I've only just moved to the city, so I don't actually know many people yet.
Well, it's a great way to meet people.
And does each ride have a leader?
sometimes, but not always.
But you don't really need one.
The group members on the ride support one another anyway.
How would we know where to go?
If you check the club website, you'll see that the route for each ride is clearly marked.
So, you can just print that out and take it along with you.
It's similar from one week to another, but it's not always exactly the same.
And what do I need to bring?
H.
Well, bring a bottle of water and your phone.
You shouldn't use it while you're cycling, but have it with you.
Right.
And in winter, it's well before sunrise when we set out, so you need to make sure your bike's got lights.
That's okay.
Well, thanks, Jim.
I'd definitely like to join.
So, what's the best way of going about it?
Oh, you can.
Thanks for coming, everyone.
Okay, so this meeting is for new staff and staff who haven't been involved with our volunteering projects yet.
So basically the idea is that we allow staff to give up some of their work time to help on various charity projects to benefit the local community.
We've [snorts] been doing this for the last 5 years and it's been very successful.
Participating doesn't necessarily involve a huge time commitment.
The company will pay for eight hours of your time that can be used over one or two days all at once or spread over several months throughout the year.
There are some staff who enjoy volunteering so much they also give up their own free time for a couple of hours every week.
It's completely up to you.
Obviously, many people will have family commitments and aren't as available as other members of staff.
Feedback from staff has been overwhelmingly positive because they felt they were doing something really useful.
Nearly everyone agreed that volunteering made them feel more motivated at work.
They also liked building relationships with the people in the local community and felt valued by them.
One or two people also said it was a good thing to have on their CVs.
One particularly successful project last year was the get working project.
This was aimed at helping unemployed people in the area get back to work.
Our staff were able to help them improve their telephone skills such as writing down messages and speaking with confidence to potential customers which they had found quite difficult.
This is something many employers look for in job applicants and something we all do without even thinking about every day at work.
We've got an exciting new project starting this year.
Up until now, we've mainly focused on projects to do with education and training, and we'll continue with our reading project in schools and our work with local charities.
But we've also agreed to help out on a conservation project in Red Fern Park.
So if any of you fancy being outside and getting your hands dirty, this is the project for you.
I also wanted to mention the annual digital inclusion day which is coming up next month.
The aim of this is to help older people keep up with technology.
And this year, instead of hosting the event in our own training facility, we're using the ICT suite at Hill College as it can hold far more people.
We've invited over 60 people from the Silver Age Community Center to take part, so we'll need a lot of volunteers to help with this event.
If you're interested in taking part, please go to the volunteering section of our website and complete the relevant form.
We won't be providing any training for this, but you'll be paired with an experienced volunteer if you've never done it before.
By the way, don't forget to tell your manager about any volunteering activities you decide to do.

The participants on the digital inclusion day really benefited.

The majority were in their 70s, though some were younger, and a few were even in their 90s.

Quite a few owned both a computer and a mobile phone, but these tended to be outdated models.

They generally knew how to do simple things like send texts, but weren't aware of recent developments in mobile phone technology.

A few were keen to learn, but most were quite dismissive at first.

They couldn't see the point of updating their skills.

But that soon changed.

The feedback was very positive.

The really encouraging thing was that participants all said they felt much more confident about using social media to keep in touch with their grandchildren who prefer this form of communication to phoning or sending emails.

A lot of them also said playing online games would help them make new friends and keep their brains active.

They weren't that impressed with being able to order their groceries online as they liked going out to the shops, but some said it would come in handy if they were ill or the weather was really bad.

One thing they asked about was using tablets for things like reading newspapers.

Some people had been given tablets as presents, but had never used them.

So, that's something we'll make sure we include this time.

Uh, come in Russ.

Thank you.

Now, you wanted to consult me about your class presentation on nanotechnology.

You were due to give it next week, aren't you?

That's right.

And I'm really struggling.

I chose the topic because I didn't know much about it and wanted to learn more.

But now I've read so much about it, in a way, there's too much to say.

I could talk for much longer than the 20 minutes I've been allocated.

Should I assume the other students don't know much and give them a kind of general introduction or should I try and make them share my fascination with a particular aspect?

You could do either but you'll need to have it clear in your own mind.

Then I think I'll give an overview.

Okay.

Now, one way of approaching this is to work through developments in chronological order.

Uh-huh.

On the other hand, you could talk about the numerous ways that nanotechnology is being applied.

You mean things like thin films on camera displays to make them water repellent and additives to make motorcycle helmets stronger and lighter?

Exactly.

Or another way would be to focus on its impact in one particular area.

Say medicine or space exploration.

That would make it easier to focus.

Perhaps I should do that.

I think that would be a good idea, right?

How important is it to include slides in the presentation?

They aren't essential by any means, and there's a danger of tailoring what you say to fit whatever slides you can find.

While it can be good to include slides, you could end up spending too long looking for suitable ones.

You might find it better to leave them out.

I see.

Another thing I was wondering about was how to start.

I know presentations often begin with first I'm going to talk about this and then I'll talk about that.

But I thought about asking the audience what they know about nanotechnology.

That would be fine if you had an hour or two for the presentation, but you might find that you can't do anything with the answers you get and it simply eats into the short time that's available.

So maybe I should mention a particular way that nanotechnology is used to focus people's attention.

That sounds sensible.

What do you think I should do next?

I really have to plan the presentation today and tomorrow.

Well, initially I think you should ignore all the notes you've made.

Take a small piece of paper and write a single short sentence that ties together the whole presentation.

It can be something as simple as nanotechnology is already improving our lives.

Then start planning the content around that.

You can always modify that sentence later if you need to.

Okay.

Okay.

Now let's think about actually giving the presentation.

You've only given one before, if I remember correctly, about an experiment you'd been involved in.

That's right.

It was pretty rubbish.

Let's say it was better in some respects than in others.

With regard to the structure, I felt that you ended rather abruptly without rounding it off.

Be careful not to do that in next week's presentation.

Okay?

And you made very little eye contact with the audience because you were looking down at your notes most of the time.
You need to be looking at the audience and only occasionally glancing at your notes.
M your body language was a little odd.
Every time you showed a slide, you turned your back on the audience so you could look at it.
You should have been looking at your laptop.
And you kept scratching your head.
So I found myself wondering when you were next going to do that instead of listening to what you were saying.
Oh dear.
What did you think of the language?
I knew that not everyone was familiar with the subject, so I tried to make it as simple as I could.
Yes, that came across.
You used a few words that are specific to the field, but you always explained what they meant, so the audience wouldn't have had any difficulty understanding.
Uh-huh.
I must say the handouts you prepared were well thought out.
They were a good summary of your presentation, which people would have been able to refer to later on.
So, well done on that.
Thank you.
Well, I hope that helps you with next week's presentation.
Yes, it will.
Thanks a lot.
I'll look forward to seeing a big improvement.
Then today we'll be continuing the series of lectures on memory by focusing on what is called episodic memory and what can happen if this is not working properly.
Episodic memory refers to the memory of an event or episode.
Episodic memories allow us to mentally travel back in time to an event from the past.
Episodic memories include various details about these events.
For example, when an event happened and other information such as the location.
To help understand this concept, try to remember the last time you ate dinner at a restaurant.
The ability to remember where you at, who you were with, and the items you ordered are all features of an episodic memory.
Episodic memory is distinct from another type of memory called semantic memory.
This is the type of factual memory that we have in common with everyone else.
That is your general knowledge of the world.
To build upon a previous example, remembering where you parked your car is an example of episodic memory.
But your understanding of what a car is and how an engine works are examples of semantic Unlike episodic memory, semantic memory isn't dependent on recalling personal experiences.
Episodic memory can be thought of as a process with several different steps of memory processing.
Encoding, consolidation, and retrieval.
The initial step is called encoding.
This involves the process of receiving and registering information which is necessary for creating memories of information or events that you experience.
The degree to which you can successfully encode information depends on the level of attention you give to an event while it's actually happening.
Being distracted can make effective encoding very difficult.
Encoding of episodic memories is also influenced by how you process the event.
For example, if you were introduced to someone called Charlie, you might make the connection that your uncle has the same name.
Future recollection of Charlie's name is much easier if you have a strategy to help you encode it.
Memory consolidation, the next step in forming an episodic memory, is the process by which memories of encoded information are strengthened, stabilized, and stored to facilitate later retrieval.
Consolidation is most effective when the information being stored can be linked to an existing network of information.
Consolidation makes it possible for you to store memories for later retrieval indefinitely.
Forming strong memories depends on the frequency with which you try to retrieve them.
Memories can fade or become harder to retrieve if they aren't used very often.
The last step in forming episodic memories is called retrieval, which is the conscious recollection of encoded information.
Retrieving information from episodic memory depends upon semantic, olfactory, auditory and visual factors.
These help episodic memory retrieval by acting as a prompt.
For example, when recalling where you parked your car, you may use the color of a sign close to where you parked.
You actually have to mentally travel back to the moment you parked.
There are a wide range of neurological diseases and conditions that can affect episodic memory.
These range from Alzheimer's to schizophrenia to autism.
An impairment of episodic memory can have a profound effect on individuals lives.
For example, the symptoms of schizophrenia can be reasonably well controlled by medication.
However, patients episodic memory may still be impaired and so they are often unable to return to university or work.
Recent studies have shown that computerassisted games designed to keep the brain active can help improve their episodic memory.
Episodic memories can help people connect with others.
For instance, by sharing intimate details about their past, something individuals with autism often have problems with.
This may be caused by an absence of a sense of self.
This is essential for the storage of episodic memory.
and has been found to be impaired in children with autism.
Research has shown that treatments that improve memory may also have a positive impact on children's social development.

One study looked at Hello, Linda speaking.
Oh, hi Linda.
This is Matt Brooks.
Alex White gave me your number.
He said you'd be able to give me some advice about moving to Bford.
Yes, Alex did mention you.
How can I help?
Well, first of all, which area to live in?
Well, I live in Dalton, which is a really nice suburb.
Not too expensive, and there's a nice park.
Sounds good.
Do you know how much it would be to rent a two-bedroom flat there?
Yeah, you should be able to get something reasonable for £850 per month.
That's what people typically pay.
You certainly wouldn't want to pay more than £900.
That doesn't include bills or anything.
No, that sounds all right.
I'll definitely have a look there.
Are the transport links easy from where you live?
Well, I'm very lucky.
I work in the city center, so I don't have to use public transport.
I go by bike.
Oh, I wish I could do that.
Is it safe to cycle around the city?
Yes, it's fine and it keeps me fit.
[snorts] Anyway, driving to work in the city center would be a nightmare because there's hardly any parking and the traffic during the rush hour can be bad.
I'd be working from home, but I'd have to go to London one or two days a week.
Oh, that's perfect.
Getting to London is no problem.
There's a fast train every 30 minutes which only takes 45 minutes.
That's good.
Yeah.
The train service isn't bad during the week and they run quite late at night.
It's weekends that are a problem.
They're always doing engineering work and you have to take a bus to Hadam and pick up the train there which is really slow.
But other than that, Banford's a great place to live.
I've never been happier.
There are some nice restaurants in the city center and a brand new cinema which has only been open a couple of months.
There's a good art center, too.
Sounds like Bford's got it all.
Yes, we're really lucky.
There are lots of really good aspects to living here.
The schools are good and the hospital here is one of the best in the country.
Everyone I know who's been there had a positive experience.
Oh, I can give you the name of my dentist too in Bridge Street if you're interested.
I've been going to him for years and I've never had any problems.
Oh, okay.
Thanks.
I'll find his number and send it to you.
Thanks.
That would be really helpful.
Are you planning to visit Banford soon?
Yes.
My wife and I are both coming next week.
We want to make some appointments with estate agents.
I could meet you if you like and show you around.
Are you sure?
We'd really appreciate that.
Either a Tuesday or Thursday is good for me after 5:30.
Thursday is preferable.
Tuesday I need to get home before 6:00 p.m.
Okay, great.
Let me know which train you're catching and I'll meet you in the cafe outside.
You can't miss it.
It's opposite the station and next to the museum.
Brilliant.
I'll text you next week then.
Thanks so much for all the advice.
No problem.

I'll see you next week.

So, if you're one of those people who hasn't found the perfect physical activity yet, here are some things to think about which might help you make the right decision for you.

The first question to ask yourself is whether you would enjoy training in a gym.

Many people are put off by the idea of having to fit a visit to the gym into their busy day.

You often have to go very early or late as some gyms can get very crowded.

But with regular training, you'll see a big difference in a relatively short space of time.

Running has become incredibly popular in recent years.

That's probably got a lot to do with the fact that it's a very accessible form of exercise.

Anyone can run, even if you can only run a few meters to begin with.

But make sure you get the right shoes.

It's worth investing in a high quality pair, and they don't come cheap.

Another great thing about running is that you can do it at any time of day or night.

The only thing that may stop you is snow and ice.

Swimming is another really good way to build fitness.

What attracts many people is that you can swim in an indoor pool at any time of year.

On the other hand, it can be quite boring or solitary.

It's hard to chat to people while you're swimming lengths.

Cycling has become almost as popular as running in recent years.

That's probably because, as well as improving their fitness, many people say being out in the fresh air in a park or in the countryside can be fun, provided the conditions are right, of course.

Only fanatics go out in the wind and rain.

Yoga is a good choice for those of you looking for exercise which focuses on developing both a healthy mind and body.

It's a good way of building strength and with the right instructor, there's less chance of hurting yourself than with other more active sports, but don't expect to find it easy.

It can be surprisingly challenging, especially for people who aren't very flexible.

Getting a personal trainer is a good way to start your fitness program.

Obviously, there can be significant costs involved, but if you've got someone there to encourage you and help you achieve your goals, you're less likely to give up.

Make sure you get someone with a recognized qualification, though, or you could do yourself permanent damage.

Whatever you do, don't join a gym unless you're sure you'll make good use of it.

So many people waste lots of money by signing up for membership and then hardly ever go.

What happens to their good intentions?

I don't think people suddenly stop caring about improving their fitness or decide they have more important things to do.

I think people lose interest when they don't think they're making enough progress.

That's when they give up hope and stop believing they'll ever achieve their goals.

Also, what people sometimes don't realize when they start is that it takes a lot of determination and hard work to keep training week after week.

And lots of people don't have that kind of commitment.

One thing you can do to help yourself is to set manageable goals.

Be realistic and don't push yourself too far.

Some people advise writing goals down, but I think it's better to have a flexible approach.

Give yourself a really nice treat every time you reach one of your goals.

And don't get too upset if you experience setbacks.

It's a journey.

There are bound to be difficulties along the way.

Okay, Jim, you wanted to see me about your textile design project.

That's right.

I've been looking at how a range of natural dyes can be used to color fabrics like cotton and wool.

Why did you choose that topic?

Well, I got a lot of useful ideas from the museum, you know, at that exhibition of textiles.

But I've always been interested in anything to do with color.

Years ago, I went to a carpet shop with my parents when we were on holiday in Turkey, and I remember all the amazing colors.

They might not all have been natural dyes.

Maybe not.

But for the project, I decided to follow it up.

And I found a great book about a botanic garden in California that specializes in plants used for dyes.

Okay.

So, in your project, you had to include a practical investigation.

Yeah.

At first, I couldn't decide on my variables.

I was going to just look at one type of fiber, for example, like cotton, and see how different types of dyes affected it.

Yes.

Then I decided to include others as well.

So, I looked at cotton and wool and nylon with just one type of dye, various types, including some that weren't natural for comparison.

Okay.

So, I did the experiments last week.

I used some ready-made natural dyes.

I found a website which supplied them.

They came in just a few days, but I also made some of my own.

That must have taken quite a bit of time.
Yes, I thought it would just be a matter of a teaspoon or so of dye.
And actually, that wasn't the case at all.
Like, I was using one vegetable, a beetroot for a red dye, and I had to chop up a whole pile of it.
So, it all took longer than I'd expected.
One possibility is to use food colorings.
I did use one that was a yellow dye, an artificial one, tartreazine.
Yeah, I used it on cotton first, it came out a great color, but when I rinsed the material, the color just washed away.
I've been going to try it out on nylon, but I abandoned that idea.
Were you worried about health issues?
I thought if it's illegal food coloring, it must be safe.
Well, it can occasionally cause allergic reactions, I believe.
So, what natural dyes did you look at?
Well, one was turmeric.
The color is great.
It's a really strong yellow.
It's generally used in dishes like curry.
It's meant to be quite good for your health when eaten, but you might find it's not permanent when it's used as a dye.
A few washes and it's gone.
Right.
I used beetroot as a dye for wool.
When I chop up beetroot to eat, I always end up with bright red hands.
But the wool ended up just a sort of watery cream shade.
Disappointing.
There's a natural dye called Tyrian purple.
Have you heard of that?
Yes.
It comes from a shellfish and it was worn in ancient times, but only by important people as it was so rare.
I didn't use it.
It fell out of use centuries ago, though one researcher managed to get hold of some recently.
But that shade of purple can be produced by chemical dyes nowadays.
Did you use any black dyes?
Logwood.
That was quite complicated.
I had to prepare the fabric so the dye would take.
I hope you were careful to wear gloves.
Yes, I know the danger with that dye.
Good.
It can be extremely dangerous if it's ingested.
Now, presumably you had a look at an insect-based dye like cochineal for example.
Yes, I didn't actually make that.
I didn't have time to start crushing up insects to get the red color and anyway, they're not available here.
But I managed to get the dye quite easily from a website, but it cost a fortune.
I can see why it's generally just used in cooking and in small quantities.
Yes, it's very effective, but that's precisely why it's not used as a dye.
I also read about using metal oxide.
Apparently, you can allow iron to rust while it's in contact with the fabric, and that colors it.
Yes, that works well for dyeing cotton, but you have to be careful as the metal can actually affect the fabric and so you can't expect to get a lot of wear out of fabrics treated in this way.
And the colors are quite subtle.
Not everyone likes them.
Anyway, it looks as if you've done a lot of work.
Last week, we started looking at reptiles, including crocodiles and snakes.
Today, I'd like us to have a look at another reptile, the lizard.
and in particular at some studies that have been done on a particular type of lizard whose Latin name is *Tiquia Rugosa*.
This is commonly known as the sleepy lizard because it's quite slow in its movements and spends quite a lot of its time dozing under rocks or lying in the sun.
I'll start with a general description.
Sleepy lizards live in Western and South Australia where they're quite common.
Unlike European lizards, which are mostly small, green, and fast-moving, sleepy lizards are brown.
But what's particularly distinctive about them is the color of their tongue, which is dark blue, in contrast with the lining of their mouth, which is bright pink.
And they're much bigger than most European lizards.
They have quite a varied diet, including insects and even small animals, but they mostly eat plants of varying kinds.
Even though they're quite large and powerful with strong jaws that can crush beetles and snail shells, they still have quite a few predators.
Large birds like casowaries were one of the main ones in the past, but nowadays they're more likely to be caught and killed by snakes.
Actually, another threat to their survival isn't a predator at all, but is man-made.
Quite a large number of sleepy lizards are killed by cars when they're trying to cross highways.
One study carried out by Michael Freak at Flinders University investigated the methods of navigation of these lizards.

Though they move slowly, they can travel quite long distances.
and he found that even if they were taken some distance away from their home territory, they could usually find their way back home as long as they could see the sky.
They didn't need any other landmarks on the ground.
Observations of these lizards in the wild have also revealed that their mating habits are quite unusual.
Unlike most animals, it seems that they're relatively monogamous, returning to the same partner year after year.
And the male and female also stay together for a long time, both before and after the birth of their young.
It's quite interesting to think about the possible reasons for this.
It could be that it's to do with protecting their young.
You'd expect them to have a much better chance of survival if they have both parents around.
But in fact, observers have noted that once the babies have hatched out of their eggs, they have hardly any contact with their parents.
So there's not really any evidence to support that idea.
Another suggestion is based on the observation that male lizards in monogamous relationships tend to be bigger and stronger than other males.
So maybe the male lizards stay around so they can give the female lizards protection from other males.
But again, we're not really sure.
Finally, I'd like to mention another study that involved collecting data by tracking the lizards.
I was actually involved in this myself.
So, we caught some lizards in the wild, and we developed a tiny GPS system that would allow us to track them, and we fixed this onto their tails.
Then we set the lizards free again and we were able to track them for 12 days and gather data not just about their location but even about how many steps they took during this period.
One surprising thing we discovered from this is that there were far fewer meetings between lizards than we expected.
It seems that they were actually trying to avoid one another.
So why would that be?
Well, again, we have no clear evidence, but one hypothesis is that male lizards can cause quite serious injuries to one another.
So maybe this avoidance is a way of preventing this, of self-preservation, if you like.
But we need to collect a lot more data before we can be sure of any of this.

Hi, Alex.
It's Martha Klein's here.
James White gave me your number.
I hope you don't mind me calling you.
Of course not.
How are you, Martha?
Good, thanks.
I'm ringing because I need a bit of advice.
Oh, yeah.
What about the training you did at JPNW a few years ago?
I'm applying for the same thing.
Oh, right.
Yes, I did mine in 2014.
Best thing I ever did.
I'm still working Really?
What are you doing?
Well, now I work in the customer services department, but I did my initial training in finance.
I stayed there for the first 2 years and then moved to where I am now.
That's the same department I'm applying for.
Did you enjoy it?
Uh, I was pretty nervous to begin with.
I didn't do well in my exams at school and I was really worried because I failed maths, but it didn't actually matter because I did lots of courses on the job.
Did you get a diploma at the end of your trainee period?
I'm hoping to do the one in business skills.
Yes, that sounds good.
I took the one on IT skills, but I wish I'd done that one instead.
Okay, that's good to know.
Um, what about the other trainees?
How did you get on with them?
There were about 20 of us who started at the same time and we were all around the same age.
I was 18 and there was only one person younger than me who was 17.
The rest were between 18 and 20.
I made some good friends.
I've heard lots of good things about the training at JPNW.
It seems like there are a lot of opportunities there.
Yeah, definitely.
Because of its size, you can work in loads of different areas within the organization.
What about pay?
I know you get a lower minimum wage than regular employees.
That's right.
Which isn't great, but you get the same number of days holiday as everyone else.

And the pay goes up massively if they offer you a job at the end of the training period.
Yeah, but I'm not doing it for the money.
It's the experience I think will be really useful.
Everyone says by the end of the year you gained so much confidence.
You're right.
That's the most useful part about it.
There's a lot of variety, too.
You're given lots of different things to do.
I enjoyed it all.
I didn't even mind the studying.
Do you have to spend any time in college?
Yes.
one day each month, so you get lots of support from both your tutor and your manager.
That's good.
And the company is easy to get to, isn't it?
Yes, it's very close to the train station, so the location's a real advantage.
Have you got a date for your interview yet?
Yes, it's on the 23rd of this month.
So long as you're well prepared, there's nothing to worry about.
Everyone's very friendly.
I am not sure what I should wear.
What do you think?
Nothing too casual, like jeans, for example.
If you've got a nice jacket, wear that with a skirt or trousers.
Okay, thanks.
Any other tips?
Um, well, I know it's really obvious, but arrive in plenty of time.
They hate people who are late, so make sure you know exactly where you have to get to.
And one other useful piece of advice my manager told me before I had the interview for this job is to smile.
Even if you feel terrified, it makes people respond better to you.
I'll have to practice doing that in the mirror.
[laughter] Yeah.
Well, good luck.
Let me know if you need any more information.
Thanks very much.
Hi everyone.
Welcome to the snow center.
My name's Annie.
I hope you enjoyed the bus trip from the airport.
We've certainly got plenty of snow today.
Well, you've come to New Zealand's premier snow and ski center, and we have a whole load of activities for you during your week here.
Most visitors come here for the cross-country skiing where you're on fairly flat ground for most of the time rather than going down steep mountain sides.
There are marked trails, but you can also leave these and go off on your own, and that's an experience not to be missed.
You can go at your own speed.
It's great aerobic exercise if you really push yourself.
Or if you prefer, you can just glide gently along and enjoy the beautiful scenery.
This afternoon, you'll be going on a dog sled trip.
You may have seen our dogs on TV recently racing in the Winter Sled Festival.
If you want, you can have your own team for the afternoon and learn how to drive them, following behind our leader on the trail.
Or if you'd prefer, you can just sit back in the sled and enjoy the ride as a passenger.
At the weekend, we have the team relay event, and you're all welcome to join in.
We have a local school coming along, and a lot of the teachers are taking part, too.
Participation rather than winning is the main focus, and there's a medal for everyone who takes part.
Participants are in teams of two to four, and each team must complete four laps of the course.
For your final expedition, you'll head off to Mount Frena, wearing a pair of special snowshoes, which allow you to walk on top of the snow.
This is an area where miners once searched for gold, though there are very few traces of their work left now.
When the snow melts in summer, the mountain slopes are carpeted in flowers and plants.
It's a long ascent, though not too steep, and walkers generally take a couple of days to get to the summit and return.
You'll spend the night in our hut halfway up the mountain.
That's included in your package for the stay.
It's got cooking facilities, firewood, and water for drinking.
For washing, we recommend you use melted snow, though, to conserve supplies.
We can take your luggage up on our snowmobile for you for just \$10 a person.
The hut has cooking facilities, so you can make a hot meal in the evening and morning, but you need to take your own food.
The weather on Mount Faer can be very stormy.
In that case, stay in the hut.
Generally, the storms don't last long.
Don't stress about getting back here to the center in time to catch the airport bus.

They'll probably not be running anyway.

We do have an emergency locator beacon in the hut, but only use that if it's a real emergency, like if someone's ill or injured.

Now, let me tell you something about the different ski trails you can follow during your stay here.

Highland Trails, directly accessible from where we are now.

This trail's been designed to give first timers an experience they'll enjoy regardless of their age or skill.

But it's also ideal for experts to practice their technique.

Then there's Pine Trail.

If you're nervous about skiing, leave this one to the experts.

You follow a steep valley looking right down on the river below.

Scary, but if you fully master the techniques needed for hills, it's great fun.

Stony Trails a good choice once you've got a general idea of the basics.

There are one or two tricky sections, but nothing too challenging.

There's a shelter halfway where you can sit and take a break and enjoy the afternoon sunshine.

And finally, losers trail.

This starts off following a gentle river valley, but the last part is quite exposed, so the snow conditions can be challenging.

If it's snowing or windy, check with us before you set out to make sure the trails open that day.

Right.

So now, if you'd like to follow me, we'll get started.

I've still got loads to do for our report on nutritional food labels.

Me, too.

What did you learn from doing the project about your own shopping habits?

Well, I've always had to check labels for traces of peanuts in everything I eat because of my allergy.

But beyond that, I've never really been concerned enough to check how healthy a product is.

This project has actually taught me to read the labels much more carefully.

I tended to believe claims on packaging like low in fat, but I now realize that the healthy yogurt I've bought for years is full of sugar and that it's actually quite high in calories.

Ready meals are the worst.

Comparing the labels on supermarket pizzas was a real eye opener.

Did you have any idea how many calories they contain?

I was amazed.

Yes, because unless you read the label really carefully, you wouldn't know that the nutritional values given are for half a pizza when most people eat the whole pizza.

Not exactly transparent, is it?

Not at all.

But I expect it won't stop you from buying pizza.

Probably not.

No.

I thought comparing the different labeling systems used by food manufacturers was interesting.

I think the kind of labeling system used makes a big difference.

Which one did you prefer?

I liked the traditional daily value system best.

The one which tells you what proportion of your required daily intake of each ingredient the product contains.

I'm not sure it's the easiest for people to use, but at least you get the full story.

I like to know all the ingredients in a product, not just how much fat, salt, and sugar they contain.

But it's good supermarkets have been making an effort to provide reliable information for customers.

Yes, there just needs to be more consistency between labeling systems used by different supermarkets in terms of portion sizes, etc.

M.

The labels on the different brands of chicken flavor crisps were quite revealing too, weren't they?

Yeah.

I don't understand how they can get away with calling them chicken flavor when they only contain artificial additives.

I know.

I'd at least have expected them to contain a small percentage of real chicken.

Absolutely.

I think having nutritional food labeling has been a good idea, don't you?

I think it will change people's behavior and stop mothers in particular buying the wrong things.

But didn't that study kind of prove the opposite?

People didn't necessarily stop buying unhealthy products.

They only said that might be the case.

Those findings weren't that conclusive and it was quite a smallscale study.

I think more research has to be done.

Yes, I think you're probably right.

What do you think of the traffic light system?

I think supermarkets like the idea of having a colorcoded system, red, orange, or green for levels of fat, sugar, and salt in a product.

But it's not been adopted universally and not on all products.

Why do you suppose that is pressure from the food manufacturers?

Hardly surprising that some of them are opposed to flagging up how unhealthy their products are.

I'd have thought it would have been compulsory.

It seems ridiculous it isn't.

I know.

And what I couldn't get over is the fact that it was brought in without enough consultation.

A lot of experts had deep reservations about it.

That is a bit weird.

I suppose there's an argument for doing the research now when consumers are familiar with this system.

Yeah, maybe.

The participants in the survey were quite positive about the traffic light system.

H but I don't think they targeted the right people.

They should have focused on people with low literacy levels because these labels are designed to be accessible to them.

Yeah.

But it's good to get feedback from all socioeconomic groups.

And there wasn't much variation in their responses.

No.

But if they hadn't interviewed participants face to face, they could have used a much bigger sample size.

I wonder why they chose that method.

Don't know.

How were they selected?

Did they volunteer or were they approached?

I think they volunteered.

The thing that wasn't stated was how often they bought packaged food.

All we know is how frequently they use the supermarket.

In my presentation, I'm going to talk about coffee and its importance both in economic and social terms.

We think it was first drunk in the Arab world, but there's hardly any documentary evidence of it before the 1500s, although of course that doesn't mean that people didn't know about it before then.

However, there is evidence that coffee was originally gathered from bushes growing wild in Ethiopia in the northeast of Africa.

In the early 16th century, it was being bought by traders and gradually its use as a drink spread throughout the Middle East.

It's also known that in 1522 in the Turkish city of Constantinople, which was the center of the Ottoman Empire, the court physician approved its use as a medicine.

By the mid 1500s, coffee bushes were being cultivated in the Yemen and for the next 100 years, this region produced most of the coffee drunk in Africa and the Arab world.

What's particularly interesting about coffee is its effect on social life.

It was rarely drunk at home, but instead people went to coffee houses to drink it.

These people, usually men, would meet to drink coffee and chat about issues of the day.

But at the time, this chance to share ideas and opinions was seen as something that was potentially dangerous.

And in 1623, the ruler of Constantinople demanded the destruction of all the coffee houses in the city.

Although after his death, many new ones opened and coffee consumption continued.

In the 17th century, coffee drinking spread to Europe, and here too, coffee shops became places where ordinary people, nearly always men, could meet to exchange ideas.

Because of this, some people said that these places performed a similar function to universities.

The opportunity they provided for people to meet together outside their own homes and to discuss the topics of the day had an enormous impact on social life.

And many social movements and political developments had their origins in coffee house discussions.

In the late 1600s, the Yemen monopoly on coffee production broke down and coffee production started to spread around the world, helped by European colonization.

Europeans set up coffee plantations in Indonesia and the Caribbean and production of coffee in the colonies skyrocketed.

Different types of coffee were produced in different areas.

And it's interesting that the names given to these different types like mocka or Java coffee were often taken from the port they were shipped to Europe from.

But if you look at the labor system in the different colonies, there were some significant differences.

In Brazil and the various Caribbean colonies, coffee was grown in huge plantations and the workers there were almost all slaves.

But this wasn't the same in all colonies.

For example, in Java, which had been colonized by the Dutch, the peasants grew coffee and passed a proportion of this onto the Dutch.

So, it was used as a means of taxation.

But whatever system was used under the European powers of the 18th century, coffee production was very closely linked to colonization.

Coffee was grown in everincreasing quantities to satisfy the growing demand from Europe and it became nearly as important as sugar production which was grown under very similar conditions.

However, coffee prices were not yet low enough for people to drink it regularly at home.

So most coffee consumptions still took place in public coffee houses and it still remained something of a luxury item.

In Britain, however, a new drink was introduced from China and started to become popular, gradually taking over from coffee, although at first it was so expensive that only the upper classes could afford it.

This was tea, and by the late 1700s, it was being widely drunk.

However, when the USA gained independence from Britain in 1776, they identified this drink with Britain and coffee remained the preferred drink in the USA as it still is today.

So, by the early 19th century, coffee was already being widely produced and consumed.

But during this century, production boomed and coffee prices started to fall.

This was partly because new types of transportation had been developed which were cheaper and more efficient.

So now working people could afford to buy coffee.

It wasn't just a drink for the middle classes.

And this was at a time when large parts of Europe were starting to work in industries.

And sometimes this meant their work didn't stop when it got dark.

They might have to continue throughout the night.

So the use of coffee as a stimulant became important.

It wasn't just a drink people drank in the morning for breakfast.
There were also changes in cultivation.

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Good morning.

What can I do for you?

I want to report a theft.

I had some things stolen out of my bag yesterday.

I'm sorry to hear that.

Right.

So, I'll need to take a few details.

Can I start with your name?

Louise Taylor.

Okay, thank you.

And are you resident in the UK?

No, I'm actually Canadian, though my mother was British.

And your date of birth?

December 14th, 1977.

So, you're just visiting this country.

That's right.

I come over most summers on business.

I'm an interior designer and I come over to buy old furniture, antiques.

You know, there are some really lovely things around here.

But you need to get out to the small towns.

I've had a really good trip this year until this happened.

Okay.

So, you've been here quite a while.

Yes, I'm here for 2 months.

I go back next week.

So, may I ask where you're staying now?

Well, at present, I've got a place at Park Apartments.

That's on King Street.

I was staying at the Riverside Apartments on the same street, but the apartment there was only available for 6 weeks, so I had to find another one.

Okay.

And the apartment number 15.

Right now, I need to take some details of the theft.

So, you said you had some things stolen out of your bag.

That's right.

And were you actually carrying the bag when the theft took place?

Yes.

I really can't understand it.

I had my backpack on and I went into a supermarket to buy a few things and when I opened it up, my wallet wasn't there.

And what did your wallet have in it?

Well, fortunately, I don't keep my credit cards in that wallet.

I keep them with my passport in an inside compartment in my backpack, but there was quite a bit of cash there.

About £250 sterling, I should think.

I withdrew £300 from my account yesterday, but I did a bit of shopping, so I must have already spent about £50 of that.

Okay.

At first, I thought, "Oh, I must have left the wallet back in the apartment." But then I realized my phone had gone as well.

It was only a week old, and that's when I realized I'd been robbed.

Anyway, at least they didn't take the keys to my rental car.

Yes.

So, you say the theft occurred yesterday?

Yes.

So, that was September the 10th.

And do you have any idea at all of where or when the things might possibly have been stolen?

Well, at first I couldn't believe it because the bag had been on my back ever since I left the apartment after lunch.

It's just a small backpack, but I generally use it when I'm traveling because it seems safer than a handbag.

Anyway, I met up with a friend and we spent a couple of hours in the museum.

But I do remember that as we were leaving there at about 4:00, a group of young boys ran up to us and they were really crowding around us and they were asking us what time it was.

Then all of a sudden they ran off.

Can you remember anything about them?

The one who did most of the talking was wearing a t-shirt with a picture of something.

Uh, let's see.

A tiger, right?

Any idea of how old he might have been?

Around 12 years old.

And can you remember anything else about his appearance?

Not much.

He was quite thin.

Color of hair?
I do remember that.
He was blonde.
All the others were dark-haired.
And any details of the others?
Not really.
They came and went so quickly.
Right.
So, what I'm going to do now is give you a crime reference number so you can contact your insurance company.
So, this is 10 digits.
87954 82361.
Thank you.
So, should I contact Good morning everyone.
My name's Janet Parker and I'm the human resources manager.
We're very happy to welcome you to your new apprenticeship.
I hope that the next 6 months will be a positive and enjoyable experience for you.
I'd like to start with some general advice about being an apprentice.
Most of you have very little or no experience of working for a big organization and the first week or so may be quite challenging.
There will be a lot of new information to take in, but don't worry too much about trying to remember everything.
The important thing is to check with someone if you're not sure what to do.
You'll find your supervisor is very approachable and won't mind explaining things or helping you out.
You're here to learn, so make the most of that opportunity.
You'll be spending time in different departments during your first week, so make an effort to talk to as many people as possible about their work.
You'll make some new friends and find out lots of useful information.
As well as having a supervisor, you'll each be assigned a mentor.
This person will be someone who's recently completed an apprenticeship, and you'll meet with them on a weekly basis.
Their role is to provide help and support throughout your apprenticeship.
Of course, this doesn't mean they'll actually do any of your work for you.
Instead, they'll be asking you about what goals you've achieved so far, as well as helping you to identify any areas for improvement.
You can also discuss your more long-term ambitions with them as well.
Now, I just want to run through a few company policies for our apprenticeship scheme with you.
Most importantly, the internet.
As part of your job, you'll be doing some research online.
So obviously you'll have unlimited access for that but please don't use it for personal use.
You'll have your own phones for that.
Some of you have already asked me about flexible working after your probationary 3month period.
Some of you will be eligible for this but it will depend on which department you're in and what your personal circumstances are.
So, please don't assume you'll automatically be permitted to do this.
I want to make sure there's no confusion about our holiday policy.
Apart from any statutory public holidays, we ask that you don't book any holidays until after your six-month apprenticeship has finished.
Time off should only be taken if you are unwell.
Please speak to your supervisor if this is going to be a problem.
You'll be expected to work a 40hour week, but there may be opportunities to do overtime during busy periods.
Although you're not required to do this, it can be a valuable experience.
So, we advise you to take it up if possible.
Obviously, we understand that people do have commitments outside work.
So, don't worry if there are times when you are unavailable.
As you know, we don't have a formal dress code here.
You may wear casual clothes as long as they are practical, and the only restriction for shoes we have is on high heels for health and safety reasons.
Comfortable shoes like trainers are preferable.
There's a heavily subsidized canteen on site where you can get hot meals or salads cheaply.
Snacks and drinks are also provided.
So, we've decided to introduce a no packed lunch policy.
This is partly to encourage healthy eating at work and partly to stop people from eating at their work station, which is unhygienic.
Okay, moving on to Okay, so what I'd like you to do now is to talk to your partner about your presentations on urban planning.
You should have done most of the reading now.
So, I'd like you to share your ideas and talk about the structure of your presentation and what you need to do next.
Okay, Rob.
I'm glad we chose quite a specific topic.
Cities built next to the sea.
It made it much easier to find relevant information.
Yeah.
And cities are growing so quickly.
I mean, we know that more than half the world's population lives in cities now.
Yeah.
Though that's all cities, not just ones on the coast.
But most of the biggest cities are actually built by the sea.
I'd not realized that before.

Nor me.

And what's more, a lot of them are built at places where rivers come out into the sea.

But apparently this can be a problem.

Why?

Well, as the city expands, agriculture and industry tend to spread further inland along the rivers.

And so agriculture moves even further in land up the river.

That's not necessarily a problem except it means more and more pollutants are discharged into the rivers.

So these are brought downstream to the cities, right?

Did you read that article about Miami on the east coast of the USA?

No.

Well, apparently back in the 1950s, they built channels to drain away the water in case of flooding.

Sounds sensible.

Yeah.

They spent quite a lot of money on them, but what they didn't take into account was global warming.

So, they built the drainage channels too close to sea level.

And now sea levels are rising.

They're more or less useless.

If there's a lot of rain, the water can't run away.

There's nowhere for it to go.

The whole design was faulty.

So, what are the authorities doing about it now?

I don't know.

I did read that they're aiming to stop disposing of waste water into the ocean over the next 10 years.

But that won't help with flood prevention now, will it?

No, really.

They just need to find the money for something to replace the drainage channels in order to protect against flooding now.

But in the long term, they need to consider the whole ecosystem.

Right.

Really though, coastal cities can't deal with their problems on their own, can they?

I mean, they've got to start acting together at an international level instead of just doing their own thing.

Absolutely.

The thing is, everyone knows what the problems are, and environmentalists have a pretty good idea of what we should be doing about them.

So, they should be able to work together to some extent, but it's going to be a long time before countries come to a decision on what principles they're prepared to abide by.

Yeah, if they ever do.

So, I think we've probably got enough for our presentation.

It's only 15 minutes.

Okay.

So, I suppose we'll begin with some general historical background about why coastal cities were established, but we don't want to spend too long on that.

The other students will already know a bit about it.

It's all to do with communications and so on.

Yes, we should mention some geographical factors.

uh things like wetlands and river estuaries and coastal erosion and so on.

We could have some maps of different cities with these features marked on a handout, you mean?

Or some slides everyone can see?

Yeah, that'd be better.

It'd be good to go into past mistakes in a bit more detail.

Did you read that case study of the problems there were in New Orleans with flooding a few years ago?

Yes, we could use that as the basis for that part of the talk.

I don't think the other students will have read it, but they'll remember hearing about the flooding at the time.

H okay, so that's probably enough background.

So then we'll go on to talk about what actions being taken to deal with the problems of coastal cities.

Okay.

What else do we need to talk about?

Maybe something on future risks, looking more at the long term if populations continue to grow.

Yeah, we'll need to do a bit of work there.

I haven't got much information.

Have you?

No, we'll need to look at some websites.

Shouldn't take too long.

Okay.

And I think we should end by talking about international implications.

Maybe we could ask people in the audience.

We've got people from quite a lot of different places.

That would be interesting if we have time.

Yes.

So, now should we go on?

Producing enough energy to meet our needs has become a serious problem.

Demand is rising rapidly because of the world's increasing population and expanding industry. Burning fossil fuels like gas, coal, and oil seriously damages the environment, and they'll eventually run out. For a number of years now, scientists have been working out how we can derive energy from renewable sources such as the sun and wind without causing pollution.

Today, I'll outline marine renewable energy, also called ocean energy, which harnesses the movement of the oceans. Marine renewable energy can be divided into three main categories. wave energy, tidal energy, and ocean thermal energy conversion. And I'll say a few words about each one.

First, wave energy.

Numerous devices have been invented to harvest wave energy with names such as wave dragon, the penguin, and mighty whale. and research is going on to try and come up with a really efficient method.

This form of energy has plenty of potential as the source is constant and there's no danger of waves coming to a standstill. Electricity can be generated using onshore systems, using a reservoir or offshore systems. But the problem with ocean waves is that they're erratic with the wind making them travel in every direction. This adds to the difficulty of creating efficient technology. Ideally, all the waves would travel smoothly and regularly along the same straight line. Another drawback is that sand and other sediment on the ocean floor might be stopped from flowing normally which can lead to environmental problems.

The second category of marine energy that I'll mention is tidal energy.

One major advantage of using the tide rather than waves as a source of energy is that it's predictable. We know the exact times of high and low tides for years to come.

For tidal energy to be effective, the difference between high and low tides needs to be at least 5 m. And this occurs naturally in only about 40 places on Earth.

But the right conditions can be created by constructing a tidal lagoon, an area of seawater separated from the sea. One current plan is to create a tidal lagoon on the coast of Wales. This will be an area of water within a bay at Swansea, sheltered by a U-shaped breakwater or dam built out from the coast. The breakwater will contain 16 hydro turbines. And as the tide rises, water rushes through the breakwater activating the turbines which turn a generator to produce electricity. Then for 3 hours as the tide goes out, the water is held back within the breakwater, increasing the difference in water level until it's several meters higher within the lagoon than in the open sea. Then in order to release the stored water, gates in the breakwater are opened. It pours powerfully out of the lagoon, driving the turbines in the breakwater in the opposite direction. and again generating thousands of megawatt of electricity.

As there are two high tides a day, this lagoon scheme would generate electricity four times a day every day for a total of around 14 hours in every 24 and enough electricity for over 150,000 homes.

This system has quite a lot in its favor. Unlike solar and wind energy, it doesn't depend on the weather. The turbines are operated without the need for fuel, so it doesn't create any greenhouse gas emissions and very little maintenance is needed. It's estimated that electricity generated in this way will be relatively cheap and that manufacturing the components would create more than 2,000 jobs. a big boost to the local economy.

On the other hand, there are fears that lagoons might harm both fish and birds, for example, by disturbing migration patterns and causing a buildup of silt affecting local ecosystems.

There are other forms of tidal energy, but I'll go on to the third category of marine energy, ocean thermal energy conversion. This depends on there being a big difference in temperature between surface water and the water a couple of kilometers below the surface. And this occurs in tropical coastal areas. The idea is to bring cold water up to the surface using a submerged pipe. The concept dates back to 1881.

Hi, come and take a seat.

Thank you.

My name's Carl Rogers and I'm one of the doctors here at the Total Health Clinic.

So, I understand this is your first visit to the clinic.

Yes, it is.

Okay.

Well, I hope you'll be very happy with the service you receive here.

So, if it's all right with you, I'll take a few details to help me give you the best possible service.

Sure.

So, can I check first of all that we have the correct personal details for you?

So, your full name is Julie Anne Garcia.

That's correct.

Perfect.

And can I have a contact phone number?

It's 2194429785.

Okay.

And then can I just check that we have the correct date of birth?

October 10th, 1992.

Oh, I actually have 1991.

I'll just correct that now.

Right.
So, that's all good.
Now, I just need a few more personal details.
Do you have an occupation either full-time or part-time?
Uh, yes.
I work full-time in Esther Hazes.
You know, the restaurant chain.
I started off as a waitress there a few years ago, and I'm a manager now.
Oh, I know them.
Yeah, they're down on 114th Street, aren't they?
Uh, that's right.
Yeah, I've been there a few times.
I just love their salads.
That's good to hear.
Right.
So, one more thing I need to know before we talk about why you're here, Julie.
And that's the name of your insurance company.
It's Collie Life Insurance.
That's C A W L E Y.
Excellent.
Thank you so much.
Now, Julie, let's look at how we can help you.
So, tell me a little about what brought you here today.
Well, I've been getting a pain in my knee, the left one.
Not very serious at first, but it's gotten worse.
So, I thought I ought to see someone about it.
That's certainly the right decision.
So, how long have you been aware of this pain?
Is it just a few days, or is it longer than that?
longer.
It's been worse for the last couple of days, but it's 3 weeks since I first noticed it.
It came on quite gradually, though, so I kind of ignored it at first.
And have you taken any medication yourself or treated it in any way?
Um, yeah, I've been taking medication to deal with the pain, Tylenol, and that works okay for a few hours, but I don't like to keep taking it.
Okay.
And what about heat treatment?
Have you tried applying heat at all?
No, but I have been using ice on it for the last few days.
And does that seem to help the pain at all?
A little.
Yes.
Good.
Now, you look as if you're quite fit normally.
I am.
Yes.
So, do you do any sport on a regular basis?
Yes.
I play a lot of tennis.
I belong to a club, so I go there a lot.
I'm quite competitive, so I enjoy that side of it as well as the exercise, but I haven't gone since this started.
Sure.
And you do any other types of exercise?
Uh yeah, I sometimes do a little swimming, but usually just when I'm on vacation, but normally I go running a few times a week, maybe three or four times.
M So your legs are getting quite a pounding, but you haven't had any problems up to now?
No, not with my legs.
I did have an accident last year when I slipped and hurt my shoulder, but that's better now.
Excellent.
And do you have any allergies?
No, none that I'm aware of.
And do you take any medication on a regular basis?
Well, I take vitamins, but that's all.
I'm generally very healthy.
Okay.
Well, let's have a closer look and see what might be causing this problem.
If you can just get up.
We'll be arriving at Branley Castle in about 5 minutes.
But before we get there, I'll give you a little information about the castle and what our visit will include.
So in fact there's been a castle on this site for over,00 years.
The first building was a fort constructed in 914 AD for defense against Danish invaders by King Alfred the Great's daughter who ruled

England at the time.

In the following century after the Normans conquered England, the land was given to a nobleman called Richard Dvere and he built a castle there that stayed in the Deve family for over 400 years.

However, when Queen Elizabeth I announced that she was going to visit the castle in 1576, it was beginning to look a bit rund down, and it was decided that rather than repair the guest rooms, they'd make a new house for her out of wood next to the main hall.

She stayed there for four nights, and apparently it was very luxurious, but unfortunately, it was destroyed a few years later by fire.

In the 17th century, the castle belonged to the wealthy Fenis family who enlarged it and made it more comfortable.

However, by 1982, the Fenis family could no longer afford to maintain the castle even though they received government support and they put it on the market.

It was eventually taken over by a company who owned a number of amusement parks.

But when we get there, I think you'll see that they've managed to retain the original atmosphere of the castle.

When you go inside, you'll find that in the stateaterooms, there are lifelike moving wax models dressed in costumes of different periods in the past, which even carry on conversations together.

As well as that, in every room, there are booklets giving information about what the room was used for and the history of the objects and furniture it contains.

The castle parks quite extensive.

At one time, sheep were kept there.

And in the 19th century, the owners had a little zoo with animals like rabbits and even a baby elephant.

Nowadays, the old zoo buildings are used for public displays of paintings and sculpture.

The park also has some beautiful trees, though the oldest of all, which dated back 800 years, was sadly blown down in 1987.

Now, you're free to wander around on your own until 4:30, but then at the end of our visit, we'll all meet together at the bottom of the great staircase.

We'll then go on to the Long Gallery where there's a wonderful collection of photographs showing the family who owned the castle 100 years ago having tea and cakes in the conservatory.

And we'll then take you to the same place where afternoon tea will be served to you.

Now, if you can take a look at your plans, you'll see Branley Castle has four towers joined together by a high wall with a river on two sides.

Don't miss seeing the great hall.

That's near the river in the main tower, the biggest one, which was extended and redesigned in the 18th century.

If you want to get a good view of the whole castle, you can walk around the walls.

The starting point's quite near the main entrance.

Walk straight down the path until you get to the south gate, and it's just there.

Don't go on to the north gate.

There's no way up from there.

There'll shortly be a show in which you can see archers displaying their skill with a bow and arrow.

The quickest way to get there is to take the first left after the main entrance and follow the path past the bridge.

Then you'll see it in front of you at the end.

If you like animals, there's also a display of hunting birds, falcans, and eagles and so on.

If you go from the main entrance in the direction of the south gate, but turn right before you get there instead of going through it, you'll see it on your right past the first tower.

At 300 p.m., there's a short performance of traditional dancing on the outdoor stage.

That's right at the other side of the castle from the entrance and over the bridge.

It's about 10 minutes walk or so.

And finally, the shop.

It's actually inside one of the towers, but the way in is from the outside.

Just take the first left after the main entrance.

Go down the path and take the first right.

It's got some lovely gifts and souvenirs.

Right.

So, we're just arriving.

So, Rosie and Martin, let's look at what you've got for your presentation on woolly mammoths.

Okay.

We've got a short outline here.

Thanks.

Uh, so it's about a research project in North America.

Yes, but we thought we needed something general about woolly mammoths in our introduction to established that they were related to our modern elephant and they lived thousands of years ago in the last ice age.

Maybe we could show a video clip of a cartoon about mammoths, but that would be a bit childish.

Or we could have a diagram.

It could be a timeline to show when they lived with illustrations.

Or we could just show a drawing of them walking in the ice.

No, let's go with your last suggestion.

Good.

Then you're describing the discovery of the mammoth tooth on St.

Paul's Island in Alaska and why it was significant.

Yes.

The tooth was found by a man called Russell Graham.

He picked it up from under a rock in a cave.

He knew it was special.

For a start, it was in really good condition, as if it had been just extracted from the animals jawbone.

Anyway, they found it was 6,500 years old.

So, why was that significant?

Well, the mammoth bones previously found on the North American mainland were much less recent than that.

So this was really amazing.

Then we're making an animated diagram to show the geography of the area in prehistoric times.

So originally St.

Paul's Island wasn't an island.

It was connected to the mainland and mammoths and other animals like bears were able to roam around the whole area.

Then the climate warmed up and the sea level began to rise and the island got cut off from the mainland.

So those mammoths on the island couldn't escape.

They had to stay on the island.

And in fact, the species survived there for thousands of years after they'd become extinct on the mainland.

So why do you think they died out on the mainland?

No one's sure.

Anyway, next we'll explain how Graham and his team identified the date when the mammoths became extinct on the island.

They concluded that the extinction happened 5,600 years ago, which is a very precise time for a prehistoric extinction.

It's based on samples they took from mud at the bottom of a lake on the island.

They analyzed it to find out what had fallen in over time.

bits of plants, volcanic ash, and even DNA from the mammoths themselves.

It's standard procedure, but it took nearly 2 years to do.

So, why don't you quickly go through the main sections of your presentation and discuss what actions needed for each part?

Okay.

So, for the introduction, we're using a visual.

So, once we've prepared that, we're done.

I'm not sure.

I think we need to write down all the ideas we want to include here, not just rely on memory.

How we begin the presentation is so important.

You're right.

The discovery of the mammoth tooth is probably the most dramatic part, but we don't have that much information, only what we got from the online article.

I thought maybe we could get in touch with the researcher who led the team and ask him to tell us a bit more.

Great idea.

What about the section with the initial questions asked by the researchers?

We've got a lot on that, but we need to make it interesting.

We could ask the audience to suggest some questions about it and then see how many of them we can answer.

I don't think it would take too long.

Yes, that would add a bit of variety.

Then the section on further research carried out on the island, analyzing the mud in the lake.

I wonder if we've actually got too much information here.

Should we cut some?

I don't think so, but it's all a bit muddled at present.

Yes.

Maybe it would be better if it followed a chronological pattern.

I think so.

The findings and possible explanations section is just about ready, but we need to practice it so we're sure it won't overrun.

I think it should be okay.

But yes, let's make sure.

H in the last section relevance to the present day you've got some good ideas but this is where you need to move away from the ideas of others and give your own viewpoint.

Okay, we'll think about that now.

Shall we show you some of the in this series of lectures about the history of weather forecasting?

I'll start by examining its early history.

That'll be the subject of today's talk.

Okay.

So, we'll start by going back thousands of years.

Most ancient cultures had weather gods and weather catastrophes such as floods played an important role in many creation myths.

Generally, weather was attributed to the whims of the gods as the wide range of weather gods in various cultures shows.

For instance, there's the Egyptian sun god Rah and Thor, the Norse god of thunder and lightning.

Many ancient civilizations developed rituals such as dances in order to make the weather gods look kindly on them.

But the weather was of daily importance.

Observing the skies and drawing the correct conclusions from these observations was really important.

In fact, their survival depended on it.

It isn't known when people first started to observe the skies, but at around 650 BC, the Babylonians produced the first short range weather forecasts based on their observations of clouds and other phenomena.

The Chinese also recognized weather patterns and by 300 BC astronomers had developed a calendar which divided the year into 24 festivals each associated with a different weather phenomenon.

The ancient Greeks were the first to develop a more scientific approach to explaining the weather.

The work of the philosopher and scientist Aristotle in the 4th century BC is especially noteworthy as his ideas held sway for nearly 2,000 years.

In 340 BC, he wrote a book in which he attempted to account for the formation of rain, clouds, wind, and storms.

He also described celestial phenomena such as halos, that is bright circles of light around the sun, the moon, and bright stars, and

comets.
Many of his observations were surprisingly accurate.
For example, he believed that heat could cause water to evaporate.
But he also jumped to quite a few wrong conclusions such as that winds are breathed out by the earth.
Errors like this were rectified from the Renaissance onwards.
For nearly 2,000 years, Aristotle's work was accepted as the chief authority on weather theory.
Alongside this though, in the Middle Ages, weather observations were passed on in the form of proverbs such as red sky at night, shepherd's delight, red sky in the morning, shepherd's warning.
Many of these are based on very good observations and are accurate, as contemporary meteorologists have discovered.
For centuries, any attempt to forecast the weather could only be based on personal observations.
But in the 15th century, scientists began to see the need for instruments.
Until then, the only ones available were weather vane to determine the wind direction and early versions of rain gauges.
One of the first invented in the 15th century was a hyrometer which measured humidity.
This was one of many inventions that contributed to the development of weather forecasting.
In 1592, the Italian scientist and inventor Galileo developed the world's first thermometer.
His student Torricelli later invented the barometer which allowed people to measure atmospheric pressure.
In 1648, the French philosopher Pascal proved that pressure decreases with altitude.
This discovery was verified by English astronomer Halley in 1686.
and he was also the first person to map trade winds.
This increasing ability to measure factors related to weather helped scientists to understand the atmosphere and its processes better and they started collecting weather observation data systematically.
In the 18th century, the scientist and politician Benjamin Franklin carried out work on electricity and lightning in particular, but he was also very interested in weather and studied it throughout most of his life.
It was Franklin who discovered that storms generally travel from west to east.
In addition to new meteorological instruments, other developments contributed to our understanding of the atmosphere.
People in different locations began to keep records and in the mid 19th century, the invention of the telegraph made it possible for these records to be collected.
This led by the end of the 19th century to the first weather services.
It was not until the early 20th century that mathematics and physics became part of meteorology.
And we'll continue from that point next week.
Hello, Flanders Conference Hotel.
Oh, hi.
I wanted to ask about conference facilities at the hotel.
Have I come through to the right person?
Mhm.
You have.
I'm the customer services manager.
My name's Angela.
So, how can I help you?
Well, I'm calling from Barit and Stansson's.
We're a medical company based in Perth.
Oh, yes.
And we're organizing a conference for our clients to be held in Sydney.
It'll be held over 2 days and we're expecting about 50 or 60 people.
When were you thinking of having it?
Sometime early next year, like the end of January.
It would have to be a weekend.
Uh, let me see.
Our conference facilities are already booked for the weekend beginning January the 28th.
We could do the first weekend in February.
How about January the 21st?
Uh oh.
I'm afraid that's book two.
Well, let's go for the February date then.
So that's the weekend beginning the 4th.
Okay.
Now, can you tell me a bit about what conference facilities you have?
Sure.
So, for talks and presentations, we have the Tesla room.
Sorry, Tesla.
That's spelled T E S L A.
It holds up to 100 people and it's fully equipped with a projector and so on.
How about a microphone?
Yes, that'll be all set up ready for you.
And there'll be one that members of the audience can use too for questions if necessary.
Fine.
And we'll also need some sort of open area where people can sit and have a cup of coffee.
And we'd like to have an exhibition of our products and services there as well.
So that'll need to be quite a big space.
Mhm.
That's fine.

There's a central atrium with all those facilities and you can come before the conference starts if you want to set everything up.
Great.
And I presume there's Wi-Fi.
Oh, yes.
That's free and available throughout the hotel.
Okay.
Would you also like us to provide a buffet lunch?
We can do a two course meal with a number of different options.
What sort of price are we looking at for that?
Well, I can send you a copy of this standard menu.
That's \$45 per person.
Or you can have the special for \$25 more.
I think the standard should be okay, but yes, send me the menu.
Now, we're also going to need accommodation on the Saturday night for some of the participants.
I'm not sure how many, but probably about 25.
So, what do you charge for a room?
Well, for conference attendees, we have a 25% reduction.
So, we can offer you rooms at \$135.
Normally, a standard room's \$180.
And does that include breakfast?
Sure.
And of course, guests can also make use of all the other facilities at the hotel.
So, we've got a spa where you can get massages and facials and so on.
And there's a pool up on the roof for the use of guests.
Oh, great.
Now, what about transport links?
The hotel's downtown, isn't it?
Yes.
It's about 12 km from the airport, but there's a complimentary shuttle bus for guests, and it's only about 10 minutes walk from the central railway station.
Okay.
Now, I don't know Sydney very well.
Can you just give me an idea of the location of the hotel?
Uh, well, it's downtown on Wilby Street.
Uh, that's quite a small street and it's not very far from the sea.
And of course, if the conference attendees want to go out on their Saturday evening, there's a huge choice of places to eat.
Then if they want to make a night of it, they can go on to one of the clubs in the area.
There are a great many to choose from.
Okay.
So, if we go ahead with this, can you give me some information about how much we're looking at?
Good morning.
My name's Lucy Kittinden and I'm the director of operations for an organization that arranges volunteering in this part of the country.
I'm hoping I can persuade one or two of you to become volunteers yourselves.
Let me start by briefly explaining what we mean by volunteering.
Volunteers are teenagers and adults who choose to spend some time unpaid helping other people in some way.
Most volunteers devote 2 or 3 hours to this every week, while a few do much more.
The people they help may have physical or behavioral difficulties, for example.
Volunteers can do all sorts of things depending on their own abilities and interests.
If they're supporting a family that's struggling, for example, they may be able to give them tips on cooking or recommend how to plan their budget or how to shop sensibly on their income.
They might even do some painting or wallpapering perhaps alongside any members of the family who were able to do it or even do some babysitting so the parents can go out for a while.
The benefit from volunteering isn't only for the people being helped.
Volunteers also gain from it.
They're using their skills to cope with somebody's mental or physical ill health.
and volunteering may be a valuable element of their CV when they're applying for jobs.
Employers usually look favorably on someone who's given up time to help others.
Significantly, most volunteers feel that what they're doing gives them a purpose in their lives.
And in my opinion, they're lucky in that respect, as many people don't have that feeling.
Now, I'd like to tell you what some of our volunteers have said about what they do to give you an idea of the range of ways in which they can help people.
Habib supports an elderly lady who's beginning to show signs of dementia.
Once a week, they along with other elderly people go to the local community center where a group of people come in and sing.
The songs take the listeners back to their youth and for a little while they can forget the difficulties that they face now.
Our volunteer Consuela is an amazing woman.
She has difficulty walking herself but she doesn't let that stop her.
She helps a couple of people with similar difficulties who had almost stopped walking altogether.
By using herself as an example, Consuela encourages them to walk more and more.
Min visits a young man who lives alone and can't leave his home on his own.
So, he hardly ever saw anyone, but together they go out to the cinema or to see friends the young man hadn't been able to visit for a long

time.

Tanya visits an elderly woman once a week.

When the woman found out that Tanya is a professional dress maker, she got interested.

Tanya showed her some soft toys she'd made, and the woman decided to try it herself.

And now she really enjoys it and spends hours making toys.

They're not perhaps up to Tanya's standard yet, but she gains a lot of pleasure from doing it.

Alexi is a volunteer with a family that faces a number of difficulties.

By calmly talking over possible solutions with family members, he's helping them to realize that they aren't helpless and that they can do something themselves to improve their situation.

This has been great for their self-esteem.

And the last volunteer I'll mention, though there are plenty more, is Juba.

She volunteers with a teenage girl with learning difficulties who wasn't very good at talking to other people.

Juba's worked very patiently with her and now the girl is far better at expressing herself and at understanding other people.

Okay, I hope that's given you an idea of what volunteering is all about.

Now, I'd like to talk to you about So, how are you getting on with your teaching practice at the high school, Joe?

Well, I've been put in charge of the school marching band, and it's quite a responsibility.

I'd like to talk it over with you.

Go ahead.

You better start by giving me a bit of background.

Okay.

Well, the band has students in it from all years, so they're aged 11 to 18, and there are about 50 of them altogether.

It's quite a popular activity within the school.

I've never worked with a band of more than 20 before, and this is very different.

I can imagine.

They aren't really good enough to enter national band competitions, but they're in a regional one later in the term.

Even if they don't win, and I don't expect them to, hopefully it'll be an incentive for them to try and improve.

Yes, hopefully.

Well, now the town council's organizing a carnival in the summer, and the band has been asked to perform.

If you ask me, they aren't really up to it yet, and I need to get them functioning better as a band and in a very short time.

Have you been doing anything with them, apart from practicing the music?

I mean, I played a recording I came across of a drummer talking about how playing in a band had changed his life.

I think it was an after-dinner speech.

I thought it was pretty inspiring because being in the band had stopped him from getting involved in crime.

The students seem to find it interesting, too.

That's good.

I'm planning to show them that old film from the 1940s, Strike Up the Band, and talk about it with the students.

What do you think?

Good idea.

As it's about a school band, it might make the students realize how much they can achieve if they work together.

That's what I've got in mind.

I'm hoping I can take some of the band to a parade that's going to take place next month.

A couple of marching bands will be performing and the atmosphere should be quite exciting.

It depends on whether I can persuade the school to hire a coach or two to take us there.

H, they sound like good ideas to me.

Thanks.

Can I tell you about a few people in the band who I'm finding it quite difficult to cope with?

I'm sure you'll have some ideas about what I can do.

Go ahead.

There's a flautist who says she loves playing in the band.

We rehearse twice a week after school, but she's hardly ever there.

Then she looks for me the next day and gives me a very plausible reason.

She says she had to help her mother or she's been ill.

But to be honest, I don't believe her.

Oh dear.

Any more students with difficulties?

Plenty.

Uh, there's a trumpeter who thinks she's the best musician in the band, though she certainly isn't.

She's always saying what she thinks other people should do, which makes my job pretty difficult.

She sounds a bit of a nightmare.

You can say that again.

One of the trombonists has got an impressive sense of rhythm and could be an excellent musician, except that he has breathing difficulties and he doesn't really have enough breath for the trombone.

He'd be much better off playing percussion, for instance, but he refuses to give up, so he ends up only playing half the notes.

I suppose you have to admire his determination.

Maybe one of the percussionists isn't too bad, but he never seems to interact with other people, and he always rushes off as soon as the rehearsal ends.

I don't know if there are family reasons or what, but it isn't good in a band where people really need to feel they're part of a group.

There are others, too.

But at least that gives you an idea of what I'm up against.

Do you have any thoughts about what I can do, Lizzy?

As you all know, the university is planning an arts festival for later this year.

And here in the music department, we've planned three concerts.
 These will be public performances and the program has just been finalized.
 The theme of the festival is links between the UK and Australia and this is reflected in the music.
 Each concert will feature both British and Australian composers.
 I'll tell you briefly about the Australian music as you probably won't be familiar with that.
 The first concert will include music by Liza Lim who was born in Perth, Western Australia in 1966.
 As a child, Lim originally learned to play the piano like so many children and also the violin.
 But when she was 11, her teachers encouraged her to start composing.
 She found this was her real strength and she studied and later taught composition both in Australia and in other countries.
 As a composer, she has received commissions from numerous orchestras, other performers, and festivals in several countries.
 Liza Lim's compositions are vibrant and full of energy and she often explores Asian and Australian Aboriginal cultural sources including the native instrument the didgeridoo.
 This is featured in a work called the compass.
 Her music is very expressive.
 So although it is complex, it has the power of connecting with audiences and performers alike.
 In the festival, we're going to give a semi-staged performance of the Orissa.
 This is an opera in seven parts based on the trilogy of ancient Greek tragedies by Euripides.
 Lim composed this when she was in her mid-20s and she also wrote the text along with Barry Koski.
 It's performed by six singers, a dancer, and an orchestra that, as well as standard orchestral instruments, includes electric guitar and a traditional Turkish stringed instrument.
 Lim wrote that because the stories in the tragedies are not easy to tell.
 The sounds she creates are also disturbing, and they include breathing, sobbing, laughing, and whistling.
 The work lasts around 75 minutes and the rest of the concert will consist of orchestral works by the British composers Rafe Vaughn Williams and Frederick Delius.
 Moving on now to our second concert.
 This will begin with instrumental music by British composers Benjamin Britten and Judith Weir.
 After the interval, we'll go to Australia for a piece by Ross Edwards, the Tower of Remoteness.
 According to Edwards, the inspiration for this piece came from nature, when he was sitting alone in the dry bed of a creek, overshadowed by the leaves of palm trees, listening to the birds and insects.
 The Tower of Remoteness is scored for piano and clarinet.
 Edwards says he realized years after writing the piece that he had subconsciously modeled its opening phrase on a bird call.
 Ross Edwards was born in 1943 in Sydney, Australia, and studied at the Sydney Conservatorium of Music and the Universities of Adelaide and Sydney.
 He's well known in Australia, and in fact, he's one of the country's most performed composers.
 He's written a wide range of music from symphonies and concertos to some composed specifically for children.
 Edwards's music has been described as being deeply connected to Australia and it can be regarded as a celebration of the diversity of cultures that Australia can be proud of.
 The last of the three Australian composers to be represented in our festival is Carl Vine.
 Born in 1954, Vine like Liza Lim comes from Perth, Western Australia.
 He took up the Cornet at the age of five, switching to the piano 5 years later.
 However, he went to university to study physics before changing to composition.
 After graduating, he moved to Sydney and worked as a freelance pianist and composer.
 Before long, he had become prominent in Australia as a composer for dance and in fact has written 25 scores of that type.
 In our third concert, Vine will be represented by his music for the flag handover ceremony of the Olympics held in 1996.
 This 7-minute orchestral piece was of course heard by millions of people worldwide and will hear it alongside works written by British composers Edward Elgar and more recently Thomas Adès.
 Good morning Clarehouse Hotel.
 Andrew speaking.
 I'm the events manager.
 Good morning Andrew.
 My name's Samantha.
 I'm arranging a party for my parents' 50th wedding anniversary, and I'm ringing to ask about hiring a room sometime next September.
 Also, my parents and several of the guests will need accommodation.
 Okay, I'm sure we can help you with that.
 Will you be having a sit-down meal or a buffet?
 Probably a sit-down.
 And do you know how many people there'll be?
 around 80, I think.
 Well, we have two rooms that can hold that number.
 One is the Adelfi room that can seat 85 or hold over a hundred if people are standing for a buffet.
 If you have live music, there's room for four or five musicians in the gallery overlooking the room.
 Our guests usually appreciate the fact that the music can be loud enough for dancing, but not too loud for conversation.
 Yes.
 I really don't like it when you can't talk.
 Exactly.
 Now, the Adelfi room is at the back of the hotel, and there are French windows leading out onto the terrace.
 This has a beautiful display of pots of roses at that time of the year.
 Which direction does it face?
 Southwest.
 So that side of the hotel gets the sun in the afternoon and early evening.

Very nice.

From the terrace, you can see the area of trees within the grounds of the hotel or you can stroll through there to the river.

That's on the far side.

So it isn't visible from the hotel.

Then another option is the Carlton room.

This is a bit bigger.

It can hold up to 110 people and it has the advantage of a stage which is useful if you have any entertainment or indeed a small band can fit onto it.

How and can you go outside from the room?

No, the Carlton room is on the first floor, but on one side the windows look out onto the lake.

Lovely.

I think either of those rooms would be suitable.

Can I tell you about some of the options we offer in addition?

Please do.

As well as a meal, you can have an MC, a master of ceremonies, who will be with you throughout the party.

What exactly is the MC's function?

I suppose they make a speech during the meal if we need one, do they?

That's right.

All our MC's are trained as public speakers, so they can easily get people's attention.

Many guests are glad to have someone who can make themselves heard above the chatter, and they're also your support.

If anything goes wrong, the MC will deal with it so you can relax.

Great.

I'll need to ask you about food, but something else that's important is accommodation.

You obviously have rooms in the hotel, but do you also have any other accommodation like cabins for example?

Yes, there are five in the grounds, all self-contained.

They each sleep two to four people and have their own living room, bathroom, and small kitchen.

That sounds perfect for what we'll need.

Now, you have various facilities, don't you?

Are they all included in the price of hiring the room?

The pool, for instance.

Normally, you'd be able to use it, but it'll be closed throughout September for refurbishment, I'm afraid.

The gym will be available, though, at no extra charge.

That's open all day from 6:00 in the morning until midnight.

And the tennis courts, but there is a small additional payment for those.

We have four courts and it's worth booking in advance if you possibly can as there can be quite a long waiting list for them.

Right now, could we discuss the food?

This would be dinner around 7:00 and Hello everyone.

I'm Jake Stevens and I'm your rep here at the hotel.

I'm sure you'll all have a great time here.

So, let me tell you a bit about what's on offer.

I'll start by telling you about some of the excursions that are available for guests.

One thing you have to do while you're here is go dolphin watching.

On our boat trips, we pretty well guarantee you'll see dolphins.

If you don't, you can repeat the trip free of charge.

We organize daily trips for just €35.

Unfortunately, there aren't any places left for this afternoon's trip, but come and see me to book for later in the week.

If you're energetic, I'd recommend our forest walk.

It's a guided walk of about 7 kilometers.

There'll be a stop halfway, and you'll be provided with a drink and sandwiches.

There's some fairly steep climbs up the hills, so you need to be reasonably fit for this one with good shoes.

Oh, and bring a waterproof in case it rains.

It's just €25 all-inclusive, and it's every Wednesday.

Then on Thursdays, we organize a cycle trip, which will give you all the fun of biking without the effort.

We'll take you and your bike up to the top of Mount Lana and leave you to bike back.

It's a 700 m drop in just 20 km.

So, this isn't really for inexperienced cyclists as you'll be going pretty fast.

And if it's a clear day, you'll have fantastic views.

On our local craft tour, you can find out about the traditional activities in the island.

And the best thing about this trip is that it's completely free.

You'll be taken to a factory where jewelry is made and also a ceramic center.

If you want, you can buy some of the products, but that's entirely up to you.

The trip starts after lunch on Thursday and you'll return by 6:00 p.m.

If you're interested in astronomy, you may already know that the islands, one of the best places in the world to observe the night sky.

We can offer trips to the observatory on Friday for those who are interested.

They cost €9 per person and you'll be shown the huge telescopes and have a talk from an expert who will explain all about how they work.

Afterwards, we'll head down to Sunset Beach where you can have a dip in the ocean if you want before we head off back to the hotel.

Finally, there's horse riding.

This is organized by the equestrian center over near Plyier Cortino and it's a great experience if you're a keen horseback rider or even if you've never been on a horse before.

They take you down to the beach and you can along the sand and through the waves.

It costs €35 and it's available every day.

So, there's plenty to do in the daytime, but what about night life?

Well, the number one attractions called Musical Favorites.

Guests enjoy a three course meal and unlimited free drinks and watch a fantastic show starting with musicals set in Paris and then crossing the Atlantic to Las Vegas and finally Copa Cabana.

At the end, the cast members come down from the stage still in their stunning costumes and you'll have a chance to chat with them.

It's hugely popular, so let me know now if you're interested because it's no good leaving it until the last minute.

It's on Friday night.

Tickets are just €50 each, but for an extra €10, you can have a table right by the stage.

If you'd like to go back in time, there's the castle feast on Saturday evening.

It's held in a 12th century castle and you eat in the great courtyard with ladies in long gowns serving your food.

You're given a whole chicken each which you eat in the medieval way using your hands instead of cutlery.

And you're entertained by competitions where the horseback riders attempt to knock one another off their horses.

Then you can watch the dancers in the ballroom and join in as well if you want.

Okay.

So now if anyone has Hello Trevor.

Hello Stephanie.

You said you wanted to talk about the course I'm taking on literature for children.

That's right.

I'm thinking of doing it next year, but I'd like to find out more about it first.

Okay.

Well, as you probably know, it's a one-year course.

It's divided into six modules, and you have to take all of them.

One of the most interesting ones for me at least was about the purpose of children's literature.

You mean whether it should just entertain children or should be educational as well.

Right.

And whether the teaching should be factual, giving them information about the world or ethical, teaching them values.

What's fascinating is that the writer isn't necessarily conscious of the message they're conveying.

For instance, a story might show a child who has a problem as a result of not doing what an adult has told them to do, implying that children should always obey adults.

Oh, I see what you mean.

That module made me realize how important stories are.

They can have a significant effect on children as they grow up.

Actually, it inspired me to have a go at it myself just for my own interest.

I know I can't compete with the really popular stories like the Harry Potter books.

They're very good.

And even young kids like my seven-year-old niece love reading them.

I'm very interested in illustrations in stories.

Is that covered in the course?

Yes, there's a module on pictures and how they're sometimes central to the story.

That's good.

I remember some frightening ones I saw as a child, and I can still see them vividly in my mind years later.

Pictures can be so powerful, just as powerful as words.

I've always enjoyed drawing, so that's the field I want to go into when I finish the course.

I bet that module will be really helpful.

I'm sure it will.

We also studied comics in that module, but I'm not convinced of their value.

Not compared with books.

One of the great things about words is that you use your imagination, but with a comic, you don't have to.

But children are so used to visual input on TV, video games, and so on.

There are plenty of kids who wouldn't even try to read a book.

So, I think comics can serve a really useful purpose.

You mean it's better to read a comic than not to read at all?

Yes, I suppose you're right.

I just think it's sad when children don't read books.

What about books for girls and books for boys?

Does the course go into that?

Yes, there's a module on it.

For years, lots of stories in English at least assumed that boys went out and did adventurous things and girls stayed at home and played with dolls.

I was amazed how many books were targeted at just one sex or the other.

Of course, this reflects society as it is when the books are written.

That's true.

So, it sounds as though you think it's a good course.

Definitely.

Have you been reading lots of children's stories to help you decide whether to take the course?

Yeah, I've gone as far back as the late 17th century, though I know there were earlier children's stories.

So, does that mean you've read Pero's fairy tales, Cinderella, The Sleeping Beauty, and so on?

Yes, they must be important because no stories of that type had been written before.

These were the first.

Then there's the Swiss Family Robinson.

I haven't read that.

The English name makes it sound as though Robinson is the family's surname, but a more accurate translation would be the Swiss Robinsons because it's about a Swiss family who are shipwrecked like Robinson Crusoe in the novel of a century earlier.

Well, I never knew that.

Have you read Hoffman's The Nutcracker in The Mouse King?

Wasn't that the basis for Tchaikovsky's ballet, The Nutcracker?

That's right.

It has some quite uh bizarre elements.

I hope you've read Oscar Wild's The Happy Prince.

It's probably my favorite children's story of all time.

A mine too.

And it's so surprising because Wild is best known for his plays and most of them are very witty, but The Happy Prince is really moving.

I struggled with Tolken's The Lord of the Rings.

Three long books and I gave up after one.

It's extremely popular though.

Yeah, but whereas something like The Happy Prince just carried me along with it, The Lord of the Rings took more effort than I was prepared to give it.

I didn't find that.

I love it.

Another one I've read is Warhorse.

Oh, yes.

It's about the First World War, isn't it?

Hardly what you'd expect for a children's story.

Exactly.

But it's been very successful.

Have you read any?

In today's class, I'm going to talk about marine archaeology.

The branch of archaeology focusing on human interaction with the sea, lakes, and rivers.

It's the study of ships, cargos, shipping facilities, and other physical remains.

I'll give you an example then go on to show how this type of research is being transformed by the use of the latest technology.

Atlid Yamam was a village on the coast of the eastern Mediterranean which seems to have been thriving until around 7,000 BC.

The residents kept cattle, caught fish, and stored grain.

They had wells for fresh water.

Many of their houses were built around a courtyard and were constructed of stone.

The village contained an impressive monument.

Seven halfton stones standing in a semicircle around a spring that might have been used for ceremonial purposes.

Atlet Yam may have been destroyed swiftly by a tsunami or climate change may have caused glacias to melt and sea levels to rise, flooding the village gradually.

Whatever the cause, it now lies 10 m below the surface of the Mediterranean, buried under sand at the bottom of the sea.

It's been described as the largest and best preserved prehistoric settlement ever found on the seabed.

For marine archaeologists, Atlet Yam is a treasure trove.

Research on the buildings, tools, and the human remains has revealed how the bustling village once functioned and even what diseases some of its residents suffered from.

But of course, this is only one small village, one window into a lost world.

For a fuller picture, researchers need more sunken settlements.

But the hard part is finding them.

Underwater research used to require divers to find shipwrecks or artifacts.

But in the second half of the 20th century, various types of underwater vehicles were developed.

Some controlled from a ship on the surface and some of them autonomous, which means they don't need to be operated by a person.

Autonomous underwater vehicles or AUVs are used in the oil industry, for instance, to create maps of the seabed before rigs and pipelines are installed.

To navigate, they use sensors such as compasses and sonar.

Until relatively recently, they were very expensive and so heavy that they had to be launched from a large vessel with a winch.

But the latest AUVs are much easier to maneuver.

They can be launched from the shore or a small ship, and they're much cheaper, which makes them more accessible to research teams.

They are also very sophisticated.

They can communicate with each other and for example work out the most efficient way to survey a site or to find particular objects on the seabed.

Field tests show the approach can work.

For example, in a trial in 2015, three AUVs searched for Rex at Marty off the coast of Sicily.

The site is the final resting place of an ancient Roman ship which sank in the 6th century AD while finging prefabricated marble elements for the construction of an early church.

The AUVs mapped the area in detail, finding other ships carrying columns of the same material.

Creating an internet in the sea for AUVs to communicate is no easy matter.

Wi-Fi networks on land use electromagnetic waves, but in water, these will only travel a few centime.

Instead, a more complex mix of technologies is required.

For short distances, AUVs can share data using light, while acoustic waves are used to communicate over long distances.

But more creative solutions are also being developed, where an AUV working on the seabed offloads data to a second AUV, which then

surfaces and beams the data home to the research team using a satellite.

There's also a system that enables AUVs to share information from seabed scans and other data.

So if an AUV surveying the seabed finds an intriguing object, it can share the coordinates of the object, that is its position, with a nearby AUV that carries superior cameras and arrange for that AUV to make a closer inspection of the object.

Marine archaeologists are excited about the huge potential of these AUVs for their discipline.

One site where they're going to be deployed is the Gulf of Baratti off the Italian coast.

In 1974, a 2,000-year-old Roman vessel was discovered here in 18 m of water.

When it sank, it was carrying medical goods in wooden or tin receptacles.

Its cargo gives us insight into the treatments available all those years ago, including tablets that are thought to have been dissolved to form a cleansing liquid for the eyes.

Other Roman ships went down nearby, taking their cargos with them.

Some held huge pots made of terra cotta.

Some were used for transporting cargos of olive oil and others held wine.

In many cases, it's only these containers that remain while the wooden ships have been buried under silt on the seabed.

Another project that's about to

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Hello, William.

This is Amber.

You said to phone if I wanted to get more information about the job agency you mentioned.

Is now a good time?

Uh, hi Amber.

Yes, fine.

So, the agency I was talking about is called Bankside.

They're based in Docklands.

I can tell you the address now.

497 East Side.

Okay, thanks.

So, is there anyone in particular I should speak to there?

The agent I always deal with is called Becky Jameson.

Let me write that down.

Becky Jameson.

J A M I E S O N.

Do you have her direct line?

Yes, it's in my contact somewhere.

Right here we are.

0786510LE3.

I wouldn't call her until the afternoon if I were you.

She's always really busy in the morning trying to fill last minute vacancies.

She's really helpful and friendly, so I'm sure it would be worth getting in touch with her for an informal chat.

It's mainly clerical and admin jobs they deal with, isn't it?

That's right.

I know you're hoping to find a full-time job in the media eventually, but Becky mostly recruits temporary staff for the finance sector, which will look good on your CV and generally pays better, too.

Yeah, I'm just a bit worried because I don't have much office experience.

I wouldn't worry.

They'll probably start you as a receptionist or something like that.

So, what's important for that kind of job isn't so much having business skills or knowing lots of different computer systems.

It's communication that really matters.

So, you'd be fine there.

And you'll pick up office skills really quickly on the job.

It's not that complicated.

Okay, good.

So, how long do people generally need temporary staff for?

It would be great if I could get something lasting at least a month.

That shouldn't be too difficult, but you're more likely to be offered something for a week at first, which might get extended.

It's unusual to be sent somewhere for just a day or two.

Right.

I've heard the pay isn't too bad.

Better than working in a shop or a restaurant.

Oh, yes, definitely.

The hourly rate is about £10, 11 if you're lucky.

That's pretty good.

I was only expecting to get 8 or 9 an hour.

Do you want me to tell you anything about the registration process?

Yes, please.

I know you have to have an interview.

The interview usually takes about an hour, and you should arrange that about a week in advance.

I suppose I should dress smartly if it's for office work.

I can probably borrow a suit from mom.

Good idea.

It's better to look too smart than too casual.

Will I need to bring copies of my exam certificates or anything like that?

No, they don't need to see those, I don't think.

What about my passport?

Oh, yes.

They will ask to see that.

Okay.

I wouldn't get stressed about the interview, though.

It's just a chance for them to build a relationship with you so they can try and match you to a job which you'll like.

So, there are questions about personality that they always ask candidates, fairly basic ones.

And they probably won't ask anything too difficult like what your plans are for the future.

Hope not.

Anyway, there are lots of benefits to using an agency.

For example, the interview will be useful because they'll give you feedback on your performance so you can improve next time.

And they'll have access to jobs which aren't advertised.

Exactly.

Most temporary jobs aren't advertised.

And I expect finding a temporary job this way takes a lot less time.

It's much easier than ringing up individual companies.

Yes, indeed.

Well, I think I've covered Good morning.

My name's Erica Matthews and I'm the owner of Matthews Island Holidays, a company set up by my parents.

Thank you for coming to this presentation in which I hope to interest you in what we have to offer.

We're a small familyrun company and we believe in the importance of the personal touch.

So we don't aim to compete with other companies on the number of customers.

What we do is build on our many years experience more than almost any other rail holiday company to ensure we provide perfect holidays in a small number of destinations which we've got to know extremely well.

I'll start with our sixth day is of man holiday.

This is a fascinating island in the Irish Sea with Wales to the south, England to the east, Scotland to the north, and Northern Ireland to the west.

Our holiday starts in Hisham where your tour manager will meet you.

Then you'll travel by ferry to the aisle of man.

Some people prefer to fly from Luton instead.

And another popular option is to go by train to Liverpool and take a ferry from there.

You have five nights in the hotel and the price covers five breakfasts and dinners and lunch on the three days when there are organized trips.

Day four is free and most people have lunch in a cafe or restaurant in Douglas.

The price of a holiday includes the ferry to the aisle of man, all travel on the island, the hotel, and the meals I've mentioned.

Incidentally, we try to make booking our holidays as simple and fair as possible.

So, unlike with many companies, the price is the same whether you book 6 months in advance or at the last minute.

And there's no supplement for single rooms in hotels.

If you make a booking then need to change the start date, for example, because of illness, you're welcome to change to an alternative date or a different tour for a small administrative fee.

Okay, so what does the holiday consist of?

Well, on day one, you'll arrive in time for a short introduction by your tour manager, followed by dinner in the hotel.

The dining room looks out at the river close to where it flows into the harbor, and there's usually plenty of activity going on.

On day two, you'll take the coach to the small town of Peele on the way, calling in at the Tinwald exhibition.

The aisle of man isn't part of the United Kingdom, and it has its own parliament called Tinwald.

It's claimed that this is the world's oldest parliament that's still functioning and that it dates back to 979.

However, the earliest surviving reference to it is from 1422.

So, perhaps it isn't quite as old as it claims.

Day three, we have a trip to the mountain Snfel.

This begins with a leisurely ride along the prominard in Douglas in a horsedrawn tram.

Then you board an electric train which takes you to the fishing village of Laxi.

From there it's an 8 km ride in the Snayfell Mountain Railway to the top.

Lunch will be in the cafe giving you spectacular views of the island.

Day four is free for you to explore using the pass which we'll give you.

So you won't have to pay for travel on local transport or for entrance to the island's heritage sites.

Or you might just want to take it easy in Douglas and perhaps do a little light shopping.

The last full day, day five, is for some people the highlight of the holiday with a ride on the steam railway from Douglas to Port Eron.

After some time to explore, a coach will take you to the headland that overlooks the calf of Man, a small island just off the coast.

From there, you continue to Castle Town, which used to be the capital of the aisle of man and its medieval castle.

And on day six, it's back to the ferry or the airport if you flew to the island and time to go home.

Now, I'd like to tell you a bit more, Ed.

How are you getting on with the reading for our presentation next week?

Well, okay, ROF, but there's so much of it.

I know.

I hadn't realized birth order was such a popular area of research.

But the stuff on birth order and personality is mostly unreliable.

From what I've been reading, a lot of the claims about how your position in the family determines certain personality traits are just stereotypes with no robust evidence to support them.

Okay.

But that's an interesting point.

We could start by outlining what previous research has shown.

There are studies going back over a hundred years.

Yeah.

So, we could just run through some of the typical traits like the consensus seems to be that oldest children are generally less well adjusted because they never get over the arrival of a younger sibling.

Right?

But on a positive note, some studies claim that they were thought to be good at nurturing.

Certainly in the past when people had large families, they would have been expected to look after the younger ones.

There isn't such a clear picture for middle children, but one trait that a lot of the studies mention is that they are easier to get on with than older or younger siblings.

H generally eager to please and helpful, although that's certainly not accurate as far as my family goes.

My middle brother was a nightmare, always causing fights and envious of whatever I had.

As I said, none of this seems to relate to my own experience.

I'm the youngest in my family and I don't recognize myself in any of the studies I've read about.

I'm supposed to have been a sociable and confident child who made friends easily, but I was actually terribly shy.

Really?

That's funny.

There have been hundreds of studies on twins, but mostly about nurture versus nature.

There was one on personality which said that a twin is likely to be quite shy in social situations because they always have their twin around to depend on for support.

My cousins were like that when they were small.

They were only interested in each other and found it hard to engage with other kids.

They're fine now though.

Only children have had a really bad press.

A lot of studies have branded them as loners who think the world revolves around them because they've never had to fight for their parents' attention.

That does seem a bit harsh.

One category I hadn't considered before was children with much older siblings.

A couple of studies mentioned that these children grow up more quickly and are expected to do basic things for themselves, like getting dressed.

I can see how that might be true, although I expect they're sometimes the exact opposite, playing the baby role and clamoring for special treatment.

What was the problem with most of these studies, do you think?

I think it was because in a lot of cases, data was collected from only one sibling per family who rated him or herself and his or her siblings at the same time.

Some of the old research into the relationship between birth order and academic achievement has been proved to be accurate though performances in intelligence tests declined slightly from the eldest child to his or her younger siblings.

This has been proved in lots of recent studies.

Yes.

Although what many of them didn't take into consideration was family size.

The more siblings there are, the likelier the family is to have a low socioeconomic status, which can also account for differences between siblings in academic performance.

The oldest boy might be given more opportunities than his younger sisters, for example.

Exactly.

But the main reason for the marginally higher academic performance of oldest children is quite surprising.

I think it's not only that they benefit intellectually from extra attention at a young age, which is what I would have expected.

It's that they benefit from being teachers for their younger siblings by verbalizing processes.

Right.

And this gives them status and confidence, which again contributes in a small way to better performance.

So would you say sibling rivalry has been a useful thing for you?

I think so.

My younger brother was incredibly annoying and we fought a lot, but I think this has made me a stronger person.

I know how to defend myself.

We had some terrible arguments and I would have died rather than apologized to him.

But we had to put up with each other and most of the time we coexisted amicably enough.

Yes, my situation was pretty similar.

But I don't think having two older brothers made me any less selfish.

I was never prepared to let my brothers use any of my stuff.

That's perfectly normal.

Whereas sometimes today I'm going to talk about the eucalyptus tree.

This is a very common tree here in Australia where it's also sometimes called the gum tree.

First, I'm going to talk about why it's important.

Then I'm going to describe some problems it faces at present.

Right.

Well, the eucalyptus tree is an important tree for lots of reasons.

For example, it gives shelter to creatures like birds and bats.

And these and other species also depend on it for food, particularly the nectar from its flowers.

So, it supports biodiversity.

It's useful to us humans, too, because we can kill germs with a disinfectant made from oil extracted from eucalyptus leaves.

The eucalyptus grows all over Australia and the trees can live for up to 400 years.

So, it's alarming that all across the country, numbers of eucalyptus are falling because the trees are dying off prematurely.

So what are the reasons for this?

One possible reason is disease.

As far back as the 1970s, the trees started getting a disease called Mandula yellows.

The tree's leaves would gradually turn yellow.

Then the tree would die.

It wasn't until 2004 that they found the cause of the problem was lime or calcium hydroxide to give it its proper chemical name which was being used in the construction of roads.

The lime was being washed away into the ground and affecting the roots of the eucalyptus trees nearby.

What it was doing was preventing the trees from sucking up the iron they needed for healthy growth.

When this was injected back into the affected trees, they immediately recovered.

But this problem only affected a relatively small number of trees.

By 2000, huge numbers of eucalyptus were dying along Australia's east coast of a disease known as bell miner associated dieback.

The bell miner is a bird and the disease seems to be common where there are high populations of bell miners.

Again, it's the leaves of the trees that are affected.

What happens is that insects settle on the leaves and eat their way around them, destroying them as they go.

And at the same time, they secrete a solution which has sugar in it.

The bell miner birds really like this solution and in order to get as much as possible, they keep away other creatures that might try to get it.

So these birds and insects flourish at the expense of other species and eventually so much damage is done to the leaves that the tree dies.

But experts say that trees can start looking sick before any sign of bell miner associated dieback.

So it looks as if the problem might have another explanation.

One possibility is that it's to do with the huge bush fires that we have in Australia.

A theory proposed over 40 years ago by ecologist William Jackson is that the frequency of bushfires in a particular region affects the type of vegetation that grows there.

If there are very frequent bushfires in a region, this encourages grass to grow afterwards.

While if the bush fires are rather less frequent, this results in the growth of eucalyptus forests.

So why is this?

Why do fairly frequent bushfires actually support the growth of eucalyptus?

Well, one reason is that the fire stops the growth of other species which would consume water needed by eucalyptus trees.

And there's another reason.

If these other quick growing species of bushes and plants are allowed to proliferate, they harm the eucalyptus in another way by affecting the composition of the soil and removing nutrients from it.

So, some bushfires are actually essential for the eucalyptus to survive as long as they are not too frequent.

In fact, there's evidence that Australia's indigenous people practiced regular burning of bush land for thousands of years before the arrival of the Europeans.

But since Europeans arrived on the continent, the number of bush fires has been strictly controlled.

Now, scientists believe that this reduced frequency of bushfires to low levels has led to what's known as dry rainforest, which seems an odd name as usually we associate tropical rainforest with wet conditions and what's special about this type of rainforest?

Well, unlike tropical rainforest, which is a rich ecosystem, this type of ecosystem is usually a simple one.

It has very thick, dense vegetation, but not much variety of species.

The vegetation provides lots of shade.

So, one species that does find it ideal is the bell miner bird, which builds its nests in the undergrowth there.

But again, that's not helpful for the eucalyptus tree.

Good morning.

You're through to the tourist information office.

Tim speaking.

How can I help you?

Oh, hello.

Could you give me some information about next month's festival, please?

My family and I will be staying in the town that week.

Of course.

Well, it starts with a concert on the afternoon of the 17th.

Oh, I heard about that.

The orchestra and singers come from the USA, don't they?

They're from Canada.

They're very popular over there.

They're going to perform a number of well-known pieces that will appeal to children as well as adults.

That sounds good.

My whole family are interested in music.

The next day, the 18th, there's a performance by a ballet company called Ustartis.

Sorry.

The name is spelled E U S T A T I S.

They appeared in last year's festival and went down very well.

Again, their program is designed for all ages.

Good.

I expect we'll go to that.

I hope there's going to be a play during the festival.

A comedy ideally.

You're in luck.

On the 19th and 20th, a local amateur group are performing one written by a member of the group.

It's called Jamaima.

That'll be on in the town hall.

They've already performed it two or three times.

I haven't seen it myself, but the review in the local paper was very good.

And is it suitable for children?

Yes.

In fact, it's aimed more at children than at adults.

So both performances are in the afternoon.

And what about dance?

Will there be any performances?

Yes, also on the 20th, but in the evening.

A professional company is putting on a show of modern pieces with electronic music by young composers.

Uh-huh.

The show is about how people communicate or fail to communicate with each other.

So, it's got the rather strange name chat.

I suppose that's because that's something we do both face to face and online.

That's right.

Now, there are also some workshops and other activities.

They'll all take place at least once every day, so everyone who wants to take part will have a chance.

Good.

We're particularly interested in cookery.

You don't happen to have a cookery workshop, do you?

We certainly do.

It's going to focus on how to make food part of a healthy lifestyle.

And it'll show that even sweet things like cakes can contain much less sugar than they usually do.

It might be worth going to.

We're trying to encourage our children to cook.

Another workshop is just for children and that's on creating posters to reflect the history of the town.

The aim is to make children aware of how both the town and people's lives have changed over the centuries.

The results will be exhibited in the community center.

Then the other workshop is in toy making and that's for adults only.

Oh, why is that?

Because it involves carpentry.

Participants will be making toys out of wood.

So there'll be a lot of sharp chisels and other tools around.

It makes sense to keep children away from it.

Exactly.

Now let me tell you about some of the outdoor activities.

There'll be supervised wild swimming.

Wild swimming?

What's that?

It just means swimming in natural waters rather than a swimming pool.

Oh, okay.

In a lake, for instance.

Yes, there's a beautiful one just outside the town and that'll be the venue for the swimming.

There'll be lifeguards on duty, so it's suitable for all ages.

And finally, there'll be a walk in some nearby woods every day.

The leader is an expert on insects.

He'll show some that live in the woods and how important they are for the environment.

So, there are going to be all sorts of different things to do during the festival.

There certainly are.

If you'd like to read about how the preparations for the festival are going, the festival organizer is keeping a blog.

Just search online for the festival website and you'll find it.

Well, thank you very much for all the information.

You're welcome.

Goodbye.

Goodbye.

I'm very pleased to welcome this evening's guest speaker, Mark Logan, who's going to tell us about the recent transformation of Minster Park.

Over to you, Mark.

Thank you.

I'm sure you're all familiar with Minster Park.

It's been a feature of the city for well over a century and has been the responsibility of the city council for most of that time.

What perhaps isn't so well known is the origin of the park.

Unlike many public parks that started in private ownership, as the garden of a large house, for instance, Minster was some wasteland which people living nearby started planting with flowers in 1892.

It was unclear who actually owned the land, and this wasn't settled until 20 years later when the council took possession of it.

You may have noticed the statue near one of the entrances.

It's of Diane Gos who played a key role in the history of the park.

Once the council had become the legal owner, it planned to sell the land for housing.

Many local people wanted it to remain a place that everyone could go to to enjoy the fresh air and natural environment.

Remember the park is in a densely populated residential area.

Diane Gosforth was one of those people and she organized petitions and demonstrations which eventually made the council change its mind about the future of the land.

Soon after this, the First World War broke out in 1914, and most of the park was dug up and planted with vegetables, which were sold locally.

At one stage, the army considered taking it over for troop exercises and got as far as contacting the city council, then decided the park was too small to be of use.

There were occasional public meetings during the war in an area that had been retained as grass.

After the war, the park was turned back more or less to how it had been before and continued almost unchanged until recently. Plans for transforming it were drawn up at various times, most recently in 2013, though they were revised in 2015 before any work had started.

The changes finally got going in 2016 and were finished on schedule last year.

Okay, let me tell you about some of the changes that have been made and some things that have been retained.

If you look at this map, you'll see the familiar outline of the park with the river forming the northern boundary and a gate in each of the other three walls.

The statue of Diane Gos has been moved.

It used to be close to the south gate, but it's now immediately to the north of the lily pond, almost in the center of the park, which makes it much more visible.

There's a new area of wooden sculptures, which are on the riverbank, where the path from the east gate makes a sharp bend.

There are two areas that are particularly intended for children.

The playground has been enlarged and improved, and that's between the river and the path that leads from the pond to the river.

Then there's a new maze, a circular series of paths separated by low hedges.

That's near the west gate.

You go north from there towards the river and then turn left to reach it.

There have been tennis courts in the park for many years and they've been doubled from 4 to 8.

They're still in the southwest corner of the park where there's a right angle bend in the path.

Something else I'd like to mention is the new fitness area.

This is right next to the lily pond on the same side as the west gate.

Now, as you're all gardeners, I'm sure you'll like to hear about the plants that have been chosen for the park.

Okay, Graeme.

So, let's check we both know what we're supposed to be doing.

So, for the university's open day, we have to plan a display on British life and literature in the mid-19th century.

That's right.

But we'll have some people to help us find the materials and set it up.

Remember, for the moment, we just need to plan it.

Good.

So, have you gathered who's expected to come and see the display?

Is it for the people studying English or students from other departments?

I'm not clear about it.

Nor me.

That was how it used to be, but it didn't attract many people.

So, this year, it's going to be part of an open day to raise the university's profile.

It'll be publicized in the city to encourage people to come and find out something of what goes on here and it's included in the information that's sent to people who are considering applying to study here next year.

Presumably some current students and lecturers will come.

I would imagine so.

But we've been told to concentrate on the other categories of people.

Right.

We don't have to cover the whole range of 19th century literature, do we?

No, it's entirely up to us.

I suggest just using Charles Dickens.

That's a good idea.

Most people have heard of him and have probably read some of his novels or seen films based on them.

So, that's a good lead-in to life in his time.

Exactly.

And his novels show the awful conditions that most people had to live in, don't they?

He wanted to shock people into doing something about it.

Did he do any campaigning other than writing?

Yes, he campaigned for education and other social reforms and gave talks, but I'm inclined to ignore that and focus on the novels.

Yes, I agree.

Okay.

So, now shall we think about a topic linked to each novel?

Yes.

I've printed out a list of Dickens's novels in the order they were published in the hope you'd agree to focus on him.

You're lucky I did agree.

Let's have a look.

Uh, okay.

The first was the Peveril of the Peak published in 1836.

It was very successful when it came out, wasn't it?

And was adapted for the theater straight away.

There's an interesting point though that there's a character who keeps falling asleep.

And that medical condition was named after the book P Wicket Syndrome.

Oh.

So, why don't we use that as the topic and include some quotations from the novel?

Right.

Next is Oliver Twist.

Mhm.

There's a lot in the novel about poverty, but maybe something less obvious.

Well, Oliver is taught how to steal, isn't he?
We could use that to illustrate the fact that very few children went to school, particularly not poor children, so they learned in other ways.
M good idea.
What's next?
Maybe Nicholas Nickleby.
Actually, he taught in a really cruel school, didn't he?
That's right.
But there's also the company of touring actors that Nicholas joins.
We could do something on theaters and other amusements of the time.
We don't want only the bad things, do we?
What about Martin Chuzzlewit?
He goes to the USA, doesn't he?
Yes.
and Dickens himself had been there a year before and drew on his experience there in the novel.
I wonder though, the main theme is selfishness, so we could do something on social justice.
No, too general.
Let's keep to your idea.
I think it would work well.
He wrote Bleak House next.
That's my favorite of his novels.
Yes, mine too.
Oh, his satire of the legal system is pretty powerful.
That's true.
But think about Esther, the heroine.
As a child, she lives with someone she doesn't know is her aunt, who treats her very badly.
Then she's very happy living with her guardian, and he puts her in charge of the household.
And at the end, she gets married, and her guardian gives her and her husband a house where, of course, they're very happy.
Yes, I like that.
What should we take next?
Uh, little Dorit.
Old Mr.
Dorret has been in a debtor's prison for years.
So was Dickens's father, wasn't he?
That's right.
What about focusing on the part when Mr.
Dorret inherits a fortune and he starts pretending he's always been rich?
Good idea.
Okay, so next we need to think about what materials we want to illustrate each issue.
That's going to be quite hard.
I'm going to report on a case study of a program which has been set up to help rural populations in Mozambique, a largely agricultural country in Southeast Africa.
The program worked with three communities in Chalaquala district near the Limpopo River.
This is a dry and arid region with unpredictable rainfall.
Because of this, people in the area were unable to support themselves through agriculture and instead they used the forest as a means of providing themselves with an income mainly by selling charcoal.
However, this was not a sustainable way of living in the long term as they were rapidly using up this resource.
To support agriculture in this dry region, the program focused primarily on making use of existing water resources from the Limpopo River by setting up systems of irrigation which would provide a dependable water supply for crops and animals.
The program worked closely with the district government in order to find the best way of implementing this.
The region already had one farmers association and it was decided to set up two more of these.
These associations planned and carried out activities including water management, livestock breeding and agriculture.
And it was notable that in general women formed the majority of the workforce.
It was decided that in order to keep the crops safe from animals, both wild and domestic, special areas should be fenced off where the crops could be grown.
The community was responsible for creating these fences, but the program provided the necessary wire for making. Once the area had been fenced off, it could be cultivated.
The land was dug so that vegetables and cereals appropriate to the climate could be grown.
And the program provided the necessary seeds for this.
The program also provided pumps so that water could be brought from the river in pipes to the fields.
However, the labor was all provided by local people and they also provided and put up the posts that supported the fences around the fields.
Once the program had been set up, its development was monitored carefully.
The farmers were able to grow enough produce not just for their own needs but also to sell.
However, getting the produce to places where it could be marketed was sometimes a problem as the farmers did not have access to transport and this resulted in large amounts of produce especially vegetables being spoiled.
This problem was discussed with the farmers associations and it was decided that in order to prevent food from being spoiled, the farmers needed to learn techniques for its preservation.
There was also an additional initiative that had not been originally planned but which became a central feature of the program.
This was when farmers started to dig holes for tanks in the fenced-off areas and to fill these with water and use them for breeding fish, an important source of protein.

After a time, another suggestion was made by local people which hadn't been part of the program's original proposal, but which was also adopted later on.

They decided to try setting up colonies of bees, which would provide honey both for their own consumption and to sell.

So what lessons can be learned from this program?

First of all, it tells us that in dry arid regions, if there is access to a reliable source of water, there is great potential for the development of agriculture.

In Chiquala, there was a marked improvement in agricultural production which improved food security and benefited local people by providing them with both food and income.

However, it's important to set realistic timelines for each phase of the program, especially for its design, as mistakes made at this stage may be hard to correct later on.

The program demonstrates that sustainable development is possible in areas where there is Good morning.

Thanks for coming in to see us here at the agency, Joe.

I'm one of the agency representatives and my name's Sally Baker.

Hi, Sally.

I think we spoke on the phone, didn't we?

That's right, we did.

So, thank you for sending in your CV.

We've had quite a careful look at it and I think we have two jobs that might be suitable for you.

The first one is in a company based in North London.

They're looking for an administrative assistant.

Okay.

What sort of company is it?

Uh they're called Home Solutions and they design and make furniture.

Oh, I don't know much about that, but it sounds interesting.

Yes.

Well, as I said, they want someone in their office.

And looking at your past experience, it does look as if you fit quite a few of the requirements.

So on your CV, it appears you've done some data entry.

Yes.

So that's one skill they want.

Then they expect the person they appoint to attend meetings and take notes there.

Okay.

I've done that before.

Yes.

And you need to be able to cope with general admin filing and keeping records and so on, but that should be okay.

And in my last job, I also had to manage the diary.

Excellent.

That's something they want here, too.

I'd suggest you add it to your CV.

I don't think you mentioned that, did you?

No.

So, as far as the requirements go, they want good computer skills, of course, and they particularly mention spreadsheets.

That should be fine.

And interpersonal skills, which would be something they'd check with your references.

I think that should be okay.

Yes.

Then they mentioned that they want someone who is careful and takes care with details.

Just looking at your CV, I'd say you're probably all right there.

I think so.

Yes.

Do they want any special experience?

I think they wanted some experience of teleconferencing.

Oh, I've got three years experience of that.

Uhhuh.

Let's see.

Yes.

Good.

In fact, they're only asking for at least one year, so that's great.

So, is that something that might interest you?

It is.

Yes.

The only thing is you said they were in North London, so it would be quite a long commute for me.

Okay.

So, the second position might suit you better as far as the location goes.

That's for a warehouse assistant and that's in South London.

Yes, that would be a lot closer.

And you've worked in a warehouse before, haven't you?

So, as far as the responsibilities for this position go, they want someone who could manage the stock, obviously, and also deliveries.

That should be okay.

You've got to keep track of stuff, but I've always been quite good with numbers.

Good.
That's their first requirement.
And they want someone who's computer literate, which we know you are.
Sure.
Then they mention organizational skills.
They want someone who's well organized.
Yes, I think I am.
And tidy.
Yes.
They go together, really, don't they?
Then the usual stuff.
They want someone who can communicate well both orally and in writing.
Okay.
And for the last warehouse job I had, one of the things I enjoyed most was being part of a team.
I found that was really essential for the job.
Excellent.
Yes, they do mention that they want someone who's used to that.
Yes.
Now, when you were working in a warehouse last time, what sorts of items were you dealing with?
It was mostly bathroom and kitchen equipment, sinks and stoves and fridges.
So, you're okay moving heavy things?
Sure, I'm quite strong and I've had the training.
Good.
Now, as far as experience goes, they mention they want someone with a license and that you have experience of driving in London so you can cope with the traffic and so on.
Yes, no problem.
And you've got experience of warehouse work.
And the final thing they mention is customer service.
I think looking at your CV, you're okay there.
Right.
So, what about pay?
Can you tell me a bit more about that, please?
My guest on the show today is Alice Riches, who started the street play scheme where she lives in Beachwood Road.
For those of you that don't already know, street play involves local residents closing off their street for a few hours so that children have a chance to play in the streets safely.
She started it in her own street, Beachwood Road, and the idea caught on, and there are now street play schemes all over the city.
So, when did you actually start the scheme, Alice?
Well, I first had the idea when my oldest child was still a toddler.
So, that's about 6 years ago now.
But it took at least 2 years of campaigning before we were actually able to make it happen.
So, the scheme's been up and running for 3 years now.
We'd love to be able to close our road for longer, for the whole weekend from Saturday morning until Sunday evening, for example.
At the moment, it's just once a week, but when we started, it was only once a month, but we're working on it.
So, what actually happens when Beachwood Road is closed?
We have volunteer wardens, mostly parents, but some elderly residents, too, who block off our road at either end.
The council have provided special signs, but there's always a volunteer there to explain what's happening to any motorists.
Generally, they're fine about it.
We've only had to get the police involved once or twice.
Now, I should explain that the road isn't completely closed to cars, but only residents' cars are allowed.
If people really need to get in or out of Beachwood Road, it's not a problem as long as they drive at under 20 km/hour.
But most people just decide not to use their cars during this time or they park in another street.
The wardens are only there to stop through traffic.
So, can anyone apply to get involved in street play?
Absolutely.
We want to include all kids in the city, especially those who live on busy roads.
It's here that demand is greatest.
Obviously, there isn't such demand in wealthier areas where the children have access to parks or large gardens or in the suburbs where there are usually more places for children to play outside.
I'd recommend that anyone listening who likes the idea should just give it a go.
We've been surprised by the positive reaction of residents all over the city.
And that's not just parents.
There are always a few who complain, but they're a tiny minority.
On the whole, everyone is very supportive and says they're very happy to see children out on the street, even if it does get quite noisy.
There have been so many benefits of street play for the kids.
Parents really like the fact that the kids are getting fresh air instead of sitting staring at a computer screen, even if they're not doing anything particularly energetic.
And of course, it's great that kids can play with their friends outside without being supervised by their parents.
But for me, the biggest advantage is that kids develop confidence in themselves to be outside without their parents.
The other really fantastic thing is that children get to know the adults in the street.
It's like having a big extended family.

It certainly does have a lot of benefits.

I want to move on now and ask you about a related project in King Street.

Right.

Well, uh, this was an experiment I was involved in where local residents decided to try and reduce the traffic along King Street, which is the busiest main road in our area, by persuading people not to use their cars for one day.

We thought about making people pay more for parking, but we decided that would be really unpopular.

So, instead, we just stopped people from parking on King Street, but left the other car parks open.

It was surprising how much of a difference all this made.

As we'd predicted, air quality was significantly better.

But what I hadn't expected was how much quieter it would be, even with the buses still running.

Of course, everyone said they felt safer, but we were actually amazed that sales in the shops went up considerably that day.

We thought there'd be fewer people out shopping, not more.

That's really interesting.

So, the fact that the street was Tom, could I ask you for some advice, please?

Yes, of course.

If you think I can help.

What's it about?

It's my first media studies assignment, and I'm not sure how to go about it.

You must have done it last year.

Is that the one comparing the coverage of a particular story in a range of newspapers?

Oh, yes.

I really enjoyed writing it.

So, what sort of things do I need to compare?

Well, there are several things.

For example, there's the question of which page of the newspaper the item appears on.

You mean because there's a big difference between having it on the front page and the bottom of page 10, for instance.

Exactly.

And that shows how important the editor thinks the story is.

Then there's the size.

how many column inches the story is given, how many columns it spreads over, and I suppose that includes the headline.

It certainly does.

It's all part of attracting the reader's attention.

What about graphics?

Whether there's anything visual in addition to the text?

Yes, you need to consider those too because they can have a big effect on the reader's understanding of the story.

Sometimes a bigger effect than the text itself.

Then you'll need to look at how the item is put together.

What structure is it given?

Bear in mind that not many people read beyond the first paragraph.

So what has the journalist put at the beginning?

And if say there are conflicting opinions about something, does one appear near the end where people probably won't read it?

And newspapers sometimes give wrong or misleading information, don't they?

Either deliberately or by accident.

Should I be looking at that too?

Yes, if you can compare what's in different versions and as far as possible, try and work out what's true and what isn't.

And that relates to a very important point.

What's the writer's purpose?

or at least the most important one, if they have several, it may seem to be to inform the public, but often it's that they want to create fear or controversy or to make somebody look ridiculous.

Gosh, I see what you mean.

And I suppose the writer may make assumptions about the reader.

That's right.

about their knowledge of the subject, their attitudes, and a level of education, which means writing so that the readers understand without feeling patronized.

All of that will make a difference to how the story is presented.

Does it matter what type of story I write about?

No.

National or international politics, the arts, anything, as long as it's covered in two or three newspapers.

though of course it'll be easier and more fun if it's something you're interested in and know something about and on that basis a national news item would be worth analyzing.

I'm quite keen on politics so I'll try and find a suitable topic.

What did you choose for your analysis Tom?

I was interested in how newspapers express their opinions explicitly.

Though I wanted to compare editorials in different papers, but when I started looking, I couldn't find two on the same topic that I felt like analyzing.

In that case, I won't even bother to look.

So, in the end, I chose a human interest story, a terribly emotional story about a young girl who was very ill and lots of other people, mostly strangers, raised money so she could go abroad for treatment.

Actually, I was surprised.

Some papers just wrote about how wonderful everyone was, but others considered the broader picture, like why treatment wasn't

available here.

H I usually find stories like that raise quite strong feelings in me.

I'll avoid that.

Perhaps I'll choose an arts topic, like different reviews of a film or something about funding for the arts.

I'll think about that.

Yes, that might be interesting.

Okay.

Well, thanks a lot for your help, Tom.

It's been really useful.

You're welcome.

Good luck with the assignment, Hazel.

Nowadays, we use different products for personal cleanliness, laundry, dishwashing, and household cleaning.

But this is very much a 20th century development.

The origins of cleanliness date back to prehistoric times.

Since water is essential for life, the earliest people lived near water and knew something about its cleansing properties, at least that it rinsed mud off their hands.

During the excavation of ancient Babylon, evidence was found that soap making was known as early as 2,800 BC.

Archaeologists discovered cylinders made of clay with inscriptions on them saying that fats were boiled with ashes.

This is a method of making soap, though there's no reference to the purpose of this material.

The early Greeks bathed for aesthetic reasons and apparently didn't use soap.

Instead, they cleaned their bodies with blocks of sand, uh, pummus, and ashes, then anointed themselves with oil and scraped off the oil and dirt with a metal instrument known as a stridel.

They also used oil mixed with ashes.

Clothes were washed without soap in streams.

The ancient Germans and Gauls are also credited with discovering how to make a substance called soap made of melted animal fat and ashes.

They used this mixture to tint their hair red.

Soap got its name, according to an ancient Roman legend, from Mount Sapo, where animals were sacrificed, leaving deposits of animal fat.

Rain washed these deposits along with wood ashes down into the clay soil along the river Tyber.

Women found that this mixture greatly reduced the effort required to wash their clothes.

As Roman civilization advanced, so did bathing.

The first of the famous Roman baths supplied with water from their aqueducts was built around 312 BC.

The baths were luxurious and bathing became very popular.

And by the 2nd century AD, the Greek physician Galen recommended soap for both medicinal and cleansing purposes.

After the fall of Rome in 476 AD and the resulting decline in bathing habits, much of Europe felt the impact of filth on public health.

This lack of personal cleanliness and related unsanitary living conditions were major factors in the outbreaks of disease in the Middle Ages and especially the black death of the 14th century.

Nevertheless, soap making became an established craft in Europe and associations of soap makers guarded their trade secrets closely.

Vegetable and animal oils were used with ashes of plants along with perfume apparently for the first time.

Gradually more varieties of soap became available for shaving and shampooing as well as bathing and laundering.

A major step toward large-scale commercial soap making occurred in 1791 when a French chemist Nicholas Leblanc patented a process for turning salt into soda ash or sodium carbonate.

Soda ash is the alkali obtained from ashes that combines with fat to form soap.

The Leblanc process yielded quantities of good quality inexpensive soda ash.

Modern soap making was born some 20 years later in the early 19th century with a discovery by Michel Eugene Chevre, another French chemist of the chemical nature and relationship of fats, glycerine and fatty acids.

His studies established the basis for both fat and soap chemistry and soap making became a science.

Further developments during the 19th century made it easier and cheaper to manufacture soap.

Until the 19th century, soap was regarded as a luxury item and was heavily taxed in several countries.

As it became more readily available, it became an everyday necessity, a development that was reinforced when the high tax was removed.

Soap was then something ordinary people could afford, and cleanliness standards improved.

With this widespread use came the development of milder soaps for bathing and soaps for use in the washing machines that were available to consumers by the turn of the 20th century.

Do you mind if I ask you some questions about your journey today?

We're doing a customer satisfaction survey.

Yes.

Okay.

I've got about 10 minutes before my train home leaves.

I'm on a day trip.

Great.

Thank you.

So, first of all, could you tell me your name?

It's Sophie Bird.

Thank you.

And would you mind telling me what you do?

I'm a journalist.

Oh, really?

That must be interesting.

Yes, it is.

So, was the reason for your visit here today?
Work?
Actually, it's my day off.
I came here to do some shopping.
Oh, right.
But I do sometimes come here for work.
Okay.
Now, I'd like to ask some questions about your journey today, if that's okay.
Right.
So, can you tell me which station you're traveling back to?
Stormf.
Ah, can I just check the spelling?
S T A U N F I R T H.
Mhm.
And you traveled from there this morning?
Okay, good.
Next, can I ask what kind of ticket you bought?
I assume it wasn't a season ticket as you don't travel every That's right.
No, I just got a normal return ticket.
I don't have a rail card, so I didn't get any discount.
I keep meaning to get one because it's a lot cheaper.
Yes.
You'd have saved 20% on your ticket today.
So, you paid the full price for your ticket.
I paid £23.70.
Okay.
Do you think that's good value for money?
Not really.
I think it's too much for a journey that only takes 45 minutes.
Yes, that's one of the main complaints we get.
So, you didn't buy your ticket in advance.
No, I know it's cheaper if you buy a week in advance, but I didn't know I was coming then.
I know you can't always plan ahead.
So, uh, did you buy it this morning?
No, it was yesterday.
Right.
And do you usually buy your tickets at the station?
Well, I do usually, but the ticket office closes early, and I hate using ticket machines.
I think ticket offices should be open for longer hours.
There's always a cue for the machines and they're often out of order.
A lot of customers are saying the same thing.
So, to answer your question, I got an e ticket online.
Okay.
Thank you.
Now, I'd like to ask you about your satisfaction with your journey.
So, what would you say you were most satisfied with today?
Well, I like the Wi-Fi on the train.
It's improved a lot.
It makes it easier for me to work if I want to.
That's the first time today anyone's mentioned that.
It's good to get some positive feedback on that.
And um is there anything you weren't satisfied with?
Well, normally the trains run on time and uh pretty reliable, but today there was a delay.
The train was about 15 minutes behind schedule.
Okay, I'll put that down.
Now, I'd also like to ask about the facilities at the station.
You've probably noticed that the whole station's been upgraded.
What are you most satisfied with?
Uh, I think the best thing is that they've improved the amount of information about train times, etc.
that's given to passengers.
It's much clearer.
Before, there was only one board and I couldn't always see it properly, which was frustrating.
That's good.
And is there anything you're not satisfied with?
Let's see.
I think things have generally improved a lot.
The trains are much more modern, and I like the new cafe.
But one thing is that there aren't enough places to sit down, especially on the platforms.
Okay.
So, I'll put seating down, shall I?

As the thing you're least satisfied with.

Yes.

Okay.

Can I ask your opinion about some of the other facilities?

Mhm.

We'd like feedback on whether people are satisfied, dissatisfied, or neither satisfied nor dissatisfied.

What about the parking at the station?

Well, to be honest, I don't really have an opinion as I never use it.

So, neither satisfied nor dissatisfied for that then.

Yes, I suppose so.

Okay.

Uh, and what about these?

As chair of the town council subcommittee on park facilities, I'd like to bring you up to date on some of the changes that have been made recently to the Croft Valley Park.

So, if you could just take a look at the map I handed out, let's begin with a general overview.

So, the basic arrangement of the park hasn't changed.

It still has two gates, north and south, and a lake in the middle.

The cafe continues to serve an assortment of drinks and snacks and is still in the same place, looking out over the lake and next to the old museum.

We're hoping to change the location of the toilets and bring them nearer to the center of the park as they're a bit out of the way at present near the adventure playground.

in the corner of your map.

The formal gardens have been replanted and should be at their best in a month or two.

They used to be behind the old museum, but we've now used the space near the south gate between the park boundary and the path that goes past the lake towards the old museum.

We have a new outdoor gym for adults and children which is already proving very popular.

It's by the glass houses just to the right of the path from the south gate.

You have to look for it as it's a bit hidden in the trees.

One very successful introduction has been our skateboard ramp.

It's in constant use during the evenings and holidays.

It's near the old museum at the end of a little path that leads off from the main path between the lake and the museum.

We've also introduced a new area for wild flowers to attract bees and butterflies.

It's on a bend in the path that goes around the east side of the lake just south of the adventure playground.

Now, let me tell you a bit more about some of the changes to Croft Valley Park.

One of our most exciting developments has been the adventure playground.

We were aware that we had nowhere for children to let off steam and decided to use our available funds to set up a completely new facility in a large space to the north of the park.

It's open year round though it closes early in the winter months and entrance is completely free.

Children can choose whatever activities they want to do irrespective of their age, but we do ask adults not to leave them on their own there.

There are plenty of seats where parents can relax and keep an eye on their children at the Lastly, the glass houses.

A huge amount of work has been done on them to repair the damage following the disastrous fire that recently destroyed their western side.

Over £80,000 was spent on replacing the glass walls and the metal supports as well as the plants that had been destroyed.

Although unfortunately the collection of tropical palm trees has proved too expensive to replace up to now.

At present, the glass houses are open from 10:00 a.m.

to 300 p.m.

Mondays to Thursdays, and it's hoped to extend this to the weekend soon.

We're grateful to all those who helped us by contributing their time and money to this achievement.

Okay, Jack, before we plan our presentation about refrigeration, let's discuss what we've discovered so far.

Fine, Annie.

Though I have to admit I haven't done much research yet, nor me.

But I found an interesting article about ice houses.

I'd seen some 18th and 19th century ones here in the UK, so I knew they were often built in a shady area or underground close to lakes that might freeze in the winter.

Then blocks of ice could be cut and stored in the ice house.

But I didn't realize that insulating the blocks with straw or sawdust meant they didn't melt for months.

The ancient Romans had refrigeration, too.

I didn't know that.

Yes.

Pits were dug in the ground and snow was imported from the mountains.

Even though they were at quite a distance, the snow was stored in the pits.

Ice formed at the bottom of it.

Both the ice and the snow were then sold.

The ice cost more than the snow, and my guess is that only the wealthy members of society could afford it.

I wouldn't be surprised.

I also came across an article about modern domestic fridges.

Several different technologies are used, but they were too complex for me to understand.

You have to wonder what happens when people get rid of old ones.

You mean because the gases in them are harmful for the environment.

Exactly.
At least there are now plenty of organizations that will recycle most of the components safely.
But of course, some people just dump old fridges in the countryside.
It's hard to see how they can be stopped.
Unfortunately, in the UK, we get rid of 3 million a year altogether.
That sounds a lot, especially because fridges hardly ever break down.
That's right.
In this country, we keep domestic fridges for 11 years on average, and a lot lasts for 20 or more.
So, if you divide the cost by the number of years you can use a fridge, they're not expensive compared with some household appliances.
True.
I suppose manufacturers encourage people to spend more by making them different colors and designs.
I'm sure when my parents bought their first fridge, they had hardly any choice.
Yes, there's been quite a change.
Right, let's make a list of topics to cover in our presentation and decide who's going to do more research on them.
Then later we can get together and plan the next step.
Okay.
How about starting with how useful refrigeration is and the range of goods that are refrigerated nowadays?
Because of course it's not just food and drinks.
No, I suppose flowers and medicines are refrigerated too.
And computers.
I could do that unless you particularly want to.
No, that's fine by me.
What about the effects of refrigeration on people's health?
After all, some of the chemicals used in the 19th century were pretty harmful.
But there have been lots of benefits, too, like always having access to fresh food.
Do you fancy dealing with that?
I'm not terribly keen, to be honest.
Nor me.
My mind just goes blank when I read anything about chemicals.
Oh, all right then.
I'll do you a favor.
But you owe me, Jack.
Okay.
What about the effects on food producers like farmers in poorer countries being able to export their produce to developed countries?
Something for you maybe?
I don't mind.
You should be quite interesting.
I think we should also look at how refrigeration has helped whole cities like Las Vegas, which couldn't exist without refrigeration because it's in the middle of a desert.
Right.
I had a quick look at an economics book in the library that's got a chapter about this sort of thing.
I could give you the title if you want to do this section.
Not particularly to be honest.
I find economics books pretty heavygoing as a rule.
Okay, leave it to me then.
Thanks.
Then there's transport and the difference that refrigerated trucks have made.
I wouldn't mind having a go at that.
Don't forget trains, too.
I read something about milk and butter being transported in refrigerated railroad cars in the USA right back in the 1840s.
I hadn't thought of trains.
Thanks.
Shall we have a separate section on domestic fridges?
After all, they're something everyone's familiar with.
What about splitting it into two?
You could investigate 19th and 20th century fridges and I'll concentrate on what's available these days and how manufacturers differentiate their products from those of their competitors.
Okay, that would suit me.
Hi everyone.
In this session, I'll be presenting my research about the social history of Britain during the industrial revolution.
I particularly looked at how ordinary lives were affected by changes that happened at that time.
This was a time that saw the beginning of a new phenomenon, consumerism, where buying and selling goods became a major part of ordinary people's lives.
In fact, it was in the 19th century that the quantity and quality of people's possessions was used as an indication of the wealth of the country.
Before this, the vast majority of people had very few possessions.
But all that was changed by the industrial This was the era from the mid 18th to the late 19th century when improvements in how goods were made as well as in technology triggered massive social changes that transformed life for just about everybody in several key areas.
First, let's look at manufacturing.
When it comes to manufacturing, we tend to think of the industrial revolution in images of steam engines and coal.

And it's true that the industrial revolution couldn't have taken place at all if it weren't for these new sources of power.

They marked an important shift away from the traditional water mills and windmills that had dominated before this.

The most advanced industry for much of the 19th century was textiles.

This meant that fashionable fabrics and lace and ribbons were made available to everyone.

Before the industrial revolution, most people made goods to sell in small workshops, often in their own homes.

But enormous new machines were now being created that could produce the goods faster and on a larger scale.

And these required a lot more space.

So large factories were built replacing the workshops and forcing workers to travel to work.

In fact, large numbers of people migrated from villages into towns.

As a result, as well as manufacturing, there were new technologies in transport, contributing to the growth of consumerism.

The horsedrawn stage coaches and carts of the 18th century, which carried very few people and goods and traveled slowly along poorly surfaced roads, were gradually replaced by the numerous canals that were constructed.

These were particularly important for the transportation of goods.

The canals gradually fell out of use though as railways were developed, becoming the main way of moving goods and people from one end of the country to the other.

And the goods they moved weren't just coal, iron, clothes, and so on.

Significantly, they included newspapers, which meant that thousands of people were not only more knowledgeable about what was going on in the country, but could also read about what was available in the shops, and that encouraged them to buy more.

So, faster forms of transport resulted in distribution becoming far more efficient.

Goods could now be sold all over the country instead of just in the local market.

The third main area that saw changes that contributed to consumerism was retailing.

The number and quality of shops grew rapidly and in particular small shops suffered as customers flocked to the growing number of department stores, a form of retailing that was new in the 19th century.

The entrepreneurs who opened these found new ways to stock them with goods and to attract customers.

For instance, improved lighting inside greatly increased the visibility of the goods for sale.

Another development that made goods more visible from outside resulted from the use of plate glass, which made it possible for windows to be much larger than previously.

New ways of promoting goods were introduced too.

Previously, the focus had been on informing potential customers about the availability of goods.

Now, there was an explosion in advertising trying to persuade people to go shopping.

Flanders claimed that one of the great effects of the industrial revolution was that it created choice.

All sorts of things that had previously been luxuries, from sugar to cutlery, became conveniences.

And before long, they turned into necessities.

Life without sugar or cutlery was unimaginable, rather like mobile phones these days.

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Hello.
Children's engineering workshops.
Oh, hello.
I wanted some information about the workshops in the school holidays.
Sure.
I have two daughters who are interested.
The younger one's Lydia.
She's four.
Do you take children as young as that?
Yes.
Our tiny engineers workshop is for four to 5y olds.
What sorts of activities do they do?
All sorts.
For example, they work together to design a special cover that goes around an egg so that when it's inside, they can drop it from a height and it doesn't break.
Well, sometimes it does break, but that's part of the fun, right?
And Lydia loves building things.
Is there any opportunity for her to do that?
Well, they have a competition to see who can make the highest tower.
You'd be amazed how high they can go, right?
But they're learning all the time as well as having fun.
For example, one thing they do is to design and build a car that's attached to a balloon.
And the force of the air in that actually powers the car and makes it move along.
They go really fast, too.
Okay.
Well, all this sounds perfect.
Now, Carly, that's my older daughter, has just had her seventh birthday.
So, presumably she'd be in a different group.
Yes, she'd be in the junior engineers.
That's for children from 6 to 8.
And do they do the same sorts of activities?
Some are the same, but a bit more advanced.
So they work out how to build model vehicles, things like cars and trucks, but also how to construct animals using the same sorts of material and technique.
And then they learn how they can program them and make them move.
So they learn a bit of coding.
They do.
They pick it up really quickly.
We're there to help if they need it, but they learn from one another, too.
Right.
And do they have competitions too?
Yes.
With the junior engineers, it's to use recycled materials like card and wood to build a bridge.
And the longest one gets a prize.
That sounds fun.
I wouldn't mind doing that myself.
Then they have something a bit different, which is to think up an idea for a 5-minute movie and then film it using special animation software.
You'd be amazed what they come up with.
And of course, that's something they can put on their phone and take home to show all their friends.
Exactly.
And then they also build a robot in the shape of a human.
And they decorate it and program it so that it can move its arms and legs.
Perfect.
So, is it the same price as the tiny engineers?
It's just a bit more.
£50 for the five weeks.
And are the classes on a Monday, too?
They used to be, but we found it didn't give our staff enough time to clear up after the first workshop, so we moved them to Wednesdays.
The classes are held in the morning from 10 to 11:00.
Okay, that's better for me, actually.
And what about the location?
Where exactly are the workshops held?
They're in building 10A.
There's a big sign on the door.
You can't miss it.
And that's in Fradston Industrial Estate.
Sorry, Fradston.

That's F R A D S T O N E.

And that's in Grassford, isn't it?

Yes.

Up past the station.

And will I have any parking problems there?

No, there's always plenty available.

So, would you like to enroll Lydia and Carly now?

Okay.

So, can I have your full name?

Good morning everyone and welcome to Stevenson's, one of the country's major manufacturers of metal goods.

Thank you for choosing us for your two weeks of work experience.

My name is Julia Simmons and since the beginning of this year, I've been the managing director.

Stevenson's is quite an old company.

Like me, the founder, Ronald Stevenson, went into the steel industry when he left school.

That was in 1923.

He set up this company when he finished his apprenticeship in 1926.

Although he actually started making plans 2 years earlier in 1924, he was a very determined young man.

Stevenson's long-term plan was to manufacture components for the machine tools industry although in fact that never came about and for the automotive industry that is cars and lorries.

However, there was a delay of 5 years before that happened because shortly before the company went into production Stevenson was given the opportunity to make goods for hospitals and other players in the healthcare industry.

So that's what we did for the first 5 years.

Over the years, we've expanded the premises considerably.

We were lucky that the site is big enough, so moving to a new location has never been necessary.

However, the layout is far from ideal for modern machinery and production methods.

So we intend to carry out major refurbishment of this site over the next 5 years.

I'd better give you some idea of what you'll be doing during your two weeks with us so you know what to expect.

Most mornings you'll have a presentation from one of the managers to learn about their department.

starting this morning with research and development.

And you'll all spend some time in each department observing what's going on and talking to people as long as you don't stop them from doing their work altogether.

In the past, a teacher from your school has come in at the end of each week to find out how the group were getting on, but your school isn't able to arrange that this year.

Okay.

Now, I'll briefly help you to orientate yourselves around the site.

As you can see, we're in the reception area, which we try to make attractive and welcoming to visitors.

There's a corridor running left from here.

And if you go along that, the door facing you at the end is the entrance to the coffee room.

This looks out onto the main road on one side and some trees on the other and that'll be where you meet each morning.

The factory is the very big room on the far side of the site.

Next to it is the warehouse which can be accessed by lries going up the road to the turning area at the end.

You can get to the warehouse by crossing to the far side of the courtyard and then the door is on your right.

Somewhere you'll be keen to find is the staff canteen.

This is right next to reception.

I can confidently say that the food's very good, but the view isn't.

The windows on one side look onto a corridor and courtyard, which aren't very attractive at all, and on the other onto the access road, which isn't much better.

You'll be using the meeting room quite often and you'll find it by walking along the corridor to the left of the courtyard and continuing along it to the end.

The meeting room is the last one on the right and I'm afraid there's no natural daylight in the room.

Then you'll need to know where some of the offices are.

The human resources department is at the front of this building.

So you head to the left along the corridor from reception and it's the second room you come to.

It looks out onto the main road.

And finally the boardroom where you'll be meeting sometimes that has quite a pleasant view as it looks out onto the trees.

Go along the corridor past the courtyard right to the end.

The boardroom is on the left next to the factory.

Okay.

Now, are there any questions before we move?

How are you getting on with your art project, Tom?

Okay.

Like, they gave us the theme of birds to base our project on, and I'm not really all that interested in wildlife, but I'm starting to get into it. I've pretty well finished the introductory stage.

So have I.

When they gave us all those handouts with details of books and websites to look at, I was really put off.

But the more I read, the more interested I got.

Me too.

I found I could research so many different aspects of birds in art.

The color, movement, texture.

So, I was looking forward to the bird park visit.

What a let down.
It pulled with rain and we hardly saw a single bird.
Much less used than the trip to the natural history museum.
Yeah, I liked all the stuff about evolution there.
The workshop sessions with Dr.
Fletcher were good, too.
Especially the brainstorming sessions.
Ah, I missed those because I was ill.
I wish we could have seen the projects last year students did.
I suppose they want us to do our own thing, not copy.
Have you drafted your proposal yet?
Yes, but I haven't handed it in.
I need to amend some parts.
I've realized the notes from my research are almost all just descriptions.
I haven't actually evaluated anything, so I'll have to fix that.
Oh, I didn't know we had to do that.
I'll have to look at that, too.
Did you do a timeline for the project?
Yes, and a mind map.
Yeah, so did I.
I quite enjoyed that, but it was hard having to explain the basis for my decisions in my action plan.
What?
You know, give a rationale.
I didn't realize we had to do that.
Okay, I can add it now.
And I've done the video diary presentation and worked out what I want my outcome to be in the project.
Someone told me it's best not to be too precise about your actual outcome at this stage, so you have more scope to explore your ideas later on.
So, I'm going to go back to my proposal to make it a bit more vague.
Really?
Okay, I'll change that, too, then.
One part of the project I'm unsure about is where we choose some paintings of birds and say what they mean to us.
Like I chose a painting of a falcon by Lancia.
I like it because the bird's standing there with his head turned to one side, but he seems to be staring straight at you.
But I can't just say it's a bit scary, can I?
You could talk about the possible danger suggested by the bird's look.
Oh, okay.
There's a picture of a fish hawk by Audubon I like.
It's swooping over the water with a fish in its talons and with great black wings which take up most of the picture.
So, you could discuss it in relation to predators and food chains.
Well, actually, I think I'll concentrate on the impression of rapid motion it gives.
Do you know that picture of a kingfisher by Van Gogh?
It's perching on a reed growing near a stream.
Yes.
It's got these beautiful blue and red and black shades.
Mhm.
I've actually chosen it because I saw a real kingfisher once when I was little.
I was out walking with my grandfather and I've never forgotten it.
Oh, so we can use a personal link.
Okay.
There's a portrait called William Wells.
I can't remember the artist, but it's a middle-aged man who's just shot a bird.
And his expression, and the way he's holding the bird in his hand suggests he's not sure about what he's done.
To me, it's about how ambiguous people are in the way they exploit the natural world.
Interesting.
There's Goan's picture via a Matty.
He did it in Tahiti.
It's a woman with a white bird behind her that is eating a lizard.
And what I'm interested in is what idea this bird refers to.
Apparently, it's a reference to the never-ending cycle of existence.
Wow.
I chose a portrait of a little boy, Javanni Demedichi.
He's holding a tiny bird in one fist.
I like the way he's holding it carefully so he doesn't hurt it.
Oh, right.
Ancient philosophy is not just about talking or lecturing or even reading long dense books.
In fact, it is something people have used throughout history to solve their problems and to achieve their greatest triumphs.
Specifically, I am referring to stoicism, which in my opinion is the most practical of all philosophies and therefore the most appealing.
Stoicism was founded in ancient Greece by Zeno of Citium in the early 3rd century BC, but was practiced by the likes of Epictetus, Kato, Seneca, and Marcus Aurelius.

Amazingly, we still have access to these ideas.

Despite the fact that the most famous Stoics never wrote anything down for publication, Kato definitely didn't.

Marcus Aurelius never intended his meditations to be anything but personal.

Seneca's letters were well, letters, and Epictetus' thoughts come to us by way of a notetaking student.

Stoic principles were based on the idea that its followers could have an unshakable happiness in this life and the key to achieving this was virtue.

The road to virtue in turn lay in understanding that destructive emotions like anger and jealousy are under our conscious control.

They don't have to control us because we can learn to control them.

In the words of Epictetus, external events I cannot control, but the choices I make with regard to them I do control.

The modern-day philosopher and writer Nassim Nicholas Taleb defines a stoic as someone who has a different perspective on experiences which most of us would see as wholly negative.

A stoic transforms fear into caution, pain into transformation, mistakes into initiation, and desire into undertaking.

Using this definition as a model, we can see that throughout the centuries, stoicism has been practiced in more recent history by kings, presidents, artists, writers, and entrepreneurs.

The founding fathers of the United States were inspired by the philosophy.

George Washington was introduced to stoicism by his neighbors at age 17 and later put on a play based on the life of Kato to inspire his men.

Thomas Jefferson kept a copy of Seneca beside his bed.

Writers and artists have also been inspired by the Stoics.

Eugene Delacroix, the renowned French romantic artist known best for his painting Liberty Leading the People, was an ardent stoic, referring to it as his consoling religion.

The economist Adam Smith's theories on capitalism were significantly influenced by the stoicism that he studied as a school boy under a teacher who had translated Marcus Aurelius's works.

Today's political leaders are no different with many finding their inspiration from the ancient texts.

Former US President Bill Clinton rereads Marcus Aurelius every single year and many have compared former President Obama's calm leadership style to that of Kato.

When Xiao Bao, the former prime minister of China, claims that meditations is one of two books he travels with and that he has read it more than 100 times over the course of his life.

Stoicism had a profound influence on Albert Ellis, who invented cognitive behavior therapy, which is used to help people manage their problems by changing the way that they think and behave.

It's most commonly used to treat depression.

The idea is that we can take control of our lives by challenging the irrational beliefs that create our faulty thinking symptoms and behaviors by using logic instead.

Stoicism has also become popular in the world of business.

Stoic principles can build the resilience and state of mind required to overcome setbacks because stoics teach turning obstacles into opportunity.

A lesson every business entrepreneur needs to learn.

I would argue that studying stoicism is as relevant today as it was 2,000 years ago, thanks to its brilliant insights into how to lead a good life.

At the very root of the thinking, there is a very simple way of living.

Control what you can and accept what you can't.

This is not as easy as it sounds and will require considerable practice.

It can take a lifetime to master.

The Stoics also believed the most important foundation for a good and happy life is not money, fame, power or pleasure, but having a disciplined and principled character.

Something which seems to resonate with many people today.

Hello, picture rep.

Can I help you?

Oh, hi.

I saw your advertisement about copying pictures to disk and I'd like a bit more information about what you do.

Sure.

What would you like to know?

Well, I've got a box full of old family photos that's been up in the attic for years.

Some of them must be 50 or 60 years old, and I'd like to get them converted to digital format.

Sure, we can do that for you.

Right.

And what about size?

The photos are all sorts of sizes.

Are there any restrictions?

Well, the maximum size of photo we can do with our normal service is 30 cm.

And each picture must be at least 4 cm.

That's the minimum we can cope with.

Well, that should be fine.

And some of them are in a frame.

Should I take them out before I send them?

Yes, please.

We can't copy them otherwise.

And also the photos must all be separate.

They mustn't be stuck into an album.

Okay, that's not a problem.

So, can you give me an idea of how much this will cost?
 I've got about 360 photos.
 I think we charge \$195 for 300 to 400 photos for the basic service.
 Okay.
 And does that include the disc?
 Yes, one disc, but you can get extra ones for £5 each.
 That's good.
 So, do I need to pay when I send you the photos?
 No, we won't need anything until we've actually copied the pictures.
 Then we'll let you know how much it is.
 And once we've received the payments, we'll send the parcel off to you.
 Is there anything else you'd like to ask about our services?
 Yes, I've roughly sorted out the photos into groups according to what they're about.
 So, can you keep them in those groups when you copy them?
 Sure.
 We'll save each group in a different folder on the disc.
 And if you like, you can suggest a name for each folder.
 So, I could have one called grandparents, for instance.
 Exactly.
 And do you do anything besides scan the photos?
 Like can you make any improvements?
 Yes, in the standard service each photo is checked and we can sometimes touch up the color a bit or improve the contrast.
 That can make a big difference.
 Okay.
 And some of the photos are actually quite fragile.
 They won't get damaged in the process, will they?
 No.
 If any look particularly fragile, we'd do them by hand.
 We do realize how precious these old photos can be.
 And another thing is we can make changes to a photo if you want.
 So if you want to remove an object from a photo or maybe alter the background, we can do Really?
 I might be interested in that.
 I'll have a look through the photos and see.
 Oh, and talking of fixing photos, I've got a few that aren't properly in focus.
 Can you do anything to make that better?
 No, I'm afraid that's one thing we can't do.
 Any other information I can give you?
 Uh, oh, how long will it take?
 We aim to get the copying done in 10 days.
 Fine.
 Right.
 Well, I'll get the photos packed up in a box and post them off to you.
 Right.
 If you've got a strong cardboard box, that's best.
 We've found that plastic ones sometimes break in the post.
 Okay.
 Right.
 Thanks for your help.
 Bye bye.
 Good morning and thank you for coming here today.
 I'd like to bring you up to date with changes in the school that will affect your children.
 As you know, the school buildings date from various times, some from the 1970s, some from the last 5 years, and of course, Dartfield House is over a century old.
 It was commissioned by a businessman, Neville Richards, and intended as his family home, but he died before it was completed.
 His heir chose to sell it to the local council who turned it into offices.
 A later plan to convert it into a tourist information center didn't come about through lack of money and instead it formed the nucleus of this school when it opened 40 years ago.
 The school has grown as the local population has increased.
 And I can now give you some news about the lower school site which is separated from the main site by a road.
 Planning permission has been granted for development of both sites.
 The lower school will move to new buildings that will be constructed on the main site.
 Developers will construct houses on the existing lower school site.
 Work on the new school buildings should start within the next few months.
 A more imminent change concerns the catering facilities and the canteen.
 The canteen is always very busy throughout the lunch period.
 In fact, it's often full to capacity because a lot of our pupils like the food that's on offer there.
 But there's only one serving point.
 So most pupils have to wait a considerable time to be served.
 This is obviously unsatisfactory as they may have hardly finished their lunch before afternoon lessons start.
 So we've had a new food hall built and this will come into use next week.

It'll have several serving areas and I'll give you more details about those in a minute.

But one thing we ask you to do to help in the smooth running of the food hall is to discuss with your children each morning which type of food they want to eat that day so they can go straight to the relevant serving point.

There won't be any junk food.

Everything on offer will be healthy and there's no change to the current system of paying for lunches by topping up your child's electronic payment card online.

You may be wondering what will happen to the old canteen.

We'll still have tables and chairs in there, and pupils can eat food from the food hall or lunch they've brought from home.

Eventually, we may use part of the canteen for storage.

But first, we'll see how many pupils go in there at lunchtime.

Okay, back to the serving points in the food hall, which will all have side dishes, desserts, and drinks on sale, as well as main courses.

One serving point we call world adventures.

This will serve a different country's cuisine each day.

Maybe Chinese one day and Lebanese the next.

The menus will be planned for a week at a time, so pupils will know what's going to be available the whole of that week.

Street life is also international with food from three particular cultures.

We'll ask pupils to make suggestions.

So perhaps sometimes there'll be food from Thailand, Ethiopia, and Mexico.

and then one of them will be replaced by Jamaican food for a week or two.

The speedy Italian serving point will cater particularly for the many pupils who don't eat meat or fish.

They can be sure that all the food served there is suitable for them.

There'll be plenty of variety, so they shouldn't get bored with the food.

Okay, that's all on the new food hall.

Now, after school lessons, these are very popular with pupils, particularly swimming.

In fact, there's a waiting list for lessons.

Cycling is another favorite, and I'm delighted that dozens of pupils make use of the chance to learn to ride in off-road conditions.

It means that more and more cycle to and from school every day.

As you know, we have a well equipped performance center and we're going to start drama classes in there, too.

Pupils will be able to join in just for fun or work up to taking part in a play.

We hope to put on at least one a year.

We already teach a number of pupils to use the sound and lighting systems in the center.

And a former pupil has given a magnificent grand piano to the school.

So a few pupils will be able to learn at the school instead of going to the local college as many of them do at the moment.

So Luke, for our next psychology assignment, we have to do something on sleep and dreams.

Right.

I've just read an article suggesting why we tend to forget most of our dreams soon after we wake up.

I mean, most of my dreams aren't that interesting anyway.

But what it said was that if we remembered everything, we might get mixed up about what actually happened and what we dreamed.

So, it's a sort of protection.

I hadn't heard that idea before.

I'd always assumed that it was just that we didn't have room in our memories for all that stuff.

Me, too.

What do you think about the idea that our dreams may predict the future?

It's a belief that you get all over the world.

Yeah.

Lots of people have a story of it happening to them.

But the explanation I've read is that for each dream that comes true, we have thousands that don't.

But we don't notice those.

We don't even remember them.

We just remember the ones where something in the real world, like a view or an action, happens to trigger a dream memory.

Right.

So, it's just a coincidence really.

Is something else I read about is what they call segmented sleeping.

That's a theory that hundreds of years ago people used to get up in the middle of the night and have a chat or something to eat, then go back to bed.

So, I tried it myself.

Why?

Well, it's meant to make you more creative.

I don't know why, but I gave it up after a week.

It just didn't fit in with my lifestyle.

But most preschool children have a short sleep in the day, don't they?

There was an experiment some students did here last term to see at what age kids should stop having naps, but they didn't really find an answer.

They spent a lot of time working out the most appropriate methodology, but the results didn't seem to show any obvious patterns.

Right.

Anyway, let's think about our assignment.

Last time I had problems with the final stage where we had to describe and justify how successful we thought we'd been.

I struggled a bit with the action plan, too.

I was okay with the planning, but I got marked down for the self assessment as well.

And I had big problems with the statistical stuff.

That's where I really lost marks.

So, shall we plan what we have to do for this assignment?

Okay.

First we have to decide on our research question.

So how about is there a relationship between hours of sleep and number of dreams?

Okay, then we need to think about who we'll do the study on.

About 12 people, right?

And shall we use other psychology students?

Let's use people from a different department.

What about history?

Yes, they might have interesting dreams or literature students.

I don't really know any.

Okay, forget that idea.

Then we have to think about our methodology.

So, we could use observation, but that doesn't seem appropriate.

No, it needs to be self-reporting, I think.

And we could ask them to answer questions online.

But in this case, paper might be better.

as they'll be doing it straight after they wake up.

In fact, while they're still half asleep, right?

And we'll have to check the ethical guidelines for this sort of research.

Because our experiment involves humans, so there are special regulations.

Yes, I had a look at those for another assignment I did.

There's a whole section on risk assessment and another section on making sure they aren't put under any unnecessary stress.

Let's hope they don't have any bad dreams.

Yeah.

Then when we've collected all our data, we have to analyze it and calculate the correlation between our two variables.

That's time sleeping and number of dreams.

and then present our results visually in a graph, right?

And the final thing is to think about our research and evaluate it.

So that seems quite straightforward.

Yeah.

So now let's get start.

Dancing is something that humans do when they want to have a good time.

It's a universal response to music found in all cultures.

But what's only been discovered recently is that dancing not only makes us feel good, it's also extremely good for our health.

Dancing, like other forms of exercise, releases hormones such as dopamine, which make us feel relaxed and happy, and it also reduces feelings of stress or anxiety.

Dancing is also a sociable activity, which is another reason it makes us feel good.

One study compared people's enjoyment of dancing at home in front of a video with dancing in a group in a studio.

The people dancing in a group reported feeling happier, whereas those dancing alone did not.

In another experiment, university researchers at York and Sheffield took a group of students and sent each of them into a lab where music was played for 5 minutes.

Each had to choose from three options.

To sit and listen quietly to the music, to cycle on an exercise bike while they listened, or to get up and dance.

All were given cognitive tasks to perform before and after.

The result showed that those who chose to dance showed much more creativity when doing problem-solving tasks.

Dr.

Love at the University of Hartfordshire believes dance could be a very useful way to help people suffering from mental health problems.

He thinks dance should be prescribed as therapy to help people overcome issues such as depression.

It's well established that dance is a good way of encouraging adolescent girls to take exercise.

But what about older people?

Studies have shown that there are enormous benefits for people in their 60s and beyond.

One of the great things about dance is that there are no barriers to participation.

Anyone can have a go, even those whose standard of fitness is quite low.

Dance can be especially beneficial for older adults who can't run or do more intense workouts or for those who don't want to.

One 2015 study found that even a gentle dance workout helps to promote a healthy heart.

And there's plenty of evidence which suggests that dancing lowers the risk of falls which could result in a broken hip, for example, by helping people to improve their balance.

There are some less obvious benefits of dance for older people, too.

One thing I hadn't realized before researching this topic was that dance isn't just a physical challenge.

It also requires a lot of concentration because you need to remember different steps and routines.

For older people, this kind of activity is especially important because it forces their brain to process things more quickly and to retain more information.

Current research also shows that dance promotes a general sense of well-being in older participants, which can last up to a week after a class.

Participants report feeling less tired and having greater motivation to be more active and do daily activities such as gardening or walking to the shops or a park.

Ballroom or country dancing.

both popular with older people have to be done in groups.

They require collaboration and often involve touching a dance partner.
All of which encourages interaction on the dance floor.
This helps to develop new relationships and can reduce older people's sense of isolation which is a huge problem in many countries.
I also looked at the benefits of Zumba.
15 million people in 180 countries now regularly take a Zumba class, an aerobic workout based on Latin American dance moves.
John Paulari, a professor of exercise and sport science at the University of Wisconsin, analyzed a group of women who were Zumba regulars and found that a class lasting 40 minutes burns about 370 calories.
This is similar to moderately intense exercises like step aerobics or kickboxing.
A study in the American Journal of Health Behavior showed that when women with obesity did Zumba three times a week for 16 weeks, they lost an average of 1.2 kilos and lowered their percentage of body fat by 1%.
More importantly, the women enjoyed the class so much that they made it a habit and continued to attend classes at least once a week.
Very unusual for an aerobic exercise program.
Dance is never going to compete with highintensity workouts when it comes to physical fitness gains, but its popularity is likely to keep on rising because it's such a fun way to keep fit.
Hello, Junior Cycle Camp.
Jake speaking.
Hi, I'm calling for some information about the cycle camp.
I'm thinking of sending my son.
Great.
Well, it's held every weekday morning over the summer vacation, and we focus on basic cycling skills and safety.
We have eight levels for children from 3 years upwards.
How old's your son, Charlie?
He's seven.
He can ride a bike, but he needs a little more training before he's safe to go on the road.
He'd probably be best in level five.
They start off practicing on the site here and we aim to get them riding on the road, but first they're taken to ride in the park away from the traffic.
Right.
And can you tell me a bit about the instructors?
Well, all our staff wear different colored shirts.
So, we have three supervisors and they have red shirts.
They support the instructors and they also stand in for me if I'm not around.
Then the instructors themselves are in blue shirts and one of these is responsible for each class.
In order to be accepted, all our instructors have to submit a reference from someone who's seen them work with children, like if they've worked as a babysitter, for example.
Then they have to complete our training course, including how to do lesson plans and generally care for the well-being of the kids in their class.
They do a great job, I have to say.
Right.
And tell me a bit about the classes.
What size will Charlie's class be?
We have a limit of eight children in each class, so their instructor really gets to know them well.
They're out riding most of the time, but they have quiet times, too, where their instructor might tell them a story that's got something to do with cycling or get them to play a game together.
It's a lot of fun.
It must be.
Now, what happens if there's rain?
Do the classes still run?
Oh, yes.
We don't let that put us off.
We just put on our waterproofs and keep cycling.
And is there anything special Charlie should bring along with him?
Well, maybe some spare clothes, especially if the weather's not so good.
And a snack for break time.
How about a drink?
No, we'll provide that.
And make sure he has shoes, not sandals.
Sure.
And just at present, Charlie has to take medication every few hours.
So, I'll make sure he has that.
Absolutely.
Just give us details of when he has to take it and we'll make sure he does.
Thanks.
Now, there are a few things you should know about day one of the camp.
The classes normally start at 9:30 every morning, but on day one, you should aim to get Charlie here by 9:20.
The finishing time will be 12:30 as usual.
We need the additional time because there are a few extra things to do.
The most important is that we have a very careful check to make sure that every child's helmet fits properly.
If it doesn't fit, we'll try to adjust it or we'll find him another one.
But he must wear it all the time he's on the bike.

Of course.

Then after that, all the instructors will be waiting to meet their classes and they'll meet up in the tent.

You can't miss it.

And each instructor will take their class away and get started.

Okay.

Well, that all sounds good.

Now, can you tell me how much the camp costs a week?

\$199.

We've managed to keep the price more or less the same as last year.

It was \$190 then, but the places are filling up quite quickly, right?

Okay.

Well, I'd like to book for the Hello everyone.

My name's Megan Baker and I'm a recruitment consultant at AVT recruitment specialists.

Now, our company specializes in positions that involve working in the agriculture and horticulture sectors.

So, that's fresh food production, garden and park maintenance and so on.

And these sectors do provide some very special career opportunities.

For a start, they often offer opportunities for those who don't want to be stuck with a 40-hour week, but need to juggle work with other responsibilities such as child care.

And this is very important for many of our recruits.

Some people like working in a rural setting surrounded by plants and trees instead of buildings, although we can't guarantee that.

But there are certainly health benefits, especially in jobs where you're not sitting all day looking at a screen.

A big plus for many people.

Salaries can sometimes be good, too, although there's a lot of variety here.

And you may have the opportunity in some types of jobs for travel overseas, although that obviously depends on the job, and not everyone is keen to do it.

Of course, working outdoors does have its challenges.

It's fine in summer, but can be extremely unpleasant when it's cold and windy.

You may need to be pretty fit for some jobs.

Though, with modern technology, that's not as important as it once was.

And standards of health and safety are much higher now than they used to be.

So, there are fewer work-related accidents.

But if you like a lively city environment surrounded by lots of people, these jobs are probably not for you.

They're often in pretty remote areas.

And some people worry about finding a suitable place to live.

But in our experience, this usually turns out fine.

Now, let me tell you about some of the exciting jobs that we have on our books right now.

One is for a fresh food commercial manager.

Our client here is a very large fresh food producer supplying a range of top supermarkets.

They operate in a very fast-paced environment with low profit margins.

The staff there work hard, but they play hard as well.

So, if you have a sociable personality, this may be for you. We have an exciting post as an agronomist advising farmers on issues such as crop nutrition, protection against pests, and the latest legislation on farming and agricultural practices.

There are good opportunities for the right person to quickly make their way up the career ladder, but a deep knowledge of the agricultural sector is expected of applicants.

A leading supermarket is looking for a fresh produce buyer who is available for a 12-month maternity cover contract.

You need to have experience in administration, planning, and buying in the fresh produce industry and in return will receive a very competitive salary.

We have also received a request for a sales manager for a chain of garden centers.

You will be visiting centers in the region to ensure their high levels of customer service are maintained.

This post is only suitable for someone who is prepared to live in the region.

There is also a vacancy for a tree technician to carry out tree cutting, forestry and conservation work.

Candidates must have a clean driving license and have training in safety procedures.

A year's experience would be preferred, but the company might be prepared to consider someone who has just completed an appropriate training course.

Finally, we have a position for a farm worker.

This will involve a wide range of farm duties, including crop sowing and harvesting, machine maintenance, and animal care.

Perks of the job include the possibility of renting a small cottage on the estate and the chance to earn a competitive salary.

A driving license and tractor driving experience are essential.

Okay, Rosie, shall we try to get some ideas together for our presentation on diet and obesity?

Sure.

I can talk about the experiment I did to see if people can tell the difference between real sugar and artificial sweeteners where you gave people drinks with either sugar or artificial sweeteners and they had to say which they thought it was.

Yeah.

It took me ages to decide exactly how I'd organize it.

Especially how I could make sure that people didn't know which drink I was giving them.

It was hard to keep track of it all, especially as I had so many people doing it.

I had to make sure I kept a proper record of what each person had had.

So, could most people tell the difference?

Yeah, I hadn't thought that they would be able to, but most people could.

Then there's that experiment I did measuring the fat content of nuts to see if the nutritional information given on the packet was accurate.

The one where you ground up the nuts and mixed them with a chemical to absorb the fat.

Yes, my results were a bit problematic.

The fat content for that type of nut seemed much lower than it said on the package, but I reckon the package information was right.

I think I should probably have ground up the nuts more than I did.

It's possible that the scales for weighing the fat weren't accurate enough, too.

I'd really like to try the experiment again sometime.

So, what can we say about helping people to lose weight?

There's a lot we could say about what restaurants could do to reduce obesity.

I read that the items at the start of a menu and the items at the end of a menu are much more likely to be chosen than the items in the middle.

So, if you put the low calorie items at the beginning and end of the menu, people will probably go for the food with fewer calories without even realizing what they're doing.

I think food manufacturers could do more to encourage healthy eating.

How?

Well, when manufacturers put calorie counts of a food on the label, they're sometimes really confusing.

And I suspect they do it on purpose because food that's high in calories taste better and so they'll sell more.

Yeah.

So, if you look at the amount of calories in a pizza, they'll give you the calories per quarter pizza, and you think, "Oh, that's not too bad."

But who's going to eat a quarter pizza?

I suppose another approach to this problem is to get people to exercise more.

Right.

In England, the current guidelines offer at least 30 minutes of brisk walking 5 days a week.

Now when you ask them about 40% of men and 30% of women say they do this but when you objectively measure the amount of walking they do with motion sensors you find that only 6% of men and 4% of women do the recommended amount of exercise.

So you can see why obesity is growing.

So how can people be encouraged to take more exercise?

Well, for example, think of the location of stairs in a train station.

If people reach the stairs before they reach the escalator when they're leaving the station, they're more likely to take the stairs.

And if you increase the width of the stairs, you'll get more people using them at the same time.

It's an unconscious process and influenced by minor modifications in their environment, right?

And it might not be a big change, but if it happens every day, it all adds up.

Yes, but actually, I'm not sure if we should be talking about exercise in our presentation.

Well, we've done quite a bit of reading about it.

I know, but it's going to mean we have a very wide focus, and our tutor did say that we need to focus on causes and solutions in terms of nutrition.

Oh, I suppose so.

And we've got plenty of information about that.

Okay.

Well, that will be simpler.

So, what shall we do now?

We've still got half an hour before our next lecture.

Let's think about what we're going to include and what will go where.

Then we can decide what slides we need.

Okay.

Fine.

Good morning everyone.

So, today we're going to look at an important creative activity and that's hand knitting.

Ancient knitted garments have been found in many different countries, showing that knitting is a global activity with a long history.

When someone says the word knitting, we might well picture an elderly person, a grandmother perhaps, sitting by the fire knitting garments for themselves or other members of the family.

It's a homely image, but one that may lead you to feel that knitting is an activity of the past.

And indeed, during the previous decade, it was one of the skills that was predicted to vanish from everyday life.

For although humans have sewn and knitted their own clothing for a very long time, many of these craftbased skills went into decline when industrial machines took over, mainly because they were no longer passed down from one generation to another.

However, that's all changing and interest in knitting classes in many countries is actually rising as more and more people are seeking formal instruction in the skill.

With that trend, we're also seeing an increase in the sales figures for knitting equipment.

So why do people want to be taught to knit at a time when a machine can readily do the job for them?

The answer is that knitting as a handicraft has numerous benefits for those doing it.

Let's consider what some of these might be.

While many people knitted garments in the past because they couldn't afford to buy clothes, it's still true today that knitting can be helpful if you're experiencing economic hardship.

If you have several children who all need warm winter clothes, knitting may save you a lot of money.

And the results of knitting your own clothes can be very rewarding, even though the skills you need to get going are really quite basic and the financial outlay is minimal.

But the more significant benefits in today's world are to do with well-being.

In a world where it's estimated that we spend up to 9 hours a day online doing something with our hands that is craftbased makes us feel good.

It releases us from the stress of a technological fastpaced life.

Now, let's look back a bit to early knitting activities.

In fact, no one really knows when knitting first began, but archaeological remains have disclosed plenty of information for us to think about.

One of the interesting things about knitting is that the earliest pieces of clothing that have been found suggest that most of the items produced were round rather than flat.

Discoveries from the 3rd and fourth centuries in Egypt show that things like socks and gloves that were needed to keep hands and feet warm were knitted in one piece using four or five needles.

That's very different from most knitting patterns today which only require two.

What's more, the very first needles people used were handcarved out of wood and other natural materials like bone.

Whereas today's needles are largely made of steel or plastic and make that characteristic clicking sound when someone's using them.

Ancient people knitted using yarns made from linen, hemp, cotton, and wool.

and these were often very rough on the skin.

The spinning wheel, which allowed people to make finer yarns and produce much greater quantities of them, led to the dominance of wool in the knitting industry, often favored for its warmth.

Another interesting fact about knitting is that because it was practiced in so many parts of the world for so many purposes, regional differences in style developed.

This visual identity has allowed researchers to match bits of knitted clothing that have been unearthed over time to the region from which the wearer came or the job that he or she did.

As I've mentioned, knitting offered people from poor communities a way of making extra money while doing other tasks.

For many centuries, it seems men, women, and children took every opportunity to knit, for example, while watching over sheep, walking to market, or riding in boats.

So, let's move on to take Hello.

Oh, hello.

I was hoping to speak to Jack Fitzgerald about renting a cottage.

I'm his wife, Shirley, and we own the cottages together, so I'm sure I can help you.

Great.

My name's Tom.

Some friends of ours rented Granary Cottage from you last year, and they thought it was great.

So, my wife and I are hoping to come in May for a week.

What date did you have in mind?

The week beginning the 14th, if possible.

I'll just check.

I'm sorry, Tom.

It's already booked that week.

It's free the week beginning the 28th though for seven nights.

In fact, that's the only time you could have it in May.

Oh, well, we could manage that.

I think we'd just need to change a couple of things.

How much would it cost?

That's the beginning of high season, so it would be £550 for the week.

Ah, that's a bit more than we wanted to pay, I'm afraid.

We've budgeted up to £500 for accommodation.

Well, we've just finished converting another building into a cottage, which we're calling Churville Cottage.

Sorry, what was that again?

Churville.

C H E R V for Victor.

I L.

Oh, that's a herb, isn't it?

That's right.

It grows fairly wild around here.

You could have that for the week you want for £480.

Okay.

So, could you tell me something about it, please?

Of course.

The building was built as a garage.

It's a little smaller than Granary Cottage.

So, that must sleep two people as well.

That's right.

There's a double bedroom.

Does it have a garden?

Yes.

You get to it from the living room through French doors and we provide two deck chairs.

We hope to build a patio in the near future, but I wouldn't like to guarantee it'll be finished by May.

The front door opens onto the old farmyard, and parking isn't a problem.

There's plenty of room at the front for that.

There are some trees and potted plants there.

What about facilities in the cottage?

It has standard things like a cooker and fridge, I presume.

In the kitchen area, there's a fridge freezer and we've just put in an electric cooker.

Is there a washing machine?

Yes.

There's also a TV in the living room which plays DVDs, too.
The bathroom is too small for a bath, so there's a shower instead.
I think a lot of people prefer that nowadays anyway.
It's more environmentally friendly, isn't it?
Unless you spend half the day in it.
What about heating?
It sometimes gets quite cool at that time of year.
There's central heating, and if you want to light a fire, there's a stove.
We can provide all the wood you need for it.
It smells so much nicer than coal, and it makes the room very cozy.
We've got one in our own house.
That sounds very pleasant.
Perhaps we should come in the winter to make the most of it.
Yes, we find we don't want to go out when we've got the fire burning.
There are some attractive views from the cottage, which I haven't mentioned.
There's a famous stone bridge.
It's one of the oldest in the region, and you can see it from the living room.
It isn't far away.
The bedroom window looks in the opposite direction.
and has a lovely view of the hills and the monument at the top.
Well, that all sounds perfect.
I'd like to book it, please.
Would you want a deposit?
Yes.

We ask for 30% to secure your booking.

So, that'll be uh £144.

And when would you like the rest of the money?

You're coming in May.

So, the last day of March, please.

Fine.

Excellent.

Could I just take your details so that I can Right.

Next on the agenda, we have traffic and highways.

Councilor Thornton.

Thank you.

Well, we now have the results of the survey carried out last month about traffic and road transport in the town.

People were generally satisfied with the state of the roads.

There were one or two complaints about potholes, which will be addressed, but a significant number of people complained about the increasing number of heavy vehicles using our local roads to avoid traffic elsewhere.

We'd expected more complaints by commuters about the reduction in the train service, but it doesn't seem to have affected people too much.

The cycle path that runs alongside the river is very well used by both cyclists and pedestrians since the surface was improved last year, but overtaking can be a problem.

So, we're going to add a bit on the side to make it wider.

At some stage, we'd like to extend the path so that it goes all the way through the town, but that won't be happening in the immediate future.

The plans to have a pedestrian crossing next to the post office have unfortunately had to be put on hold for the time being.

We'd budgeted for this to be done this financial year, but then there were rumors that the post office was going to move, which would have meant there wasn't really a need for a crossing.

Now, they've confirmed that they're staying where they are, but the highways department have told us that it would be dangerous to have a pedestrian crossing where we'd originally planned it as there's a bend in the road there.

So, that'll need some more thought.

On station road near the station and level crossing, drivers can face quite long waits if the level crossings closed, and we've now got signs up requesting them not to leave their engines running at that time.

This means pedestrians waiting on the pavement to cross the railway line don't have to breathe in car fumes.

We've had some problems with cyclists leaving their bikes chained to the railings outside the ticket office, but the station has agreed to provide bike racks there.

So, next on the agenda is proposals for improvements to the recreation ground.

Councilor Thornton again.

Well, since we managed to extend the recreation ground, we've spent some time talking to local people about how it could be made a more attractive and useful space.

If you have a look at the map up on the screen, you can see the river up in the north and the community hall near the entrance from the road.

At present, cars can park between the community hall and that line of trees to the east, but this is quite dangerous for pedestrians.

So, we're suggesting a new car park on the opposite side of the community hall right next to it.

We also have a new location for the cricket pitch.

As we've now purchased additional space to the east of the recreation ground beyond the trees, we plan to move it away from its current location, which is rather near the road into this new area beyond the line of trees.

This means there's less danger of stray balls hitting cars or pedestrians.

We've got plans for a children's playground, which will be accessible by a footpath from the community hall and will be alongside the

river.

We'd originally thought of having it close to the road, but we think this will be a more attractive location.

The skateboard ramp is very popular with both younger and older children.

We had considered moving this up towards the river, but in the end, we decided to have it in the southeast corner near the road.

The pavilion is very well used at present by both football players and cricketers.

It will stay where it is now, to the left of the line of trees and near to the river, handy for both the football and cricket pitches.

And finally, we'll be getting a new notice board for local information.

And that will be directly on people's right as they go from the road into the recreation ground.

Now that we've done all the research into bike sharing schemes in cities around the world, we need to think about how we're going to organize our report, right?

I think we should start by talking about the benefits.

I mean, it's great that so many cities have introduced these schemes where anyone can pick up a bike from dozens of different locations and hire it for a few hours.

It makes riding a bike very convenient for people.

Yes, but the costs can add up and that puts people on low incomes off in some places.

I suppose so.

But if it means more people in general are cycling rather than driving, then because they're increasing the amount of physical activity they do, it's good for their health.

Okay.

But isn't that of less importance?

I mean, doesn't the impact of reduced emissions on air pollution have a more significant effect on people's health?

Certainly in some cities, bike sharing has made a big contribution to that and also helped to cut the number of cars on the road significantly, which is the main point.

Exactly.

But I'd say it's had less of an impact on noise pollution because there are still loads of buses and lorries around.

Shall we quickly discuss the recommendations we're going to make in order to ensure bike sharing schemes are successful?

Yes.

Well, while I think it's nice to have really state-of-the-art bikes with things like GPS, I wouldn't say they're absolutely necessary.

But some technical things are really important, like a fully functional app so people can make payments and book bikes easily.

Places which haven't invested in that have really struggled.

Good point.

Some people say there shouldn't be competing companies offering separate bike sharing schemes, but in some really big cities, competition's beneficial.

And anyway, one company might not be able to manage the whole thing, right?

Deciding how much to invest is a big question.

Cities which have opened loads of new bike lanes at the same time as introducing bike sharing schemes have generally been more successful.

But there are examples of successful schemes where this hasn't happened.

What does matter though is having a big publicity campaign.

Definitely.

If people don't know how to use the scheme or don't understand its benefits, they won't use it.

People need a lot of persuasion to stop using their cars.

Shall we look at some examples now and say what we think is good or bad about them?

I suppose we should start with Amsterdam as this was one of the first cities to have a bike sharing scheme.

Yes, there was already a strong culture of cycling here.

In a way, it's strange that there was such a demand for bike sharing because you'd have thought most people would have used their own bikes and yet it's one of the best used schemes.

Dublin's an interesting example of a success story.

It must be because the public transport system's quite limited.

Not really.

There's no underground, but there are trams and a good bus network.

I'd say price has a lot to do with it.

It's one of the cheapest schemes in Europe to join, but the buses are really slow.

Anyway, the weather certainly can't be a factor.

No, definitely not.

The London scheme's been quite successful.

Yes, it's been a really good thing for the city.

The bikes are popular and the whole system is wellmaintained, but it isn't expanding quickly enough.

Basically, not enough's been spent on increasing the number of cycle lanes.

Hopefully, that'll change.

Yes.

Now, what about outside Europe?

Well, bike sharing schemes have taken off in places like Buenaziris.

H they built a huge network of cycle lanes to support the introduction of the scheme there, didn't they?

It attracted huge numbers of cyclists where previously there were hardly any.

An example of good planning.

Absolutely.

New York is a good example of how not to introduce a scheme.

When they launched it, it was more than 10 times the price of most other schemes.

More than it costs to take a taxi.

Crazy.

I think the organizers lacked vision and ambition there.

I think so, too.

Sydney would be a good example to use.

I would have expected it to have grown pretty quickly here.

Yes.

I can't quite work out why it hasn't been an instant success like some of the others.

It's a shame really.

I know.

Okay, so now we've thought about all the One of the most famous cases of extinction is that of a bird known as the dodo.

In fact, there's even a saying in English, as dead as the dodo, used to refer to something which no longer exists.

But for many centuries, the dodo was alive and well, although it could only be found in one place, the island of Maitius in the Indian Ocean.

It was a very large bird, about 1 m tall, and over the centuries it had lost the ability to fly.

But it survived happily under the trees that covered the island.

Then in the year 1507, the first Portuguese ships stopped at the island.

The sailors were carrying spices back to Europe and found the island a convenient stopping place where they could stock up with food and water for the rest of the voyage.

But they didn't settle on Maitius.

However, in 1638, the Dutch arrived and set up a colony there.

These first human inhabitants of the island found the dodo birds a convenient source of meat.

Although not everyone liked the taste, it's hard to get an accurate description of what the dodo actually looked like.

We do have some written records from sailors and a few pictures, but we don't know how reliable these are.

The best known picture is a Dutch painting in which the bird appears to be extremely fat, but this may not be accurate.

An Indian painting done at the same time shows a much thinner bird.

Although attempts were made to preserve the bodies of some of the birds, no complete specimen survives.

In the early 17th century, four dried parts of a bird were known to exist.

Of these, three have disappeared.

So only one example of soft tissue from the dodo survives, a dodo head.

Bones have also been found.

But there's only one complete skeleton in existence.

This single dodo skeleton has recently been the subject of scientific research which suggests that many of the earlier beliefs about dodos may have been incorrect.

For example, early accounts of the birds mention how slow and clumsy it was.

But scientists now believe the bird's strong knee joints would have made it capable of movement, which was not slow, but actually quite fast.

In fact, one 17th century sailor wrote that he found the birds hard to catch.

It's true that the dodo's small wings wouldn't have allowed it to leave the ground, but the scientists suggest that these were probably employed for balance while going over uneven ground.

Another group of scientists carried out analysis of the dodo's skull.

They found that the reports of the lack of intelligence of the dodo were not borne out by their research, which suggested the bird's brain was not small, but average in size.

In fact, in relation to its body size, it was similar to that of the pigeon, which is known to be a highly intelligent bird.

The researchers also found that the structure of the bird's skull suggested that one sense which was particularly well developed was that of smell.

So the dodo may also have been particularly good at locating ripe fruit and other food in the island's thick vegetation.

So it looks as if the dodo was better able to survive and defend itself than was originally believed.

Yet less than 200 years after Europeans first arrived on the island, they had become extinct.

So what was the reason for this?

For a long time, it was believed that the dodos were hunted to extinction.

But scientists now believe the situation was more complicated than this.

Another factor may have been the new species brought to the island by the sailors.

These included dogs, which would have been a threat to the dodos, and also monkeys, which ate the fruit that was the main part of the dodo's diet.

These were brought to the island deliberately.

But the ships also brought another type of creature, rats, which came to land from the ships and rapidly overran the island. These upset the ecology of the island, not just the dodos, but other species, too.

However, they were a particular danger to the dodos because they consumed their eggs.

And since each dodo only laid one at a time, this probably had a devastating effect on populations.

However, we now think that probably the main cause of the bird's extinction was not the introduction of non-native species, but the introduction of agriculture.

This meant that the forest that had once covered all the island and that had provided a perfect home for the dodo was cut down so that crops such as sugar could be grown.

So although the dodo had survived for thousands of years, suddenly it was gone.

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Hello.
Oh, hello.
My name's Jan.
Are you the right person to talk to about the Buckworth Conservation Group?
Yes.
I'm Peter.
I'm the secretary.
Good.
I've just moved to this area and I'm interested in getting involved.
I was in a similar group where I used to live.
Could you tell me something about your activities, please?
Of course.
Well, we have a mixture of regular activities and special events.
One of the regular ones is trying to keep the beach free of litter.
A few of us spend a couple of hours a month on it, and it's awful how much there is to clear.
I wish people would be more responsible and take it home with them.
I totally agree.
I'd be happy to help with that.
Is it okay to take dogs?
I'm afraid not, as they're banned from the beach itself.
You can take them along the cliffs though and children are welcome.
Right.
We also manage a nature reserve and there's a lot to do there all year round.
For example, because it's a popular place to visit, we spend a lot of time looking after the paths and making sure they're in good condition for walking.
I could certainly help with that.
Good.
And we have a program of creating new habitats there.
We've just finished making and installing nesting boxes for birds to use.
And next, we're going to work on encouraging insects.
They're important for the biodiversity of the reserve.
They certainly are.
Oh, and we're also running a project to identify the different species of butterflies that visit the reserve.
You might be interested in taking part in that.
Sure.
I was involved in something similar where I used to live, counting all the species of moths.
I'd enjoy Another job we're doing at the reserve is replacing the wall on the southern side between the parking area and our woodshed.
It was badly damaged in a storm last month.
Okay.
Then, as I said, we have a program of events as well, both at the weekend and during the week.
Right.
I presume you have guided walks.
I'd like to get to know the local countryside as I'm new to the area.
Yes, we do.
The next walk is to Rustin Island a week on Saturday.
We'll be meeting in the car park at Dunmore Beach at low tide.
That's when the sands are dry enough for us to walk to the island without getting wet.
Sounds good.
The island's a great place to explore.
It's quite small and it's got a range of habitats.
It's also an ideal location for seeing seals just off the coast or even on the beach.
Okay.
And is there anything we should bring like a picnic for instance?
Yes, do bring one as it's a full day walk.
And of course, it'll be wet walking across and back.
So, make sure your boots are waterproof.
I must buy a new pair.
There's a hole in one of my current ones.
Well, I'd definitely like to come on the walk.
Great.
Then later this month, we're having a one-day woodwork session in Hopton Wood.
I've never tried that before.
Is it okay for beginners to take part?
Definitely.
There'll be a couple of experts leading the session and we keep the number of participants down so you'll get as much help as you need.
Excellent.
I'd love to be able to make chairs.
Huh?

That's probably too ambitious for one day.
You'll be starting with wooden spoons and of course learning how to use the tools.
And anything you make is yours to take home with you.
That sounds like fun.
When is it?
It's on the 17th from 10:00 a.m.
until 3.
There's a charge of £35 including lunch or £40 if you want to camp in the wood.
I should think I'll come home the same day.
Well, I'd certainly like to join the group.
It sounds like you.
So, hello everyone.
My name's Lou Miller and I'm going to be your tour guide today as we take this fantastic boat trip around the Tasmanian coast.
Before we set off, I just want to tell you a few things about our journey.
Our boats aren't huge, as you can see.
We already have three staff members on board.
And on top of that, we can transport a further 15 people, that's you, around the coastline.
But please note, if there are more than nine people on either side of the boat, we'll move some of you over.
Otherwise, all 18 of us will end up in the sea.
We've recently upgraded all our boats.
They used to be jet black, but our new ones now have these comfortable dark red seats and a light green exterior in order to stand out from others and help promote our company.
This gives our boats a rather unique appearance, don't you think?
We offer you a free lunchbox during the trip, and we have three types.
Lunchbox one contains ham and tomato sandwiches.
Lunchbox two contains a cheddar cheese roll.
And lunchbox three is saladbased and also contains eggs and tuna.
All three lunchboxes also have a packet of crisps and chocolate bar inside.
Please let staff know which lunchbox you prefer.
I'm sure I don't have to ask you not to throw anything into the sea.
We don't have any bins to put litter in, but Jess, myself, or Ry, our other guide, will collect it from you after lunch and put it all in a large plastic sack.
The engine on the boat makes quite a lot of noise.
So, before we head off, let me tell you a few things about what you're going to see.
This area is famous for its ancient lighthouse, which you'll see from the boat as we turn past the first little island.
It was built in 1838 to protect sailors as a number of shipwrecks had led to significant loss of life.
The construction itself was complicated as some of the original drawings kept by the local council show.
It sits right on top of the cliffs in a very isolated spot.
In the 19th century, there were many jobs there, such as polishing the brass lamps, chopping firewood, and cleaning windows that kept lighthouse keepers busy.
These workers were mainly prison convicts until the middle of that century when ordinary families willing to live in such circumstances took over.
Some of you have asked me what creatures we can expect to see.
I know everyone loves the penguins, but they're very shy and unfortunately tend to hide from passing boats.
But you might see birds in the distance, such as sea eagles flying around the cliff edges where they nest.
When we get to the rocky area inhabited by furs seals, we'll stop and watch them swimming around the coast.
They're inquisitive creatures, so don't be surprised if one pops up right in front of you.
Their predators, orca whales, hunt along the coastline, too, but spotting one of these is rare.
Dolphins, on the other hand, can sometimes approach on their own or in groups as they ride the waves beside us.
Lastly, I want to mention the caves.
Tasmania is famous for its caves, and the ones we'll pass by are so amazing that people are lost for words when they see them.
They can only be approached by sea, but if you feel that you want to see more than we're able to show you, then you can take a kayak into the area on another day, and one of our staff will give you more information on that.
What we'll do is to go through a narrow channel past some incredible rock formations and from there we'll be able to see the openings to the caves and at that point we'll talk to you about what lies beyond.
So Tim, we have to do a short summary of our work experience on a farm.
Right.
My farm was great but arranging the work experience was hard.
One problem was it was miles away and I don't drive.
And also I'd really wanted a placement for a month, but I could only get one for 2 weeks.
I was lucky.
The farmer let me stay on the farm, so I didn't have to travel.
But finding the right sort of farm to apply to wasn't easy.
No, they don't seem to have websites, do they?
I found mine through a friend of my mother's, but it wasn't easy.
No, my farm was mostly livestock, especially sheep.
I really enjoyed helping out with them.
I was up most of one night helping a sheep deliver a lamb on your own.
No, the farmer was there, and he told me what to do.
It wasn't a straightforward birth, but I managed.

It was a great feeling to see the lamb stagger to its feet and start feeding almost straight away, and to know that it was okay. M then another time a lamb had broken its leg and they got the vet in to set it and he talked me through what he was doing. That was really useful.

Yes, my farm had sheep, too.

The farm was in a valley and they had a land breed called Suffix, although the farmer said they'd had other breeds in the past. So, were they bred for their meat?

Mostly, yes.

They're quite big and solid.

My farm was up in the hills and they had a different breed of sheep.

They were chevots.

Oh, I heard their wolves really sought after.

Yes, it's very hardwearing and they use it for carpets.

I was interested in the amount of supplements they add to animals feed nowadays.

Like even the chickens got extra vitamins and electrolytes in their feed.

Yes, I found that too.

And they're not cheap.

But my farmer said some are overpriced for what they are.

And he didn't give them as a matter of routine, just at times when the chickens seem to particularly require them.

Yes, mine said the same.

He said certain breeds of chickens might need more supplements than the others.

But the cheap and expensive ones are all basically the same.

So, did your farm have any other livestock, Diana?

Yes, dairy cows.

I made a really embarrassing mistake when I was working in the milkshed.

Some cows had been treated with antibiotics, so their milk wasn't suitable for human consumption, and it had to be put in a separate container.

But I got mixed up and I poured some milk from the wrong cow in with the milk for humans.

So, the whole lot had to be thrown away.

The farmer wasn't too happy with me.

I asked my farmer how much he depended on the vet to deal with health problems.

I'd read reports that the livestock's health is being affected as farmers are under pressure to increase production.

Well, he didn't agree with that, but he said that actually some of the stuff the vets do like minor operations, he'd be quite capable of doing himself.

Yeah, my farmer said the same, but he reckons vet skills are still needed.

Now we've got to give a bit of feedback about last term's modules.

Just short comments apparently.

Shall we do that now?

Okay.

So medical terminology.

Well, my heart sank when I saw that especially right at the beginning of the course and I did struggle with it.

I thought it would be hard but actually I found it all quite straightforward.

What did you think about diet and nutrition?

Okay, I suppose.

Do you remember what they told us about pet food and the fact that there's such limited checking into whether or not it's contaminated?

I mean, in comparison with the checks on food for humans.

I thought that was terrible.

H I think the module that really impressed me was the animal disease one when we looked at domesticated animals in different parts of the world like camels and water buffalo and alpaca.

The economies of so many countries depend on these but scientists don't know much about the diseases that affect them.

Yes.

I thought they'd know a lot about ways of controlling and eradicating those diseases, but that's not the case at all.

I loved the wildlife medication unit.

Things like helping birds that have been caught in oil spills.

That's something I hadn't thought about before.

Yeah.

I thought I might write my dissertation on something connected with Right.

So, actually, I was thinking something along the same lines.

Labyrinths have existed for well over 4,000 years.

Labyrinths and labyrinthine symbols have been found in regions as diverse as modern-day Turkey, Ireland, Greece, and India.

There are various designs of labyrinth, but what they all have in common is a winding spiral path which leads to a central area.

There is one starting point at the entrance and the goal is to reach the central area.

Finding your way through a labyrinth involves many twists and turns, but it's not possible to get lost as there is only one single path.

In modern times, the word labyrinth has taken on a different meaning and is often used as a synonym for a maze.

A maze is quite different as it is a kind of puzzle with an intricate network of paths.

Mazes became fashionable in the 15th and 16th centuries in Europe and can still be found in the gardens of great houses and palaces.

The paths are usually surrounded by thick high hedges so that it's not possible to see over them.

Entering a maze usually involves getting lost a few times before using logic to work out the pattern and find your way to the center and then out again.

There are lots of dead ends and paths which lead you back to where you started.

The word maze is believed to come from a Scandinavian word for a state of confusion.

This is where the word amazing comes from.

Labyrinths, on the other hand, have a very different function.

Although people now often refer to things they find complicated as labyrinths, this is not how they were seen in the past.

The winding spiral of the labyrinth has been used for centuries as a metaphor for life's journey.

It served as a spiritual reminder that there is purpose and meaning to our lives and helped to give people a sense of direction.

Labyrinths are thought to encourage a feeling of calm and have been used as a meditation and prayer tool in many cultures over many centuries.

The earliest examples of the labyrinth spiral pattern have been found carved into stone from Sardinia to Scandinavia, from Arizona to India to Africa.

In Europe, these spiral carvings date from the late Bronze Age.

The Native American Puma tribe wove baskets with a circular labyrinth design that depicted their own cosmology.

In ancient Greece, the labyrinth spiral was used on coins around 4,000 years ago.

Labyrinths made of mosaics were commonly found in bathous, villas, and tombs throughout the Roman Empire.

In northern Europe, there were actual physical labyrinths designed for walking on.

These were cut into the turf or grass, usually in a circular pattern.

The origin of these walking labyrinths remains unclear, but they were probably used for fertility rights, which may date back thousands of years.

11 examples of turf labyrinths survive today, including the largest one at Saffron Walden, England, which used to have a large tree in the middle of it.

More recently, labyrinths have experienced something of a revival.

Some believe that walking a labyrinth promotes healing and mindfulness, and there are those who believe in its emotional and physical benefits, which include slower breathing and a restored sense of balance and perspective.

This idea has become so popular that labyrinths have been laid into the floors of spas, wellness centers, and even prisons in recent years.

A pamphlet at Colorado Children's Hospital informs patients that walking a labyrinth can often calm people in the midst of a crisis.

And apparently, it's not only patients who benefit.

Many visitors find walking a labyrinth less stressful than sitting in a corridor or waiting room.

Some doctors even walk the labyrinth during their breaks.

In some hospitals, patients who can't walk can have a paper finger labyrinth brought to their bed.

The science behind the theory is a little sketchy, but there are dozens of smallscale studies which support claims about the benefits of labyrinths.

For example, one study found that walking a labyrinth provided short-term calming, relaxation, and relief from anxiety for Alzheimer's patients.

So, what is it about labyrinths that makes their appeal so universal?

Hello, Jane Fairbank speaking.

Oh, good morning.

My name's Frank Pritchard.

I've just retired and moved to South.

I'd like to become a volunteer.

And I gather you coordinate voluntary work in the village.

That's right.

What sort of thing could I do?

Well, we need help with the village library.

We borrow books from the town library and individuals also donate them.

So, one thing you could do is get involved in collecting them.

If you've got a car, that is.

Yes, that's no problem.

The times are pretty flexible, so we can arrange it to suit you.

Oh, another thing is the records that we keep of the books we're given and those we borrow and need to return to the town library.

It would be very useful to have another person to help keep them up to date.

Right.

I'm used to working on a computer.

I presume they're computerized.

Oh, yes.

Is the library purposebuilt?

I haven't noticed it when I've walked around the village.

No, we simply have the use of a room in the village hall.

The west room.

It's on the left as you go in.

I must go and have a look inside the hall.

Yes, it's a nice building.

Do you run a lunch club in the village for elderly people?

I know a lot of places do.

Yes, we have a very successful club.

I could help with transport if that's of any use.

Oh, definitely.

People come to the club from neighboring villages, and we're always in need of more drivers.

And does the club have groups that focus on a particular hobby, too?

I could get involved in one or two, particularly if there are any art groups.

Excellent.

I'll find out where we need help and get back to you.

Fine.

What about help for individual residents?

Do you arrange that at all?

Yes, we do it as a one-off.

Oh, in fact, there's Mrs.

Carol.

She needs a lift to the hospital next week and we're struggling to find someone.

When's her appointment?

On Tuesday.

It would take the whole morning.

I could do that.

Oh, that would be great.

Thank you.

And also next week, we're arranging to have some work done to Mr.

Salsbury's house before he moves as he isn't healthy enough to do it himself.

We've got some people to decorate his kitchen, but if you could do some weeding in his garden, that would be wonderful.

Okay, I'd enjoy that.

And presumably the day and time are flexible.

Oh, yes.

Just say when would suit you best and we'll let Mr.

Ssbury know.

Good.

The volunteers group also organizes monthly social events, which is a great way to meet other people, of course.

Uhhuh.

So, next month on the 19th of October, we're holding a quiz.

A couple of residents are great at planning unusual ones, and we always fill the village That sounds like fun.

Can I do anything to help?

Well, because of the number of people, we need plenty of refreshments for halfway through.

So, if you could provide any, we'd be grateful.

I'm sure I could.

I'll think about what to make and let you know.

Thank you.

Then, on November the 18th, we're holding a dance also in the village hall.

We've booked a band that specializes in music of the 1930s.

They've been before and we've had a lot of requests to bring them back.

I'm not really a dancer, but I'd like to do something to help.

Well, we sell tickets in advance, and having an extra person to check them at the door as people arrive would be good.

It can be quite a bottleneck if everyone arrives at once.

Okay, I'm happy with that.

We're also arranging a New Year's Eve party.

We're expecting that to be a really big event.

So instead of the village hall, it'll be held in the Mountfort Hotel.

The Mountfort M O U N T F O R T Hotel.

It isn't in South itself, but it's only a couple of miles away.

The hotel will be providing dinner and we've booked a band.

The one thing we haven't got yet is a poster.

That isn't something you could do by any chance, is it?

Well, actually, yes.

Before I retired, I was a graphic designer, so that's right up my street.

Oh, perfect.

I'll give you the details and then perhaps you could send me a draft.

Of course.

Good morning and welcome to Onitton Hall, one of the largest estates in the area.

My name's Nick and I'm one of the guides.

I'll give you a brief introduction to the estate while you're sitting down and then we'll walk round.

The estate consists of the house, gardens, parkland, and farm, and it dates back to the 14th century.

The original house was replaced in the late 17th century and of course it has had a large number of owners.

Almost all of them have left their mark generally by adding new rooms like the ballroom and conservatory or by demolishing others.

The farm looks much as it's always done, although the current owner has done a great deal of work to the flower beds.

In the 17th century, the estate was owned by a very wealthy man called Sir Edward DS.

His intention was to escape from the world of politics after years as an active politician and to build a new house worthy of his big collection of books, paintings, and sculptures.

He broke off contact with his former political allies and hosted meetings of creative and literary people like painters and poets.

Unusually for his time, he didn't care whether his guests were rich or poor as long as they had talent.

Big houses like Onitton had dozens of servants until the 1920s or 30s, and we've tried to show what their working lives were like.

Photographs, of course, don't give much of an idea.

So instead, as you go around the house, you'll see volunteers dressed up as 19th century servants going about their work.

They'll explain what they're doing and tell you their recipes or what tools they're using.

We've just introduced this feature to replace the audio guide we used to have available.

I see there are a number of children here with you today.
Well, we have several activities especially for children, like dressing up in the sorts of clothes that children wore in the past.
And as it's a fine day, some of you will probably want to play in the adventure playground.
Our latest edition is child-sized tractors that you can drive around the grounds.
We'll also be going into the farm that's part of the estate where there's plenty to do.
Most of the buildings date from the 18th century, so you can really step back into an agricultural past.
Until recently, the dairy was where milk from the cows was turned into cheese.
It's now the place to go for lunch or afternoon tea or just a cup of coffee and a slice of homemade cake.
The big stone building that dominates the farm is the large barn.
And in here is our collection of agricultural tools.
These were used in the past to plow the earth, sow seeds, make gates, and much more.
There's a small barn, also made of stone, where you can groom the donkeys and horses to keep their coats clean.
They really seem to enjoy having it done, and children love grooming them.
The horses no longer live in the stables, which instead is the place to go to buy gifts, books, our own jams and pickles, and clothes and blankets made of wool from our sheep.
Outside the shed, which is the only brick building, you can climb into a horse-drawn carriage for a lovely relaxing tour of the park and farm.
The carriages are well over a 100 years old.
And finally, the parkland, which was laid out in the 18th century with a lake and trees that are now well established.
You'll see types of cattle and sheep that are hardly ever found on farms these days.
We're helping to preserve them to stop their numbers falling further.
Okay.
Well, if you'd like to come with me, we will start.
Did you make notes while you were watching the performances of Romeo and Juliet, Gemma?
Yes, I did.
I found it quite hard, though.
I kept getting too involved in the play.
Me, too.
I ended up not taking notes.
I wrote down my impressions when I got home.
Do you mind if I check a few things with you in case I've missed anything?
And I've also got some questions about our assignment.
No, it's good to talk things through.
I may have missed things, too.
Okay, great.
So, first of all, I'm not sure how much information we should include in our reviews.
Right.
Well, I don't think we need to describe what happens, especially as Romeo and Juliet is one of Shakespeare's most well-known plays.
Yeah, everyone knows the story.
In an essay, we'd focus on the poetry and Shakespeare's use of imagery, etc., but that isn't really relevant in a review.
We're supposed to focus on how effective this particular production is.
H, we should say what made it a success or a failure.
And part of that means talking about the emotional impact the performance had on us.
I think that's important.
Yes.
And we should definitely mention how well the director handled important bits of the play, like when Romeo climbs onto Juliet's balcony and the fight between Mercutio and Tybalt.
Yes.
It would also be interesting to mention the theater space and how the director used it, but I don't think we'll have space in 800 words.
No.
Okay, that all sounds quite straightforward.
So, what about the Emporium Theatre's production of the play?
I thought some things worked really well, but there were some problems, too.
Yeah.
What about the set, for example?
I think it was visually really stunning.
I'd say that was probably the most memorable thing about this production.
You're right.
The set design was really amazing, but actually, I have seen similar ideas used in other productions.
H.
What about the lighting?
Some of the scenes were so dimly lit, it was quite hard to see.
I didn't dislike it.
It helped to change the mood of the quieter scenes.
That's a good point.
What did you think of the costumes?
I was a bit surprised by the contemporary dress, I must say.
Yeah, I think it worked well, but I'd assumed it would be more conventional.
Me, too.
I liked the music at the beginning and I thought the musicians were brilliant, but I thought they were wasted because the music didn't

have much impact in acts two and three.

Yes, that was a shame.

One problem with this production was that the actors didn't deliver the lines that well.

They were speaking too fast.

It was a problem.

I agree.

But I thought it was because they weren't speaking loudly enough, especially at key points in the play.

I actually didn't have a problem with that.

It's been an interesting experience watching different versions of Romeo and Juliet, hasn't it?

Definitely.

It's made me realize how relevant the play still is.

Right.

I mean, a lot's changed since Shakespeare's time, but in many ways, nothing's changed.

There are always disagreements and tension between teenagers and their parents.

Yes, that's something all young people can relate to, more than the violence and the extreme emotions in the play.

How did you find watching it in translation?

Really interesting.

I expected to find it more challenging, but I could follow the story pretty well.

I stopped worrying about not being able to understand all the words and focused on the actor's expressions.

The ending was pretty powerful.

Yes, that somehow intensified the emotion for me.

Did you know Shakespeare's been translated into more languages than any other writer?

What's the reason for his international appeal, do you think?

I was reading that it's because his plays are about basic themes that people everywhere are familiar with.

Yeah.

And they can also be understood on different levels.

The characters have such depth, right?

which allows directors to experiment and find new angles.

That's really important because right everyone, let's make a start.

Over the past few sessions, we've been considering the reasons why some world languages are in decline.

And today, I'm going to introduce another factor that affects languages and the speakers of those languages.

And that's technology and in particular digital technology.

In order to illustrate its effect, I'm going to focus on the Icelandic language, which is spoken by around 321,000 people, most of whom live in Iceland, an island in the North Atlantic Ocean.

The problem for this language is not the number of speakers, even though this number is small, nor is it about losing words to other languages such as English.

In fact, the vocabulary of Icelandic is continually increasing because when speakers need a new word for something, they tend to create one rather than borrowing from another language.

All this makes Icelandic quite a special language.

It's changed very little in the past millennium.

Yet, it can handle 21st century concepts related to the use of computers and digital technology.

Take for example the word for web browser.

This is vafri in Icelandic which comes from the verb to wander.

I can't think of a more appropriate term because that's exactly what you do mentally when you browse the internet.

Then there's an Icelandic word for podcast which is too hard to pronounce and so on.

Icelandic then is alive and growing but and it's a big butt.

Young Icelanders spend a great deal of time in the digital world and this world is predominantly English.

Think about smartphones.

They didn't even exist until comparatively recently, but today young people use them all the time to read books, watch TV or films, play games, listen to music, and so on.

Obviously, this is a good thing in many respects because it promotes their bilingual skills, but the extent of the influence of English in the virtual world is staggering and it's all happening really fast.

For their parents and grandparents, the change is less concerning because they already have their native speaker skills in Icelandic.

But for young speakers, well, the outcome is a little troubling.

For example, teachers have found that playground conversations in Icelandic secondary schools can be conducted entirely in English, while teachers of much younger children have reported situations where their classes find it easier to say what is in a picture using English rather than Icelandic.

The very real and worrying consequence of all this is that the young generation in Iceland is at risk of losing its mother tongue.

Of course, this is happening to other European languages, too.

But while internet companies might be willing to offer, say, French options in their systems, it's much harder for them to justify the expense of doing the same for a language that has a population the size of a French town such as Nice.

The other drawback of Icelandic is the grammar, which is significantly more complex than in most languages.

At the moment, the tech giants are simply not interested in tackling this.

So, what is the Icelandic government doing about this?

Well, large sums of money are being allocated to a language technology fund that it is hoped will lead to the development of Icelandic sourced apps and other social media and digital systems.

But clearly, this is going to be an uphill struggle.

On the positive side, they know that Icelandic is still the official language of education and government.

It has survived for well over a thousand years, and the experts predict that its future in this nation state is sound and will continue to be so.

However, there's no doubt that it's becoming an inevitable second choice in young people's lives.

This raises important questions.

When you consider how much of the past is tied up in a language, will young Icelanders lose their sense of their own identity?

Another issue that concerns the government of Iceland is this.

If children are learning two languages through different roots, neither of which they are fully fluent in, will they be able to express themselves properly?

Jack, I'm thinking of taking the kids to the seaside on a surfing holiday this summer and I wanted to ask your advice as I know you're such an expert.

Well, I don't know about that, but yes, I've done a bit of surfing over the years.

I'd thoroughly recommend it.

I think it's the kind of holiday all the family can enjoy together.

The thing about surfing is that it's great for all ages and all abilities.

My youngest started when he was only three.

Wow.

But it's quite physically demanding, isn't it?

I've heard you need to be pretty fit.

Yes.

You'll certainly learn more quickly and won't tire as easily.

Well, that should be okay for us.

You've been surfing a few times in Ireland, haven't you?

Yes, there's some great surfing there, which people don't always realize.

And which locations would you recommend?

There seem to be quite a few.

Yes, there are loads.

Last year we went to County Dona.

There are several great places to surf there.

What about in County Clare?

I read that's also really good for surfing.

Yes, it is.

I've been there a few times.

Most people go to La Hinch.

My kids love it there.

The waves aren't too challenging and the town is very lively.

Are there good hotels there?

Yes, some very nice ones.

And there are also a few basic host and campsites.

It's great if you need lessons, as the surf schools are excellent.

Yes, and there's lots to see in the area, like those well-known cliffs.

I've forgotten the name of them.

Oh, don't worry.

I can look them up.

I've also been surfing in County Mayo, which is less well known for surfing, but we had a really good time.

That was a few years ago when the kids were younger.

There's a good surf school at Karaniski Beach.

How do you spell that?

C A R O W N I S K E Y.

Okay.

I put the kids into the surf camp they run during the summer for 10 to 16 year olds.

Oh, right.

How long was that for?

3 hours every day for a week.

It was perfect.

They were so tired out after I can imagine.

One thing we did while the kids were surfing was to rent some kayaks to have a look around the bay which is nearby.

It's really beautiful.

Oh, I'd love to do that.

Now, the only time I went to Ireland, it rained practically every day.

H yes, that can be a problem.

But you can surf in the rain.

You know, it doesn't have the same appeal somehow.

Well, the weather's been fine the last couple of years when I've been there.

But actually, it tends to rain more in August than in the spring or autumn.

September's my favorite month because the water is warmer.

Then the only problem is that the kids are back to school then.

I know.

But one good thing about Irish summers is that it doesn't get too hot.

The average temperature is about 19° and it usually doesn't go above 25°.

That sounds all right.

Now, what about costs?

Surfing is a pretty cheap holiday, really.

The only cost is the hire of equipment.

You can expect to pay a daily rate of about €30 for the hire of a wet suit and board, but you can save about €40 if you hire by the week. That's not too bad.

No, it's important to make sure you get good quality wetsuits.

You'll all get too cold if you don't.

And make sure you also get boots.

They keep your feet warm and it's easier to surf with them on, too.

Okay.

Well, thanks very much for that.

Good afternoon.

My name's Mrs.

Carter and I run the before and after school extended hours child care service.

I hope you've had a chance to have a good look around the school and talk to staff and pupils.

I know that many of you are interested in using our child care service when your child joins the school.

And perhaps you already know something about it.

But for those that don't, I'll go through the main details now.

We offer child care for children from the ages of 4 to 11, both before and after school.

I know that many parents who work find this service invaluable.

You can leave your child with us safe in the knowledge that they will be extremely well cared for.

We're insured to provide care for up to 70 children, although we rarely have this many attending at any one session.

I think we generally expect around 50 to 60 children for the afternoon sessions and about half that number for the breakfast sessions.

Although we currently do have 70 children registered with us, not all of these attend every day.

It's 10 years since we began offering an extended hours service and we've come a long way during that time.

When we first opened, we only had about 20 children attending regularly.

We try to keep our costs as low as we can, and we think we provide very good value for money.

For the afternoon sessions, which run from 3:30 until 6:00 p.m., it's £720.

But if you prefer, you can pay for 1 hour only, which costs £350 or 2 hours, which costs £5.70.

The cost of the child care includes food and snacks.

They'll be given breakfast in the morning and in the afternoon, a healthy snack as soon as they finish school.

At 5:00 p.m., children are given something more substantial, such as pasta or a casserole.

Please inform us of any allergies that your child might have and we'll make sure they're offered a suitable alternative.

As you may know, the child care service runs through the school holidays from 8:00 a.m.

to 6:00 p.m.

We offer a really varied and exciting program to keep the children entertained.

We don't want them to feel as if they are still at school.

It will also feel different because they'll get the chance to make new friends with children from other schools.

Spaces are available for them because a lot of our termtime children don't always attend during the holiday.

In the past, parents have asked if children over the age of 11 are allowed to come with their younger brothers and sisters, but I'm afraid we're unable to do this because of the type of insurance we have.

So now let me tell you about some of the activities that your child can do during the after school sessions.

As well as being able to use the playground equipment, computers, and the library, there is usually at least one special activity that children can do each day.

For example, Spanish.

We have a specialist teacher coming in every Thursday to give a basic introduction to the language through games and songs.

She does two sessions, one for the over eights and one for the younger children.

This is the only activity which we have to make an extra charge for, but it's well worth it.

Once a week, the children have the opportunity to do some music.

We're very lucky that one of our staff is a member of a folk band.

On Mondays, she teaches singing and percussion to groups of children.

We do rely on parental support for this.

So, if any of you sing or play an instrument and would be prepared to help out at these sessions, we'd be delighted.

Painting continues to be one of the most popular activities.

To begin with, we weren't keen on offering this because of the extra mess involved, but children kept asking if they could do some art, and so we finally gave in.

Art is great for helping the children to relax after working hard at school all day.

Yoga is something that we've been meaning to introduce for some time, but haven't been able to find anyone available to teach it until now.

That is, so we'll see how this goes.

Hopefully, children will benefit in all sorts of ways from this.

Cooking is another popular activity.

They make a different sort of cake or pizza or bread each week.

Although the younger children love doing it, we found that the mess was just too much.

So, we've decided to restrict this to the overates as they are better able to clean up after themselves.

Hello, Dr.

Green.

I'm here to talk to you about my work placement.

Oh, yes.

It's Holly, isn't it?

Yes.

So, which work placement have you chosen?

I decided to go for the Orion Stadium placement.

The event I'll be managing is one where I'm helping to set up a sports competition for primary school children.
Yeah, that's always a popular placement.
Even though it can be tougher than you think, working with children.
I know, but it's the fresh air that attracts me.
Organizing something indoors doesn't have the same appeal, even though it might be fun.
Okay.
So, obviously safety is going to be one of your key concerns for this event.
Yes, I've already thought about that.
I'll need to make sure none of the equipment's damaged.
Ah, well, you'll be working with schools, so the equipment will be their responsibility.
However, the grounds and what goes on there will be yours.
Oh, I see.
That'll include keeping everyone within the boundary once they're in their kit and on the field.
Exactly.
You'll need to inspect areas like changing rooms as well for anything someone can trip over, but your main priority will be not to lose anyone.
Right.
I'll need staff to help with And don't forget about the spectators.
I was thinking that many of them will be parents who could help run the event.
I wouldn't rely on that.
They'll be more interested in filming their children than volunteering.
I'll need to make sure they don't interfere with events doing that.
And that's not always easy, especially when a proud parents trying to get a snap of their child and you want them to move elsewhere.
Okay.
Uh what about theuling?
With sporting events, there are all sorts of things that can alter the timetable like rain for instance.
though.
So far, we've always been lucky with that.
Yeah.
And I was thinking about what to do if someone got hurt as well.
I know that last year that caused a terrible delay.
You have to be prepared for such things.
Oh, what if a match ends in a draw?
Do you let the teams keep going until someone wins?
That'll be up to you.
And again, you need to plan for it.
Right now, the aim of your work placement is to give you the opportunity to develop the skills that an events manager needs.
So, let's talk about those a bit.
Well, I think my communication skills are pretty good.
I can talk on the phone to people and book venues and that kind of thing.
Good.
Just remember, it isn't only about what you say.
If you meet someone face to face and want to persuade them to be a sponsor, for example.
Oh, I'll dress up for that.
Sure.
Good.
Let's go on to think about your organizational skills.
You're working in a very peoplebased industry, and that means things won't always go to plan.
I guess it's being prepared to make changes that matters.
That's right.
You may have to make an on the spot change to a timetable because of a problem you hadn't anticipated.
Just do it.
Okay.
How's your time management these days?
I'm working on it.
I'm certainly better when I have a deadline, which is why this work suits me.
Yes, but it's how you respond as that deadline approaches.
I know I've got to look calm even if I'm in a panic.
Just think to yourself, "No one must know I'm under pressure." Yeah, even though I'm multitasking like crazy.
Another skill that events managers need is creativity.
Often your client has what we call the big picture idea, but it's up to the events manager to think of all the fine points that go to making it work.
Right.
So, I need to listen carefully to that idea and then fill in all the gaps.
That's right.
and you will have a team working under you.
So, another key skill is leadership.
Your team may have lots of ideas, too, but you've got to make the ultimate choices.
Do we have refreshments inside or out?
For example, isn't it better to be democratic?

It's a nice idea, but you have the ultimate responsibility.

So, believe in what you think best.

be prepared to say, "Yes, that's a good idea, but it won't work here." I see what you mean.

What about the networking side of things?

I know it's an area that a lot of students worry about because we don't have much experience to offer others.

But even without it, you can still be an interesting person with useful ideas.

And the more people you impress, the better.

I guess that will help me when I apply for a real job.

Exactly.

Think ahead.

Remember what your ambitions are and keep them in mind.

Definitely.

Scientists believe that a majority of the Earth's bird population migrate in some fashion or other.

Some travel seasonally for relatively short distances, such as birds that move from their winter habitats in lowlands to mountaintops for the summers.

Others, like the Arctic turn, travel more than 25,000 m seasonally between the northern and southern poles.

Bird migration has been studied over many centuries through a variety of observations.

But until relatively recently, where birds went to in the winter was considered something of a mystery.

The lack of modern science and technology led to many theories that we now recognize as errorfilled and even somewhat amusing.

Take hibernation theory for example.

2,000 years ago, it was commonly believed that when birds left an area, they went underwater to hibernate in the seas and oceans.

Another theory for the regular appearance and disappearance of birds was that they spent winter hidden in mud till the weather changed and food became abundant again.

The theory that some birds hibernate persisted until experiments were done on caged birds in the 1940s, which demonstrated that birds have no hibernation instinct.

One of the earliest naturalists and philosophers from ancient Greece was Aristotle, who was the first writer to discuss the disappearance and reappearance of some bird species at certain times of year.

He developed the theory of transmutation, the seasonal change of one species into another by observing red starts and robins.

He observed that in the autumn, small birds called red starts began to lose their feathers, which convinced Aristotle that they changed into robins for the winter and back into red starts in the summer.

These assumptions are understandable given that this pair of species are similar in shape but are a classic example of an incorrect interpretation based on correct observations.

The most bizarre theory was put forward by an English amateur scientist Charles Morton in the 17th century.

He wrote a surprisingly well-regarded paper claiming that birds migrate to the moon and back every year.

He came to this conclusion as the only logical explanation for the total disappearance of some species.

One of the key moments in the development of migration theory came in when a white stalk was shot in Germany.

This particular stalk made history because of the long spear in its neck, which incredibly had not killed it.

Everyone immediately realized this spear was definitely not European.

It turned out to be a spear from a tribe in central Africa.

This was a truly defining moment in the history of ornithology because it was the first evidence that stalks spend their winters in subsaharan Africa.

You can still see the arrow stalk in the zoological collection of the university of roaststock in Germany.

People gradually became aware that European birds moved south in autumn and north in summer, but didn't know much about it until the practice of catching birds and putting rings on their legs became established.

Before this, very little information was available about the actual destinations of particular species and how they traveled there.

People speculated that larger birds provided a kind of taxi service for smaller birds by carrying them on their backs.

This idea came about because it seemed impossible that small birds weighing only a few grams could fly over vast oceans.

This idea was supported by observations of bird behavior such as the harassment of larger birds by smaller birds.

The development of bird ringing by a Danish school teacher, Hans Christian Cornelius Mortensson, made many discoveries possible.

This is still common practice today and relies upon what is known as recovery.

This is when ringed birds are found dead in the place they have migrated to and identified.

Huge amounts of data were gathered in the early part of the 20th century and for the first time in history, people understood where birds actually went to in winter.

In 1931, an atlas was published showing where the most common species of European birds migrated to.

More recent theories about bird migration very w.

Hello, Easy Life Cleaning Services.

Justinta speaking.

Oh, hello.

I'm looking for a cleaning service for my apartment.

Do you domestic cleaning?

Sure.

Well, it's just a one-bedroom flat.

Do you have a basic cleaning package?

Yes.

For a one-bedroom flat, we're probably looking at about 2 hours for a clean.

So, we do a thorough clean of all surfaces in each room and polish them where necessary.

Does your apartment have carpets?

No, I don't have any.

But the floor would need cleaning.

Of course, we do that in every room and we do a thorough clean of the kitchen and bathroom.

Then we have some additional services which you can request if you want.

So, for example, we can clean your oven for you every week.

Actually, I hardly ever use that, but can you do the fridge?

Sure.

Would you like that done every week?

Yes, definitely.

And would ironing clothes be an additional service you can do?

Yes, of course.

It wouldn't be much.

Just my shirts for work that week.

That's fine.

And we could also clean your microwave if you want.

No, I wipe that out pretty regularly.

So, there's no need for that.

We also offer additional services that you might want a bit less often, say every month.

So, for example, if the inside of your windows need cleaning, we could do that.

Yes, that would be good.

I'm on the 15th floor, so the outside gets done regularly by specialists, but the inside does get a bit grubby.

And we could arrange for your curtains to get cleaned if necessary.

No, they're okay.

But would you be able to do something about the balcony?

It's quite small, and I don't use it much, but it could do with a wash every month or so.

Yes, we can get the pressure washer onto that.

Now, if you're interested, we do offer some other possibilities to do with general maintenance.

For example, if you have a problem with water and you need a plumber in a hurry, we can put you in touch with a reliable one who can come out straight away.

And the same thing if you need an electrician, right, that's good to know.

I've only just moved here, so I don't have any of those sorts of contacts.

And I don't know if this is of interest to you, but we also offer a special vacuum cleaning system which can improve the indoor air quality of your home by capturing up to 99% of all the dust in the air.

So, if you're troubled by allergies, this can make a big difference.

Right.

In fact, I don't have that sort of problem, but I'll bear it in mind.

Now, can you tell me a bit about your cleaning staff?

Of course.

So all our cleaners are very carefully selected.

When they apply to us, they have to undergo a security check with the police to make sure they don't have any sort of criminal background.

And of course, they have to provide references as well.

Then if we think they might be suitable for the job, we give them training for it.

That lasts for 2 weeks, so it's very thorough.

And at the end of it, they have a test.

If they pass that, we take them on, but we monitor them very carefully.

We ask all our clients to complete a review of their performance after every visit and to email it to us so we can pick up any problems straight away and deal with them.

Okay.

Well, that all sounds good.

And will I always have the same cleaner?

Yes, we do our best to organize it that way and we usually manage it.

Good.

That's fine.

Right.

So, I'd like to go ahead and as many of you here today have worked in the hotel industry for some time, I'm sure you have experienced the problem of high staff turnover in your hotels.

Every hotel relies on having loyal and experienced members of staff who make sure that everything runs smoothly.

If staff are constantly changing, it can make life difficult for everyone.

But why do staff leave frequently in many hotels?

Of course, many hotel jobs, such as cleaning, are low-skilled, and are not well paid.

A lot of managers think it's this and the long hours that are the main causes of high staff turnover.

But what they don't realize is that it's the lack of training in many hotel jobs which is a huge factor.

So what kind of problems does a high turnover of staff cause?

Well, having to recruit new staff all the time can be very time consuming and managers may have to cover some duties while waiting for new staff to arrive.

This means they don't have time to think about less immediate problems such as how to improve their service.

When staff leave, it can also severely affect the colleagues they leave behind.

It has a negative effect on remaining staff who may start to feel that they too should be thinking about leaving.

So what can be done to change this situation?

Firstly, managers should stop making basic errors which leave their staff feeling upset and resentful.

When organizing shifts, for example, make sure you never give certain staff preferential treatment.

All staff should be given some choice about when they work, and everyone should have to work some evening and weekend shifts.

If you treat staff fairly, they'll be more likely to step in and help when extra staff are needed.

Keeping staff happy has other tangible benefits for the business.

Take the Donnage Hotel as an example.

It had been experiencing a problem with staff complaints and in order to deal with this invested in staff training and improved staff conditions.

Not only did the level of complaints fall, but they also noticed a significant increase in the amount each customer spent during their stay.

They have now introduced a customer loyalty scheme, which is going really well.

Now, I'd like to look at some ways you can reduce staff turnover in your hotels.

And I'll do this by giving some examples of hotels where I've done some training recently.

The Sun Club received feedback which showed that staff thought managers didn't value their opinions.

They weren't made to feel they were partners who are contributing to the success of the business as a whole.

This situation has changed.

Junior staff at all levels are regularly invited to meetings where their ideas are welcomed.

A year ago, the Portland recognized the need to invest in staff retention.

Their first step was to introduce a scheme for recognizing talent amongst their employees.

The hope is that organizing training for individuals with management potential will encourage them to stay with the business.

At Bluewater, managers decided to recognize 50 high achievers from across the company's huge hotel chain.

As a reward, they're sent on an all expenses paid trip abroad every year.

Fun is an important element in the trips, but there's also the opportunity to learn something useful.

This year's trip included a visit to a brewery where staff learned about the new beer that would be served in the hotel.

Pentlo identified that retention of junior reception staff was an issue.

In order to encourage them to see that working in a hotel could be worthwhile and rewarding with good prospects, they introduced a management program.

These staff were given additional responsibilities and the chance to work in various roles in the hotel.

Green Planet wanted to be seen as a caring employer to make life easier for staff, many of whom had child care responsibilities.

The hotel began issuing vouchers to help cover the cost of child care.

Louise Marsh at the Aimsbury has one of the best staff retention rates in the business.

Since she joined the company, she's made a huge effort to achieve this by creating a cooperative and supportive environment.

For her, the staff are part of a large family where everyone is valued.

Okay, now I'd like to move.

Hi, Thomas.

How are you enjoying the course so far?

Yeah, I think it's good.

Remind me, why did you decide to study sports science?

Didn't you want to be a professional athlete when you were at school?

Yeah, that was my goal.

And all my classmates assumed I would achieve it.

They thought I was brilliant.

That must have been a nice feeling.

I thought I could win anything.

There was no one who could run faster than me.

Exactly.

So, what happened?

Did your mom and dad want you to be more academic?

Not at all.

Perhaps they should have pushed me harder though.

What do you mean?

I think I should have practiced more.

What makes you say that?

Well, I went out to Kenya for a couple of weeks to train.

Really?

I didn't know that.

I was chosen to go there out of loads of kids and run with some of the top teenage athletes in the world.

And I was so calm about it.

I just kept thinking how fortunate I was.

What a great chance this was.

Everyone back home was so proud of me.

But once we started competing, I very quickly realized I wasn't good enough.

That must have been a huge shock.

I thought this can't be happening.

I was used to winning.

I'm sorry to hear that.

It's okay.

I'm over it now.

And I think it's much better to do a university course.

And this one has such a variety of sports related areas.

It's going to be good.

Well, I agree.

I chose it because of that.

So, Gianne, have you thought of any ideas for the discussion session next week on technology and sport?

We have to cover more than one sport, don't we?

Yeah.

You know, we always think technology is about the future, but we could gather some ideas about past developments in sport.

Look at early types of equipment perhaps.

Uh, I remember reading something about table tennis bats once, how they ended up being covered with pimpled rubber cuz they were just wooden at first, I'd imagine.

Yeah.

In about the 1920s, a factory was making rolls of the rubber in bulk for something like horse harnesses.

Really?

Yeah.

And someone realized that it would make a perfect covering for the wooden bats.

So, what about cricket?

that's had a few innovative changes.

Maybe the pads they wear on their legs.

I don't think they've changed much, but I'm just looking on the internet and it says that when the first cricket helmet came in in 1978, the Australian batsman who first wore it was booed and jered by people watching because it was so ugly.

Wow.

Players have to protect themselves from getting hurt.

I mean, everyone wears one now.

H unlike the cycle helmet.

Well, unless you're a professional, but you're right.

Many ordinary bikers don't wear a helmet.

Hey, look at these pictures of original helmet designs.

This one looks like an upside down bowl.

Yet, the woman's laughing.

She's so proud to be wearing it.

It says serious cyclists ended up with wet hair from all the hard exercise.

I guess that's why they have large air vents in them now, so that the skin can breathe more easily.

Okay, so we've done helmets.

What about golf balls?

Or better still, golf clubs?

They've changed a lot.

Yeah, I remember my greatgrandfather telling me that because a club was made entirely of wood, it would easily break and players had to get another.

There's no wood at all in them now, is there?

No, they're much more powerful.

The same must be true of hockey sticks.

I don't think so because players still use wooden sticks today.

What it does say here though is that when the game started, you had to produce a stick yourself.

I guess they just weren't being manufactured.

So, one more perhaps.

What about football?

Well, I know the first balls were made of animal skin.

Yeah.

They covered them with pieces of leather that was stitched together, but the balls let in water when it rained.

Oh, that would have made them much heavier.

That's right.

You can imagine the damage to players's necks when the ball was headed.

Oh, how painful that must have been.

Yeah.

Well, I think we can put together some useful ideas.

Hello everyone.

Today we're going to look at another natural food product and that's maple syrup.

What is this exactly?

Well, maple syrup looks rather like clear honey, but it's not made by bees.

It's produced from the plant fluid or sap inside the maple tree.

And that makes maple syrup a very natural product.

Maple syrup is a thick, golden, sweet tasting liquid that can be bought in bottles or jars and poured onto food such as waffles and ice cream or used in the baking of cakes and pastries.

It contains no preservatives or added ingredients and it provides a healthy alternative to refined sugar.

Let's just talk a bit about the maple tree itself, which is where maple syrup comes from.

So, there are many species of maple tree, and they'll grow without fertilizer in areas where there's plenty of moisture in the soil.

However, they'll only do this if another important criterion is fulfilled, which is that they must have full or partial sun exposure during the day.

and very cool nights and I'll talk more about that in a minute.

There are only certain parts of the world that provide all these conditions.

One is Canada and by that I mean all parts of Canada and the other is the northeastern states of North America.

In these areas the climate suits the trees perfectly.

In fact, Canada produces over 2/3 of the world's maple syrup, which is why the fiveointed maple leaf is a Canadian symbol and has featured on the flag since 1964.

So, how did maple syrup production begin?

Well, long before Europeans settled in these parts of the world, the indigenous communities had started producing maple sugar. They bored holes in the trunks of maple trees and used containers made of tree bark to collect the liquid sap as it poured out. As they were unable to keep the liquid for any length of time, they didn't have storage facilities in those days. They boiled the liquid by placing pieces of rock that had become scorching hot from the sun into the sap. They did this until it turned into sugar and they were then able to use this to sweeten their food and drinks. Since that time, improvements have been made to the process, but it has changed very little overall.

So, let's look at the production of maple syrup today.

Clearly, the maple forests are a valuable resource in many Canadian and North American communities.

The trees have to be well looked after and they cannot be used to make syrup until the trunks reach a diameter of around 25 cm.

This can take anything up to 40 years.

As I've already mentioned, maple trees need the right conditions to grow and also to produce sap.

Why is this?

Well, what happens is that during a cold night, the tree absorbs water from the soil and that rises through the tree's vascular system.

But then in the warmer daytime, the change in temperature causes the water to be pushed back down to the bottom of the tree.

This continual movement up and down leads to the formation of the sap needed for maple syrup production.

When the tree is ready, it can be tapped and this involves drilling a small hole into the trunk and inserting a tube into it that ends in a bucket.

The trees can often take several taps, though the workers take care not to cause any damage to the healthy growth of the tree itself.

The sap that comes out of the trees consists of 98% water and 2% sugar and other nutrients.

It has to be boiled so that much of that water evaporates.

And this process has to take place immediately using what are called evaporators.

These are basically extremely large pans.

The sap is poured into these.

A fire is built and the pans are then heated until the sap boils.

As it does this, the water evaporates and the syrup begins to form.

The evaporation process creates large quantities of steam and the sap becomes thicker and denser.

And at just the right moment, when the sap is thick enough to be called maple syrup, the worker removes it from the heat.

After this process, something called sugar sand has to be filtered out as this builds up during the boiling and gives the syrup a cloudy appearance and a slightly gritty taste.

Once this has been done, the syrup is ready to be packaged so that it can be used for a whole variety of products.

It takes 40 L of sap to produce one liter of maple syrup.

So you can get an idea of how much is needed.

So that's the basic process.

In places like Quebec,

09-cam18 full cut.en

Excuse me.

Would you mind if I asked you some questions?

We're doing a survey on transport.

Yes, that's okay.

First of all, can I take your name?

Yes, it's Sadi Jones.

Thanks very much.

And could I have your date of birth?

Just the year will do, actually.

Is that all right?

Yes, that's fine.

It's 1991.

So, next your post code please.

It's DW30 7 Y Zed.

Great.

Thanks.

Is that in Wells?

No, it's actually in Harborne.

Wells isn't far from there, though.

I really like that area.

My grandmother lived there when I was a kid.

Yes, it is nice.

Right.

So now I want to ask you some questions about how you traveled here today.

Did you use public transport?

Yes, I came by bus.

Okay.

And that was today.

It's the 24th of April, isn't it?

Isn't it the 25th?

Uh, no.

Actually, you're right.

And what was the reason for your trip today?

I can see you've got some shopping with you.

Yes, I did some shopping.

But the main reason I came here was to go to the dentist.

That's not much fun.

Hope it was nothing serious.

No, it was just a checkup.

It's fine.

Good.

Do you normally travel by bus into the city center?

Yes.

I stopped driving in ages ago because parking was so difficult to find and it costs so much.

I see the bus is much more convenient, too.

It only takes about 30 minutes.

That's good.

So, where did you start your journey?

At the bus stop on Claxby Street.

Is that C L A XB Y?

That's right.

And how satisfied with the service are you?

Do you have any complaints?

Well, as I said, it's very convenient and quick when it's on time, but this morning it was late.

Only about 10 minutes, but still.

Yes, I understand that's annoying.

And what about the timet?

Do you have any comments about that?

M I suppose I mainly use the bus during the day, but anytime I've been in town in the evening for dinner or at the cinema, I've noticed you have to wait a long time for a bus.

There aren't that many.

Okay, thanks.

So, now I'd like to ask you about your car use.

Well, I have got a car, but I don't use it that often.

mainly just to go to the supermarket, but that's about it really.

My husband uses it at the weekends to go to the golf club.

And what about a bicycle?

I don't actually have one at the moment.

What about the city bikes you can rent?

Do you ever use those?

No, I'm not keen on cycling there because of all the pollution, but I would like to get a bike.

It would be good to use it to get to work.

So, why haven't you got one now?

Well, I live in a flat on the second floor, and it doesn't have any storage, so we'd have to leave it in the hall outside the flat.

I see.

Okay.

Well, I think that's all we need from you today, S.

Good evening, everyone.

Let me start by welcoming you all to this talk and thanking you for taking the time to consider joining ACCE voluntary organization.

ACCE offers support to people and services in the local area and we're now looking for more volunteers to help us do this.

By the way, I hope you're all comfortable.

We have brought in extra seats so that no one has to stand, but it does mean that the people at the back of the room may be a bit squashed.

We'll only be here for about half an hour, so hopefully that's okay.

One of the first questions we're often asked is how old you need to be to volunteer.

Well, you can be as young as 16 or you can be 60 or over.

It all depends on what type of voluntary work you want to do.

Other considerations such as reliability are crucial in voluntary work and age isn't related to these in our experience.

Another question we get asked relates to training.

Well, there's plenty of that and it's all face to face.

What's more, training doesn't end when you start working for us.

It takes place before, during, and after periods of work.

Often it's run by other experienced volunteers as managers tend to prefer to get on with other things.

Now I would ask you to consider a couple of important issues before you decide to apply for voluntary work.

We don't worry about why you want to be a volunteer.

People have many different reasons that range from getting work experience to just doing something they've always wanted to do.

But it is critical that you have enough hours in the day for whatever role we agree is suitable for you.

If being a volunteer becomes stressful, then it's best not to do it at all.

You may think that your income is important, but we don't ask about that.

It's up to you to decide if you can work without earning money.

What we value is dedication.

Some of our most loyal volunteers earn very little themselves but still give their full energy to the work they do with us.

Okay, so let's take a look at some of the work areas that we need volunteers for and the sort of things that would help you in those.

You may wish simply to help us raise money.

If you have the creativity to come up with an imaginative or novel way of fundraising, we'd be delighted as standing in the local streets or shops with a collection box can be rather boring.

One outdoor activity that we need volunteers for is litter collection.

And for this, it's useful if you can walk for long periods, sometimes uphill.

Some of our regular collectors are quite elderly but very active and keen to protect the environment.

If you enjoy working with children, we have three vacancies for what are called playmates.

These volunteers help children learn about staying healthy through a range of out of school activities.

You don't need to have children yourself, but it's good if you know something about nutrition and can give clear instructions.

If that doesn't appeal to you, maybe you would be interested in helping out at our story club for disabled children, especially if you have done some acting.

We put on three performances a year based on books they have read, and we're always looking for support with the theatrical side of this.

The last area I'll mention today is first aid.

Volunteers who join this group can end up teaching others in vulnerable groups who may be at risk of injury.

Initially though, your priority will be to take in a lot of information and not forget any important steps or details.

Right.

So, does anyone have any questions for me about Hi, Shantel.

What did you think of the talk then?

Hi, Hugo.

I thought it was good once I'd moved seats.

Oh, were the people beside you chatting or something?

It wasn't that.

I went early so that I'd get a seat and not have to stand, but then this guy sat right in front of me and he was so tall.

It's hard to see through people's heads, isn't it?

Impossible.

Anyway, to answer your question, I thought it was really interesting, especially what the speaker said about the job market.

Me, too.

I mean, we know we're going into a really competitive field, so it's obvious that we may struggle to get work.

That's right.

And we know we can't all have that dream job.

Yeah.

But it looks like there's a whole range of areas of work that we hadn't even thought of, like fashion journalism, for instance.

Yeah, I wasn't expecting so many career options.

Hi overall, she had quite a strong message, didn't she?

She did.
She kept saying things like, "I know you all think this, but" and then she'd tell us how it really is.
Perhaps she thinks students are a bit narrow-minded about the industry.
It was a bit harsh, though.
We know it's a tough industry.
Yeah.
And we're only first years after all.
We've got a lot to learn.
Exactly.
Do you think our secondary school education should have been more career focused?
Well, we had numerous talks on careers, which was good, but none of them were very inspiring.
They could have asked more people like today's speaker to talk to us.
I agree.
We were told about lots of different careers just when we needed to be, but not by the experts who really know stuff.
So, did today's talk influence your thoughts on what career you'd like to take up in the future?
Well, I promised myself that I'd go through this course and keep an open mind till the end, but I think it's better to pick an area of the industry now and then aim to get better and better at it.
Well, I think we'll just have to differ on that issue.
One thing's for certain, though.
From what she said, we'll be unpaid assistants in the industry for quite a long time.
H I'm prepared for that.
Aren't you?
Actually, I'm not going to accept that view.
Really?
But she knows it's the case and everyone else says the same.
That doesn't mean it has to be true for me.
Okay.
Well, I hope you're right.
I thought the speaker's account of her first job was fascinating.
Yeah.
She admitted she was lucky to get work being a personal dresser for a musician.
She didn't even apply for the job and there she was getting paid to choose all his clothes.
It must have felt amazing though.
She said all she was looking for back then was experience, not financial reward.
And then he was so mean telling her she was more interested in her own appearance than his.
But she did realize he was right about that, which really made me think.
I'm always considering my own clothes, but now I can see you should be focusing on your client.
She obviously regretted losing the job.
Well, as she said, she should have hidden her negative feelings about him, but she didn't.
It was really brave the way she picked herself up and took that job in retail.
Fancy working in a shop after that.
Yeah, well, she recommended we all do it at some point.
I guess as a designer, you'd get to find out some useful information, like how big or small the average shopper is.
I think that's an issue for manufacturers, not designers.
However, it would be useful to know if there's a gap in the market.
You know, an item that no one's stocking, but that consumers are looking for.
Yeah.
People don't give up searching.
They also take things back to the store if they aren't right.
Yeah.
Imagine you worked in an expensive shop and you found out the garments sold there were being returned because they fell apart in the wash.
Yeah, it would be good to know that kind of thing.
Yeah.
For my presentation today, I want to tell you about how groups of elephants have been moved and settled in new reserves.
This is known as translocation and has been carried out in Malawi in Africa in recent years.
The reason this is being done is because of overpopulation of elephants in some areas.
Overpopulation is a good problem to have and not one we tend to hear about very often.
In Malawi's Majetti National Park, the elephant population had been wiped out by poachers who killed the elephants for their ivory.
But in 2003, the park was restocked and effective law enforcement was introduced.
Since then, not a single elephant has been poached.
In this safe environment, the elephant population boomed.
breeding went so well that there were more elephants than the park could support.
This led to a number of problems.
Firstly, there was more competition for food, which meant that some elephants were suffering from hunger.
As there was a limit to the amount of food in the national park, some elephants began looking further a field.
Elephants were routinely knocking down fences around the park which then had to be repaired at a significant cost.
To solve this problem, the decision was made to move dozens of elephants from Majetti National Park to Inkotakakota Wildlife Park where there were no elephants.
But obviously attempting to move significant numbers of elephants to a new home 300 km away is quite a challenge.

So how did this translocation process work in practice?

Elephants were moved in groups of between 8 and 20 all belonging to one family.

Because relationships are very important to elephants, they all had to be moved at the same time.

A team of vets and park rangers flew over the park in helicopters and targeted a group which were rounded up and directed to a designated open plane.

The vets then used darts to immobilize the elephants.

This was a tricky maneuver as they not only had to select the right dose of tranquilizer for different sized elephants, but they had to dart the elephants as they were running around.

This also had to be done as quickly as possible so as to minimize the stress caused.

As soon as the elephants began to flop onto the ground, the team moved in to take care of them.

To avoid the risk of suffocation, the team had to make sure none of the elephants were lying on their chests because their lungs could be crushed in this position.

So, all the elephants had to be placed on their sides.

One person stayed with each elephant while they waited for the vets to do checks.

It was very important to keep an eye on their breathing.

If there were fewer than six breaths per minute, the elephant would need urgent medical attention.

Collars were fitted to the matriarch in each group so their movements could be tracked in their new home.

Measurements were taken of each elephant's tusks.

Elephants with large tusks would be at greater risk from poachers and also of their feet.

The elephants were then taken to a recovery area before being loaded onto trucks and transported to their new home.

The elephants translocated to Enkot settled in very well and the project has generally been accepted to have been a huge success and not just for the elephants.

Employment prospects have improved enormously contributing to rising living standards for the whole community.

Poaching is no longer an issue as former poachers are able to find more reliable sources of income.

In fact, many of them volunteered to give up their weapons as they were no longer of any use to them.

More than two dozen elephants have been born at Enkot since relocation.

With an area of more than 1,800 square kilometers, there's plenty of space for the elephant population to continue to grow.

Their presence is also helping to rebalance Enkot's damaged ecosystem and providing a sustainable conservation model which could be replicated in other parks.

All this has been a big draw for tourism which contributes five times more than the illegal wildlife trade to GDP and this is mainly because of the elephants.

There's also been a dramatic rise in interest from So I understand you're interested in restaurant work.

Yes, I've got a bit of experience and I can provide references.

That's good.

I can check all that later.

Now, Milo's Restaurants have some vacancies at the moment.

They're a really good company to work for.

Lots of benefits.

All right.

Yes.

They've got a very good reputation for looking after staff.

For example, all employees get training, even temporary staff.

Oh, really?

That's quite unusual, isn't it?

Certainly is.

And do staff get free uniforms, too?

Um, you just need to wear a white t-shirt and black trousers, it says here.

So, I guess not.

But another benefit of working for a big company like this is that you can get a discount at any of their restaurants, even at weekends.

No, but you'll be working then anyway.

Oh, yes, I suppose so.

Um, most of their restaurants are in the city center, aren't they?

So, easy to get to by bus.

Yes, that's right.

But if you have to do a late shift and finish work after midnight, the company will pay for you to get a taxi home.

I probably won't need one.

I think I'd use my bike.

Okay.

Now, they do have some quite specific requirements for the kind of person they're looking for.

Milo is a young, dynamic company, and they're really keen on creating a strong team.

It's really important that you can fit in and get on well with everyone.

Yeah, I've got no problem with that.

It sounds good, actually.

The last place I worked for was quite demanding, too.

We had to make sure we gave a really high level of service.

That's good to hear because that will be equally important at Milo's.

I know they want people who have an eye for detail.

That's fine.

I'm very used to working in that kind of environment.

Perfect.

So, the only other thing that's required is good communication skills.

So, you'll need to have a certificate in English.

Sure.

Okay.

Let's have a look at the current job vacancies at Milo's.

The first one is in Wanho Street.

Sorry, where?

Wanho.

W I V E N H O E.

It's quite central, just off Cork Street.

Oh, right.

They're looking for a breakfast supervisor.

That would be okay.

So you're probably familiar with the kind of responsibilities involved.

Obviously checking that all the portions are correct etc.

And then things like checking all the procedures for cleaning the equipment are being followed.

Okay.

And what about the salary?

In my last job I was getting £950 per hour.

I was hoping to get a bit more than that.

Well, to begin with, you'd be getting £9.75, but that goes up to £1125 after 3 months.

That's not too bad.

And I suppose it's a very early start.

That's the only unattractive thing about this job, but then you have the afternoons and evenings free.

So, the restaurant starts serving breakfast from 7:00 a.m.

and you'd have to be there at 5:30 to set everything up, but you'd be finished at 12:30.

H well, as you say, there are advantages to that.

Now, you might also be interested in the job at the City Road branch.

That's for a junior chef.

So, again, a position of responsibility.

I might prefer that actually.

Right.

Well, obviously this role would involve supporting the sue chef and other senior staff, and you'd be responsible for making sure there's enough stock each week and sorting out all the deliveries.

I've never done that before, but I imagine it's fairly straightforward once you get the hang of it.

Yes, and you'd be working alongside more experienced staff to begin with, so I'm sure it wouldn't be a problem.

The salary is slightly higher here.

It's an annual salary of £23,000, right?

I know that if they like you, it's likely you'll be promoted quite quickly, so that's worth thinking about.

Yes, it does sound interesting.

What are the hours like?

The usual, I think.

There's a lot of evening and weekend work, but they're closed on Mondays, but you do get one Sunday off every four weeks.

So, would you like me to send off your CV?

Hello everyone.

It's good to see that so many members of the public have shown up for our presentation on the new housing development planned on the outskirts of Nunston.

I'm Mark Reynolds and I'm communications manager at the development.

I'll start by giving you a brief overview of our plans for the So, one thing I'm sure you'll want to know is why we've selected this particular site for a housing At present, it's being used for farming.

like much of the land around Nunston.

But because of the new industrial center in Nunston, there's a lot of demand for housing for employees in the region as many employees are having to commute long distances at present.

Of course, there's also the fact that we have an international airport just 20 minutes drive away.

But although that's certainly convenient, it wasn't one of our major criteria for choosing the site.

We were more interested in the fact that there's an excellent hospital just 15 km away and a large secondary school even closer than that.

One drawback to the site is that it's on quite a steep slope, but we've taken account of that in our planning, so it shouldn't be a major problem.

We've had a lot of positive feedback about the plans.

People like the wide variety of accommodation types and prices and the fact that it's only a short drive to get out into the countryside from the development.

We were particularly pleased that so many people liked the designs for the layout of the development with the majority of people saying it generally made a good impression and blended in well with the natural features of the landscape with provision made for protecting trees and wildlife on the site.

Some people have mentioned that they'd like to see more facilities for cyclists, and we'll look at that.

But the overall feedback has been that the design and facilities of the development make it seem a place where people of all ages can live together happily.

Okay.

So, I'll put a map of the proposed development up on the screen.

You'll see it's bounded on the south side by the main road which then goes on to Nunston. Another boundary is formed by London Road on the western side of the development. Inside the development, there'll be about 400 houses and three apartment blocks. There'll also be a school for children up to 11 years old. If you look at the south entrance at the bottom of the map, there's a road from there that goes right up through the development. The school will be on that road at the corner of the second turning to the left. A large sports center is planned with facilities for indoor and outdoor activities. This will be on the western side of the development just below the road that branches off from London Road. There'll be a clinic where residents can go if they have any health problems. Can you see the lake towards the top of the map? The clinic will be just below this to the right of a street of houses. There'll also be a community center for people of all ages. On the northeast side of the development, there'll be a row of specially designed houses specifically for residents over 65 and the community center will be adjoining this. We haven't forgotten about shopping. There'll be a supermarket between the two entrances to the development. We're planning to leave the three large trees near London Road and it'll be just to the south of these. It's planned to have a playground for younger children. If you look at the road that goes up from the south entrance, you'll see it curves round to the left at the top and the playground will be in that curve with nice views of the lake. Okay. So now, does anyone have any question? So Michelle, shall we make a start on our presentation? We haven't got that much time left. No, Adam, but at least we've done all the background reading. I found it really interesting. I'd never even heard of the Lark eruption before this. Me neither. I suppose 1783 is a long time ago. But it was a huge eruption and it had such devastating consequences. I know. It was great. There were so many primary sources to look at. It really gives you a sense of how catastrophic the volcano was. People were really trying to make sense of the science for the first time. That's right. But what I found more significant was how it impacted directly and indirectly on political events as well as having massive social and economic consequences. I know that should be the main focus of our presentation. The observations made by people at the time were interesting, weren't they? I mean, they all gave a pretty consistent account of what happened, even if they didn't always use the same terminology. Yeah, I was surprised there were so many weather stations established by that time. So, you know, you can see how the weather changed often by the hour, right? Writers at the time talked about the lucky haze to describe the volcanic fog that spread across Europe. They all realized that this wasn't the sort of fog they were used to. And of course, this was in pre-industrial times, so they hadn't experienced sulfur smelling fog before. No, that's true. Reports from the period blamed the haze for an increase in headaches, respiratory issues, and asthma attacks. And they all describe how it covered the sun and made it look a strange red color. Must have been very weird. It's interesting that Benjamin Franklin wrote about the haze. Did you read that? He was the American ambassador in Paris at the time. Yeah. At first, no one realized that the haze was caused by the volcanic eruption in Iceland. It was Benjamin Franklin who realized that before anyone else. He's often credited with that, apparently. But a French naturalist beat him to it. I can't remember his name. I'd have to look it up. Then other naturalists had the same idea, all independently of each other. Oh, right. We should talk about the immediate impact of the eruption which was obviously enormous especially in Iceland where so many people died. You'd expect that and the fact that the volcanic ash drifted so swiftly but not that the effects would go on for so long or that 2 years after the eruption strange weather events were being reported as far away as North America and North Africa. No, I found all that hard to believe, too. It must have been terrible. And there was nothing anyone could do about it. Even if they knew the ash cloud was coming in their direction.

We should run through some of the terrible consequences of the eruption experienced in different countries.

There's quite a varied range.

Starting with Iceland, where the impact on farming was devastating.

One of the most dramatic things there was the effect on livestock as they grazed in the fields.

They were poisoned because they ate vegetation that had been contaminated with fluorine as a result of the volcanic fallout.

That was horrible.

In Egypt, the bizarre weather patterns led to a severe drought and as a result, the Nile didn't flood which meant the crops all failed.

It's so far from where the eruption happened and yet the famine there led to more people dying than any other country.

It was worse than the plague.

Okay.

Then in the UK, the mortality rate went up a lot, presumably from respiratory illnesses.

According to one report, it was about double the usual number and included an unusually high percentage of people under the age of 25.

Oh I think people will be surprised to hear that the weather in the USA was badly affected too.

George Washington even makes a note in his diary that they were snowbound until March in Virginia.

That was before he became president.

Yes.

And there was ice floating down the Mississippi, which was unprecedented.

Astonishing, really.

Anyway, what do you think we should include next?

Good morning.

Now, we've been asked to choose an aspect of European clothing or fashion and to talk about its development over time.

I decided to focus on a rather small area of clothing and that's pockets.

I chose pockets for two reasons really.

We all have them in jeans, jackets, coats, for example.

And even though we often carry bags or briefcases as well, nothing is quite as convenient as being able to pop your phone or credit card into your pocket.

Yet, I suspect that other than that, people don't really think about pockets too much, and they're rather overlooked as a fashion item.

It's certainly very interesting to go back in time and see how pockets developed for men and women.

In the 18th century, fashions were quite different from the way they are now.

And pockets were too.

If we think about male fashion first, that was the time when suits became popular.

Trousers were knee-length only and referred to as breeches.

The waist coats were short and the jackets were long, but all three garments were lined with material and pockets were sewn into this cloth by whichever tailor the customer used.

The wearer could then carry small objects such as pencils or coins on their person and reach them through a gap in the lining.

Coat pockets became increasingly decorative on the outside for men who wanted to look stylish, but they were often larger but flatter if the wearer was someone with a profession who needed to carry medical instruments, a doctor or physician, for example.

The development of women's pockets was a little different.

For one thing, they weren't nearly as visible or as easy to reach as men's.

In the 18th and 19th centuries, women carried numerous possessions on their person, and some of these could be worth a lot of money.

Women were more vulnerable to theft, and wealthy women in particular worried constantly about pickpockets.

So, what they did was to have a pair of pockets made that were tied together with string.

The pockets were made of fabric, which might be recycled cloth if the wearer had little money, or something more expensive, such as linen, sometimes featuring very delicate embroidery.

Women tied the pockets around their waist so that they hung beneath their clothes.

Remember, skirts were long then, and there was plenty of room to hide a whole range of small possessions between the layers of petticoats that were commonly worn.

They would have an opening in the folds of their skirts through which they could reach whatever they needed, like their perfume.

Working women, of course, also needed to carry around items that they might use for whatever job or trade they were involved in.

But their pairs of pockets still remained on the inside of their clothing.

They just got bigger or longer, sometimes reaching down to their knees.

So the tie-on pockets went well into the 19th century and only changed when fashion altered towards the end of that period.

That's when dresses became tighter and less bulky and the pairs of pockets became very noticeable.

They stood out too much and detracted from the woman's image.

Women who had been used to carrying around a range of personal possessions and still wanted to need somewhere to carry these items about their person.

That was when small bags or pouches as they were known came into fashion.

And of course they inevitably led onto the handbag of more modern times particularly when fashion removed pockets altogether.

Hello, Wayside Camera Club.

Breeder speaking.

Oh, hello.

Um, my name's Dan and I'd like to join your club.

That's great, Dan.

We have an application form.

Would you like to complete it over the phone, then you can ask any questions you might have.

Oh, yes.

Thanks.

Okay.

So, what's your family name?

It's Green.

Dan Green.
So, can I take your email address?
Yes, it's dan1068@market.com.
Thanks.
And what about your home address?
Well, I'm about 10 miles away from your club in Peacetown.
I live in a house there.
Okay.
So, what's the house number and street?
It's 52 Marfield Street.
Is that M A R O W F I E L D?
And that's Peace Town, you said.
Uh-huh.
So, how did you hear about our club?
Did you look on the internet?
I usually do that, but this time, well, I was talking to a relative the other day and he suggested it.
Oh, is he a member, too?
He belongs to another club, but he'd heard good things about yours.
Okay.
So, what do you hope to get from joining?
Well, one thing that really interests me is the competitions that you have.
I enjoy entering those.
Right.
Anything else?
Well, I also like to socialize with other photographers.
That's great.
So, what type of membership would you like?
What are the options?
It's £30 a year for full membership or £20 a year if you're an associate.
I think I'll go for the full membership then.
That's a good idea because you can't vote in meetings with an associate membership.
If I could just find out a bit more about you.
Okay.
So, you said you wanted to compete.
Have you ever won any photography competitions?
Not yet, but I have entered three in the past.
Oh, that's interesting.
So, why don't you tell me something about those?
Let's start with the first one.
Well, the theme was entitled domestic life.
I see.
So, it had to be something related to the home.
Yeah.
I chose to take a photo of a family sitting around the dinner table having a meal and um I didn't win, but I did get some feedback.
Oh, what did the judges say?
That it was too busy as a picture.
Aha.
So, it was the composition of the picture that they criticized.
That's right.
And once they'd told me that, I could see my mistake.
So, what was the theme of the second competition?
Well, my university was on the coast and that area gets a lot of beautiful sunsets.
So, that was the theme.
Oh, sunsets.
That's a great theme.
Yes.
The instructions were to capture the clouds as well.
It couldn't just be blue sky and a setting sun.
Sure.
Because they give you all those amazing pinks and purples.
Yeah.
And I thought I'd done that well, but the feedback was that I should have waited a bit longer to get the shot.
I see.
So, the timing wasn't right.
Yes.
I took it too soon, basically.
And then the third competition I entered was called animal magic.
Well, that's a difficult subject.
I know.
I had to take hundreds of shots.

I'm sure because animals move all the time.
That's what we had to show.
There had to be some movement in the scene.
I got a great shot of a fox in the end, but I took it at night and well, I suspected that it was a bit dark, which is what I was told.
Well, Dan, you seem to be really keen, and we'd be delighted to have you in our club.
I'm sure we can help with all those areas that you've outlined.
Thanks.
That's great.
This evening, we're delighted to welcome Dan Beagle, who's just written a book on looking for and finding food in the wild.
He's going to tell us everything we need to know about picking wild mushrooms.
Thank you very much.
Well, I need to start by talking about safety.
You really need to know what you're doing because some mushrooms are extremely poisonous.
Having said that, once you know what to look for, it's really worth doing for the amazing variety of mushrooms available, which you can't get in the shops.
But of course, you have to be very careful.
And that's why I always say you should never consume mushrooms picked by friends or neighbors.
Always remember that some poisonous mushrooms look very similar to edible ones and it's easy for people to get confused.
The other thing to avoid is mushrooms growing beside busy roads for obvious reasons.
But nothing beats the taste of freshly picked mushrooms.
Don't forget that the ones in the shops are often several days old and past their best.
There are certain ideas about wild mushrooms that it's important to be aware of.
Don't listen to people who tell you that it's only okay to eat mushrooms that are pale or dull.
This is completely untrue.
Some edible mushrooms are bright red, for example.
Personally, I prefer mushrooms cooked, but it won't do you any harm to eat them uncooked in salads.
It's not necessary to peel them.
Another thing you should remember is that you can't tell if a mushroom is safe to eat by its smell.
Some of the most deadly mushrooms have no smell and taste quite nice apparently.
Finally, just because deer or squirrels eat a particular mushroom doesn't mean that you can.
Of course, mushroom picking is associated with the countryside, but if you haven't got a car, your local park can be a great place to start.
There are usually a range of habitats where mushrooms grow, such as playing fields and wooded areas, but you need to be there first thing in the morning, as there's likely to be a lot of competition, not just from people, but wildlife, too.
The deer often get the best mushrooms in my local park.
If you're a complete beginner, I wouldn't recommend going alone or relying on photos in a book, even the one I've written.
There are some really good phone apps for identifying mushrooms, but you can't always rely on getting a good signal in the middle of a wood.
If possible, you should go with a group led by an expert.
You'll stay safe and learn a lot that way.
Conservation is a really important consideration and you must follow a few basic rules.
You should never pick all the mushrooms in one area.
Collect only enough for your own needs.
Be very careful that you don't trample on young mushrooms or other plants.
and make sure you don't pick any mushrooms that are endangered and protected by law.
There's been a decline in some varieties of wild mushrooms in this part of the country.
Restaurants are becoming more interested in locally sourced food like wild mushrooms.
But the biggest problem is that so many new houses have been built in this area in the last 10 years.
And more water is being taken from rivers and reservoirs because of this and mushroom habitats have been destroyed.
Anyway, a word of advice on storing mushrooms.
Collect them in a brown paper bag and as soon as you get home, put them in the fridge.
They'll be fine for a couple of days, but it's best to cook them as soon as possible after washing them really carefully first, of course.
So, everybody knows what a mushroom tastes like, right?
Well, you'll be surprised by the huge variety of wild mushrooms there are.
Be adventurous.
They're great in so many dishes.
stir fries, risotos, pasta, but just be aware that some people can react badly to certain varieties.
So, it's a good idea not to eat huge quantities to begin with.
Okay.
So, now I'm going to show you a few examples of poisonous.
That seminar yesterday on automation and the future of work was really good, wasn't it?
Looking at the first industrial revolution in Britain in the 19th century and seeing how people reacted to massive change was a real eye opener.
Yes, it was interesting to hear how people felt about automation then and what challenges they faced.
I didn't know that first started with workers in the textile industry with those protesting workers called the lites destroying their knitting machines because they were so worried about losing their jobs.
Yes.
And ultimately they didn't achieve anything.
And anyway, industrialization created more jobs than it destroyed.
Yes, that's true.
But it probably didn't seem a positive thing at the time.

I can see why the Lites felt so threatened.

I know.

I'm sure I would have felt the same.

The discussion about the future of work was really optimistic for a change.

I like the idea that work won't involve doing boring repetitive tasks as robots will do all that.

Normally, you only hear negative stuff about the future.

Bit too optimistic, don't you think?

For example, I can't see how people are about to have more leisure time when all the evidence shows people are spending longer than ever at work.

No, that's true.

And what about lower unemployment?

I'm not so sure about that.

Perhaps in the long term, but not in the foreseeable future.

And I expect most people will be expected to work until they're much older, as everyone's living much longer.

That's already happening.

I enjoyed all that stuff on how technology has changed some jobs and how they're likely to change in the near future.

Yeah, incredible.

like accountants.

You might think all the technological innovations would have put them out of a job, but in fact, there are more of them than ever.

They're still really in demand and have become far more efficient.

Right.

That was amazing.

20 times more accountants in this country compared to the 19th century.

I know.

I'd never have thought the demand for hairdressing would have gone up so much in the last 100 years.

One hairdresser for every 287 people now compared to one for over 1,500.

Yeah, because people's earning power has gone up so they can afford to spend more on personal services like that.

But technology hasn't changed the actual job that much.

No, they've got hair dryers etc.

But it's one job where you don't depend on a computer.

The kind of work that administrative staff do has changed enormously thanks to technology.

Even 20 years ago, there were secretaries doing dictation and typing.

Yes, really boring compared to these days when they're given much more responsibility and higher status.

A lot of graduates go in for this kind of work now.

I'd expected there to be a much bigger change in the number of agricultural workers in the 19th century, but the 1871 census showed that roughly 25% of the population worked on the land.

Yeah, I'd have assumed it would be more than 50%.

Now it's less than 0.2%.

What about care workers?

They barely existed in the 19th century as people's lifespan was so much shorter.

But now, of course, this sector will see huge growth.

Yeah, and it's hard enough to meet current demand.

The future looks quite bleak for bank clerks.

They've been in decline since ATMs were introduced in the 80s.

and technology will certainly make most of the jobs they do now redundant.

I think I agree.

Although the situation may change, it's very hard to predict what will happen.

In today's astronomy lecture, I'm going to talk about the need for a system to manage the movement of satellites and other objects in orbit around the Earth.

In other words, a space traffic management system.

We already have effective air traffic control systems that are used internationally to ensure that planes navigate our skies safely.

Well, space traffic management is a similar concept, but focusing on the control of satellites.

The aim of such a system would be to prevent the danger of collisions in space between the objects in orbit around the Earth.

In order to do this, we'd need to have a set of legal measures and we'd also have to develop the technical systems to enable us to prevent such accidents.

But unfortunately, at present, we don't actually have a space traffic management system that works.

So why not?

What are the problems in developing such a system?

Well, for one thing, satellites are relatively cheap these days compared with how they were in the past, meaning that more people can afford to put them into space.

So, there's a lot more of them out there, and people aren't just launching single satellites, but whole constellations consisting of thousands of them designed to work together.

So, space is getting more crowded every day.

But in spite of this, one thing you may be surprised to learn is that you can launch a satellite into space and once it's out there, it doesn't have to send back any information to Earth to allow its identification.

So, while we have international systems for ensuring we know where the planes in our skies are and to prevent them from colliding with one another, when it comes to the safety of satellites, at present, we don't have anything like enough proper ways of tracking them.

And it isn't just entire satellites that we need to consider.

A greater threat is the huge amount of space debris in orbit around the Earth.

broken bits of satellite and junk from space stations and so on.

And some of these are so small that they can be very hard to identify, but they can still be very dangerous. In addition, some operators may be unwilling to share information about the satellites they've launched. For example, a satellite may be designed for military purposes or it may have been launched for commercial reasons and the operators don't want competitors to have information about it. And even if the operators are willing to provide it, the information isn't easy to collect. Details are needed about the object itself as well as about its location at a particular time. And remember that a satellite isn't very big and it's likely to be moving at thousands of kilometers an hour. We don't have any sensors that can constantly follow something moving so fast. So all that the scientists can do is to put forward a prediction concerning where the satellite is heading next. So those are some of the problems that we're facing. Let's consider now some of the solutions that have been suggested. One key issue is the way in which information is dealt with. We need more information, but it also needs to be accessible at a global level. So we need to establish shared standards that we can all agree on for the way in which this information is presented. We already do this in other areas of science. So although this is a challenge, it's not an impossible task. Then as all this information is collected, it needs to be put together so it can be used. And that will involve creating a single database on which it can be entered. As we continue to push forward new developments, congestion of the space environment is only going to increase. To cope with this, we need to develop a system like the one I've described to coordinate the work of the numerous spacecraft operators. But it's also essential that this system is one that establishes trust in the people that use it both nationally and at a global level. One interesting development that I'll mention at this Hello.

Oh, hello.

Is that Julie Davidson?

Yes.

This is Greg Preston from the employment agency.

We met last week when you came in to inquire about office Oh, that's right.

Now, we've just had some details come in of a job which might interest you.

So, this is a position for a receptionist.

I believe you've done that sort of work before.

Yes, I have.

I worked in a sports center for a couple of years before I got married and had the children.

Right.

Well, this job's in Forom, so not too far away for you.

And it's at the medical center there.

Okay.

So, where exactly is that?

It's quite near the station on Chaston's Road.

Sorry.

Chaston's Road.

That's C H A S T O N S.

Okay.

Thanks.

So, what would the work involve?

Dealing with inquiries from patients.

Yes.

And you'd also be involved in making appointments, whether face to face or on the phone, and rescheduling them if necessary.

Fine, that shouldn't be a problem.

And another of your duties would be keeping the cent's database up to date.

Then you might have other general administrative duties as well, but those would be the main ones.

Now, when the details came in, I immediately thought of you because one thing they do require is someone with experience.

And you did mention your work at the sports center when you came in to see us.

Yes, in fact, I enjoyed that job.

Is there anything else they're looking for?

Well, they say it's quite a high pressure environment.

They're always very busy and patients are often under stress.

So they want someone who can cope with that and stay calm and at the same time be confident when interacting with the public.

Well, after dealing with three children all under five, I reckon I can cope with that.

I'm sure you can.

And then another thing they mention is that they're looking for someone with good IT skills.

Not a problem.

So, you'd be interested in following this up?

Sure.

When would it start?

Well, they're looking for someone from the beginning of next month, but I should tell you that this isn't a permanent job. It's temporary.

So, the contract would be just at the end of September.

But they do say that there could be further opportunities after Okay.

And what would the hours be?

Well, they want someone who can start at a/4 to 8:00 in the morning.

Could you manage that?

Yes.

My husband would have to get the kids up and off to my mother's.

She's going to be looking after them while I'm at work.

What time would I finish?

1:15.

That should work out all right.

I can pick the kids up on my way home and then I'll have the afternoon with them.

Oh, one thing.

Is there parking available for staff at the center?

Yes, there is.

And it's also on a bus route.

Right.

Well, I expect I'll have the car, but it's good to know that.

Okay.

So, where do I go from here?

Well, if you are happy for me to do so, I'll forward your CV and references.

And then the best thing would probably be for you to phone them so they can arrange for an interview.

Great.

Well, thank you very much.

You're welcome.

Bye now.

Bye.

Good morning everyone and welcome to the Museum of Farming Life.

I understand it's your first visit here.

So, I'd like to give you some background information about the museum and then explain a little about what you can see during your visit.

So, where we're standing at the moment is the entrance to a large building that was constructed in 1880 as the home of a local businessman Alfred Palmer of the Palmer Biscuit Factory.

It was later sold and became a hall of residence for students in 1911 and a museum in 1951.

In 2005, a modern extension was built to accommodate the museum's collections.

The museum's owned by the university and apart from two rooms that are our offices, the university uses the main part of the building.

You may see students going into the building for lessons, but it's not open to museum visitors, I'm afraid.

It's a shame because the interior architectural features are outstanding, especially the room that used to be the library.

Luckily, we've managed to keep entry to the museum free.

This includes access to all the galleries, outdoor areas, and the rooms for special exhibitions.

We run activities for children and students such as the museum club for which there is no charge.

We do have a donation box just over there.

So feel free to give whatever amount you consider appropriate.

We do have a cloak room if you'd like to leave your coats and bags somewhere.

Unlike other museums, photography is allowed here.

So, you might like to keep your cameras with you.

You might be more comfortable not carrying around heavy rucksacks, though.

Keep your coats and jackets on as it's quite cold in the museum garden.

Today, I'd like to tell you about the different areas of the museum.

Just inside and outside the main gallery, we have an area called Four Seasons.

Here you can watch a 4-minute animation of a woodland scene.

It was designed especially for the museum by a group of young people on a film studies course and it's beautiful.

Children absolutely love it, but then so do adults.

The main gallery is called Town and Country.

It includes a photographic collection of prize-winning sheep and shepherds.

Leaving town and country, you enter farmhouse kitchen, which is well self-explanatory.

Here we have the oldest collection of equipment for making butter and cheese in the country.

And this morning, a specialist cheese maker will be giving demonstrations of how it's produced.

You may even get to try some.

After that, you can go in two directions.

To the right is a staircase that takes you up to a landing from where you can look down on the galleries.

To the left is a room called a year on the farm.

There is lots of seating here as sometimes we use the room for school visits, so it's a good place to stop for a rest.

If you're feeling competitive, you can take our memory test in which you answer questions about things you've seen in the museum.

The next area is called Wagon Walk.

This contains farm carts from nearly every part of the country.

It's surprising how much regional variation there was.

Beside the carts are display boards with information about each one.

The carts are old and fragile, so we ask you to keep your children close to you and ensure they don't climb on the carts.

From Wagon Walk, you can either make your way back to reception or go out into the garden or even go back to take another look in the galleries.

In the far corner of the garden is Bees are Magic, but we're redeveloping this area, so you can't visit that at the moment.

You can still buy our honey in the shop, though.

Finally, there's the pond, which contains all kinds of interesting wildlife.

There are baby ducks that are only a few days old as well as tiny frogs.

The pond isn't deep and there is a fence around it, so it's perfectly safe for children.
 So now I want you to discuss the lesson we've just been watching on the video and think about the ways in which origami can be a useful educational tool.
 Can you all work with the person sitting next to you?
 I had no idea that such a simple thing like folding squares of paper to make the shape of something like a bird could be such an amazing tool.
 It's made me see origami in a whole new light.
 I know.
 It was interesting to see the educational skills the children were developing by doing origami.
 On the video, you could see them really listening hard to make sure they did all the steps in the right order to make the bird.
 That's right.
 In this lesson, they were working individually, but it would also be interesting to see if the children could work out how to make something simple without being given any direction.
 That would help with building teamwork as well.
 Yes, but much more of a challenge.
 One thing that really stood out for me was that the children were all having fun while being taught something new, which is a key aim of any lesson with this age group.
 And although these kids had no problems with folding the paper, with younger children, you could do origami to help practice fine motor skills.
 Absolutely.
 Shall we talk about the individual children we saw on the video?
 I wrote all their names down and took some notes.
 Yes, I did too.
 Okay, good.
 Let's start with Sid.
 He was interesting because before they started doing the origami, he was being quite disruptive.
 Yes, he really benefited from having to use his hands.
 It helped him to settle down and start concentrating.
 Yes, I noticed that, too.
 What about Jack?
 I noticed he seemed to want to work things out for himself.
 You could see him trying out different things rather than asking the teacher for help.
 What did you make of Naomi?
 She seemed to be losing interest at one point, but then she decided she wanted her mouse to be the best, and that motivated her to try harder.
 She didn't seem satisfied with hers in the end, though.
 No, Anna was such a star.
 She listened so carefully and then produced the perfect bird with very little effort.
 H I think the teacher could have increased the level of difficulty for her.
 Maybe I think it was the first time Zara had come across origami.
 She looked as if she didn't really get what was going on.
 She seemed unsure about what she was supposed to do.
 But in the end, hers didn't turn out too badly.
 Yeah, I'm sure it was a positive learning experience for her.
 M I think one reason why the origami activity worked so well in this class was that the teacher was well prepared.
 Right.
 I think it would have taken me ages to prepare examples showing each of the steps involved in making the bird.
 But that was a really good idea.
 The children could see what they were aiming for and much better for them to be able to hold something rather than just looking at pictures.
 M those physical examples supported her verbal explanations really well.
 It's strange that origami isn't used more widely.
 Why do you think that is?
 Well, teachers may just feel it's not that appealing to children who are used to doing everything on computers, especially boys, even if they're aware of the benefits.
 Oh, I don't know.
 It's no different to any other craft activity.
 I bet it's because so many teachers are clumsy like me.
 That's true.
 Too much effort required if you're not good with your hands.
 Well, anyway, I think we should try it out in our maths teaching practice with year three.
 I can see using origami is a really engaging way of reinforcing children's knowledge of geometric shapes like they were doing in the video, but I think it would also work really well for presenting fractions, which is coming up soon.
 Good idea.
 That's something most of the kids in that class might struggle with.
 Origami would also be good practice for using symmetry, but I think they did that last term.
 Okay.
 Well, let's try and get some ideas together and plan the lesson next week.
 Okay.
 If you could all The person I've chosen to talk about is the French writer Victor Hugo.

Many people have heard of him because his novel *Les Misérables* which he wrote in is famous around the world. It became a stage musical in the 1980s and a film version was also released in 2012.

So some of us, I'm sure, have a pretty general idea of the plot, but we know much less about the author. Today I'm going to provide a little more insight into this talented man and I'm going to talk particularly about the home he had on the island of Gernzi in the British Channel Islands.

But first, his early career.

As I've said, he was a writer.

He was at the height of his career in Paris and he was very highly regarded by his colleagues.

As far as literature was concerned, he was the leading figure of the romantic movement.

However, as well as being a literary genius, he also gave many speeches about issues like the level of poverty in his society. He felt very strongly about this and about other areas where change was needed like education.

This kind of outspoken criticism was not well liked by the rulers of France and eventually the emperor Napoleon III told Victor Hugo to leave Paris and not return.

In other words, he sent him into exile.

So Victor Hugo was forced to reside in other parts of Europe.

Gernzi was actually his third place of exile and he landed there in 1855.

He produced a lot while on Gernzi including *Les Misérables* and to do this he had to spend a great deal of time in the home that he had there.

This was a property that he bought using the money he'd made in France from the publication of a collection of his poetry. It was the only property he ever owned and he was very proud of it.

The property Victor Hugo bought on Gernzie was a large five-story house in the capital town of St. Peterport and he lived there for 15 years, returning to France in 1870 when Napoleon's empire collapsed.

He decorated and furnished each level or floor of the house in unique and wonderful ways.

And many people consider the inside of the house to be a work of art.

Today, it's a museum that attracts 200,000 visitors a year.

He lived in the house with his family and portraits of its members still hang in rooms on the ground floor along with drawings that he did during his travels that he felt were important to him.

In other ground floor rooms, there are huge tapestries that he would have designed and loved.

The walls are covered in dark wood paneling that Victor Hugo created himself using wooden furniture that he bought in the market.

The items were relatively inexpensive and he used them to create intricate carvings.

They gave an atmosphere on the lower level that was shadowy and rather solemn.

On the next level of the house, there are two impressive lounges where he entertained his guests.

One lounge has entirely red furnishings such as sofas and wall coverings and the other blue.

There's a strong Chinese influence in these areas in things like the wallpaper pattern and the lamps which he would have made himself by copying original versions.

His library, where he left many of his favorite books, forms the hallway to the third floor and was a comfortable area where he could relax and enjoy his afternoons.

And then at the very top of the house, there's a room called the lookout.

Called that because it looks out over the harbor.

In contrast to the rather dark lower levels, it's full of light and was like a glass office where he would write until lunchtime, often at his desk.

So, Victor Hugo was a man of many talents, but he was also true to his values.

While living in his house on Gernzi, he entertained many other famous writers, but he also invited a large group of local children from the deprived areas of the island to dinner once a week.

What's more, he served them their food, which was an extraordinary gesture for the time period.

In 1927, the house was owned by his relatives and they decided to donate it to the city of Paris.

It has since been restored using photographs from the period and as I mentioned earlier is now a museum that is open to the public.

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Okay.
What would you like to know?
Well, I'm new to this area, so perhaps you could tell me something about the park first, please.
Of course.
Altogether, the park covers 170 acres.
That's 69 hectares.
There are three main types of habitat: wetland, grassland, and woodland.
The woods are well established and varied with an oak plantation and other areas of mixed species.
Right.
The wetland is quite varied, too.
The original farmland was dug up around 40 years ago to extract gravel.
Once this work was completed, the gravel pits filled with water, forming the two large lakes.
There are also several smaller ones, ponds, and a stream that flows through the park.
Okay.
So, I suppose with these different habitats, there's quite a variety of wildlife.
There certainly is a lot of different species of birds and insects and also animals like deer and rabbits.
And I understand you organize educational visits for school parties.
That's right.
We can organize a wide range of activities and adapt them to suit all ages.
Can you give me some examples of the activities?
Well, one focus is on science where we help children to discover and study plants, trees, and insects.
They also collect and analyze data about the things they see.
Uh-huh.
Another focus is on geography.
The park is a great environment to learn and practice reading a map and using a compass to navigate around the park.
Do you do anything connected with history?
Yes, we do.
For instance, the children can explore how the use of the land has changed over time.
Then there's leisure and tourism.
That focuses on your visitors, I would imagine.
Yes, mostly.
The children find out about them, their requirements, the problems they may cause and how we manage these.
And another subject we cover is music.
Here the children experiment with natural materials to create sounds and explore rhythm and tempo.
That must be fun.
Most children really enjoy it.
And of course, all the activities are educational, too.
Learning outside the classroom encourages children to be creative and to explore and discover for themselves.
I would imagine they get a sense of freedom that might not be a normal part of their lives.
That's right.
And very often the children discover that they can do things they didn't know they could do and they develop new skills.
This gives them greater self-confidence.
It sounds great.
So, what about the practical side of it?
How much does it cost for a full day visit?
We would expect to bring between 30 and 40 children.
If there are over 30, it costs £4.95 for each child who attends on the day.
We invoice you afterwards so you don't pay for children who can't come because of sickness, for example.
There's no charge for leaders and other adults.
As many as you want to bring.
That sounds very fair.
Well, thanks for all the information.
I'll need to discuss it with my colleagues and I hope to get back to you soon to make a booking.
We'll look forward to hearing from you.
Goodbye.
Goodbye.
And thank you.
It's great to see so many members of the twinning association here tonight.
Since the twinning link between our two towns, Stanthorp here in England and Malat in France was established.
The relationship between the towns has gone from strength to strength.
Last month, 25 members of the association from Stanthorp spent a weekend in Malat.
Our hosts had arranged a great program.
We learned how cheese is produced in the region and had the chance to taste the products.
The theme park trip had to be cancelled, but we all had a great time on the final boat trip down the river.
That was the real highlight.
This is a special year for the association because it's 25 years since we were founded in Malat.
They're planning to mark this by building a foot bridge in the municipal park.
We've been discussing what to do here and we've decided to plant a poppler tree in the museum gardens.

We considered buying a garden seat to put there, but the authorities weren't happy with that idea.
In terms of fundraising to support our activities, we've done very well.
Our pancake evening was well attended and made record profits, and everyone enjoyed the demonstration of French cookery, which was nearly as successful.
numbers for our film show were limited because of the venue.
So, we're looking for somewhere bigger next year.
We're looking forward to welcoming our French visitors here next week.
And I know that many of you here will be hosting individuals or families.
The coach from France will arrive at 5:00 p.m.
on Friday.
Don't try to do too much that first evening as they'll be tired.
So have dinner in the house or garden rather than eating out.
The weather looks as if it'll be okay, so you might like to plan a barbecue.
Then the next morning's market day in town, and that's always a good place to stroll around.
On Saturday evening, we'll all meet up at the football club where once again we'll have Toby Sharp and his band performing English and Scottish country songs.
Toby will already be well known to many of you as last year he organized our special quiz night and presented the prizes.
Now on Sunday we'll be taking our visitors to Farley House.
You may not all be familiar with it, so here's a map to help you.
You can see the car park at the bottom of the map.
There's an excellent farm shop in the grounds where our visitors can buy local produce.
It's in the old stables, which is the first building you come to.
They're built round a courtyard and the shops in the far corner on the left.
There's also a small cafe on the right as you go in.
I know that one or two of our visitors may not be all that mobile.
The main entrance to the house has a lot of steps, so you might want to use the disabled entry.
This is on the far side of the house from the car park.
Children will probably be most interested in the adventure playground.
That's at the northern end of the larger lake in a bend on the path that leads to the lake.
There's lots for children to do there.
There are a number of lovely gardens near the house.
The kitchen gardens are rectangular and surrounded by a wall.
They're to the northeast of the house, quite near the smaller lake.
They're still in use and have a great collection of fruit and vegetables.
The Temple of the Four Winds is a bit more of a walk, but it's worth it.
Take the path from the car park and go past the western sides of the stables and the house.
Then, when the path forks, take the right hand path.
Go up there with the woods on your left and the temple is right at the end.
There are great views over the whole area.
Okay.
So, that's I haven't seen you for a bit, Marie.
No, I've been busy with my project.
You're making a vegan alternative to eggs, aren't you?
Something that doesn't use animal products.
Yes, I'm using chickpeas.
I had two main aims when I first started looking for an alternative to eggs, but actually, I found chickpeas have got more advantages.
Right.
But how about your project on reusing waste food?
You were looking at bread, weren't you?
Yes.
It's been hard work, but I've enjoyed it.
The basic process was quite straightforward.
Breaking the stale bread down to a paste, then reforming it.
But you were using 3D printing, weren't you?
To make the paste into biscuits.
Yeah, I'd used that before, but in this project, I had time to play around with different patterns for the biscuits and finding how I could add fruit and vegetables to make them a more appetizing color.
And I was really pleased with what I managed to produce.
It must have been a great feeling to make something appetizing out of bits of old bread that would have been thrown away otherwise.
It was.
and I'm hoping that some of the restaurants in town will be interested in the biscuits.
I'm going to send them some samples.
I came across something on the internet yesterday that might interest you.
It was a company that's developed touch sensitive sensors for food labels.
Mhm.
It's a special sort of label on the food package.
When the label's smooth, the food is fresh.
And then when you can feel bumps on the label, that means the food's gone bad.
It started off as a project to help visually impaired people know whether food was fit to eat or not.
Interesting.

So just solid food.

No, things like milk and juice as well.

But actually, I thought it might be really good for drug storage in hospitals and pharmacies.

Right.

And coming back to food, maybe it would be possible to use it for other things besides freshness, like how many kilograms a joint of meat is, for example.

Yes, there's all sorts of possibilities.

I was reading an article about food trends predicting how eating habits might change in the next few years.

Oh, things like more focus on local products.

That seems so obvious, but the shops are still full of imported foods.

Yes, they need to be more proactive to address that and somehow motivate consumers to change.

Yes.

One thing everyone's aware of is the need for a reduction in unnecessary packaging.

But just about everything you buy in supermarkets is still covered in plastic.

The government needs to do something about it.

Absolutely.

It's got to change.

Do you think there'll be more interest in gluten and lactose-free food for people with allergies or food intolerances?

I don't know.

Lots of people I know have been buying that type of food for years now.

Yes.

Even if they haven't been diagnosed with an allergy.

That's right.

One thing I've noticed is the number of branded products related to celebrity chefs.

People watch them cooking on TV and then buy things like spice mixes or frozen foods with the chef's name on.

I bought something like that once, but I won't again.

Yeah, I bought a ready-made spice mix for chicken which was supposed to be used by a chef I'd seen on television and it didn't actually taste of anything.

H Did the article mention ghost kitchens used to produce takeaway food?

No.

What are they?

Well, they might have the name of a restaurant, but actually they're a cooking facility just for delivery meals.

The public don't ever go there.

But people aren't aware of that.

It's all kept very quiet.

So people don't realize the food's not actually from the restaurant.

Did you know more and more people are using all sorts of different mushrooms now to treat different health concerns?

Things like heart problems.

H they might be taking a big risk there.

Yes.

It's hard to know which varieties are safe to eat.

Anyway, maybe now we for my presentation today, I'm going to talk about the Ki Fields in the northwest of Ireland, one of the largest Neolithic sites in the world.

I recently visited this site and observed the work that is currently being done by a team of archaeologists there.

The site was first discovered in the 1930s by a local teacher, Patrick Kfield.

He noticed that when local people were digging in the bog, they were constantly hitting against what seemed to be rows of stones.

He realized that these must be walls and that they must be thousands of years old for them to predate the bog which subsequently grew over them.

He wrote to the National Museum in Dublin to ask them to investigate, but no one took him seriously.

It wasn't until 40 years later when Patrick Kfield's son Sheamus, who had become an archaeologist by then, began to explore further.

He inserted iron probes into the bog to map the formation of the stones, a traditional method which local people had always used for finding fuel buried in the bog for thousands of years.

Carbon dating later proved that the site was over 5,000 years old and was the largest Neolithic site in Ireland.

Thanks to the bog which covers the area, the remains of the settlement at KG Fields, which is over 5,000 years old, are extremely well preserved.

A bog is 90% water.

Its soil so saturated that when the grasses and heathers that grow on its surface die, they don't fully decay, but accumulate in layers.

Objects remain so well preserved in these conditions because of the acidity of the peat and the deficiency of oxygen.

At least 175 days of rain a year are required for this to happen.

This part of Ireland gets an average of 225 days.

The Neolithic farmers at Ki would have enjoyed several centuries of relative peace and stability.

Neolithic farmers generally lived in larger communities than their predecessors with a number of houses built around a community building.

As they lived in permanent settlements, Neolithic farmers were able to build bigger houses.

These weren't round as people often assume, but rectangular with a small hole in the roof that allowed smoke to escape.

This is one of many innovations and indicates that the Neolithic farmers were the first people to cook indoors.

Another new technology that Neolithic settlers brought to Ireland was pottery.

Fragments of Neolithic pots have been found in Ki and elsewhere in Ireland.

The pots were used for many things as well as for storing food.

Pots were filled with a small amount of fat.

And when this was set a light, they served as lamps.
It's thought that the ki fields were mainly used as paddocks for animals to graze in.
Evidence from the ki fields suggests that each plot of land was of a suitable size to sustain an extended family.
Uh they may have used a system of rotational grazing in order to prevent overgrazing and to allow for plant recovery and regrowth.
This must have been a year round activity as no structures have been found which would have been used to shelter animals in the winter.
However, archaeologists believe that this way of life at Ki ceased abruptly.
Why was this?
Well, several factors may have contributed to the changing circumstances.
The soil would have become less productive and led to the abandonment of farming.
The crop rotation system was partly responsible for this as it would have been very intensive and was not sustainable.
But there were also climatic pressures too.
The farmers at Ki would have enjoyed a relatively dry period.
But this began to change and the conditions became wetter as there was a lot more rain.
It was these conditions that encouraged the bog to form over the area which survives today.
So, now I'd like to show you some Hi, Coleman.
How are you?
Good, thanks.
I wanted to have a chat with you because our friend Josh told me that you've joined a guitar group and it sounds interesting.
I'd really like to learn myself.
Why don't you come along?
I'm sure there's room for another person.
Really?
So, who runs the classes?
He's called a coordinator.
His name's Gary Mat.
Let me note that down.
Gary, how do you spell his surname?
It's M A T H I E S O N.
Right.
Thanks.
He's retired actually, but he's a really nice guy and he used to play in a lot of bands.
Thanks.
So, how long have you been going?
About a month now.
And could you play anything before you started?
I knew a few chords, but that's all.
I'm sure everyone will be better than me.
That's what I thought, too.
When I first spoke to Gary on the phone, he said it was a class for beginners, but I was still worried that everyone would be better than me, but we were all equally hopeless.
Oh, that's reassuring.
So, where do you meet?
Well, when I joined the group, they were meeting in Gary's home, but as the group got bigger, he decided to book a room at the college in town.
I prefer going there.
I know that place.
I used to go to tap dancing classes there when I was at secondary school.
I haven't been since, though, and I can't remember what road it's in.
Is it Lock Street?
It's just beyond there at the bottom of New Street near the city roundabout.
Yes, of course.
The guitar club is on the first floor in room T347.
And when do you meet?
Is it at the weekend?
We meet on Thursdays.
It used to be 10:30 and that suited me well, but now we meet at 11.
The class that's in there before us asked if they could have the room for another 30 minutes.
Oh, I see.
Well, I'd love to come, but I don't have a guitar.
Well, you can always buy a secondhand one.
There's a website called The Perfect Instrument that sells all kinds of guitars, violins, and so on.
I'm sure you'll find something there.
So, what's a typical lesson like with Gary?
Well, he always starts by getting us to tune our guitars.
That takes about 5 minutes.
Some people have an app they use, but others do it by ear.
Gary goes round and helps them.
And while he's doing that, he tells us what he's going to do during the lesson.
Right.

First, we usually spend about 10 minutes doing some strumming.

So, is that using Oh, what are they called?

Perrums.

No, we just use our thumbs.

Oh, much easier.

Gary reminds us where to put our fingers for each chord and then we play them together.

Sometimes we all just start laughing because we're so bad at keeping time.

So Gary starts clapping to help us.

Do you learn to play any songs?

Yes, we do at least one song with words and chords.

I mean, that's harder than you think.

Oh, I'm sure it is.

That part of the lesson takes about 15 He often brings a recording of the song and plays it to us first.

Then he hands out the song and if there's a new chord in it, we practice that before we play it together, but really slowly.

Do you do any finger picking?

That's the last 10 minutes of the lesson.

When we pick out the individual notes from a tune he's made up, it's always quite simple.

That must be hard though.

It is.

But people like it because they can really concentrate.

And if we're all playing well, it sounds quite impressive.

The only trouble is that he sometimes gets us to play one at a time, you know, alone.

Oh, that's scary.

It is, but I've got used to it now.

At the end, he spends about 5 minutes telling us what to practice for the following week.

Well, thanks, Coleman.

I'll go and have a look at that website, I think.

I never really planned to be a lifeboat volunteer when I came to live in North Sea.

I'd been working in London as a website designer, but although that was interesting, I didn't like city life.

I'd been really keen on boats as a teenager, and I thought if I went to live by the sea, I might be able to pursue that interest a bit more in my free time.

Then I found that the lifeboat institution was looking for volunteers.

So I decided to apply.

The lifeboat institution building here in North Sea is hard to miss.

It's one of the largest in the country.

It was built 15 years ago with funds provided by a generous member of the public who'd lived here all her life.

As the Lifeboat Institution is a charity that relies on that kind of donation rather than funding provided by the government, that was a huge help to us.

When I applied, I had to have a health assessment.

The doctors were particularly interested in my vision.

I used to be shortsighted, so I'd had to wear glasses, but I'd had laser eye surgery 2 years earlier, so that was okay.

They gave me tests for color blindness, and they thought I might have a problem there, but it turned out I was okay.

When the Coast Guard gets an alert, all the volunteers are contacted and rushed to the lifeboat station.

Our targets to get there in 5 minutes.

Then we tried to get the boat off the dock and out to sea in another 6 to 8 minutes.

Our team's proud that we usually achieve that.

The average time across the country is 8 1/2 I've recently qualified as what's called a helmsman, which means I have the ultimate responsibility for the lifeboat.

I have to check that the equipment we use is in working order.

The crew have special life jackets that can support up to four people in the water, and it's ultimately my decision whether it's safe to launch the boat.

But it's very rare not to launch it, even in the worst weather.

As well as going out on the lifeboat, my work involves other things, too.

A lot of people underestimate how quickly conditions can change at sea.

So, I speak to youth groups and sailing clubs in the area about the sorts of problems that sailors and swimmers can have if the weather suddenly gets bad.

We also have a lot of volunteers who organize activities to raise money for us and we couldn't manage without them.

The training we get is a continuous process focusing on technical competence and safe handling techniques and it's given me the confidence to deal with extreme situations without panicking.

I was glad I'd done a first aid course before I started as that's a big help with the casualty care activities we do.

We've done a lot on how to deal with ropes and tie knots.

That's an essential skill.

After a year, I did a one-week residential course led by specialists.

They had a wave tank where they could create extreme weather conditions so we could get experience at what to do if the boat turned over in a storm at night, for example.

Since I started, I've had to deal with a range of emergency situations.

But the work's hugely motivating.

It's not just about saving lives.

I've learned a lot about the technology involved.

My background in IT's been useful here, and I can use my expertise to help other volunteers.

They're a great group.

We're like a family, really, which helps when you're dragging yourself out of bed on a cold, stormy night.

But actually, it's the colder months that can be the most rewarding time.

That's when the incidents tend to be more serious and you realize that you can make a huge difference to the outcome.

So, if any of you listeners are interested, why don't you give us a call?

Hi, Don.

Did you get the copy of the article on recycling footwear that I emailed you?

Yeah, it's here.

I've had a look at it.

So, do you think it's a good topic for our presentation?

Well, before I started reading it, I thought recycling footwear.

Well, although it's quite interesting, perhaps there isn't enough to say about it cuz we put shoes in recycling bins, they go to charity shops, and that's about it.

But there's much more to it than that.

I realize that now, and I'm keen to research the topic more.

That's great.

One of the things I didn't realize until I read the article was just how many pairs of trainers get recycled.

Well, a lot of young people wear them all the time now.

They've become more popular than ordinary shoes.

I know.

I guess they are very hardwearing.

But don't they look a bit casual for school uniform?

I don't think they're right for that.

Actually, I think some of them look quite smart on pupils.

Better than a scruffy old pair of shoes.

So, do you keep shoes a long time?

Yes, though I do tend to wear my old pairs for doing dirty jobs like cleaning my bike.

I must admit, I've recycled some perfectly good shoes that haven't gone out of fashion and still fit just because they don't look great on me anymore.

That's awful, isn't it?

I think it's common because there's so much choice.

The article did say that recent sales of footwear have increased enormously.

That didn't surprise me.

No.

But then it said that the amount of recycled footwear has fallen.

It's 6% now compared to a previous level of 11%.

That doesn't seem to make sense.

That's because not everything goes through the recycling process.

Some footwear just isn't good enough to resell for one reason or another and gets rejected.

So, let's find some examples in the article of footwear that was rejected for recycling.

Okay, I think there are some in the interview with the recycling manager.

Um, yeah, here it is.

Let's start with the lady's high heeled shoes.

What did he say about those?

He said they were probably expensive.

The material was suede and they were beige in color.

It looked like someone had only worn them once, but in a very wet field, so the heels were too stained with mud and grass to resell them.

Okay.

And the leather ankle boots, what was wrong with them?

Apparently, the heels were worn, but that wasn't the problem.

One of the shoes was a much lighter shade than the other one.

It had obviously been left in the sun.

I suppose even secondhand shoes should look the same.

Sure.

Then there were the red baby shoes.

Oh, yes.

We're told to tie shoes together when we put them in a recycling bin, but people often don't bother.

You'd think it would have been easy to find the other, but it wasn't.

That was a shame because they were obviously new.

The trainers were interesting.

He said they looked like they'd been worn by a marathon runner.

Yeah.

Weren't they split?

Not exactly.

One of the soles was so worn under the foot that you could put your finger through it.

Well, we could certainly use some of those examples in our presentation to explain why 90% of shoes that people take to recycling centers or bins get thrown into landfill.

What did you think about the project his team set up to avoid this by making new shoes out of the good parts of old shoes?

It sounded like a good idea.

They get so many shoes, they should be able to match parts.

I wasn't surprised that it failed, though.
I mean, who wants to buy secondhand shoes?
Really?
Think of all the germs you could catch.
Well, people didn't refuse them for that reason, did they?
It was because the pairs of shoes weren't identical.
They still managed to ship them overseas, though.
That's another area we need to discuss.
You know, I used to consider this topic just from my own perspective by thinking about my own recycling behavior without looking at the bigger picture.
So much happens once shoes leave the recycling area.
It's not as simple as you first think.
And we can show that by taking a very different approach to it.
Absolutely.
So let's discuss how we're going to split up the present.
For my project on invertebrates, I chose to study tardigrades.
These are microscopic or to be more precise near microscopic animals.
There are well over a thousand known species of these tiny animals which belong to the phylum tardigrada.
Most tardigrades range in length from 0.05 to 1 mm.
Though the largest species can grow to be 1.2 millimeters in length, they are also sometimes called water bears.
Water because that's where they thrive best and bear because of the way they move.
Moss piglet is another name for tardigrades because of the way they look when viewed from the front.
They were first discovered in Germany in by Johan Gertzer who coined the name. As I say, there are many different species of tardigrade.
Too many to describe here, but generally speaking, the different species share similar physical traits.
They have a body which is short and also rounded, a bit like a barrel.
And the body comprises four segments.
Each segment has a pair of legs at the end of which are between four and eight sharp claws.
I should also say that some species don't have any claws.
What they have are discs and these work by means of suction.
They enable the tardigrade to cling onto surfaces or to grip its prey.
Within the body, there are no lungs or any organs for breathing at all.
Instead, oxygen and also blood are transported in a fluid that fills the cavity of the body.
As far as the tardigrade's head is concerned, the best way I can describe this is that it looks rather strange, a bit squashed even.
Though many of the websites I looked at described its appearance as cute, which isn't exactly very scientific.
The tardigrade's mouth is a kind of tube that can open outwards to reveal teethlike structures known as stylots.
These are sharp enough to pierce plant or animal cells.
So where are tardigrades found?
Well, they live in every part of the world in a variety of habitats, most commonly on the bed of a lake or on many kinds of plants or in very wet environments.
There's been some interesting research which has found that tardigrades are capable of surviving radiation and very high pressure.
And they're also able to withstand temperatures as cold as -200° centigrade or highs of more than 148° centigrade which is incredibly hot.
It has been said that tardigrades could survive long after human beings have been wiped out.
Even in the event of an asteroid hitting the Earth, if conditions become too extreme and tardigrades are at risk of drying out, they enter a state called cryptobiosis.
They curl into a ball called a tun, that's tun by retracting their head and legs, and their metabolism drops to less than 1% of normal levels.
They can remain like this until they're reintroduced to water when they will come back to life in a matter of a few hours.
While in a state of cryptobiosis, tardigrades produce a protein that protects their DNA.
In 2016, scientists revived two tardigrades that had been tuns for more than 30 years.
There was a report that in 1948 a 120-year-old tun was revived, but this experiment has never been repeated.
There are currently several tests taking place in space to determine how long tardigrades might be able to survive there.
I believe the record so far is 10 days.
So um moving on in terms of their diet, tardigrades consume liquids in order to survive.
Although they have teeth, they don't use these for chewing.
They suck the juices from moss or extract fluid from seaweed.
But some species prey on other tardigrades, from other species or within their own.
I suppose this isn't surprising given that tardigrades are mainly comprised of liquid and are coated with a type of gel.
Finally, I'd like to mention the conservation status of tardigrades.
It is estimated that they have been in existence for approximately half a billion years and in that time they have survived five mass extinctions.
So, it will probably come as no surprise to you that tardigrades have not been evaluated by the International Union for Conservation of Nature and are not on any endangered list.
Some researchers have described them as thriving.
Does anyone have any questions they'd like to ask?
Hi, Shannon.
How are you settling into your new flat?
Really well, thanks.
You look like you're going shopping.
Yes, I am.

My cousins are coming to stay for a couple of days and I have to cook for them.

Well, there are plenty of places to buy food in Kite Place.

It's the area by the harbor.

Oh, okay.

I'll find that on the map.

Thanks.

What sort of food do you need to get?

Well, neither of them eats meat, but they both like fish.

Well, there's a really good fish market there.

Oh, great.

Where is it exactly?

It's at the far end of Kite Place, so you have to go over the bridge and then it's on the right.

Okay.

Is it open all day?

It doesn't close until 4:00, but I'd recommend going earlier than that.

It does run out of some things.

Oh, I don't want that to happen.

As long as you get there by 3:30, you should be fine.

It's only 11:00 now, so plenty of time, right?

Do you need to buy vegetables, too?

I do, and I want to avoid all the plastic packaging in the supermarket.

Well, there's a really nice organic shop there.

Now, what's it called?

It's the name of a flower.

I know.

It's rose.

Oh, that's a nice name.

Yeah, it sells vegetables and quite a lot of other stuff.

And where's that?

Well, as you reach the market, you'll see a big gray building on your left.

I think it used to be a warehouse.

Anyway, now it's a restaurant upstairs, but the ground floor has two shops either side of the entrance, and it's the one on the left.

That's easy enough.

You can't miss it.

There's also a big sign on the pavement, so you can look for that.

Fine.

I guess if I need anything else, I'll have to go to the supermarket.

Yeah, you should be able to get everything you need.

But there's a mini bus that goes to the supermarket if you need it.

It's purple and the number is 289.

Thanks.

That's great.

So, what do you need to get at the fish market?

The salmon is always very good and the shellfish.

I'm going to make a curry, I think.

And I need about 12 prawns for that.

They'll have plenty of those.

Okay.

Have you ever tried samfire?

No.

What's that?

It's a type of seaweed.

Oh, I just ask for a handful and you fry it in butter.

It's delicious.

Oh, I might try that.

How do you spell it?

It's S A M P H I R E.

Great.

It's always good to try something different.

Yeah, I'll see what beans they have in the organic shop.

And I think I'll get something for dessert there.

How about a mango?

I'm not sure.

They are not always ripe.

I'd prefer a melon.

It's bigger, too.

Good idea.

The owner also sells a lot of spices there that you can put in a curry and things like coconut.

Oh, that's very helpful.

I'll have a look.

No problem.

I know bread doesn't really go with curry, but I always like to have some in case.

As I said, all the bread is homemade and there's lots of variety.

I like the brown bread myself.

M sounds good.

They sell other things there, too.

Like cakes?

I love chocolate cake.

Well, not that, but they have a whole range of tarts.

And the best are the strawberry ones.

Perfect.

Hopefully, I won't even have to go to the supermarket.

The Children's Book Festival is coming up again soon, and here to tell us all about it is the festival's organizer, Jenny Morgan.

So, tell us what we can expect this year, Jenny.

Well, as usual, we've got 5 days of action-packed, exciting events for children with writers coming from all over the country getting involved.

Just to give you an idea of what's on offer in the workshops.

First of all, there's a very special event called Superheroes.

This is a chance for deaf children to share their reading experiences with author Meline Gordon, who is herself hearing impaired.

Just do it is a practical workshop led by the well-known illustrator Mark Keane.

He'll take participants on a magical journey to far away lands with an opportunity for aspiring actors to do some roleplay.

Count on Me is an inspiring and entertaining look at the issues of friendship for 13 to 14 year olds.

It looks at some of the friendships described in popular books and asks participants to compare these with their own experiences.

Speak up is part of a series of workshops on the subject of mental health.

This is a creative writing workshop encouraging children to describe situations where young people experience loneliness.

A recent survey revealed that children can be lonely even when they're at home with their families.

Jump for Joy, as many of you will know, is the heartwarming, best-selling story by Nenah Karen about a young girl's trip to visit her relatives in India.

It recently received the gold medal at the Waterford Awards.

Nenina will get children to celebrate the word joy by writing a poem.

Sticks and Stones is the beautifully illustrated picture book for young readers about a community who organize an African Caribbean festival to help local children learn about their Jamaican roots.

This will be a musical event where children will have the chance to play steel drums.

This is bound to be very popular, so please book as soon as possible.

Thanks, Jenny.

That all sounds really interesting.

I'm just wondering if you have a favorite book you could recommend for our readers.

It's hard to choose, but Alive and Kicking is definitely worth mentioning.

You won't have heard of the writer as it's her first book, which is really impressive.

It's basically the teenage diary of a boy from Somalia who comes to live in the UK.

It deals with the serious issue of immigration and all the challenges the boy has to face at school and with the language barrier etc.

Usually books like this are quite sad but this one actually made me cry with laughter.

On each page there are simple but hilarious black and white stick drawings of the boy with his friends and teachers.

At the end of each diary entry, there are new English words the boy learns each day, which may help develop some children's vocabulary.

I think my kids would enjoy that.

What about any advice for parents on how to encourage their children to read more?

Well, this is something I get asked about a lot.

There are so many distractions for kids these days that it can be hard to find time for reading.

One thing I'd say is to make time to sit down with your child and share books with them.

A lot of parents give up reading aloud to their children as soon as they learn to read independently, but this is a mistake.

It's good to read more advanced books to them as it helps to develop their vocabulary.

If you don't have time for this, then let them listen to audio books.

Often they'll want to read books they've listened to for themselves.

I think it's a good idea to make a mental note of the type of books your child is reading.

Often they just read the same genre all the time, which can get a bit boring.

You can introduce new authors and genres to them.

Librarians should be able to help you with this.

Well, Jenny, I think that's really useful.

Hi, Jake.

How are you getting on with the practical teaching?

Oh, it's harder than I expected, but I've got some great classes.

How about you?

Not brilliant.

I'm really struggling with my year 12 science class.

Are they hard to control?

Well, I don't have discipline problems as such.

It's just that they don't seem to think that science has anything to do with their lives.

It's depressing.

They listen to what I say and I gave them a test last week and the results weren't too bad, but there's no real engagement.

Right.

And as part of my teaching practice, I have to design an experiment for them to do.

I was wondering about something on the children's diets.

You know, asking them to record what they eat and maybe linking it to their state of health.

Let's think.

So, your methodology would involve the children recording what they eat.

Okay.

But you'd also need to have access to the children's medical records.

And I don't think people would be happy about that.

Confidentiality would be an issue.

If you could get the right data, the conclusions might be significant, but I suspect it's not going to be easy. Have you thought about doing an experiment using animals?

Wouldn't that be upsetting for the children?

Well, the animals don't have to be harmed in any way.

It could just be an experiment where they're given a certain diet and the effects are observed.

Would I have to get permission to use animals?

Yes, you'd have to submit an outline of the experiment and fill in a form, but it's quite straightforward.

But if we found out that, say, a particular diet affects the health of animals, the same thing wouldn't necessarily be true for people, would it?

No, that's true.

But the findings for any experiment are going to be limited.

It's inevitable.

I suppose so.

So, what animals could I use to investigate the effects of diet?

Mice.

Yes.

You'd need experimental mice, ones that have been specially bred for experiments.

Okay.

So, what will your experiment be investigating exactly?

Well, uh, something to do with nutrition.

So, maybe we could look at food supplements, things like extra iron and extra protein, and their impact on health.

That might be rather broad.

Maybe just look at the effects of one supplement like sugar on the health of the mice.

In fact, maybe the focus could be on whether mice can control their own diet.

So, what happens when they have access to more sugar than they don't really need?

Exactly.

Do they eat it or do they decide to leave it?

Great.

Then later on, you could do a follow-up experiment, adding another variable, like you could give some of the mice the chance to be more active, running on a wheel or something, and the others just sit around and don't do much.

Or I could repeat the experiment, but change the type of food I provided, or use mice with a different genetic structure.

But I think your idea would be more interesting.

I might think about that some more.

So, can I talk through a possible procedure for the experiment where mice are given a sugar supplement?

Sure.

I did a similar experiment in college, actually.

Great.

So, how many mice would I need?

I'd say about 12 and all young ones, not a mixture of old and young.

Okay.

And I'd need two groups of equal sizes, so six in each group.

And how would I tell them apart?

I suppose I could put some sort of tag on one group or just mark them in some way.

You could use food coloring.

That wouldn't hurt them.

Perfect.

Then each group would go into a separate cage and one group, let's call them group A, would be the control group.

So they'd just have ordinary mouse food.

I suppose you can buy that.

Yes, it comes in dry pellets.

And the other group would have the same as the first group, but they'd also have the extra sugar.

Would you just give them straight sugar?

It might be better to give them something like cereal with it.

Then you'd need to weigh the mice.

I should think once a week.

And you'd need an electronic balance.

But we can't hold them on the balance or it would affect the reading.

Exactly.

So you need something called a weighing chamber to stop the mice from running away.

It sounds complicated, but actually you can just use a plastic box with holes in the top.

Okay.

So, once we've measured the weight gain of each mouse, we can work out the average for each group as well as the standard deviation and then see where we go from there.

That sounds cool.

I think the students will enjoy it.

Yes.

One thing you must remember.

In today's lecture, I'm going to be talking about microplastics.

Microplastics are tiny pieces of plastic, smaller than 5 mm in size.

Recently, there's been a greater awareness that there are large quantities of plastic waste, big and small, in the environment.

The amount of plastic waste in the oceans has received widespread attention, but far less is known about the effects of microplastics in fresh water and particularly in soil.

Microplastics can enter the environment via a number of different sources.

Threads and microfibers detach from synthetic clothing every time they're put in a washing machine, and these find their way into the water system.

Other sources include big pieces of plastic waste that are already in the environment, and these break down into microscopic particles over a period of time.

On a larger scale, factory waste is another route as our tires which wear down as cars, lorries, and so on travel along road surfaces.

We already understand some of the impacts of microplastics from studies involving fish and other animals.

There is evidence that microplastics harm small creatures in a variety of ways, such as by damaging their mouths or by impairing their ability to feed.

For example, when microplastics get lodged in their digestive system.

Surprisingly, perhaps, it is likely that humans consume microplastics, as these have been detected in a wide range of food and drink products, including bottled water, as well as in water that comes direct from the tap.

What's more, salt and kinds of seafood, have also been found to contain microplastics.

However, it's important to underline that there is not yet conclusive proof that microplastics cause significant harm to people.

In many countries, including here in the UK, there is legislation which prevents manufacturers from adding plastic micro beads to shower gels, facial cleansers, and toothpaste.

It is very difficult to accurately estimate the total amount of microplastic particles in the soil as they can be hard to detect.

But we do know they are carried in the air and deposited in the soil by rain.

What's more, many of the fertilizers used by both farmers and gardeners contain A team from the Anglia Ruskin University in Cambridge has carried out a study of the effects of microplastics on the digestive tracts of earthworms.

These worms which live in top soil are an essential component of our agricultural system.

By feeding on soil, they mix nutrients into it, thereby making it more fertile.

The researchers set out to discover whether the introduction of microplastics into the soil and the subsequent ingestion of these by earthworms would impact soil quality and ultimately inhibit plant growth.

The short answer was yes, it did.

After placing three different types of microplastic particles into the soil, they planted perennial ryegrass.

The particles of microplastic which included biodegradable PLA and conventional highdensity polyethylene or HDPE were then ingested by the earthworms in the soil.

The result was that the worms lost weight rapidly.

What's more, a lower percentage than normal of the ryegrass seeds germinated, and the researchers concluded that this was a direct result of the earthworms being unable to fulfill their normal role in making soil more fertile.

The team also discovered that there was an increase in the amount of acid found in the soil and this was attributed mainly to the microplastic particles from conventional HDP plastic.

The conclusions of the study make for very interesting reading.

I've included the reference in the notes to give you at the end of this session.

To summarize, the authors proposed the idea that we need to regard soil as we would regard any other process in nature.

This means we should accept the implications of soil being dependent on decaying and dead matter constantly being passed through the bodies of earthworms.

That is when soil becomes impoverished by the presence of microplastics, not only ecosystems but also the whole of society are negatively impacted.

Hello Charlotte, I'm Kaden, one of the supervisors.

Welcome to the team.

Hi Aiden.

Uh, it's Kaden.

I'm so sorry.

Don't worry.

People often get my name wrong.

They never know how to spell it.

It's K A D E N in case you ever need to write it.

I'll try and remember.

So, there are a few practical things you need to sort out this morning.

Then I'll show you what you're going to do today.

The email I received said to go to the front desk to show my letter of appointment and pick up my badge.

You'll need that for the staff room and other areas of the supermarket where shoppers aren't allowed.

So, after you've finished at the front desk, I'll take you to the staff room.

Put your coat and rucksack in one of the lockers there.

Uh, take whichever one is free.

Will I have a key?

Yes.
Try not to lose it.
At the end of the day, leave it in the door for the next person to use.
We'll do.
You also need to go to the HR department to see Tiffany.
She's really helpful.
I was told to bring my passport with me.
HR need to take a note of the number in That's right.
Or you can show your ID card.
I don't have one of those.
Okay.
Tiffany will give you a uniform.
Uh they have lots in different sizes, so you just tell her what you need.
I won't come with you to HR.
I've got to go and sort something else out.
Um, problem with a bread slicer.
Is the HR office near the staff room?
The staff room's on the first floor and HR are a couple of floors above that on the third floor.
There's a staircase outside the staff room.
When you're finished with HR, come and find me in the bakery section of the shop.
I'm looking forward to getting started.
Uh, I'll just give you my phone number in case you can't find me.
Uh, have you got your phone there?
Yes.
Uh, okay.
Ready?
It's 04126 5903.
Okay.
Done.
So, Charlotte, your tasks today are in the bakery section, on the sushi counter, and on the meat and fish counters.
The first job is to check sell by dates on the bread and cakes.
If any of the dates are today's, put a new price label on the packaging.
What if any of the labels are yesterday's dates or older?
Do I throw those items away?
Yes, but that shouldn't happen.
We check the stock every day.
When something needs a new price label, put a yellow one on the package next to the original price.
After that, you'll go to the sushi takeaway counter.
Will I be preparing boxes of food?
Uh, for today, you'll just be helping the staff.
Yes, of course.
You'll see lots of flat cardboard boxes at one end of the counter.
Beneath those is where we keep the plastic boxes.
We run out of those really quickly, so you should bring more from the store room.
Is that my only task on the sushi counter?
No, you also need to clean the area where they prepare the dishes.
There are cloths and bottles of spray by the sink.
Oh, and please make sure you clean that, too.
Sure, that's important, isn't it?
Absolutely.
But you mustn't wash up knives.
Uh, you have to do some training before you're allowed to touch sharp objects.
What should I do if there are any?
Ask someone to put them in the dishwasher.
Okay, thanks.
I don't want to get anything wrong.
Don't worry, you'll be fine, and I'll be around to help.
Uh, finally, the meat and fish counters.
You need to clean the area where staff serve customers, including wiping the weighing scales.
Okay.
Anything else?
Um, the fish is laid on ice, but when that starts to melt, you'll need to get more from the cauldron.
I know the staff on the food counters wear a hat.
Will that be the same for me?
You won't be serving customers directly, so no.
But make sure you put on thermal gloves when you take anything out of the cold room.
The temperature is low enough in there to get frostbite from touching things.
Understood.
My name's Liz Fuller and I'm a running coach with Compton Park Runners Club.
Welcome to my podcast.

If you're thinking about taking up running, I'm here to help.

There are many training programs available online which aim to help people build up to running 5 km.

Some of them are great and thousands of people of all ages are taking part in 5 km races across the country as a result.

People like them because they're easy to follow and don't push them too hard.

However, they don't work for everyone, especially if you suffer from something like a heart condition or asthma because they're aimed at people with average fitness and running ability.

Another thing is that everyone is different and if you have any specific questions related to your needs, there's no one to provide any answers.

I have a couple of simple tips I always give to new runners.

I expect you've been told to run very slowly until your fitness increases.

Well, I find that can prevent progress.

You should run at a speed that feels comfortable, but time yourself and try to run a bit faster each time.

Listening to music can be very helpful.

It takes your mind off things and helps your body get into a rhythm.

I'd say that is better than running with a friend, especially as most people are competitive, and that's not what you want when you're just starting.

I don't think the time of day is especially important.

Some people are better in the evening while others are morning people.

But you need to be consistent.

So aim to train regularly.

Twice a week is enough to begin with.

New members often say to me that they've been put off running either because they lack confidence or they don't have time or they think they dislike running.

Carrie, for example, joined the club 2 years ago at the age of 40.

She'd always enjoyed running at school, but wasn't sure if she'd be able to do it.

She was worried about being left behind and being the slowest runner.

But she says she was made to feel so welcome, she soon forgot all about that.

James had always hated the idea of running, but a friend encouraged him to come along for a taster session, and he hasn't looked back.

He never misses a training session despite having a really demanding job.

Leo was worried about having to commit himself to training sessions every week and wasn't sure he'd be able to fit training into his busy schedule.

But after experiencing a lot of stress at work, he came along to us and gave it a go.

Now he says he feels much more relaxed and he looks forward to his weekly run.

Mark is quite typical of our new members.

He's never considered himself to be a sporty person and it was only when he retired that he decided to take up the challenge of trying to run 5 km.

It took him months to find the courage to contact us but felt reassured immediately as there were other people his age who were only just taking up running for the first time.

My own journey hasn't been easy.

I did my first marathon when I was 37 after having had two kids.

My husband had been running marathons for years, but I never dreamed I'd be doing one with him.

I managed to complete it in 4 hours, but I felt like giving up halfway through.

It was only the support of the spectators that kept me going.

I do think signing up for a race of whatever length is motivating whether it's 5k or 25k because it's good to have something to work towards and it gives you a sense of achievement.

I did my first 10k after only 6 months which was certainly very challenging and not something I'd necessarily recommend but after you've been training for a few weeks it's worth putting your aim down for a 5k.

Some people find they only need a few practice runs before taking part in a race, but I'd give yourself a couple of months at least.

Well, I hope that's given you a really good idea of So, Jane, you'll be off to Denmark soon to do your work placement.

Yes, I'm really looking forward to it.

And I've just started packing up all my books to put in storage.

Well, I hope they don't get spoiled.

It's okay.

My grandfather works in a bookshop and he told me how to pack them.

Oh, that's helpful.

He says you have to support the spine otherwise the paper can come away from the cover.

Yeah, that's obvious.

He also told me to pack them flat in the box, not on their side again because they can bend and if you leave them like that for say a year, it's quite hard to get them back to their normal shape.

Well, it's pretty clear that ruins them.

But a lot of people just can't be bothered to protect their books.

He always says it's such a shame that publishers don't use better quality paper.

It's the acid in the paper that causes the problem, isn't it?

Yeah.

That's why old books go yellow.

You know, some of the books my grandfather's given me are like that already.

I should dump them really if they're going to deteriorate further.

But I'd feel bad.

They'll always remind me of him.

He's quite a collector, you know.

Well, if they're important to you.
Yeah, I'd regret just throwing them away.
You know, maybe it's because I was taught to treasure books, but I hate seeing students force open the pages of paperbacks.
They press so hard they end up breaking the spine.
I know, but unfortunately, paperbacks aren't designed to last a long time, and people know that.
Hardbacks aren't quite as weak.
Yeah, they're different, I suppose.
But I still don't think people value hardbacks like they used to.
Well, they aren't decorative, are they?
Like other objects.
Plus, nowadays, people don't keep them out on shelves as much as they used to.
That's such a pity.
When I visit someone, if they have, say, a colorful book on a table, it's the first thing I'm drawn to.
I agree.
And book covers can be a work of art in themselves.
Some are really eye-catching.
I've always been taught to handle books carefully.
If you watch someone take a book off a shelf, well, they usually do it wrong.
My grandfather says you should put your hand right over the top of the book.
Or if you can't do that, pull the other books on the shelf aside so that you can hold the whole cover.
When did you learn all this?
He watched me pull a heavy book off the shelf when I was small and it fell on the floor and broke apart.
Oh dear.
I can still remember it.
You know what I really like?
What?
The smell of new books.
Me too.
My parents used to laugh at me when I was a kid because I loved putting books up to my nose almost as much as reading New books
aren't cheap though, are they?
I guess we're lucky we can buy them.
My grandfather stocks secondhand books as well as new ones, and they don't smell quite as good.
I'd love to have a bookshop like your grandfather.
What's it like?
Well, it's quite big.
It's got two floors and an attic.
And he stocks all kinds of books, really.
I guess he treasures things like first editions and other rare books.
Yeah, you might think he'd keep those in the attic or somewhere.
So they'd be hidden.
Yeah.
But he likes people to know that he has them.
So he puts them out in the shop, but makes sure you need a ladder to get them.
Right.
That would prevent any thefts.
Does he stock books for children?
He does.
He particularly likes to encourage kids to read.
He always says that he used to sit under the stairs as a child with a pile of books and read them all.
Is that where he keeps them then?
Not exactly.
He's got a dedicated area on the ground floor with cushions so that parents can enter with their toddlers, go there and spend some time
reading to Oh, cool.
And then there's a place for push chairs by the front door and a cafe if anyone needs refreshments.
That's good to know.
As I said, it's a big shop.
And there's a storage area out the back as well.
Oh, what does he keep there?
books he wants to throw away.
He hardly ever throws anything away.
He just leaves unwanted books by the front door for customers to take.
Well, that's very nice.
Yeah.
And books people or institutions have requested, they all go at the far end.
he thinks it's best to keep these out of the main shopping area as they're boxed and new.
Did you get your course books from him?
Naturally, he stocks books for a lot of the colleges.
He used to keep these books on the first floor, but now there's a new university in my hometown.
He's moved them downstairs to attract the students.
They're actually part of the coffee shop on low shelves all around it.

Pretty central, then.

You'll have to take me there sometime.

Tree planting now dominates political and popular agendas and is often presented as an easy answer to the climate crisis as well as a way for business corporations to offset their carbon emissions.

But unfortunately, tree planting isn't as straightforward as some people think.

When the wrong trees are planted in the wrong place, it can do considerably more damage than good, failing to help either people or the environment.

Reforestation projects are currently being undertaken on a huge scale in many countries, and it's crucial that the right trees are selected. A mix of species should always be planted, typical of the local natural forest ecosystem and including rare and endangered species in order to create a rich ecosystem.

It's important to avoid non-native species that could become invasive.

Invasive species are a significant contributor to the current global biodiversity crisis and are often in competition with native species and may threaten their long-term survival.

Restoring biodiversity that will maximize carbon capture is key when reforesting an area, but ideally any reforestation project should have several goals.

These could include selecting trees that can contribute to wildlife conservation, improve the availability of food for the local community, and maintain the stability of soil systems.

Meeting as many of these goals as possible whilst doing no harm to local communities, native ecosystems, and vulnerable species is the sign of a highly successful tree planting scheme.

To ensure the survival and resilience of a planted forest, it's vital to use tree seeds with appropriate levels of genetic diversity.

The amount of genetic variation found within a species essential for their survival.

Using seeds with low genetic diversity generally lowers the resilience of restored forests, which can make them vulnerable to disease and unable to adapt to climate change.

Choosing the right location for reforestation projects is as important as choosing the right trees.

Ultimately, the best area for planting trees would be informally forested areas that are in poor condition.

It's better to avoid non-forested landscapes such as natural grasslands, savas, or wetlands as these ecosystems already contribute greatly to capturing carbon.

It would also be advantageous to choose an area where trees could provide other benefits such as recreational spaces.

Reforesting areas which are currently exploited for agriculture should be avoided as this often leads to other areas being deforested.

Large-scale reforestation projects require careful planning.

Making the right decisions about where to plant trees depends on having the right information.

Having detailed and up-to-date maps, identifying high priority areas for intervention is essential.

Drone technology is a useful tool in helping to prioritize and monitor areas of degraded forest for restoration.

In Brazil, it's being used to identify and quantify how parts of the Amazon are being devastated by human activities, such as rearing cattle and illegal logging.

A good example of where the right trees were picked to achieve a restored forest is in Lampang Province in northern Thailand.

A previously forested site which had been degraded through mining was reforested by a cement company together with Chiang Mai University.

After spreading 60 cm of top soil, they planted 14 different native tree species which included several species of fig.

Figs are a keystone species because of the critical role they play in maintaining wildlife populations.

They are central to tropical reforestation projects as they accelerate the speed of the recovery process by attracting animals and birds which act as natural seed dispersers.

This helps to promote diversity through the healthy regrowth of a wide range of plant species.

Unlike the majority of fruit trees, figs bear fruit all year round, providing a reliable food source for many species.

At this site, for example, after only three rainy seasons, monkeys started visiting to eat the fig fruits, naturally dispersing seeds through defecation.

Reforestation projects should always aim to make sure that local communities are consulted and involved in the decision-making process.

The restoration of mangrove forests in Madagascar is an example of a project which has succeeded in creating real benefits for the community.

Destruction of the mangrove forests had a terrible impact on plant and animal life and also badly affected the fishing industry which was a major source of employment for local people living in coastal areas.

The reforestation project involved hiring local people to plant and care for the new mangrove trees.

Millions of mangrove trees have now been planted which has resulted in the return of a healthy aquatic ecosystem.

The mangroves also act as a defense against the increased threat of flooding caused by climate change.

What's more, the local economy is more stable and thousands more Madagascans are now able to send their children to school.

One other important point to consider

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I've been meaning to ask you for some advice about restaurants.

I need to book somewhere to celebrate my sister's 30th birthday, and I liked the sound of that place you went to for your mom's 50th.

The junction?

Yeah, I definitely recommend that for a special occasion.

We had a great time there.

Everyone really enjoyed it.

Where is it again?

I can't remember.

It's on Grayson Street, only about a 2minute walk from the station.

Oh, that's good.

I'd prefer not to have to drive anywhere, but I don't want to have to walk too far either.

Yes, the location's perfect, but that's not necessarily why I'd recommend it.

The food's amazing.

If you like fish, it's probably the best restaurant in town for that.

It's always really fresh, and there are lots of interesting dishes to choose from.

But all the food is good there.

Is it really expensive?

It's certainly not cheap, but for a special occasion, I think it's fine.

It's got a great atmosphere, and before dinner, you can go up on the roof and have a drink.

It's really nice up there, but you need to book.

It's very popular as the views are spectacular.

Sounds good.

So, that's definitely a possibility then.

Is there anywhere else you can think of?

Um, if you want somewhere a bit less formal, then you could try Paloma.

Where's that?

I haven't heard of it.

No, it's quite new.

It's only been open a few months, but it's got a great reputation already.

It's in a really beautiful old building on Bow Street.

Oh, I think I know where you mean.

Right beside the cinema.

Yes, that's it.

I've only been there a couple of times, but I was really impressed.

The chef used to work at Don Filipes, apparently.

I was really sorry when that closed down.

So, is all the food they serve Spanish, then?

Yeah, you can get lots of small dishes to share, which always works really well if you're in a group.

M worth thinking about.

Yeah, there's a lively atmosphere and the waiters are really friendly.

The only thing is that you need to pay a £50 deposit to book a table.

A lot of restaurants are doing that these days.

I should have a look at the menu to check there's a good choice of vegetarian dishes.

A couple of my friends have stopped eating meat.

Not sure.

I'd say the selection of those would be quite limited.

I've just thought of another idea.

Have you been to the orderly?

No, don't think I've heard of it.

How's it spelled?

A U D L E Y.

You must have heard of it.

There's been a lot about it in the press.

I don't tend to pay much attention to that kind of thing.

So, where is it exactly?

It's in that hotel near Baxter Bridge on the top floor.

Oh, the views would be incredible from up there.

Yeah, I'd love to go.

I can't think of the chef's name, but she was a judge on that TV cookery show recently, and she's written a couple of cookery books.

Oh, Angela Fra.

That's the one.

Anyway, it's had excellent reviews from all the newspapers.

That would be a memorable place for a celebration.

Definitely.

Obviously, it's worth going there just for the view, but the food is supposed to be really special.

She only likes cooking with local products, doesn't she?

Yes.

Everything at the restaurant has to be sourced within a short distance and absolutely nothing flown in from abroad.

I imagine it's really expensive, though.
Well, you could go for the set lunch.
That's quite reasonable for a top class restaurant.
£30 ahead.
In the evening, I think it would be more like £50.
At least that, I should think.
But I'm sure everyone would enjoy it.
It's not the kind of place you leave feeling hungry though, is it?
With tiny portions.
No, the reviews I've read didn't mention that.
I imagine they'd be average.
Well, that's all great.
Thanks.
So, hello and welcome.
My name's Heather McCallum and I'm one of the potters who work here at Adelman Pottery.
Before we go into the workshop, I just want to say a bit about the craft of pottery.
Then we'll have a look at the equipment and you can try making a pot of your own.
Like many people, I'm sure you know that pottery as an art form is tens of thousands of years old.
And we know this because it stands the test of time.
Things like baskets and pictures don't survive in the earth in the same way that pots do.
And even if ancient pots are found in small pieces, they still provide a lot of information about the past.
There's no doubt that pottery has given archaeologists a fascinating insight into how ancient humans lived.
The shape of an artifact may have been lost.
But archaeologists can tell whether the pots were for, say, storage or cooking by examining the impressions on the clay, the scratches from tools.
And the clay itself can reveal where the pots came from.
When I ask people why they want to take a pottery class with me, they sometimes talk about these things.
Like our ancestors, they hope that something they create will also last longer than they do.
that their work, whether it is good or not, might say something about humanity many years after their death.
Of course, you will all have your own reasons for coming here.
As far as I'm concerned, what I love most is the concentration you need to make a good pot.
That focus takes you away from the stresses of everyday life.
if you're elderly.
It's also good exercise for hands and wrists and helps with arthritis.
And of course, it's a fun activity for children because it's so messy.
Here at Adelman Pottery, we show you some of the basic pottery techniques so that you can use these to create whatever you wish.
A gift for a friend perhaps.
Like nearly everyone who comes here, I'm sure this is the first time you will have tried the art.
So, we'll keep things simple today.
Now, before we move on, can I just say a word about what you're wearing?
As we said in our email, please remove any watches, necklaces, etc., and put them somewhere safe.
If you have long hair, do tie it back.
Now, we'll provide aprons later, but I trust your clothes are old, but comfortable.
Not your favorite t-shirt or jeans.
So, now we're in the workshop.
Have a look around.
There's a lot going on.
To make pottery that will last, you need a potter's wheel, a kiln, which is basically a very hot oven where you fire the pottery, and some tools.
So, first, the kil.
If you look over in the far corner, you'll see one of ours.
Since their invention, kilns have changed very little.
Though in the past 20 years, a lot of progress has been made in temperature control.
Basically, a kiln removes the water from clay at temperatures of around 1,000° C.
This allows anything you've made to set permanently in shape.
It's a pretty ugly, heavy object that's hard to keep in a house or flat.
So, most people don't have one.
You may think, "Can't I use my oven?" Well, that's possible, but domestic ovens don't really get hot enough, and eventually the clay will crack and fall apart.
Some people fire pottery in a fire pit outside, but bear in mind that can be dangerous.
You also need to know about safety procedures for kilns as they release toxic compounds into the air.
Every potter needs a potter's wheel.
This machine is used to shape the clay into an object with circular walls or sides such as a bowl.
Its invention revolutionized the pottery industry, allowing multiple items to be produced in a day.
Lastly, there are a number of different tools that potters use depending on what they want to make.
When you start, your hands can make all kinds of shapes and curves without relying on a sculpting tool.
However, there are some basic tools that you will need to handle the clay on the wheel.
Some look very strange and have even odd names that you may find hard to remember.
Rather than go through them all now, I'll just name a few tools as we go along.
We can provide these and I wouldn't recommend spending money on them yet.

So, let's try making a pot of your own.
If you sit down, shall we go through the notes we've made from our research into loneliness now, Dev?
Okay, Samara, it's been a real eye opener.
I had no idea that loneliness has been increasing steadily for the last 20 years.
I know.
And it's the same all over the world.
the downside of a modern lifestyle, I guess.
Did you come to any conclusions about what the reasons for the increase are?
Well, I'd assumed it was mainly an issue for the elderly, but in fact, it's something which affects young people just as much.
So, nothing really to do with longer lifespans.
Um, what about social media?
In my case, far from making me feel isolated, it actually does the opposite.
It definitely does more good than harm.
I'd say loneliness has a lot to do with the way cities are designed.
People living in high-rise flats with not much opportunity to speak to their neighbors.
I think you're right.
Another possible reason is that people are having fewer children and don't live in large extended family groups.
But in this country anyway, that all changed decades ago.
And yet loneliness is a more recent problem.
I suppose so.
A more plausible explanation is that people are having to move around for work and often end up living miles away from their family and friends.
That's true.
Looking at the studies on health risks and loneliness, there are claims that loneliness has as much impact as smoking 15 cigarettes a day, or similar to the risks caused by obesity.
But I'm not sure there's enough evidence for some of these claims.
Well, what about that one in Finland which showed that loneliness increased the risk of cancer by about 10%.
And those findings have been supported by other studies, too.
You're right about that one.
I was actually thinking of the studies on dementia.
Some found no association between loneliness and dementia, and others found the opposite.
Not exactly reliable, then.
There's been a lot of research on cardiovascular disease and whether loneliness contributes to that.
Yes, I read that it was hard to reach a judgment as the definition of loneliness varied quite a lot and the responses from participants were too subjective.
But there's no doubt that loneliness contributes to a weakened immune system.
Unquestionably, the data on that is sound.
What did you think about the evolutionary theory of loneliness?
Well, I thought the idea that loneliness evolved because it motivated people to be with other people quite convincing.
Survival often depended on group cooperation.
But I don't think there's enough evidence to claim that there must be a group of neurons in our brains which influence social behavior by making us feel bad when we're alone.
There are a few studies which support the theory but not conclusively enough.
More evidence is needed.
And anyway, this theory is not really useful when it comes to solving the problem of loneliness today.
True.
Should we look at the relationship between loneliness and mental health now?
Okay.
So loneliness and depression are clearly related and that's been recognized by various governments around the world.
But unlike depression, loneliness has no recognized clinical form.
There's no available diagnosis or effective treatment and that's not likely to change.
I don't think so either.
I was thinking we should start our presentation with an example from our own experience.
I'd like to talk about how lonely I was when I started university, being away from home for the first time and all that.
Good idea.
Everyone will be able to relate to that.
Although a lot of students were probably too embarrassed to admit to it.
Yeah, we could discuss ways of dealing with loneliness as well, like just talking to strangers.
Loads of studies have shown that interactions with shop assistants and bar staff make people feel more optimistic and relaxed.
I don't know about that, but it must make people feel more connected with their community.
True.
Although you need to be a certain kind of person to be able to just strike up a conversation.
Good point.
We should say something about solitude and how being alone and being lonely aren't the same thing.
It's strange the way some people can't stand being by themselves while others love it.
Yeah, the research shows a certain amount of solitude is beneficial for well-being, which I appreciate, but being alone isn't something I actually like.
I'd never choose to go on holiday alone, for example.
Me neither.
Well, let's not.

It's quite hard to think of a city that doesn't have a big river running through it. If you think about the major cities in the world, Shanghai, New York, Mumbai, London, they're nearly all built on rivers. When these cities were established hundreds or even thousands of years ago, the rivers were a big part of people's lives. In 16th century London, the quickest way to get from one part of the city to another was by river. But people also used the river for fishing, as the water then was relatively clean. And they would also go on boat trips up and down the river just for pleasure. As a relaxing escape from the noise and bustle of the city streets. But as industries developed and populations increased, city rivers suffered. The rising number of people meant there was a huge increase in the amount of sewage discharged into the rivers. Rivers had always been used for this purpose, but when the number of inhabitants was so small, that wasn't such a problem. However, as cities grew to over a million inhabitants, the impact on the rivers became more serious. In addition, other types of pollution increased as factories were built beside the river and discharged their waste materials into the water. This got worse over time. As recently as 1957, scientists at London's Natural History Museum declared that the river Tempames was dead in biological terms as the water was too filthy to support any kind of life. But in recent years, as rivers lost their industrial function, cities have begun to recognize their true value and to take steps to clean them up. For example, the river tempames is now cleaner than it's been for 150 years. These days, you can see seals swimming in the water. And recently, people had to try to rescue a whale which had got lost and swam up the river from the sea by mistake. Unfortunately, they didn't succeed. But the problem was disorientation rather than the quality of the water. Then all around the world, riverside areas are now seen as prime sites for development. Warehouses that were once used for storing goods are now being converted into expensive restaurants and also into apartments with river views which are in great demand and sell for astronomical prices. In Los Angeles on the west coast of the USA, an architect has plans to revitalize the banks of the river and to make a park there which can provide facilities for sports as well as a natural environment for relaxing in. It's also hoped that the riverside can be used for other purposes. It's been proposed that facilities could be provided for displaying projects related to various kinds of art that have been produced by local people. For example, in the city of Paris, during the summer months of July and August, all the traffic is banned from the roads by the sides of the river, and the banks are transformed into beaches where people can relax in deck chairs under potted palm trees, sunbathe, or buy a drink or a snack while enjoying the view. But to make the most of our rivers in our increasingly crowded cities, we need to allow them to regain their original purpose and be used as a means of transport. Reclaiming our streets from cars and lorries. To do this, we'll have to shift more traffic back to the river, but this time cleanly and silently, making the most of modern technology. Already more than 2 billion passengers use the ferry to travel in cities around the world like Istanbul, San Francisco, and New York. And these numbers are set to rise further. Admittedly, it's not a fast way of traveling, but neither is a car when it's stuck in traffic. Of course, passenger traffic on roads might decrease as more people start working from home. But another recent development, the huge rise in online shopping has meant that another form of urban traffic just keeps on growing and that's deliveries. Trucks and vans in the city pollute and double park while dropping off parcels. Imagine using the immense capacity of shipping to take these trucks off the road. One freight barge can replace 44 large trucks, uses far less energy, and causes less pollution. When the barge docks at the riverside, the parcels could be taken the last few kilometers to their final destination on cargo bikes, electric ones. Of course, this is already happening in the Dutch city of Amsterdam. And in future, the final stage could even be carried out by drone, although at present this isn't allowed. Wouldn't it be great to unblock our city centers in this way? Looking further ahead, good morning everyone. I'm Steve Wright from Elmley Town Council and I organize the town's volunteer scheme. I'm delighted you're all interested in joining the scheme. Uh, our volunteers help to create a sense of community among the many people who live in our historic town of Elmley and make residents and visitors feel welcome at local events. Uh, first I'll mention just a few of the activities that volunteers carry out. One is to walk around the town center streets wearing our volunteer t-shirt. Tourists often ask how to get to a particular shop and they might also be grateful for recommendations about what to visit. The town holds a large number of concerts each year and part of the volunteers role is to get everyone in the audience to the right place as smoothly as possible. You'd be surprised how many people buy tickets then don't check them and head for the wrong section of the hall. Volunteers may get involved with community groups such as sports clubs or gardeners associations. Here the volunteers talk about how groups can help each other. For instance, a writing group might want to travel to another town to hear a talk by a well-known author, but may not know that another club has a coach they could travel in. The town produces a monthly magazine and anyone who lives in the town can send in articles. It's free to residents and is paid for by local businesses. That's the responsibility of the council's advertising department. We depend on volunteers though to find out what people think of events they've attended and any suggestions they have for the future.

The volunteers then send a summary to the editors.
There are a number of clubs for retired people and every year the council arranges lunch for all the members.
The volunteers welcome the guests and when everyone's sitting down and relaxing after the meal, some volunteers put on a show, usually around half an hour of songs and short plays.
The club members really welcome the chance to chat to the volunteers at these events.
The town council has a website of course and volunteers are asked to help by making sure residents know about it.
It's updated every day with information about future activities and we want as many people as possible to use it.
Okay, now you know some of the things our volunteers do.
I'll go on to some practical matters.
As you probably know, the town arranges three major festivals every year, and they all depend on a large number of volunteers.
The book festival lasts three days and uses several venues, which all need volunteers.
More are needed for the music festival because that lasts a whole week, and even more help is required for the science festival, even though it's only 2 days long.
It involves quite a lot of venues, though.
It's a good idea to help at the festival you're most interested in because you can attend most of the events for free.
We try to use volunteers who are flexible though because some festival events are held outdoors and the weather may affect the size of the audience and even whether the event can take place.
So there can be changes at short notice.
What's essential though is being able to get on well with other people and also to deal with someone who's behaving badly as occasionally happens.
Our plan is to get your work in in September after a week's training starting on the 2nd.
So, we'll be timetabling you for duties the following week from the 9th onward.
Later in the week beginning September 23rd, we have a chat with each of you to find out how you feel about being a volunteer and what extra support you need.
As a thank you to the volunteers, we arranged an annual event.
In recent years, we've had a party in the town hall and last year a barbecue in Chamber Park.
Our fourthcoming event is a trip along the canal from here to Duehurst and back.
It's on Saturday, September the 28th.
And if you'd like to attend, you can sign up once you start work.
Now, this is the volunteers t-shirt.
Colin, I'm really struggling to think of a topic for our human geography assignment.
Me too, Rosie.
I'll tell you what, let's think about the different aspects of human geography and see if we can narrow the topic down a bit to help us decide.
Okay.
So, one aspect is population.
That would be all about population density and migration and so on.
Lots of facts and statistics.
Maybe a bit boring.
Yeah, but quite straightforward to find on the internet.
Suppose so.
How about health?
I'd never thought about the links between that and geography until Professor Lee gave us that lecture on cholera.
How in the 19th century a physician used street plans and plans of water supplies to find the source of a colera epidemic.
Yes.
Fascinating, wasn't it?
Or we could do something more general like economies.
So how financial and commercial factors are linked to the physical environment.
Yeah.
I thought that had been taken off the syllabus for this year.
Has it?
I'm not sure, but it might be best to avoid it.
Okay.
Maybe we could do something on culture.
We had that lecture about culture and geography last week.
I didn't get much out of that.
It was also general and the lecturer didn't give any useful examples.
Yeah, I hardly took any notes.
It didn't seem worth it.
We could focus on poverty.
That's something that's a global problem.
The trouble is, Dr.
Lee was saying that you have to be careful with some of the figures relating to poverty.
They're sometimes deliberately manipulated.
You mean the information gets changed for political reasons?
That sort of thing.
Yes.
So, what are we going to do our assignment on?
I'll tell you what, one of the possibilities we haven't discussed is urbanization.
And now over half the world's population lives in cities.

That's really important.
Okay, good idea, Rosie.
Let's do that.
I love living in a big city, but of course, there are problems.
Things like theft and robbery.
Yes, but where I come from, that's linked to another more serious issue, which is that a lot of people don't have jobs.
That's getting worse and worse.
And we also still have quite a lot of people who are homeless, though that's not quite so bad as it was.
It would be nice to talk about some positive developments, like some of the new developments on the outskirts of cities.
Yeah, they've opened some massive new shopping centers outside my city.
Yeah, the same with mine.
But it's meant a lot of the shops in the city center are closing down.
But the outskirts are ideal for buildings that need a lot of space, like for conferences.
They've opened a couple of big ones.
Yes, we've got some, too.
It'd be nicer to have more facilities for things like football, too.
But that's not happening where I live.
Same in my area.
We could include something about developing disused industrial sites.
It seems like a good idea because you're not doing any harm to the natural environment.
Yeah, but aren't the buildings architecturally significant?
Not really.
And what people forget is that they often used quite dangerous materials, chemicals, and things which haven't been properly cleared away.
So, the whole site has to be made safe.
That can't be cheap.
And I bet it's often not budgeted for.
You're right.
Have you read about Mazdar City in Abu Dhabi?
Yes.
It was designed to be a green city, wasn't it?
That might be a good example of a city which set out to depend entirely on renewable energy.
Yes, we should say something about that.
It was designed to be totally pedestrianized, too, wasn't it?
With the transport underground.
Yes.
And they had big plans for recycling to reduce waste to the lowest possible level.
But let's stick to talking about power sources.
Then there's that eco town in England, Green Hill Abbots.
It set out to conform to the usual principles, sustainability and so on.
A lot of people were against it at first.
They said the plans were unrealistic.
I'm not sure how far they've got with it.
Um, I'll check.
Right.
So it looks as if we have a sort of plan and we can work from there are trends in food much as there are trends in clothing.
Interest in food fashions has risen rapidly since the birth of the smartphone when people first began taking photos of their food and instantly sharing them with their friends.
The food industry in the UK in particular is obsessed with finding and exploiting the next big food trend.
Marketeteers aim to create a huge demand for a food item which was previously unknown or not needed.
One of the most effective ways of promoting a new food product is by using social media influencers as brand ambassadors.
In return for free samples, many influencers will post content about a product.
Although there are influencers with hundreds of thousands of followers who can command large fees for their services.
Companies which sell vegan produce were pioneers in being able to increase sales really quickly in this way.
For a food item to become really popular, it has to be readily available.
So supermarkets have a role to play in creating a new food trend.
They have dedicated teams closely following which new products or ingredients are trending on social media and are particularly interested in what well-known chefs are putting on their menus.
I'd like to look at a few examples of marketing campaigns which were really successful in launching a new fashion trend.
Starting in the 1990s with the avocado, a British PR company was hired to raise its profile and stimulate demand.
They paid for a group of journalists to travel out to South Africa to meet avocado farmers.
Articles written following this visit helped to educate the British public about the avocado, which at this time was certainly not the daily staple it's since become.
Advertisements were designed to promote the avocado as a superfood rich in nutrients and therefore beneficial for health.
Avocados became hugely fashionable and within a few years UK avocado sales had grown from 13 million pounds annually to around 150 million making it one of the most successful fresh produce campaigns in UK history.
Oat milk is a recent example of a new product which became fashionable very quickly.
Now there are many brands available but one company which had early success was the Swedish brand Oatley.
They attracted a lot of attention with a media campaign which used provocation as a way of getting their message across effectively.
The fact that this campaign aggravated competitors producing milk from dairy cows was seen as a plus as it helped to make oat milk

seem cool.

In the USA, the brand decided against a big retail launch in favor of getting the product into coffee chains, which removed the need for a big advertising budget.

This proved far more effective than offering samples in supermarkets.

Oat milk had an advantage over other alternative milk products such as almond milk.

Many consumers prefer it because it has less of an impact on the It requires significantly less water to produce than other alternative milk products and it also has a relatively low carbon footprint.

Norwegian scray, a rarely available seasonal fish delicacy, otherwise known as Arctic cod, is now found on the menus of Michelin starred restaurants throughout Europe.

The demand for scray has been used by a food marketing agency to build the reputation of Norway's fisheries in general.

Marketing surveys have shown that a significant number of shoppers now associate Norway with excellent seafood.

Food trends can be considered a good thing in some ways as they can benefit farmers and food producers enormously.

The public can also be encouraged to buy things which are more sustainably produced.

But ethical concerns have been raised about the effects a surge in demand can cause.

Quinoa is a classic example.

This plant is native to Peru and when demand peaked some years ago, the price soared, making it unaffordable for local people.

While the popularity of quinoa has benefited farmers financially, there have been other negative consequences.

As demand grew, farmers began working the land all year round in order to produce more quinoa.

One issue has been that the fertility of the soil decreased dramatically which could potentially lead to desertification in some areas.

Another example would be the case of Good morning.

Hi, this is Shelley Meyer.

I'm renting your house on Archwood Avenue.

I'm due to move in next week.

Oh yes.

Hello, Miss Meyer.

What can I do for you?

When I viewed the house, I told you I'll most probably need to rent some furniture.

At least until I know whether my temporary work contract is going to be made permanent.

Yes, of course.

I remember.

And I said I could give you some information about furniture rental companies in the city.

That's right.

Well, the biggest company is called Peak Rentals.

I've recommended them to other people and I've always heard positive reports about them.

Could you give me an idea of their costs?

Sure.

I actually have one of their brochures here.

It says the monthly price per room starts at \$15 and goes up to \$239.

That depends on which rooms you need furniture for, of course.

Sure.

It's just to get a general idea of how much it's going to cost.

And you said you had some positive feedback about this company.

Yes, people have mentioned that the furniture from Peak Rentals is more modern than any of the other companies.

And also, once you place an order, the furniture will be delivered to you in just one or two days.

That would be really helpful.

Oh, and the brochure says that there's a special offer at the moment.

If you rent living room furniture, I believe that's a set of chairs and a TV table, you'll also get a lamp at no extra cost.

Okay, but you know that price range you gave is more than I was hoping to pay.

Then you could try Aaron and Oliver.

Sorry, what and Oliver?

Aaron.

A r o n.

Okay.

Are they cheaper?

I'd say they're a mid-pric company, but if you chose them, you need to be aware that they charge an extra 12% every month in case of damage.

Oh, I see.

I'd have to do the math carefully then, right?

But one helpful thing is that they also do cleaning for customers for the furniture.

For the house.

Oh, I see.

I probably won't need that.

There's another company called Larch Furniture.

It's quite new and it has the lowest prices in town.

That's for both furniture and also electronic equipment.

Well, that would be good.

I'm not bringing much with me and I won't have much time to go shopping after I start my job.

There are two things you need to know about large furniture.

First of all, you have to take out insurance on the furniture, and you need to organize that yourself.

That wouldn't be too hard.

Also, you can't take out a contract for less than 6 months, but I figure that might not be a problem for you.

You're renting the house for 12 months after all, aren't you?

Yes.

Okay.

Well, sorry to interrupt.

I just thought of another furniture rental company.

It's called Space Rentals and it's located very near to the house.

Okay.

I don't have any information about their charges, so it's best to use their app to find out what it would cost you to use them.

Okay, thanks.

I'll do that.

One good thing about that company is that if you don't like the furniture once it's delivered, you can request exchanges as long as you do that within a week of receiving it.

That sounds really great.

Okay.

Well, thanks very much.

That's so helpful.

Hello, I'm Hayden.

I'm one of the archaeologists investigating the site here at Bidcaster.

This is the third summer for this community project, and most of the people digging here are volunteers.

I'm a full-time archaeologist for the town council, but I was asked to join the project by NHA, a charity which sets up projects like this up and down the country.

As you can see, we're next to Bidcaster Castle, which is great because the owners let us use their facilities.

So, how did we get to where we are today?

Many archaeology projects happen when an ancient object is found.

And in our case, that object was a gold coin.

Coins are often found by people using metal detectors to look for things buried in the ground.

Or coins are uncovered when wild animals like rabbits have been digging tunnels.

Here, a walker found it on the ground after a rainstorm washed away some of the earth and sand.

When the story of the gold coin hit the news, Peter Swift, an amateur historian, contacted me to say he believed there had been a village on this site centuries before the castle was built.

Just by chance, the team found some old maps and documents in our library, which showed 500-year-old drawings of ruined buildings on the grassy area between the outer stone walls of the castle and the river.

We knew then we were on to something.

Over the three summers the team has been here, we found the remains of several buildings and more broken pots than you can count.

Normally, you'd expect to find brooches and other jewelry, but we're still waiting to uncover any such items.

The people who once lived here were skilled at making tools from animal bones, as you'll see when you visit the exhibition.

Besides the discovery of the village, we've also found evidence of human activity on the other side of the river.

No other houses or huts so far, but we can see the borders of an ancient field system.

At one point, we found a long wall and thought it was an ancient palace, but it turned out to be a modern wall.

This summer's work will end soon, but we'll be back next summer.

In the meantime, we're putting on a series of guided tours for school groups this autumn.

Oh, and um maybe you saw the TV documentary about our project that suggested the objects we found are going to the town's museum, but we don't know that for sure yet.

When you enter the site, please make sure you keep to the paths at all times.

There are a few other things um the highlights of the site if you like that I want to mention.

Take a look at the map.

Our present location is marked at the bottom.

This year, we've identified the foundations of an ancient bridge.

And it's really exciting today because a team of divers are in the river searching for lost objects.

To reach the bridge, take the main path ahead of you.

Go straight on and keep going till the path bends to the left.

You'll see a smaller track leading off to the right.

Follow that to take you to the river where the divers are.

You might be interested to see the rubbish pit.

This is very near the castle walls in the northwest corner of the site.

It actually dates to the time of the castle and not the ancient village.

We found oyster shells and fishbones, and we assumed they were thrown from the castle kitchen above.

One area we excavated in the first summer uncovered the sight of a meeting hall.

We knew it was an important building because it had two rows of post holes deep enough to support a large roof.

It's the largest structure in the central area of the site next to the current excavation area.

Last year, we discovered a fish pond in the ancient village.

Normally, these were beside a river.

The pond here is further away, but it's possible the river has moved slightly.

Anyway, to get there from here, you turn right at the first information board you come to and follow the path into the trees.

Before you come out of the trees, you'll see it on your right.

If you reach the river, you've gone too far.

So, does anyone have So, Finn, I've done as much as I can for our project on theater programs.

How's your research coming along?

Okay, Maya.

I didn't know theater programs are called play bills in the USA till I started looking into the topic.

Even though I struggled to find many useful websites, I'm glad we picked this subject.
No one else on the course is doing the same as us, although it is one of the research areas of the module convenor.
That might actually put some people off, I suppose.
though.
Anyway, I hadn't realized there are actually companies specializing in creating theater programs.
Yes, they're quite common nowadays.
Contrary to what many people think, theaters don't hire people to do the programs.
In fact, companies buy the rights to publish programs on the theater's behalf and then make their money selling advertising space within the program booklet.
It must be easier for theaters to do it that way.
Yes, I remember reading something about programs in early British theater.
It said that the cast was always very important.
Yeah, audiences were very familiar with leading actors and big names would draw huge crowds.
But I hadn't realized that if the program named a famous actor, that's who the public expected to perform.
And if that didn't happen, people accused the theater of breaking their agreement with the audience.
They would demand refunds and if they didn't get them, there were riots.
Outrageous.
That would never happen now.
No, people are too polite.
Even when they're disappointed if the star of the show misses a performance, we should definitely include that information about early audiences in our project.
I also think it's important to mention that lots of ordinary people at that time were illiterate.
So theater programs were of limited value in advertising plays.
When a company of actors arrived in a town, they'd parade around the streets in their costumes, beating drums and announcing their upcoming performances.
Interesting.
I couldn't imagine that happening now either.
There's also an interesting comparison to make between 18th and 19th century programs.
Wasn't it in the 19th century that theater programs started to resemble programs today?
Yes.
And unlike programs from the 18th century, they always used color and there was a greater variety of designs.
But personally, I think 18th century programs were superior because they told the theatergoers so many things, including about the actors and about the writer, the plot, and sometimes the history of the play.
That's right.
What should we say about theater programs in the 20th century?
I reckon the most important thing is the dramatic change they underwent during World War II when the government imposed restrictions on the use of paper.
Yeah, but that was only in the UK.
In the USA, programs, or rather play bills, continued to be published in the same format, while here in the UK, programs became merely a single sheet of paper folded to create four pages for text.
H what I don't really get is that after the war they didn't go back to being more than one sheet or change in any way for over 25 years.
I know there were paper shortages after the war but only for 5 or 10 years.
H strange.
I've got some pictures of programs we could include on the slides for our presentation.
I found a couple too, Maya.
Let's go through and see what we think.
Um, oh, this is an old one for a play called Ruy Blass.
H, never heard of that, but the program looks very decorative.
Good enough to put in a frame on the wall.
The images are just beautiful.
Finn, what did you find?
I've got some pages from a program for Man of Lancha.
I thought this was a good program to show, not because of the pictures, but because it contains articles written by members of the theater company, so we can learn how the production was created and the thoughts and feelings of the cast.
Good.
I've got a copy of a program that's now in a museum.
It's for the tragedy of Jane Shaw, and it's said to be the earliest surviving document to have been printed on Australia's first printing press.
Fantastic.
Another program to talk about is for the Sailors Festival.
It comes from the British Library's digitized collection of programs that was started a few years ago.
It already comprises over 200,000 programs, which is amazing.
Huh.
Wish I'd known about it while I was doing my research.
It's only relatively recently that designers have become aware of the need to be inclusive when designing products.
But what does that mean exactly?
Well, it simply means designing products that span economic, social, and cultural barriers.
It means making sure products are accessible so that as many different types of people as possible can use them without any type of adaptation having to be made to the original design.
Inclusive design is often linked with universal design, although they are not quite the same thing.

Universal design aims to make products that work for everyone and that includes considering the needs of people who have cognitive difficulties which can present quite a challenge.

Today, examples of successful inclusive design can be seen all around us.

In workplaces, it is common to see desks which can be adjusted to suit people of different heights or for wheelchair users.

This still isn't always the case, however, and is one reason why office workers often suffer from back or neck problems.

You'll find another example in the public toilets of countless hotels, airports, and offices.

Taps that you activate by sensor require no pressing or twisting movements.

These are not only more hygienic, they're also easier for people with dexterity or mobility issues.

The tech industry has been criticized in the past for focusing too much on young consumers, but this is changing.

Many products are now designed with the elderly in mind.

For example, it's well known that vision declines with age and that we also become worse at distinguishing between similar colors, in particular shades of blue, which is why software designers rarely create interfaces with this color.

Motor skills also decline with age and some people have difficulty doing everyday things like picking up a cup or opening a door.

This can also affect their ability to use a mouse or keyboard.

So, voice access is now a routine way of making commands.

It's worth looking at the problems non-inclusive designs cause when not enough consideration is given to a range of users as it can have a serious impact on people's lives.

Access is one obvious example because it has such a huge impact on disabled people's independence.

Not being able to access public transport because buses or trains are not wheelchair friendly means many disabled people can't go out unless someone goes with them.

Safety is another issue.

Inexplicably, most cars are still crash tested using a dummy based on an averagesized male.

This has safety implications for all women, particularly those who are pregnant, as the seat belts worn by the dummy are not adapted to accommodate them.

Over the past 100 years, workplaces in the UK have on the whole become considerably safer.

Employers are legally required to provide well-maintained personal protective equipment or PPE.

Anything from goggles to full bodysuits to workers who need it free of charge.

But most PPE is designed to fit men.

A recent report found that employers often think that when it comes to female workers, all they need to do to comply with this legal requirement is to buy jackets.

for example, designed for a small man.

The problem with this is that women can be tall and still have much smaller shoulders than the average man.

Ill-fitting PPE such as high viz jackets, vests, and body armor can put women at risk.

The report found that 95% of women said that their PPE often hampered their work and that this problem was worst in the emergency services, particularly the police.

Another problem is related to comfort at work.

A very common scenario in offices in summertime is to see women wrapped in blankets or wearing sweaters while the air conditioning is on high.

Meanwhile, the men are in shorts and t-shirts.

This is due to differences in metabolic rates for men and women.

There is a standard setting for air conditioning to be at a temperature of 21° designed to suit men.

And in most modern offices, it's not possible to turn the air conditioning up or down.

This means that many offices which mainly employ women are wasting energy by having the air conditioning set too high.

As you can see from the examples I've just mentioned, there are serious consequences for designs which don't consider the needs of all users.